Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Full book compiled from Chapters 1-20

With a table of contents before Chapter 1

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Machine Learning and AI for Environmental Economics and Choice Modelling

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Chapter 1

From Prediction to Policy: Why Machine Learning and Choice Modelling Need Each Other

This chapter introduces the book's central distinction: machine learning is often optimized for prediction, while environmental economics and choice modelling often require interpretation, causal reasoning, welfare measurement, and counterfactual policy analysis. The chapter uses the uploaded reference papers as a core base and supplements them with selected peer-reviewed sources, major textbooks, and official software documentation.

Prepared for self-study and research training

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How to study this chapter

This book uses an evidence-informed self-study design. Each chapter contains learning objectives, short explanations, worked examples, coding labs, common mistakes, retrieval questions, and transfer exercises. The design is based on three principles.

First, do not only reread. After each major section, close the text and answer the checkpoint questions. Retrieval practice and distributed practice have stronger evidence than passive rereading or highlighting for durable learning (Dunlosky et al., 2013).

Second, learn by comparing models. Many confusions in this field disappear when the same data are analysed with a transparent baseline model, a flexible predictive model, and a hybrid model.

Third, move from passive to active, constructive, and interactive engagement. The ICAP framework argues that deeper engagement generally produces stronger learning outcomes than passive exposure alone (Chi and Wylie, 2014). In this book, "active" means running code, "constructive" means modifying assumptions and explaining results, and "interactive" means discussing diagnostics with peers, supervisors, or future users of the model.

Before you start

You do not need a previous background in AI or machine learning. You should be comfortable with basic regression, probability, and the idea of a likelihood function. If the terms utility, likelihood, cross-validation, and marginal willingness to pay are unfamiliar, that is fine. They are introduced gradually.

1 Learning objectives

By the end of this chapter, you should be able to:

1. Distinguish prediction, explanation, causal inference, structural modelling, and welfare analysis.
2. Explain why machine learning is useful in environmental economics and choice modelling.
3. Explain why high predictive accuracy is not sufficient for welfare analysis or policy counterfactuals.
4. Write the basic random utility model and the multinomial logit probability.
5. Understand the first practical difference between training a classifier and estimating a choice model.
6. Run a minimal R workflow that simulates an environmental choice experiment, estimates a multinomial logit model, and compares it to a predictive machine learning classifier.

2 The map of the territory

Economists, data scientists, and choice modellers often use the word "model" to mean different things. In this book, a model may be used for at least five distinct purposes:

Purpose	**Typical question**	**Typical success criterion**
Prediction	What will happen for new observations drawn from a similar environment?	Low out-of-sample prediction error, good probability calibration, stable performance across relevant subgroups.
Explanation	Which variables are associated with observed behaviour, and in what direction?	Interpretable patterns that are stable, plausible, and consistent with domain knowledge.
Causal inference	What is the effect of changing a policy, treatment, price, attribute, or exposure?	Credible identification assumptions, valid uncertainty quantification, robustness to confounding.
Structural modelling	What behavioural mechanism generated the choices?	Parameters with behavioural meaning, identification, interpretable heterogeneity, credible counterfactual structure.
Welfare analysis	What are the monetary or welfare consequences of a change in attributes or policy?	Valid marginal rates of substitution, willingness-to-pay estimates, compensating variation, and policy-relevant uncertainty.

Machine learning is extremely powerful for prediction and representation learning. It is also increasingly useful for causal inference when combined with credible identification strategies. However, environmental economics and choice modelling often require more than prediction. They often require answering questions such as:

- How much are households willing to pay for an improvement in water quality?
- How will commuting behaviour change if a city introduces a low-emission zone?
- Which conservation programme produces the highest welfare gain per unit of public spending?
- How does taste heterogeneity vary by income, location, perceived risk, or environmental attitudes?
- Which policy would be preferred by different segments of the population under scenarios not yet observed in the data?

These are prediction questions only in part. They are also questions about causal mechanisms, counterfactuals, substitution patterns, and welfare. This is why the integration of machine learning and choice modelling must be done carefully.

3 Machine learning in one page

A supervised machine learning problem usually starts with a dataset

$$\{(x_i, y_i)\}_{i=1}^{n},$$

where x_i contains predictors and y_i is an outcome. The goal is to learn a function $\hat{f}(x)$ that predicts y well for new observations. A typical estimator can be written as

$$\hat{f} = \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} L\left(y_i, f(x_i)\right) + \lambda \Omega(f),$$

where L is a loss function, $\Omega(f)$ is a complexity penalty, and λ controls the amount of regularization.

Examples of supervised learning include:

- predicting energy demand from weather, building, and price data;
- predicting transport mode choice from trip, traveller, and built-environment variables;
- predicting household adoption of solar panels;
- classifying land cover from satellite imagery;
- predicting whether a respondent will choose a conservation policy in a stated choice experiment.

A large part of ML practice is built around the bias-variance tradeoff. Very simple models may underfit because they cannot capture the true pattern. Very flexible models may overfit because they adapt to noise in the training data. Cross-validation, regularization, early stopping, and test-set evaluation are practical tools for controlling this tradeoff (Hastie et al., 2009; Bishop, 2006; Kuhn and Silge, 2022).

Athey and Imbens argue that economists should know ML methods not because they replace econometrics, but because they provide useful tools for prediction, classification, unsupervised learning, matrix completion, causal inference, policy estimation, and counterfactual demand analysis (Athey and Imbens, 2019). Chernozhukov and coauthors show that ML can also be used for valid inference on low-dimensional causal or structural parameters, but only when combined with tools such as orthogonal scores and cross-fitting that address regularization bias and overfitting (Chernozhukov et al., 2018).

4 Choice modelling in one page

Choice modelling starts from a different data structure. The decision maker does not simply have an outcome label. The decision maker faces a **choice set**. For person n, alternative i, and choice set C_n, the random utility model writes

$$U_{ni} = V_{ni} + \varepsilon_{ni},$$

where U_{ni} is total utility, V_{ni} is observed or systematic utility, and ε_{ni} is unobserved utility.

The individual chooses alternative i if

$$U_{ni} > U_{nj} \quad \forall j \neq i,\ j \in C_n.$$

A simple environmental utility function might be

$$V_{ni} = \beta_{cost} cost_{ni} + \beta_{bio} biodiversity_{ni} + \beta_{co2} co2_{ni} + \beta_{access} access_{ni}.$$

If the unobserved utility terms are independently and identically distributed Type I extreme value, the choice probability is the multinomial logit probability

$$P_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in C_n} \exp(V_{nj})}.$$

This expression is mathematically equivalent to the softmax function used in multiclass machine learning. But the interpretation is different. In a random utility model, the systematic utility usually comes from behavioural theory, the coefficients are interpreted as marginal utilities, and ratios of coefficients can produce economic trade-offs.

For example, if cost is measured in money, a marginal willingness to pay for biodiversity can be estimated as

$$WTP_{bio} = -\frac{\beta_{bio}}{\beta_{cost}}.$$

This is one reason discrete choice models remain central in environmental valuation, transport, energy, health, marketing, and public policy. They can do more than predict a class. They can estimate trade-offs, elasticities, substitution patterns, and welfare measures (McFadden, 1974; Ben-Akiva and Lerman, 1985; Train, 2009; Hess and Daly, 2024).

5 The central distinction: prediction is not policy analysis

The strongest single idea in this chapter is simple:

Core principle

Prediction is about learning patterns that generalize to similar data. Policy analysis is about what changes when the environment, attributes, incentives, or constraints change. A model can be excellent for prediction and still be unsuitable for welfare analysis.

Consider a model that predicts household adoption of electric vehicles. It may learn that adoption is high in rich urban neighbourhoods with charging infrastructure. This could be a good predictive model. But if we ask whether a purchase subsidy causes adoption, the model's predictive success is not enough. Income, infrastructure, local policy, household preferences, and peer effects may all confound the association.

Similarly, a model may predict that respondents choose a climate adaptation programme when it is described with positive images and low cost. That does not automatically tell us the welfare effect of the programme, the causal effect of the imagery, or the compensating variation associated with the policy.

Aboutaleb, Danaf, Xie, and Ben-Akiva explicitly frame this problem for discrete choice: traditional ML is built for prediction, while policy analysis requires causal and behavioural assumptions. They argue that ML capabilities are valuable, especially for systematic model selection, but direct off-the-shelf ML is insufficient for many policy settings (Aboutaleb et al., 2021). Van Cranenburgh and coauthors similarly argue that ML can enrich choice modelling, especially through model building, large data, and unstructured data, but that theory-driven choice models still have strengths in interpretability, economic outputs, and forecasting in new settings (van Cranenburgh et al., 2022).

6 Why environmental economics needs both ML and choice models

Environmental economics is a natural meeting point for ML, causal inference, and choice modelling because environmental problems involve complex systems, heterogeneous preferences, non-market values, and policy counterfactuals.

6.1 Prediction problems

ML can be valuable when the task is prediction:

- pollution exposure prediction from sensors, monitoring stations, weather, land use, and satellite data;
- land-cover and deforestation classification from remote sensing data;
- forecasting energy consumption under weather and price variation;
- predicting wildfire risk, flood exposure, heat vulnerability, or ecosystem stress;
- predicting uptake of green technologies or conservation programmes.

In these settings, flexible ML methods such as random forests, gradient boosting, and neural networks may outperform linear models because environmental processes are nonlinear, spatially structured, and interaction-rich. However, the resulting predictions still require careful validation: spatial leakage, temporal leakage, and grouped observations can make performance appear better than it is.

6.2 Causal and policy problems

Many environmental policy questions are causal:

- What is the effect of a carbon tax on emissions and welfare?
- What is the effect of pollution regulation on health, employment, or firm behaviour?
- Which households reduce energy demand in response to dynamic pricing?
- Which farmers respond to payments for ecosystem services?
- Do conservation campaigns change actual behaviour or only stated intentions?

ML helps here by estimating nuisance functions flexibly, discovering heterogeneous treatment effects, or improving targeting. Double/debiased ML is useful when a low-dimensional treatment or policy parameter is the target and high-dimensional controls must be learned flexibly (Chernozhukov et al., 2018). Generalized random forests are useful when the analyst wants to estimate treatment-effect heterogeneity with local moment conditions (Athey et al., 2019). But causal ML does not remove the need for identification. It helps estimate causal objects after a credible design is in place.

6.3 Valuation and welfare problems

Environmental goods are often not bought and sold in ordinary markets. Examples include biodiversity, landscape quality, clean air, flood protection, recreation access, and ecosystem resilience. Choice experiments and random utility models allow researchers to estimate preferences for these attributes and convert trade-offs into WTP measures.

Recent environmental DCE texts emphasize the full workflow: survey design, experimental design, implementation, econometric analysis, and interpretation of environmental values (Mariel et al., 2025, 2021). ML can support this workflow by improving sampling, detecting attention patterns, modelling heterogeneity, incorporating text and images, and building predictive baselines. But welfare measures still require a bridge from observed or stated choices to utility.

7 Why choice modelling needs ML

The traditional choice-modelling workflow is powerful but not flawless. It often involves manual specification search, limited functional forms, difficult treatment of high-dimensional variables, and challenges with text, image, GPS, smartphone, or other passively collected data.

Van Cranenburgh and coauthors identify several opportunities for ML in choice modelling: helping with model specification, using large and continuous data streams, handling unstructured data such as text and images, and developing hybrid models that retain theory-driven structure while adding data-driven flexibility (van Cranenburgh et al., 2022).

7.1 Specification search

Choice modellers often test many alternative utility functions: linear cost, log cost, income interactions, piecewise time effects, alternative-specific coefficients, nonlinear waiting-time penalties, class-specific preferences, and more. This process is slow, subjective, and vulnerable to researcher degrees of freedom.

ML can assist by ranking variables, detecting nonlinearities, proposing interactions, or formalizing specification search as an optimization problem. Ortelli and coauthors propose assisted specification as a multi-objective combinatorial optimization problem (Ortelli et al., 2021). Hernandez and coauthors use association rules and random forests to support the specification of complex portfolio choice models (Hernandez et al., 2023). These methods do not remove the need for theory; they help structure the search.

7.2 Flexible heterogeneity

Mixed logit and latent class models are central tools for preference heterogeneity. ML can add flexibility to these models. For example, neural networks can map contextual variables into preference shifts, as in context-aware mixed logit models (Lukawska et al., 2024). Neural networks can also be used to construct latent variables from psychometric indicators in latent class choice models (Torres Lahoz et al., 2023).

The main challenge is not flexibility itself. The main challenge is disciplined flexibility. Flexible heterogeneity models should preserve behavioural meaning, identify relevant parameters, and avoid implausible implied trade-offs.

7.3 Unstructured data

Traditional choice models are best suited to structured attributes such as cost, time, distance, and quality levels. ML can process text, images, audio, trajectories, and high-dimensional spatial data. This opens new possibilities:

- using street-view images in bicycle route or residential location choice;
- using satellite imagery in land-use and conservation choices;
- using social media or open-ended survey text to measure beliefs and attitudes;
- using GPS traces to reconstruct revealed preference data;
- using embeddings to represent neighbourhoods, products, or policy descriptions.

The challenge is to connect these representations to choice theory. An image embedding may predict cycling route choice, but the analyst must still ask what behavioural attribute the embedding represents, whether it is stable under policy changes, and whether it can support welfare analysis.

8 A comparison of modelling cultures

Breiman famously distinguished between two statistical cultures: data modelling and algorithmic modelling (Breiman, 2001). In economics and choice modelling, this distinction appears as theory-driven versus data-driven modelling.

Dimension	**Theory-driven choice modelling**	**Machine learning**
Main question	What behavioural process generated the choices?	What function predicts well on new data?
Model form	Often specified from utility theory, decision rules, and domain knowledge.	Often learned from data with weak assumptions about functional form.
Validation	Statistical inference, likelihood, signs, elasticities, behavioural plausibility, predictive checks.	Cross-validation, test performance, calibration, robustness, interpretability tools.
Interpretability	Usually central. Parameters and ratios should have behavioural meaning.	Sometimes secondary, though explainable AI and interpretable ML are growing fields.
Counterfactuals	Central use case: simulate policy, welfare, substitution, demand forecasts.	Safe mainly within the support of the training environment unless causal or structural assumptions are added.
Software style	Transparent equations and specialized packages such as Apollo, mlogit, gmnl, logitr, and Biogeme.	Modular pipelines using packages such as tidymodels, scikit-learn, xgboost, PyTorch, and TensorFlow.

Strength	Interpretation, welfare, policy counterfactuals, choice-set structure.	Prediction, flexible functions, high-dimensional data, unstructured data, automation.
Risk	Misspecification, manual search, restrictive assumptions.	Data leakage, black-box behaviour, weak causal interpretation, implausible economic outputs.

The point is not to declare one culture superior. The point is to select a model according to the scientific purpose. If the task is short-run prediction of observed choices in a stable environment, flexible ML may be excellent. If the task is policy appraisal, welfare estimation, or extrapolation to new alternatives, a theory-consistent choice model or a carefully constrained hybrid model is usually safer.

9 The first notation bridge: softmax and multinomial logit

Many students first learn multiclass classification through the softmax model. Suppose a classifier assigns a score $s_k(x)$ to each class k. The predicted class probability is

$$P(Y = k \mid x) = \frac{\exp(s_k(x))}{\sum_{m=1}^{K} \exp(s_m(x))}.$$

In multinomial logit, the corresponding probability is

$$P_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in C_n} \exp(V_{nj})}.$$

The equations look the same if $s_k(x) = V_{nk}$. The distinction is conceptual:

- In ML, the scores are usually learned to minimize predictive loss.
- In choice modelling, the systematic utilities are usually specified to reflect attributes, constraints, and behavioural hypotheses.
- In ML, classes are often treated as labels.
- In choice modelling, alternatives are part of a choice set and have attributes that may change under policy.

This bridge is important because many hybrid models use ML to parameterize some part of V_{ni}, while retaining the logit probability structure. For example, a neural network might learn nonlinear interactions in tastes, but the final probabilities remain logit probabilities. Such models can improve prediction while retaining some choice-theoretic structure.

10 Worked example: a conservation policy DCE

Suppose a regional government is designing a conservation programme. Households are asked to choose among two policy alternatives and a status quo. Each policy is described by four attributes:

- annual household cost;
- biodiversity improvement;
- carbon reduction;
- recreation access restrictions.

A choice model might estimate the marginal utility of each attribute and derive WTP for biodiversity and carbon reduction. An ML classifier might predict which option each household will choose. Both are useful, but they answer different questions.

10.1 Research questions

Question	Recommended starting model	Reason
Which option will each respondent choose?	Random forest, gradient boosting, neural net, MNL baseline.	Prediction is the direct target.
How much are respondents willing to pay for biodiversity?	MNL, mixed logit, latent class, WTP-space model.	WTP requires interpretable utility-cost trade-offs.
Does WTP vary by income and environmental attitudes?	Mixed logit, latent class, hybrid choice, ML-assisted heterogeneity.	Heterogeneity must be interpretable and uncertainty should be quantified.
Which attributes should be included in the final utility specification?	Theory-driven baseline plus ML-assisted specification search.	ML can reduce manual search but theory should discipline final specification.
Can a policy not shown in the survey be appraised?	Theory-consistent choice model, possibly hybrid and constrained.	Extrapolation requires assumptions about utility under new attribute combinations.

10.2 A minimal R simulation and MNL estimation

The following R code simulates a simple stated choice experiment and estimates a multinomial logit model using `mlogit`. This is not yet a full environmental valuation study. It is a first demonstration of the data shape and the difference between a choice model and a classifier.

```
# Packages
# install.packages(c("tidyverse", "mlogit"))
library(tidyverse)
library(mlogit)

set.seed(2026)

# Design dimensions
N  <- 500     # respondents
Tn <- 8       # choice tasks per respondent
alts <- c("Policy_A", "Policy_B", "Status_Quo")

# True utility parameters used for simulation
beta_cost   <- -0.060
```

```r
beta_bio    <-  0.650
beta_co2    <-  0.300
beta_access <- -0.250
asc <- c(Policy_A = 0.15, Policy_B = 0.10, Status_Quo = 0.00)

# Long-format choice data: one row per alternative per choice task
dce_long <- expand_grid(
  id = 1:N,
  task = 1:Tn,
  alt = alts
) |>
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = if_else(alt == "Status_Quo", 0,
                   sample(c(10, 25, 50, 75), n(), replace = TRUE)),
    biodiversity = if_else(alt == "Status_Quo", 0,
                           sample(c(1, 2, 3), n(), replace = TRUE)),
    co2 = if_else(alt == "Status_Quo", 0,
                  sample(c(1, 2, 3), n(), replace = TRUE)),
    access_restriction = if_else(alt == "Status_Quo", 0,
                                 sample(c(0, 1, 2), n(), replace = TRUE)),
    V = asc[alt] +
      beta_cost * cost +
      beta_bio * biodiversity +
      beta_co2 * co2 +
      beta_access * access_restriction
  ) |>
  group_by(chid) |>
  mutate(
    p = exp(V) / sum(exp(V)),
    chosen_alt = sample(alt, size = 1, prob = p),
    choice = as.integer(alt == chosen_alt)
  ) |>
  ungroup()

# Prepare data for mlogit
mdata <- mlogit.data(
  dce_long,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

# Estimate MNL without alternative-specific intercepts for simplicity
mnl_fit <- mlogit(
  choice ~ cost + biodiversity + co2 + access_restriction | 0,
  data = mdata
)

summary(mnl_fit)

# Marginal willingness to pay estimates
b <- coef(mnl_fit)
wtp_biodiversity <- -b["biodiversity"] / b["cost"]
wtp_co2          <- -b["co2"] / b["cost"]
wtp_access       <- -b["access_restriction"] / b["cost"]

c(
  WTP_biodiversity = wtp_biodiversity,
  WTP_co2 = wtp_co2,
  WTP_access_restriction = wtp_access
```

```
)
```

10.3 Interpretation

The MNL model is estimated from long-format data because each row represents an alternative within a choice situation. The coefficients are utility parameters. If the cost coefficient is negative and the biodiversity coefficient is positive, then the WTP ratio $-\beta_{bio}/\beta_{cost}$ is positive.

The important lesson is not the numerical output. The important lesson is that the MNL model can be interpreted as a model of trade-offs. This is why it can support environmental valuation.

10.4 A predictive ML comparison in R

A random forest classifier usually requires one row per choice task, not one row per alternative. We therefore create a wide dataset. The outcome is the chosen alternative, and the predictors are the attributes of the alternatives.

A crucial point: because each respondent answers multiple choice tasks, the train/test split should be grouped by respondent. If the same respondent appears in both training and test data, the model can partly learn respondent-specific patterns and overstate out-of-sample performance. Hillel and coauthors warn that this kind of leakage is common in ML mode choice studies and can bias performance estimates (Hillel et al., 2021a; Hillel, 2021).

```
# Packages
# install.packages(c("tidyverse", "tidymodels", "ranger"))
library(tidyverse)
library(tidymodels)
library(ranger)

# Convert long choice data to wide ML data
ml_wide <- dce_long |>
  select(id, task, chid, alt, cost, biodiversity, co2,
         access_restriction, chosen_alt) |>
  pivot_wider(
    names_from = alt,
    values_from = c(cost, biodiversity, co2, access_restriction)
  ) |>
  distinct(id, task, chid, .keep_all = TRUE) |>
  mutate(chosen_alt = factor(chosen_alt))

set.seed(2026)

# Grouped split: whole respondents go either to training or testing
split <- group_initial_split(ml_wide, group = id, prop = 0.80)
train_data <- training(split)
test_data  <- testing(split)

# Grouped cross-validation inside the training data
folds <- group_vfold_cv(train_data, group = id, v = 5)

rf_spec <- rand_forest(
  trees = 500,
  mtry = tune(),
  min_n = tune()
) |>
  set_engine("ranger", importance = "permutation") |>
  set_mode("classification")
```

```
rf_wf <- workflow() |>
  add_formula(chosen_alt ~ . - id - task - chid) |>
  add_model(rf_spec)

rf_grid <- grid_regular(
  mtry(range = c(2, 10)),
  min_n(range = c(2, 25)),
  levels = 4
)

rf_tuned <- tune_grid(
  rf_wf,
  resamples = folds,
  grid = rf_grid,
  metrics = metric_set(accuracy, mn_log_loss)
)

best_rf <- select_best(rf_tuned, metric = "mn_log_loss")

rf_fit <- finalize_workflow(rf_wf, best_rf) |>
  fit(train_data)

rf_pred <- predict(rf_fit, test_data) |>
  bind_cols(test_data |> select(chosen_alt))

metrics(rf_pred, truth = chosen_alt, estimate = .pred_class)
```

10.5 What this comparison teaches

The MNL and random forest do not compete on one universal scale. They answer different questions.

- The MNL gives interpretable marginal utilities and WTP estimates, but it may underfit nonlinear patterns or interactions.
- The random forest may predict well, but it does not automatically provide valid WTP or welfare measures.
- A careful applied paper should often report both: a theory-consistent model for interpretation and a strong predictive benchmark for validation.

11 Python counterpart: grouped validation for a classifier

Python is not the main language of this book, but it is important for deep learning, NLP, image analysis, and some discrete choice packages. Biogeme is a major Python package for maximum likelihood estimation of parametric models, with special emphasis on discrete choice models (Bierlaire, 2026). Scikit-learn is a widely used package for supervised and unsupervised ML (Pedregosa et al., 2011).

The following code sketch shows the grouped-validation idea in Python. It assumes that a wide dataset like `ml_wide` has been created and exported to CSV.

```
import pandas as pd

from sklearn.model_selection import GroupShuffleSplit
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score, log_loss

# One row per choice task
# Columns include respondent id, chosen alternative, and alternative attributes.
df = pd.read_csv("conservation_choice_wide.csv")

y = df["chosen_alt"]
X = df.drop(columns=["chosen_alt", "id", "chid", "task"])
groups = df["id"]

splitter = GroupShuffleSplit(n_splits=1, test_size=0.20, random_state=2026)
train_idx, test_idx = next(splitter.split(X, y, groups=groups))

X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]
y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]

rf = RandomForestClassifier(
    n_estimators=500,
    min_samples_leaf=5,
    random_state=2026
)

rf.fit(X_train, y_train)

pred_class = rf.predict(X_test)
pred_prob = rf.predict_proba(X_test)

print("Accuracy:", accuracy_score(y_test, pred_class))
print("Log loss:", log_loss(y_test, pred_prob, labels=rf.classes_))
```

This code illustrates a general lesson: validation is part of the model. A badly validated flexible model is not evidence of good predictive performance.

12 Model evaluation: not one metric but a dashboard

For ML classifiers, the default metrics are often accuracy and log loss. For choice models, the default outputs often include log-likelihood, likelihood ratio tests, standard errors, signs, elasticities, WTP, and market shares. A modern workflow should combine both traditions.

Metric or diagnostic	What it tells you	Why it matters
Accuracy or hit rate	Fraction of choices correctly classified.	Easy to understand, but can be misleading under class imbalance or poor probability calibration.
Log loss / cross-entropy	Quality of predicted probabilities.	More informative than accuracy for probabilistic choice prediction.
Calibration	Whether predicted probabilities match observed frequencies.	Choice forecasts and policy simulations often need calibrated probabilities, not only class labels.
Grouped test performance	Generalization to new individuals, households, firms, or locations.	Prevents leakage in repeated-choice or panel data.

Coefficient signs	Whether marginal utilities have plausible directions.	A positive cost coefficient usually signals misspecification, coding error, or identification problems.
Elasticities	Sensitivity of probabilities to attribute changes.	Central for demand forecasting and policy analysis.
WTP and confidence intervals	Monetary trade-offs between attributes and cost.	Central for environmental valuation and cost-benefit analysis.
Counterfactual plausibility	Whether predictions under policy changes are behaviourally credible.	Essential for policy appraisal.
Subgroup performance	Performance across income, gender, region, vulnerability, or exposure groups.	Prevents models from working only for the majority group.

13 Common mistakes in early ML-choice applications

13.1 Mistake 1: Treating alternatives as ordinary labels

In many ML tasks, labels are just categories. In choice modelling, alternatives are meaningful objects in a choice set. Their attributes can change. The same alternative can appear with different cost, time, risk, or environmental quality. Ignoring this structure can produce models that predict labels but cannot analyse policy changes.

13.2 Mistake 2: Reporting only accuracy

Accuracy is not enough. A model that predicts the majority class can have high accuracy but poor usefulness. A model can also have good accuracy but poor log loss if it is overconfident. For policy simulation, predicted probabilities often matter more than predicted class labels.

13.3 Mistake 3: Splitting repeated choice data row by row

If the same person contributes multiple rows, a row-wise random split can place some of that person's choices in training and some in testing. Flexible ML models can exploit this dependence. Test performance then overstates true generalization. Use respondent-level, household-level, firm-level, route-level, spatial, or temporal splits when the research question requires it.

13.4 Mistake 4: Confusing feature importance with welfare importance

A variable may be important for prediction because it is correlated with many other variables. That does not mean it has a causal effect or a welfare interpretation. Feature importance is not a WTP estimate.

13.5 Mistake 5: Trusting black-box elasticities without plausibility checks

A flexible model can imply nonmonotonic or implausible responses to cost, time, or risk. In environmental valuation, a model that predicts well but implies positive marginal utility of cost

is usually unsuitable for WTP analysis unless the reason is explicitly understood and justified.

13.6 Mistake 6: Ignoring sample size requirements for neural choice models

Neural networks can have many more weights than a comparable MNL model has coefficients. Alwosheel, van Cranenburgh, and Chorus propose a conservative rule of thumb for ANN-based discrete choice analysis: use at least 50 observations per network weight (Alwosheel et al., 2018). The exact requirement depends on architecture and data complexity, but the warning is important: many stated choice datasets are small relative to deep networks.

14 A first workflow for applied research

For most projects in this book, use the following workflow.

1. **Define the estimand or target.** Is the goal prediction, WTP, causal effect, welfare, segmentation, or policy simulation?
2. **Draw the data structure.** Identify individuals, choice occasions, alternatives, attributes, repeated choices, locations, time periods, and clusters.
3. **Estimate a transparent baseline.** For choice data, start with MNL. For prediction, start with penalized logistic regression or a simple tree-based model.
4. **Validate properly.** Use grouped, temporal, or spatial splits if needed. Evaluate probabilities, not only class labels.
5. **Estimate a stronger flexible model.** Use random forests, gradient boosting, neural networks, or regularized models depending on sample size and structure.
6. **Compare model purpose by purpose.** Prediction, interpretability, WTP, and counterfactual validity may lead to different rankings.
7. **Use ML to improve choice modelling.** Let ML suggest nonlinearities, interactions, segmentation, or latent representations, but impose behavioural constraints when welfare analysis is required.
8. **Report uncertainty and limitations.** Especially for WTP, elasticities, heterogeneous effects, and policy counterfactuals.

15 Software map for the book

R is the primary language because it has strong ecosystems for econometrics, statistics, reproducible reporting, and choice modelling. Python is used when it is especially strong: deep learning, NLP, images, embeddings, and some flexible ML workflows.

Software	Main use	Role in this book
`tidymodels`	General ML workflows in R.	Data splitting, resampling, recipes, model tuning, evaluation, and reproducible pipelines (Kuhn and Silge, 2022).

`mlogit`	Random utility models in R.	MNL and extensions, especially for teaching the long-format structure (Croissant, 2020).
`gmnl`	Mixed, generalized, scaled, and latent class multinomial logit.	Preference heterogeneity and latent classes in R (Sarrias and Daziano, 2017).
`logitr`	Fast MNL and MXL in preference and WTP space.	Practical WTP-space estimation and fast applied models (Helveston, 2023).
`apollo`	Flexible choice model estimation in R.	Advanced choice models, panel data, latent variables, and custom likelihoods (Hess and Palma, 2019).
`grf`	Generalized random forests in R.	Heterogeneous treatment effects and causal forests (Athey et al., 2019).
`scikit-learn`	General ML in Python.	Baseline ML, preprocessing, validation, and model comparison (Pedregosa et al., 2011).
`Biogeme`	Choice modelling in Python.	Discrete choice estimation and teaching Python-based choice workflows (Bierlaire, 2026).
`PyTorch` or `TensorFlow`	Deep learning.	Neural choice models, text/image embeddings, and hybrid architectures.

16 Concept checkpoint

Close the chapter and answer these questions without looking back.

1. What is the difference between prediction and policy analysis?
2. Why can a model with high predictive accuracy be inappropriate for WTP estimation?
3. In a random utility model, what is the difference between U_{ni}, V_{ni}, and ε_{ni}?
4. Why is the multinomial logit formula related to the softmax function?
5. What is one way ML can improve choice-model specification?
6. What is one way ML can mislead choice modellers?
7. Why should repeated choice data often be split by respondent rather than by row?
8. What is the role of the cost coefficient in WTP estimation?
9. Why does causal ML still require identification assumptions?
10. Give one example where a random forest is useful and one example where a mixed logit model is more appropriate.

17 Exercises

Exercise 1: Prediction versus welfare

A city estimates an XGBoost model that predicts cycling route choice with very high accuracy using distance, slope, traffic speed, street greenery, and neighbourhood income. The city wants to use the model to estimate residents' WTP for additional protected bike lanes.

(a) What information does the XGBoost model provide directly?

(b) What additional assumptions or modelling structure are needed for WTP?

(c) Which variables might create confounding or interpretation problems?

(d) Propose a hybrid workflow that uses the XGBoost model without treating it as a direct welfare model.

Exercise 2: Data leakage

A travel diary contains 10 trips per person. A researcher randomly splits trips into training and test sets and finds that a neural network strongly outperforms MNL.

(a) Explain the leakage problem.

(b) Redesign the validation split.

(c) Which performance metrics would you report?

(d) How would your interpretation change if the neural network still outperforms MNL under grouped validation?

Exercise 3: Environmental DCE utility

Write a utility function for a DCE on wetland restoration with attributes: annual cost, flood-risk reduction, bird species richness, recreation access, and implementation delay. Decide expected signs for each coefficient. Explain which coefficients can be used for WTP calculations and which model diagnostics you would check before interpreting them.

Exercise 4: R modification

Modify the R simulation in this chapter so that high-income respondents have a smaller marginal disutility of cost. Then answer:

(a) Does the simple MNL recover the average cost effect?

(b) What interaction would you add to the utility function?

(c) How might a random forest detect this pattern?

(d) Why does detecting the pattern differ from estimating a welfare-relevant parameter?

18 Chapter summary

Machine learning and choice modelling overlap, but they do not have identical goals. ML is strongest for prediction, high-dimensional data, flexible nonlinear functions, automation, and unstructured data. Choice modelling is strongest for behavioural interpretation, choice-set structure, substitution patterns, welfare analysis, and policy counterfactuals.

The frontier is not "ML versus choice models." The frontier is disciplined integration. Use ML to improve prediction, representation, specification search, and heterogeneity discovery. Use economic and choice theory to discipline interpretation, WTP estimation, welfare analysis, and counterfactual policy simulations.

Takeaway

For environmental economics and choice modelling, the right question is not "Which model is best?" The right question is "Best for what purpose: prediction, causal effect, utility interpretation, WTP, or policy counterfactual?"

Suggested readings

Start with Athey and Imbens (2019) for the economist's ML map, Train (2009) for discrete choice foundations, van Cranenburgh et al. (2022) for the choice-modelling and ML research agenda, and Mariel et al. (2025) for an R-based environmental DCE workflow. Then read Hillel et al. (2021a) before comparing ML classifiers in panel choice data, because validation errors can otherwise dominate substantive conclusions.

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A Self-Study Textbook for PhD Students

Machine Learning and AI for Environmental Economics and Choice Modelling

Chapter 2

Data, Validation, and the Machine Learning Workflow

Chapter role. Chapter 1 explained why prediction, causal inference, structural interpretation, and welfare analysis are different targets. This chapter turns that distinction into a practical workflow. You will learn how to structure data, split data, tune models, avoid leakage, select metrics, and validate models for environmental economics and choice modelling.

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How to use this chapter

This chapter is meant to be used with a running R session. First read each section, then answer the short checkpoint before looking at the answer key, then run the code lab. The chapter is deliberately cumulative: the same simulated choice-experiment data are reused to teach data structure, row-wise and group-wise splitting, resampling, tuning, and diagnostic reporting.

The main habit to develop is this: before fitting any model, write down the *prediction target*, the *unit of generalization*, the *data structure*, and the *validation design*. Many unreliable ML applications in economics and choice modelling do not fail because the algorithm is weak. They fail because the validation design does not match the scientific question.

Learning objectives

After completing this chapter, you should be able to:

1. Define generalization error and explain why training performance is not evidence of out-of-sample performance.
2. Distinguish holdout validation, k-fold cross-validation, repeated cross-validation, nested cross-validation, bootstrap validation, temporal validation, spatial block validation, and grouped validation.
3. Identify common forms of data leakage in environmental economics and choice modelling.
4. Explain why panel choice data usually require respondent-, household-, market-, site-, route-, or time-blocked resampling rather than naive row-wise splitting.
5. Choose suitable performance metrics for regression, classification, probability prediction, and discrete choice applications.
6. Implement a basic leak-resistant ML workflow in R using `tidymodels`, `rsample`, `recipes`, `yardstick`, and `ranger`.
7. Build a minimal validation report that is useful for prediction, interpretation, and policy appraisal.

Core message

In applied ML, validation is not a technical afterthought. It is part of research design. The split defines what kind of future, population, market, place, time period, or policy context the model is being asked to generalize to.

1 The modelling workflow as a research design

A basic supervised learning problem observes data

$$\{(y_i, x_i) : i = 1, \ldots, n\},$$

where y_i is the target and x_i is a vector of predictors. The analyst chooses a class of functions $\mathcal{F}$, a loss function L, and often a regularization penalty Ω. The fitted model is often written as

$$\hat{f} = \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} L(y_i, f(x_i)) + \lambda \Omega(f).$$

The conceptual target is not the training loss. The target is the expected loss on new cases drawn from the relevant population:

$$R(f) = \mathbb{E}[L(Y, f(X))].$$

This is called *generalization risk* or *test risk*. The training data are only a means to estimate a function that will work well on cases not used to build the function.

For environmental economics and choice modelling, the phrase "new cases" is ambiguous. It may mean new households, new survey respondents, new choice tasks from known respondents, new travel diaries, new locations, new years, new conservation sites, new technologies, new policy packages, or new attribute levels. Different meanings require different validation designs.

1.1 The six-step workflow

The complete workflow used throughout this book is:

1. **Question and target.** Is the goal prediction, explanation, causal inference, welfare, forecasting, segmentation, design, or model discovery?
2. **Data audit.** What is the unit of observation? Are there repeated observations, choice sets, spatial clusters, time periods, markets, or sites?
3. **Data budget.** Which observations are used for training, tuning, calibration, model comparison, final testing, and external validation?
4. **Preprocessing pipeline.** Which transformations are learned from training data and then applied unchanged to assessment data?
5. **Model fitting and tuning.** Which model classes are compared, how are hyperparameters selected, and what constraints are imposed?
6. **Validation and reporting.** Which metrics, diagnostics, uncertainty summaries, and behavioural checks support the research claim?

A bad workflow can make a good algorithm look scientific

A flexible model can exploit even subtle leakage. It can memorize people, repeated trips, spatial neighbours, future information, or survey design artifacts. When this happens, a model can appear impressive in cross-validation and fail in policy use.

2 Data structures in environmental economics and choice modelling

A standard ML textbook often starts with independent rows. Our applications rarely look that clean.

2.1 Common data structures

Data structure	Example	Validation implication
Independent cross-section	One household survey response per household	Random split may be acceptable if the future target is a similar population.
Panel or repeated measures	Repeated choice tasks by respondent; multi-day travel diary; smart-meter observations by household	Split by respondent, household, device, or panel unit when predicting new units.
Choice-set data	Each choice task has multiple alternatives in long format	Keep whole choice sets together. Never put alternatives from the same task in different folds.

Data structure	Example	Validation implication
Spatial data	Pollution monitors, parcels, farms, biodiversity sites, raster cells	Use spatial blocks or leave-location-out validation when predicting new locations.
Temporal data	Energy demand, climate adaptation, policy adoption over years	Use forward-chaining or rolling-origin validation when predicting the future.
Clustered markets	Products nested in markets; households nested in regions; trips nested in networks	Hold out entire markets, regions, routes, or clusters when transfer is the target.
Linked multimodal data	Survey plus GPS traces, satellite images, text, weather, and policy indicators	Prevent leakage through identifiers, time alignment, spatial joins, and duplicated records.

Roberts et al. (2017) show why random cross-validation can be inappropriate for spatial, temporal, hierarchical, and phylogenetic data, and recommend blocked approaches when dependence structures exist. This is directly relevant for environmental economics, where spatial autocorrelation and temporal dependence are common. Hillel (2021) makes an analogous point for travel mode choice: trip-wise sampling in panel travel data can leak information from repeated, return, and shared trips into validation folds.

2.2 Long and wide choice data

Choice data are usually stored in one of two forms.

Long format has one row per alternative per choice task. This is natural for random utility models.

id	task	chid	alt	choice	cost	time	co2
1	1	1_1	A	0	20	30	4
1	1	1_1	B	1	35	20	2
1	1	1_1	SQ	0	0	40	7

Wide format has one row per choice task, with alternative-specific columns. This is often more convenient for generic ML classifiers.

id	task	chid	chosen	cost_A	cost_B	cost_SQ	co2_A	co2_B	co2_SQ
1	1	1_1	B	20	35	0	4	2	7

The transformation is not innocent. If a model is trained as an ordinary classifier, the data need to preserve the choice-set structure. Otherwise the model may learn artifacts of column order, status quo coding, or impossible choice sets.

Rule for choice data

Keep the entire choice task together. If the prediction target is new respondents, keep all tasks from the same respondent together. If the target is new markets or routes, keep all observations from the same market or route together.

2.3 A data audit checklist

Before splitting data, answer these questions.

1. What is the decision maker? Person, household, firm, farmer, traveller, expert, community, or authority?
2. What is the choice occasion? A DCE task, trip, purchase occasion, route choice, time interval, site visit, or policy vote?
3. What is the choice set? Is it fixed, sampled, personalized, generated, or endogenous?
4. Are alternatives repeated across tasks or people?
5. Are observations clustered by person, household, route, market, site, country, survey wave, or interviewer?
6. Are features measured before the choice, at the time of the choice, or after the choice?
7. Is the goal interpolation within observed support or extrapolation to new policies, new places, or new technologies?

3 Validation designs

Validation estimates performance on data not used to fit the model. Different designs answer different questions.

3.1 Holdout validation

In a simple holdout design, the data are split into training and test sets. The model is built on the training set and evaluated once on the test set. The split might be 70/30 or 80/20.

Holdout validation is simple and useful when the dataset is large and observations are close to independent. It is often unstable for small DCEs and survey datasets. In a DCE with 500 respondents and 8 tasks per respondent, the number of rows can look large, but the number of independent decision makers is still only 500.

3.2 k-fold cross-validation

In k-fold cross-validation, the training data are split into K folds. Each fold is held out once while the model is trained on the other $K-1$ folds. The estimated risk is

$$\widehat{R}_{CV} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{|A_k|}\sum_{i\in A_k} L(y_i, \hat{f}^{(-k)}(x_i)),$$

where A_k is the assessment fold and $\hat{f}^{(-k)}$ is trained without fold k.

The `rsample` package implements v-fold cross-validation through `vfold_cv`; the official documentation describes it as randomly splitting data into V roughly equal groups, with each assessment set containing one fold and each analysis set containing the remaining folds.

3.3 Grouped cross-validation

Grouped cross-validation forms folds by groups rather than rows. In choice modelling, the group is often respondent, household, route, market, or choice set. The `rsample` documentation for `group_vfold_cv` notes that grouped resampling is commonly used when there are repeated measures of the same subject.

For mode choice, Hillel (2021) shows why this matters. Travel diary data contain repeated, return, and shared trips. Trip-wise sampling can put matching trips by the same person or household into both training and validation data. Flexible nonlinear classifiers can exploit this

leakage, producing inflated cross-validation performance and selecting high-variance models that do not generalize.

3.4 Nested cross-validation

Model tuning creates another source of bias. If the same cross-validation scores are used both to choose hyperparameters and to report final performance, performance estimates can be optimistic. Cawley and Talbot (2010) discuss overfitting in model selection and the resulting selection bias in performance evaluation. Varma and Simon (2006) show that when classifiers are tuned using cross-validation, all tuning steps should be repeated inside each cross-validation loop; nested cross-validation gives a much less biased estimate of error.

Nested cross-validation uses two loops:

1. The **outer loop** estimates final performance.
2. The **inner loop** tunes hyperparameters using only the training portion of each outer split.

This is computationally expensive, but it is the cleanest design when datasets are small and model selection is aggressive.

3.5 Temporal validation

When the target is future prediction, do not randomly mix years, weeks, or days. Use a temporal split. A simple example trains on years 1–4 and tests on year 5. Rolling-origin validation repeats this over several origins:

$$\text{train } 1{:}t, \quad \text{test } t+1,$$

for several values of t.

Temporal validation is central for energy demand, technology adoption, environmental forecasting, air-quality prediction, and climate adaptation data.

3.6 Spatial validation

When the target is prediction in new places, use spatial blocking or leave-region-out validation. Randomly splitting spatial cells can lead to very optimistic performance because nearby cells share covariates and unobserved conditions. This matters for land-use change, pollution exposure, biodiversity, wildfire risk, crop yields, conservation targeting, and remote sensing.

3.7 External validation

External validation tests a model on a dataset collected in a different study, region, time, population, market, survey mode, or policy context. It is the strongest practical check for applied policy use. It is also often the hardest to obtain.

External validity is not a metric

A model can have excellent accuracy on a held-out sample from the same survey and still fail in a new country, a new policy regime, a new transport network, or a new environmental market. External validation requires a meaningful external setting, not just a different random seed.

3.8 Which split answers which question?

Research target	Appropriate validation design	Usually inappropriate design
Predict new tasks from known respondents	Hold out tasks within respondent, with respondent history available by design	Claiming this predicts new respondents
Predict choices of new respondents	Group split by respondent	Row-wise split over tasks or alternatives
Predict new households in smart-meter data	Group split by household	Random split over hourly observations
Forecast next year adoption	Train on past years, test on future year	Random split over all years
Predict pollution in new locations	Spatial block or leave-site-out validation	Random split over spatially autocorrelated grid cells
Test transfer to another country	Leave-country-out or external country validation	Random split pooling countries
Tune a complex neural model and report unbiased performance	Nested cross-validation or untouched test set after all tuning	Reporting the same CV used for tuning
Estimate WTP from a DCE	Holdout log-likelihood plus behavioural diagnostics and uncertainty	Accuracy alone

4 Data leakage

Data leakage occurs when information unavailable at prediction time enters model training or evaluation. Leakage is not a small technical flaw. It changes the meaning of the evidence.

4.1 Types of leakage

Leakage type	How it appears	Prevention
Target leakage	Predictors include variables measured after the outcome, or direct transformations of the outcome	Audit variable timing and causal order
Preprocessing leakage	Scaling, imputation, PCA, feature selection, or encoding is fit on the full dataset before splitting	Estimate preprocessing only on training data; apply to assessment data
Panel leakage	Same person, household, device, trip, or route appears in both training and test data	Use grouped splitting
Choice-set leakage	Alternatives from the same choice task are split across folds	Split by choice task or respondent
Temporal leakage	Future weather, prices, policy conditions, or outcomes are used to predict the past	Use temporal splits and lagged features
Spatial leakage	Nearby observations share unobserved conditions across folds	Use spatial blocks or leave-site-out validation

Leakage type	How it appears	Prevention
Survey-design leakage	Model learns design blocks, task order, version IDs, or labels that reveal choice probabilities	Audit design variables and randomization structure
Duplicate leakage	Repeated, copied, or oversampled records appear in both training and test data	Deduplicate or split before oversampling
Model-selection leakage	Test set is repeatedly used to choose features, tune hyperparameters, or select models	Keep a final test set untouched, or use nested CV

4.2 Leakage in choice experiments

In DCEs, leakage is easy to create because each respondent answers multiple tasks. A row-wise split may put tasks from respondent 17 in both train and test. This may be acceptable only if the target is predicting new tasks from already observed respondents and the model is explicitly allowed to use respondent history. It is not acceptable if the target is predicting choices of new respondents.

Leakage can also occur through experimental design identifiers. Suppose a design has 20 blocks and block 7 accidentally contains more high-cost status quo options. A flexible classifier may learn the block ID rather than preferences. In a policy application, the block ID will not exist.

4.3 Leakage in environmental data

Environmental data have their own leakage mechanisms.

- In air-pollution prediction, monitors close to each other can make random splits overly optimistic.
- In land-use prediction, raster cells from the same landscape patch can appear in train and test.
- In climate-impact forecasting, future climate variables can be joined to outcomes from earlier years if date alignment is careless.
- In conservation targeting, protected-area status may be used as a predictor even though it is partly determined by the conservation policy under evaluation.
- In energy demand, household fixed signatures can leak if hourly readings from the same household are randomly split.

The leakage question

Ask: would this exact piece of information be available, in this exact form, at the moment when the model is supposed to make the prediction? If the answer is no, the variable is suspect.

5 Performance metrics

A metric is not just a number. It expresses what kind of error matters.

5.1 Regression metrics

For continuous outcomes, common metrics are:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2},$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|.$$

RMSE penalizes large errors more strongly than MAE. MAE is easier to interpret in original units. For environmental valuation, the right metric may depend on policy loss: a large error for high-exposure households or threatened species may be more costly than the same numerical error elsewhere.

5.2 Classification metrics

For classification, accuracy is

$$Accuracy = \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\{\hat{y}_i = y_i\}.$$

Accuracy is intuitive but can be misleading under class imbalance. If 90 percent of trips are car trips, a classifier can achieve 90 percent accuracy by always predicting car.

For probability prediction, log loss is usually more informative. In multiclass classification,

$$LogLoss = -\frac{1}{n}\sum_{i=1}^{n}\log \hat{p}_{i,y_i},$$

where $\hat{p}_{i,y_i}$ is the predicted probability assigned to the observed class. Van Cranenburgh et al. (2022) note the close vocabulary link between choice modelling and ML: log-likelihood in choice modelling corresponds closely to log loss in ML, while the multinomial logit function corresponds to the softmax function.

Gneiting and Raftery (2007) provide the general statistical argument for proper scoring rules: probability forecasts should be evaluated by scores that reward honest probabilistic prediction. This is crucial for choice modelling because predicted probabilities, not just chosen labels, feed market-share forecasts, welfare simulations, and policy appraisal.

5.3 Choice-model metrics

Choice models are usually evaluated using likelihood-based and behavioural diagnostics:

- log-likelihood on training and holdout data;
- average negative log-likelihood or cross-entropy;
- pseudo-R^2, AIC, and BIC for in-sample model comparison;
- hit rate or discrete choice accuracy, if label prediction is relevant;
- predicted market shares and segment-specific shares;
- signs and magnitudes of coefficients;
- standard errors, confidence intervals, and credible intervals;
- elasticities and marginal effects;
- WTP estimates and their uncertainty;
- calibration of predicted probabilities;
- performance by respondent group, income, region, route, and alternative.

For choice modelling, log loss is usually more useful than accuracy

A model that assigns 0.49 probability to the chosen alternative and 0.51 to another alternative is counted as wrong by accuracy. A model that assigns 0.01 probability to the chosen alternative is also counted as wrong. Log loss distinguishes these cases. Welfare and demand forecasts need probabilities, not just labels.

5.4 Calibration

A probabilistic model is calibrated when predicted probabilities match observed frequencies. If a model predicts 0.70 probability of choosing public transport for 1,000 cases, about 700 of those cases should choose public transport, conditional on being in comparable prediction bins.

Calibration matters in policy. A mode choice model used for ridership forecasts, an energy model used for demand response, or a conservation adoption model used for targeting should not merely rank cases correctly. It should produce probabilities with interpretable frequencies.

6 Preprocessing as part of the model

Preprocessing includes imputation, scaling, transformations, encoding categorical variables, text cleaning, feature extraction, balancing, and feature selection. The important rule is simple: preprocessing must be estimated from training data only.

6.1 Examples of preprocessing leakage

Suppose a pollutant exposure variable has missing values. If the mean used for imputation is computed from the full dataset before splitting, the test set has influenced the training pipeline. Suppose PCA is fit on all satellite-image features before splitting. The test set has helped define the feature space. Suppose LASSO feature selection is run on the full dataset before cross-validation. The assessment folds have helped choose variables.

The `recipes` package in tidymodels is designed to reduce this risk by defining preprocessing steps as a recipe. The recipe is prepared on the training set and then applied to new data. The official recipes documentation describes recipes as pipeable preprocessing sequences whose statistical parameters are estimated from an initial dataset and then applied to other datasets.

6.2 Preprocessing and choice data

Choice data require additional care:

- Alternative-specific variables should be transformed consistently across alternatives.
- Cost transformations should preserve welfare interpretation where needed.
- Status quo indicators should be handled explicitly.
- Availability variables should not be confused with preferences.
- Attributes generated by the experimental design should be documented and reproducible.
- Sampling weights and choice-set sampling corrections should be treated as part of the estimation design.

7 Worked example: simulate a panel DCE and build validation designs

The rest of this chapter uses a simulated environmental DCE. Respondents repeatedly choose among two policy alternatives and a status quo. The goal is not realism; the goal is to learn validation mechanics.

7.1 Setup

```
# Install once if needed:
# install.packages(c("tidyverse", "tidymodels", "mlogit", "ranger", "janitor"))

library(tidyverse)
library(tidymodels)
library(mlogit)
library(ranger)
library(janitor)

set.seed(2026)
```

7.2 Simulate long-format DCE data

```
simulate_env_dce <- function(N = 600, Tn = 8) {
  alts <- c("A", "B", "SQ")

  respondent_data <- tibble(
    id = 1:N,
    income = rlnorm(N, meanlog = log(45000), sdlog = 0.35),
    green_attitude = rnorm(N),
    rural = rbinom(N, size = 1, prob = 0.35)
  )

  dce <- expand_grid(
    id = 1:N,
    task = 1:Tn,
    alt = alts
  ) %>%
    left_join(respondent_data, by = "id") %>%
    mutate(
      chid = paste(id, task, sep = "_"),
      cost = case_when(
        alt == "SQ" ~ 0,
        TRUE ~ sample(c(10, 25, 50, 75, 100), n(), replace = TRUE)
      ),
      biodiversity = case_when(
        alt == "SQ" ~ 0,
        TRUE ~ sample(c(0, 1, 2, 3), n(), replace = TRUE)
      ),
      co2_reduction = case_when(
        alt == "SQ" ~ 0,
        TRUE ~ sample(c(0, 1, 2, 3), n(), replace = TRUE)
      ),
      access_restriction = case_when(
        alt == "SQ" ~ 0,
        TRUE ~ sample(c(0, 1), n(), replace = TRUE)
      ),
      asc = case_when(
        alt == "A" ~ 0.10,
        alt == "B" ~ 0.05,
        alt == "SQ" ~ 0
      ),
      beta_cost = -0.075 + 0.00000045 * income,
      beta_bio = 0.45 + 0.20 * green_attitude,
      beta_co2 = 0.25 + 0.10 * green_attitude,
```

```
      beta_access = -0.35 - 0.10 * rural,
      V = asc +
        beta_cost * cost +
        beta_bio * biodiversity +
        beta_co2 * co2_reduction +
        beta_access * access_restriction
    ) %>%
    group_by(chid) %>%
    mutate(
      p = exp(V) / sum(exp(V)),
      chosen_alt = sample(alt, size = 1, prob = p),
      choice = as.integer(alt == chosen_alt)
    ) %>%
    ungroup()

  dce
}

dce_long <- simulate_env_dce()

glimpse(dce_long)
```

The data-generating process includes heterogeneity: higher income reduces cost sensitivity, environmental attitude increases sensitivity to biodiversity and CO2 reduction, and rural respondents dislike access restrictions more. In real data, we do not know the data-generating process. Here we know it so that diagnostics can be interpreted.

7.3 Check choice-set integrity

Before any modelling, verify that each choice task has exactly one chosen alternative.

```
choice_audit <- dce_long %>%
  group_by(chid) %>%
  summarise(
    n_alternatives = n(),
    n_chosen = sum(choice),
    .groups = "drop"
  )

choice_audit %>% count(n_alternatives, n_chosen)
```

A well-formed three-alternative DCE should show `n_alternatives = 3` and `n_chosen = 1` for every task.

7.4 Create wide-format data for ML

```
dce_wide <- dce_long %>%
  select(
    id, task, chid, alt, chosen_alt,
    income, green_attitude, rural,
    cost, biodiversity, co2_reduction, access_restriction
  ) %>%
  pivot_wider(
    names_from = alt,
    values_from = c(cost, biodiversity, co2_reduction, access_restriction)
  ) %>%
  distinct(id, task, chid, .keep_all = TRUE) %>%
  mutate(chosen_alt = factor(chosen_alt))
```

```
glimpse(dce_wide)
```

This format treats a choice task as one classification instance. That is usually the right shape for random forests, boosted trees, or neural networks, but it is not the natural shape for MNL estimation.

8 Naive splitting versus grouped splitting

8.1 Naive row-wise split

The following code creates a naive row-wise split over choice tasks. Because each respondent contributes eight tasks, many respondents will appear in both training and test sets.

```
set.seed(2026)
naive_split <- initial_split(dce_wide, prop = 0.8, strata = chosen_alt)
naive_train <- training(naive_split)
naive_test  <- testing(naive_split)

length(intersect(naive_train$id, naive_test$id))
```

The intersection is the number of respondents whose tasks appear in both sets. If the goal is predicting new respondents, this is leakage.

8.2 Grouped split by respondent

```
set.seed(2026)
group_split <- group_initial_split(dce_wide, group = id, prop = 0.8)

group_train <- training(group_split)
group_test  <- testing(group_split)

length(intersect(group_train$id, group_test$id))
```

The intersection should be zero. This split asks the model to predict choices for respondents not seen during training.

8.3 Grouped cross-validation

```
set.seed(2026)
folds_grouped <- group_vfold_cv(group_train, group = id, v = 5)
folds_grouped
```

Each fold holds out complete respondents. This is the right tuning design when the final test set also contains new respondents.

Match the inner and outer target

If the final test set holds out respondents, the tuning folds should also hold out respondents. Otherwise hyperparameters may be selected for the wrong generalization problem.

9 A leak-resistant tidymodels workflow

9.1 Define the recipe

```
choice_recipe <- recipe(chosen_alt ~ ., data = group_train) %>%
  update_role(id, chid, task, new_role = "id") %>%
  step_zv(all_predictors()) %>%
  step_normalize(all_numeric_predictors())
```

The recipe says: use all variables except the outcome, but treat `id`, `chid`, and `task` as identifiers rather than predictors. Remove zero-variance predictors and normalize numeric predictors. The normalization parameters are estimated inside resampling folds, not on the full dataset.

For tree-based models, normalization is not necessary, but including it here demonstrates leak-resistant preprocessing. For linear models, SVMs, k-nearest neighbours, and neural networks, scaling is often important.

9.2 Specify a random forest model

```
rf_spec <- rand_forest(
  trees = 800,
  mtry = tune(),
  min_n = tune()
) %>%
  set_engine("ranger", importance = "permutation", probability = TRUE) %>%
  set_mode("classification")

rf_workflow <- workflow() %>%
  add_recipe(choice_recipe) %>%
  add_model(rf_spec)
```

9.3 Tune with grouped folds

```
rf_grid <- grid_regular(
  mtry(range = c(2, 10)),
  min_n(range = c(2, 30)),
  levels = 4
)

choice_metrics <- metric_set(accuracy, mn_log_loss)

set.seed(2026)
rf_tuned <- tune_grid(
  rf_workflow,
  resamples = folds_grouped,
  grid = rf_grid,
  metrics = choice_metrics,
  control = control_grid(save_pred = TRUE)
)

collect_metrics(rf_tuned) %>%
  arrange(mean)
```

For probability models, select using log loss rather than accuracy unless the research target is genuinely label accuracy.

```
best_rf <- select_best(rf_tuned, metric = "mn_log_loss")
best_rf
```

9.4 Fit final model and evaluate on the untouched group test set

```
final_rf <- finalize_workflow(rf_workflow, best_rf) %>%
  fit(group_train)

rf_class <- predict(final_rf, group_test)
rf_prob  <- predict(final_rf, group_test, type = "prob")

rf_results <- bind_cols(
  group_test %>% select(chosen_alt),
  rf_class,
  rf_prob
)

accuracy(rf_results, truth = chosen_alt, estimate = .pred_class)
mn_log_loss(rf_results, truth = chosen_alt, .pred_A, .pred_B, .pred_SQ)
conf_mat(rf_results, truth = chosen_alt, estimate = .pred_class)
```

The final test set is used once. If you repeatedly inspect this result and change the model, the test set becomes part of model selection. Then it is no longer an honest final test.

10 A baseline multinomial logit model

ML workflows should be compared with appropriate econometric baselines. For a first baseline, estimate a pooled MNL using the long data.

```
# Keep only respondents in the grouped training set.
train_ids <- unique(group_train$id)
test_ids  <- unique(group_test$id)

mnl_train_long <- dce_long %>% filter(id %in% train_ids)
mnl_test_long  <- dce_long %>% filter(id %in% test_ids)

mnl_train <- mlogit.data(
  mnl_train_long,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

mnl_fit <- mlogit(
  choice ~ cost + biodiversity + co2_reduction + access_restriction | 0,
  data = mnl_train
)

summary(mnl_fit)
```

This pooled MNL is deliberately simple. It ignores the simulated heterogeneity in income, green attitude, and rural status. It is a useful baseline, but not a fair final benchmark against a flexible ML model. Later chapters will introduce mixed logit, latent class, hybrid choice models, and ML-assisted specifications.

10.1 Behavioural diagnostics

For the MNL, inspect signs and implied WTP:

```
b <- coef(mnl_fit)

wtp <- tibble(
  attribute = c("biodiversity", "co2_reduction", "access_restriction"),
  wtp = -c(b["biodiversity"], b["co2_reduction"], b["access_restriction"]) / b["cost"]
)

wtp
```

The expected signs are: cost negative, biodiversity positive, CO2 reduction positive, and access restriction negative. If a flexible ML model has higher accuracy but implies that choices become more likely as cost rises, the model may be useful for short-run prediction but dangerous for welfare analysis.

Two baselines, two purposes

The random forest baseline asks: can a flexible model predict choices for new respondents? The MNL baseline asks: can a simple utility model recover interpretable trade-offs? A strong applied paper often reports both.

11 Validation for environmental economics

The validation design should mirror the environmental policy use case.

11.1 Prediction within existing support

Examples include short-term air-quality nowcasting, short-term energy demand forecasting, or predicting which households will respond to a rebate in the same program year. Random or grouped validation may be sufficient if dependence is handled correctly and the future population resembles the training population.

11.2 Prediction in new places

Examples include predicting deforestation risk in a region not used for model training or transferring a biodiversity model to a new conservation area. Use spatial blocks, leave-region-out validation, or external validation. Report performance by region, not only pooled performance.

11.3 Prediction in new times

Examples include forecasting EV adoption, heat-pump adoption, energy demand, drought damages, or climate adaptation. Use forward validation. Do not let future variables or future outcomes enter training features.

11.4 Policy counterfactuals

Policy counterfactuals require more than predictive validation. A model used to simulate a new carbon tax, road pricing scheme, conservation policy, or green tariff must be credible under interventions. This often requires causal identification, structural restrictions, or a choice model with interpretable policy variables.

Athey and Imbens (2019) argue that ML methods are important for economists but must be adapted for causal inference, policy estimation, and counterfactual demand. Chernozhukov et al. (2018) show one formal route: use ML for high-dimensional nuisance functions while protecting inference on target parameters through orthogonal scores and cross-fitting.

12 Validation for choice modelling

Choice models need both predictive and behavioural validation.

12.1 Predictive validation

Predictive validation uses holdout or cross-validated log-likelihood, log loss, hit rate, calibration, and predicted market shares. It asks whether the model predicts choices not used for estimation.

12.2 Behavioural validation

Behavioural validation asks whether model outputs are plausible:

- Is cost negative?
- Are time, risk, or inconvenience effects signed correctly?
- Are environmental improvements valued positively where theory expects it?
- Are WTP estimates plausible relative to income and prior studies?
- Are cross-elasticities plausible?
- Are substitution patterns sensible?
- Are class memberships or random coefficients interpretable?
- Do predictions behave sensibly under monotone changes in cost or quality?

Van Cranenburgh et al. (2022) emphasize that theory-driven choice models use behavioural theory to validate parameter signs and relative magnitudes; ML models often require additional explainability tools or structural constraints to establish that learned relationships are plausible.

12.3 Policy validation

Policy validation asks whether counterfactual predictions are credible. It includes:

- sensitivity to alternative specifications;
- uncertainty in WTP and welfare measures;
- robustness to sample weights and survey design;
- plausibility of predictions outside observed attribute support;
- comparison with external studies or revealed preference evidence;
- checks for dominance, monotonicity, and sign constraints.

Do not confuse fit with policy validity

A model can fit observed DCE choices well because respondents avoid expensive options. That does not mean the same model is credible for a counterfactual with costs and environmental improvements outside the experimental design range.

13 Model comparison: fair baselines and fair claims

Many ML-versus-choice comparisons compare a highly tuned ML model against a simple MNL. This can answer a useful question, but it should not be overstated. The fair benchmark depends on the claim.

Claim	Minimum fair comparison	Stronger comparison
ML predicts better than a simple linear utility model	MNL with sensible variables and validation	Mixed logit, latent class, tuned ML, and hybrid model under same split
ML helps specification search	Manual expert model versus ML-assisted model	Pre-registered search space, nested validation, and behavioural diagnostics
ML handles high-dimensional data	Baseline choice model using hand-crafted features	Representation model plus choice layer, external validation
ML supports welfare analysis	Demonstrate monotonicity and interpretable cost trade-offs	RUM-consistent or constrained architecture with WTP uncertainty
Choice model is preferable for policy	Compare predictive and behavioural outputs	Stress tests under policy counterfactuals and external validation

The uploaded literature repeatedly supports a complementary view. Hillel et al. (2021) show that ML mode choice papers often used fragmented methodologies and had common validation limitations. Van Cranenburgh et al. (2022) argue that ML can enrich choice modelling through specification search, large data, unstructured data, and predictive practices, while choice theory remains valuable for behavioural and economic outputs. Ortelli et al. (2021) and Hernandez et al. (2023) show concrete ways to use ML for assisted specification rather than replacing choice models.

14 Uncertainty in validation

A single test score is an estimate. It has uncertainty.

14.1 Sources of uncertainty

- sampling uncertainty in the original data;
- variation across resampling folds;
- hyperparameter tuning uncertainty;
- random initialization and stochastic optimization;
- simulation error in mixed logit or Bayesian models;
- data revision, geocoding, imputation, or measurement error;
- distribution shift between training and application settings.

Report uncertainty where possible: fold standard errors, bootstrap intervals, repeated cross-validation summaries, confidence intervals for WTP, posterior intervals for Bayesian models, and sensitivity to alternative splits.

14.2 The bootstrap

The bootstrap resamples observations or clusters to estimate uncertainty. For independent rows, ordinary bootstrap resamples rows. For choice panels, resample respondents or households. For spatial data, use block bootstrap or other dependence-aware methods.

Bootstrap validation is common in prediction modelling and can estimate optimism: the gap between apparent training performance and expected test performance. It is useful when data are small, but it must respect dependence.

15 Reproducibility and research hygiene

Machine learning workflows are computational experiments. Reproducibility is part of credibility.

15.1 Minimum reproducibility practices

1. Store raw data separately from processed data.
2. Write scripts that rebuild every derived dataset.
3. Use deterministic random seeds where possible.
4. Save train/test split identifiers, not just random seeds.
5. Record package versions with `sessionInfo()` or `renv`.
6. Use Quarto or R Markdown for analysis reports.
7. Use Git for code, but do not commit sensitive data.
8. Save model objects only when the code to recreate them also exists.
9. Keep an analysis log explaining changes to data, variables, tuning grids, and exclusions.
10. Separate exploratory analysis from final confirmatory validation.

Van Cranenburgh et al. (2022) highlight open-science practices as an area where choice modelling can learn from ML, including shared datasets, open software, and reproducibility culture.

15.2 A minimal project folder

```
project/
  data_raw/
  data_processed/
  scripts/
    01_clean_data.R
    02_make_splits.R
    03_fit_baselines.R
    04_tune_ml_models.R
    05_validate_models.R
  reports/
    chapter2_lab.qmd
  models/
  outputs/
  renv.lock
  README.md
```

16 R lab: compare naive and grouped validation

This lab shows how a split can change the claim. It uses the simulated DCE created earlier. The random forest should usually perform better under naive splitting than under grouped splitting, because naive splitting lets information about respondents enter both train and test sets. The size of the gap will vary by simulation seed and data-generating process.

16.1 Function to fit and evaluate a random forest

```
fit_evaluate_rf <- function(split_obj, v = 5) {
  train <- training(split_obj)
  test  <- testing(split_obj)

  # If there are overlapping respondent IDs, report them.
  overlap <- length(intersect(train$id, test$id))

  folds <- group_vfold_cv(train, group = id, v = v)

  rec <- recipe(chosen_alt ~ ., data = train) %>%
    update_role(id, chid, task, new_role = "id") %>%
    step_zv(all_predictors())

  spec <- rand_forest(
    trees = 500,
    mtry = tune(),
    min_n = tune()
  ) %>%
    set_engine("ranger", probability = TRUE) %>%
    set_mode("classification")

  wf <- workflow() %>%
    add_recipe(rec) %>%
    add_model(spec)

  grid <- grid_regular(
    mtry(range = c(2, 10)),
    min_n(range = c(2, 25)),
    levels = 3
  )

  tuned <- tune_grid(
    wf,
    resamples = folds,
    grid = grid,
    metrics = metric_set(accuracy, mn_log_loss)
  )

  best <- select_best(tuned, metric = "mn_log_loss")

  final <- finalize_workflow(wf, best) %>%
    fit(train)

  pred <- bind_cols(
    test %>% select(chosen_alt),
    predict(final, test),
    predict(final, test, type = "prob")
  )

  tibble(
    overlapping_ids = overlap,
    accuracy = accuracy(pred, truth = chosen_alt, estimate = .pred_class)$.estimate,
    log_loss = mn_log_loss(pred, truth = chosen_alt, .pred_A, .pred_B,
    .pred_SQ)$.estimate
  )
}
```

16.2 Run the comparison

```
set.seed(2026)
naive_split <- initial_split(dce_wide, prop = 0.8, strata = chosen_alt)

group_split <- group_initial_split(dce_wide, group = id, prop = 0.8)

naive_result <- fit_evaluate_rf(naive_split)
group_result <- fit_evaluate_rf(group_split)

bind_rows(
  naive = naive_result,
  grouped = group_result,
  .id = "split_type"
)
```

16.3 Interpretation

Do not expect the same numbers every time. Focus on the logic. If the naive split has overlapping respondents and the grouped split does not, they answer different questions. The naive split estimates performance for new tasks from a population that includes already observed respondents. The grouped split estimates performance for new respondents.

In a thesis or paper, report the split design explicitly. Do not write only "we used five-fold cross-validation." Write what the folds were made of.

17 A small Python equivalent

Python is useful when you use scikit-learn, PyTorch, TensorFlow, JAX, or Biogeme. The same validation principles apply.

```
from sklearn.model_selection import GroupKFold, cross_val_score
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import log_loss, accuracy_score
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer

# X: one row per choice task in wide format
# y: chosen alternative
# groups: respondent IDs

gkf = GroupKFold(n_splits=5)

rf = RandomForestClassifier(
    n_estimators=500,
    min_samples_leaf=5,
    random_state=2026
)

# For random forests, scaling is optional. A pipeline is shown to make
# the anti-leakage principle explicit: transformations are fit inside folds.
pipe = Pipeline([
    ("model", rf)
])

scores = cross_val_score(
```

```
    pipe,
    X,
    y,
    groups=groups,
    cv=gkf,
    scoring="neg_log_loss"
)

print(-scores.mean())
```

scikit-learn also provides examples of nested versus non-nested cross-validation. The concept is the same as in R: separate the loop that tunes hyperparameters from the loop that estimates final performance.

18 Reporting template for applied papers

A validation report should be short but complete. Use the following template.

Validation report template

Research target. State the target: new respondents, new tasks, new households, future years, new locations, new markets, or policy counterfactuals.
Data structure. State the unit of observation, grouping structure, time structure, spatial structure, and choice-set structure.
Split design. State exactly how training, tuning, validation, and test sets were created. Name the grouping variable if grouped splitting was used.
Preprocessing. State which transformations were estimated inside training folds and which variables were excluded as identifiers or leakage risks.
Model classes. State baselines, ML models, hyperparameter grids, and tuning criteria.
Metrics. Report probability metrics, label metrics if relevant, calibration, subgroup performance, and choice-specific diagnostics.
Behavioural checks. For choice models and hybrid models, report signs, elasticities, WTP, predicted shares, and plausibility under counterfactual changes.
Uncertainty. Report standard errors, fold variability, confidence intervals, posterior intervals, or sensitivity to alternative splits.
Limitations. State what the validation design does not test.

19 Checkpoint questions

Checkpoint 2.1: Choose the split

For each case, choose a validation design.

1. A DCE with 1,000 respondents and 12 tasks per respondent, used to predict choices of new respondents.
2. A travel diary with repeated trips by people in households, used to compare ML and logit mode choice models.
3. A satellite-based model predicting deforestation in districts not used for training.
4. An energy demand model forecasting next winter's peak load.
5. A survey model tuned over 200 hyperparameter combinations and reported in a journal article.

Checkpoint 2.2: Identify leakage

A student normalizes all numeric variables, uses PCA on all predictors, selects the 20 most important variables using the full dataset, then runs five-fold cross-validation. What is wrong?

Checkpoint 2.3: Accuracy versus log loss

Two models are used to predict three-alternative choices. Model A has higher accuracy. Model B has lower log loss. Which is better for demand forecasting and why?

20 Common mistakes

Mistake 1: Splitting rows when the target is new people

Repeated tasks, trips, purchases, or observations by the same decision maker should usually stay together. Row-wise splitting is one of the fastest ways to overstate ML performance in panel choice data.

Mistake 2: Saying "cross-validation" without saying what was crossed

A cross-validation report should identify the fold unit: row, task, respondent, household, route, market, site, country, or time block.

Mistake 3: Letting preprocessing see the test set

Any operation that learns from the data must be inside the resampling loop: imputation, scaling, PCA, feature selection, text vocabulary, class balancing, and encoding.

Mistake 4: Optimizing accuracy for probability decisions

Policy simulations need probabilities. A model can have slightly lower accuracy but much better probability calibration and log loss.

Mistake 5: Comparing ML with a weak econometric baseline

A simple MNL is useful, but it is not always state-of-practice. A serious benchmark may need mixed logit, latent class, panel structure, and a careful utility specification.

Mistake 6: Treating the final test set as a dashboard

Every time you inspect the final test score and change the model, you train yourself on the test set. Keep a final test set untouched or use nested validation.

21 Exercises

Exercise 2.1: Audit your own dataset

Choose one dataset from your research or from a public source. Write a one-page data audit: unit of observation, grouping variables, time variables, spatial variables, outcome timing, possible leakage variables, and the target population for prediction.

Exercise 2.2: Three validation designs for one DCE

Using the simulated DCE, create three splits: row-wise by task, grouped by respondent, and grouped by design block. Explain which scientific question each split answers.

Exercise 2.3: Re-run the R lab

Change the simulation so that each respondent has a strong individual-specific preference for the status quo. Compare naive and grouped validation. What happens to the gap between naive and grouped performance?

Exercise 2.4: Add calibration

Using the random forest predictions, create calibration bins by predicted probability of the chosen alternative. Plot mean predicted probability against observed frequency. Is the model overconfident or underconfident?

Exercise 2.5: Design a validation report

Write a validation report for a hypothetical ML model predicting adoption of household solar panels. Include temporal, spatial, and policy-counterfactual limitations.

Exercise 2.6: Nested cross-validation

Explain nested cross-validation in your own words. Then describe when you would avoid it because the computational cost is not justified.

22 Answers to checkpoints

Checkpoint 2.1

1. Group split by respondent; use grouped cross-validation for tuning and a respondent-held-out test set.
2. Group by person or household. In household travel data, household grouping may be safer because shared trips can create leakage across household members.
3. Spatial block or leave-district-out validation.
4. Temporal validation, such as train on earlier winters and test on the next winter; rolling-origin validation if several years are available.
5. Nested cross-validation or a strictly untouched final test set after all tuning and model selection.

Checkpoint 2.2

The preprocessing and feature selection are learned from the full dataset, so the assessment folds influence the training pipeline. Normalization, PCA, and variable selection must be estimated inside the training portion of each resample and then applied to the assessment portion.

Checkpoint 2.3

For demand forecasting, Model B may be better because demand forecasts use probabilities and shares, not just the most likely label. Log loss rewards calibrated probability assignments to the actually chosen alternatives. Accuracy ignores the difference between a near miss and a confident false prediction.

23 Chapter summary

The main lesson of this chapter is that the validation design is the scientific claim. Random holdout, grouped cross-validation, temporal validation, spatial blocking, nested cross-validation, and external validation answer different questions. In environmental economics and choice modelling, dependence is the rule rather than the exception: respondents answer multiple tasks, households make repeated decisions, trips repeat, locations are spatially autocorrelated, and policies unfold over time.

A trustworthy ML workflow therefore starts with a data audit, chooses a split that matches the target of generalization, keeps preprocessing inside the resampling loop, tunes models without using the final test set, reports probability-based metrics, and adds behavioural diagnostics when the model is used for choice, welfare, or policy counterfactuals. The next chapter introduces supervised learning methods, starting with linear and generalized linear models as the bridge between econometrics and machine learning.

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Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 3

Supervised Learning for Economists

Regression, classification, regularization, trees, forests, boosting, SVMs, and neural networks

Core theme. Supervised machine learning estimates a function that maps observed information to outcomes. In economics, this is useful only when the target, validation design, loss function, and interpretation are aligned with the scientific or policy question.

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How to study this chapter

This chapter is written for readers who know basic statistics and econometrics but do not yet have a machine learning background. The material is cumulative. Do not try to memorize algorithm names. Instead, repeatedly ask four questions:

1. What prediction target is being learned?
2. What loss function is the algorithm trying to reduce?
3. What regularization or tuning choice controls overfitting?
4. What economic interpretation survives after the model is fitted?

The chapter uses retrieval practice, worked examples, comparison tables, and coding tasks. The goal is not only to understand algorithms, but to know when each algorithm is appropriate for environmental economics and choice modelling.

Learning objectives. By the end of this chapter you should be able to:

1. define supervised learning as empirical risk minimization;
2. distinguish regression, binary classification, and multiclass classification;
3. explain the bias-variance tradeoff in practical rather than purely mathematical terms;
4. fit and tune penalized regression, random forests, gradient boosting, SVMs, and simple neural networks in R;
5. choose performance metrics for regression and classification problems;
6. identify validation pitfalls in panel choice data;
7. explain why high predictive accuracy is not enough for welfare or policy counterfactuals;
8. translate supervised ML outputs into useful inputs for choice model specification, segmentation, and prediction.

1 The supervised learning problem

Supervised learning starts with examples. Each observation has predictors X and an observed target Y. We want to learn a rule $\hat{f}$ that predicts Y from X for future observations.

$$Y = f(X) + \varepsilon. \tag{1}$$

The function f is unknown. The algorithm searches inside a class of possible functions $\mathcal{F}$ and chooses the one that performs best according to a loss function. The general form is:

$$\hat{f} = \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} L(y_i, f(x_i)) + \lambda \Omega(f). \tag{2}$$

The first term rewards fit to the training data. The second term penalizes complexity. The tuning parameter λ determines how strongly complexity is penalized. This simple equation is the common language behind most of this chapter.

1.1 Regression and classification

Supervised learning has two major task types.

- **Regression:** Y is numeric. Examples include electricity demand, annual vehicle kilometres, household carbon emissions, crop yields, recreational trip counts, or willingness-to-pay values.
- **Classification:** Y is categorical. Examples include adoption or non-adoption of rooftop solar, support or opposition to a policy, travel mode, car ownership category, chosen alternative in a discrete choice experiment, or whether a household belongs to a high-risk flood adaptation group.

Choice modelling often looks like classification, but it has more structure. In an ordinary multiclass classifier, the labels might be $\{\text{car}, \text{bus}, \text{bike}\}$. In a random utility choice model, the alternatives have attributes, decision makers compare alternatives in choice sets, and the analyst often needs elasticities, welfare measures, and policy counterfactuals. This distinction will return throughout the book.

1.2 Supervised learning as risk minimization

The ideal target is the expected prediction loss on new data:

$$R(f) = \mathbb{E}[L(Y, f(X))]. \tag{3}$$

Because the data-generating process is unknown, we minimize empirical risk on observed data, then use validation methods to estimate how well the model generalizes. Chapter 2 explained why this validation design matters. In panel choice data, the train-test split should often be grouped by person, household, route, location, or time period rather than by individual rows.

Choice modelling warning. A row-wise train-test split can be misleading when the same person appears many times. Hillel's experimental review of mode choice classifiers shows how trip-wise sampling can leak information from training to validation sets when repeated, return, or shared trips appear across splits. Flexible nonlinear models are especially able to exploit that leakage. For panel choice data, use grouped splitting unless your scientific question truly concerns prediction for new rows from the same individuals.

2 Why economists should care about supervised ML

Economists have long estimated predictive relationships. Linear regression, logistic regression, probit, and multinomial logit can all be viewed as supervised learning methods. What distinguishes modern ML is not a different philosophical universe, but a stronger emphasis on out-of-sample performance, algorithmic tuning, flexible function classes, high-dimensional data, and regularization.

Athey and Imbens argue that empirical economists should learn ML methods because economic data increasingly include large cross-sections, many predictors, linked administrative records, text, images, sensors, and complex panel structures. They also emphasize that economists should preserve their traditional strengths: identification, uncertainty, welfare interpretation, and policy relevance.

For environmental economics and choice modelling, supervised ML is useful in at least six ways.

Use	**Examples**
Forecasting	Predict household electricity demand, pollution peaks, transport mode shares, or conservation programme uptake.
High-dimensional controls	Use many weather, spatial, demographic, and behavioural variables without hand-selecting a small subset.
Specification discovery	Find nonlinearities and interactions that may later be translated into interpretable utility functions.
Segmentation	Identify groups with similar predicted behaviour, risk, or policy responsiveness.
Data fusion	Combine survey data with geospatial, remote-sensing, text, image, or smartphone tracking features.
Benchmarking	Compare a theory-driven choice model with strong predictive alternatives to learn where the theory-based model is weak.

The uploaded choice-modelling literature repeatedly supports this balanced view. Van Cranenburgh and co-authors describe ML as an opportunity for cross-fertilization with choice modelling rather than a simple replacement. Aboutaleb and co-authors argue that off-the-shelf ML is limited for policy analysis unless structural assumptions and interpretability are retained. Paredes and co-authors show in car ownership demand that ML performance depends strongly on feature engineering; with discrete-choice-style features ML can underperform, while with ML-oriented features it can outperform. These examples teach a key lesson: supervised ML performance is not only about the algorithm; it is also about the representation of the problem.

3 The bias-variance tradeoff

A model that is too simple has high bias. A model that is too flexible can have high variance. This is the classic bias-variance tradeoff.

For squared-error regression, at a point x, the expected prediction error can be decomposed as:

$$\mathbb{E}[(Y - \hat{f}(x))^2] = \sigma^2 + \Big(\mathbb{E}[\hat{f}(x)] - f(x)\Big)^2 + \mathrm{Var}(\hat{f}(x)). \tag{4}$$

The three terms are:

- **irreducible noise:** randomness that no model can predict;
- **bias:** systematic error from an overly restrictive model;
- **variance:** instability from fitting noise in the training data.

A linear model may have high bias if the true relationship is nonlinear. A large decision tree may have high variance if small changes in training data produce very different trees. Regularization, pruning, bagging, boosting, early stopping, and cross-validation are different ways to manage this tradeoff.

Retrieval practice 3.1. Without looking back, answer in one sentence each:

1. What does regularization penalize?
2. Why can a flexible model have worse test performance than a simpler model?
3. Why is bias not always bad in prediction problems?

4 Linear and generalized linear models as ML baselines

A good applied ML workflow begins with simple baselines. The baseline is not a formality. It tells us whether a complex model is doing useful work.

4.1 Linear regression

For numeric targets, the simplest supervised learner is linear regression:

$$Y_i = \beta_0 + X_i'\beta + \varepsilon_i. \tag{5}$$

Estimated by least squares, it minimizes:

$$\sum_{i=1}^{n} \left(y_i - \beta_0 - X_i'\beta\right)^2. \tag{6}$$

In environmental economics, linear regression might predict household energy demand from income, dwelling size, heating system, weather, and prices. In choice modelling, linear regression is less often the final model for a categorical choice, but it is useful for continuous outcomes such as travel time, exposure, expenditure, WTP estimates, or simulated welfare changes.

4.2 Logistic regression

For a binary target $Y \in \{0, 1\}$, logistic regression models:

$$\mathbb{P}(Y_i = 1 \mid X_i) = \frac{1}{1 + \exp[-(\beta_0 + X_i'\beta)]}. \tag{7}$$

The fitted model minimizes negative log likelihood, also called cross-entropy or log loss:

$$-\sum_{i=1}^{n} \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right]. \tag{8}$$

Logistic regression is the natural baseline for adoption and participation problems: whether a farmer adopts a soil conservation practice, whether a household accepts a green electricity tariff, whether a commuter switches to cycling, or whether a respondent supports a carbon tax.

4.3 Multinomial logistic regression and choice data

For multiclass classification, the softmax model is:

$$\mathbb{P}(Y_i = k \mid X_i) = \frac{\exp(\eta_{ik})}{\sum_{l=1}^{K} \exp(\eta_{il})}. \tag{9}$$

A multinomial logit classifier and a multinomial logit choice model are related, but not identical in typical use. A generic classifier often uses one row per choice situation and predicts a label. A choice model uses one row per alternative within each choice situation and models utilities of alternatives. For final welfare analysis, the choice model structure is usually needed.

Do not skip the baseline. In applied research, a random forest or XGBoost model should normally be compared with at least one transparent baseline: OLS for regression, logistic regression for binary classification, multinomial logit for multiclass classification, and MNL or mixed logit for discrete choice. A complex model that improves accuracy by 0.2 percentage points but destroys interpretability is usually not worth the cost.

5 Regularization: ridge, lasso, and elastic net

Regularization intentionally shrinks a model to improve generalization. It is central to modern ML and also highly relevant for applied economics, where researchers often have many candidate controls, interactions, transformations, and survey variables.

5.1 Ridge regression

Ridge regression minimizes squared error plus an L_2 penalty:

$$\hat{\beta}^{ridge} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}. \tag{10}$$

Ridge shrinks coefficients toward zero but usually does not set them exactly to zero. It is useful when many correlated predictors each carry some signal. In environmental applications, this is common with weather variables, land-use variables, or correlated socioeconomic indicators.

5.2 Lasso

The lasso uses an L_1 penalty:

$$\hat{\beta}^{lasso} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}. \tag{11}$$

The L_1 penalty can set coefficients exactly to zero, so lasso performs variable selection. Tibshirani's original paper emphasizes that the lasso combines shrinkage and selection, producing models that can be easier to interpret than dense high-dimensional regressions.

For choice modelling, lasso is useful for screening large sets of candidate interactions, nonlinear transformations, attitudinal indicators, and alternative-specific variables. But the selected model is not automatically a valid welfare model. Economic signs, scale, substitution patterns, and policy relevance still need to be checked.

5.3 Elastic net

Elastic net combines L_1 and L_2 penalties:

$$\hat{\beta}^{EN} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (y_i - x_i'\beta)^2 + \lambda \left[\alpha \sum_{j=1}^{p} |\beta_j| + (1-\alpha) \sum_{j=1}^{p} \beta_j^2 \right] \right\}. \tag{12}$$

When predictors are highly correlated, elastic net can be more stable than lasso because it tends to keep groups of correlated variables together. This matters in environmental economics because spatial, climatic, and socioeconomic variables are often highly collinear.

5.4 Applied example: high-dimensional predictors for green tariff adoption

Suppose an electricity provider offers a green tariff. You observe whether each household adopts the tariff and have 200 predictors: demographics, billing history, weather exposure, dwelling characteristics, distance to public transport, political area indicators, and survey attitudes. Ordinary logistic regression may be unstable. Lasso or elastic net can help identify a sparse predictive model.

Listing 1: Penalized logistic regression in R with tidymodels and glmnet.

```r
# install.packages(c("tidymodels", "glmnet"))
library(tidymodels)
library(glmnet)

set.seed(301)

# dat must contain adoption as a factor with levels "no", "yes"
# and many numeric/categorical predictors.
split <- initial_split(dat, prop = 0.8, strata = adoption)
train <- training(split)
test  <- testing(split)

folds <- vfold_cv(train, v = 5, strata = adoption)

rec <- recipe(adoption ~ ., data = train) %>%
  step_zv(all_predictors()) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_normalize(all_numeric_predictors())

elastic_spec <- logistic_reg(
  penalty = tune(),
  mixture = tune()
) %>%
  set_engine("glmnet") %>%
  set_mode("classification")

wf <- workflow() %>%
  add_recipe(rec) %>%
  add_model(elastic_spec)

grid <- grid_regular(
  penalty(range = c(-5, 0)),   # 10^-5 to 10^0
  mixture(range = c(0, 1)),    # ridge to lasso
  levels = c(20, 5)
)

res <- tune_grid(
  wf,
  resamples = folds,
  grid = grid,
  metrics = metric_set(roc_auc, mn_log_loss, accuracy)
)

best <- select_best(res, metric = "mn_log_loss")
final_fit <- finalize_workflow(wf, best) %>% fit(train)

pred <- predict(final_fit, test, type = "prob") %>%
```

```
  bind_cols(predict(final_fit, test)) %>%
  bind_cols(test %>% select(adoption))

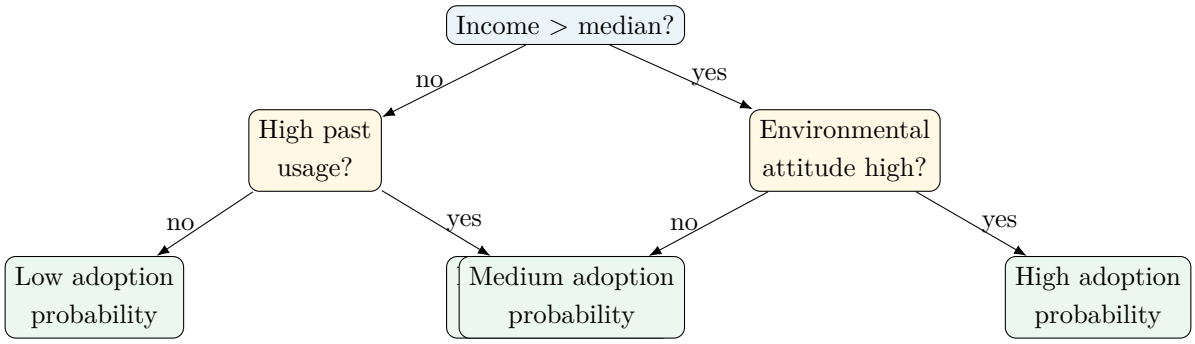
roc_auc(pred, truth = adoption, .pred_yes)
mn_log_loss(pred, truth = adoption, .pred_no, .pred_yes)
conf_mat(pred, truth = adoption, estimate = .pred_class)
```

Interpretation discipline. If the lasso selects environmental attitudes as strong predictors of adoption, that does not prove that changing attitudes will cause adoption. It means attitudes are predictive in this dataset. Causal interpretation requires research design or causal ML methods, which are introduced in Chapter 5.

6 Decision trees

A decision tree partitions the predictor space into regions and assigns a prediction to each region. For regression, the prediction is usually the mean outcome in the terminal node. For classification, it is the class probability or majority class.

A simple tree for green tariff adoption might learn rules such as:



Trees are attractive because they automatically handle interactions and nonlinear thresholds. Their weakness is instability: a small change in the data can produce a different tree. Single trees are therefore often useful for explanation and teaching but are usually not the strongest predictive model.

7 Random forests

Random forests reduce tree instability by averaging many decorrelated trees. Each tree is trained on a bootstrap sample of the data, and each split considers only a random subset of predictors. Breiman's random forest paper emphasizes that the prediction error depends on the strength of individual trees and the correlation among them. Good forests make individual trees strong while keeping them diverse.

For regression:

$$\hat{f}_{RF}(x) = \frac{1}{B}\sum_{b=1}^{B} \hat{f}_b(x), \tag{13}$$

where B is the number of trees. For classification, the forest averages predicted class probabilities or votes.

7.1 Why forests are useful in environmental economics

Random forests are a good default for tabular environmental and transport data because they:

- capture nonlinearities and interactions automatically;
- tolerate many predictors;
- are relatively robust to outliers and monotonic transformations;
- require less preprocessing than linear models or SVMs;
- provide variable importance diagnostics;
- often perform well without extensive tuning.

Examples include predicting high pollution days, identifying parcels at high deforestation risk, predicting energy consumption, predicting adoption of green technologies, and benchmarking mode choice prediction.

7.2 R implementation

Listing 2: Random forest classification with grouped resampling.

```r
# install.packages(c("tidymodels", "ranger"))
library(tidymodels)
library(ranger)

set.seed(302)

# Suppose dat has one row per observed choice occasion.
# person_id is a grouping variable and mode is the target.
split <- group_initial_split(dat, group = person_id, prop = 0.8)
train <- training(split)
test  <- testing(split)

folds <- group_vfold_cv(train, group = person_id, v = 5)

rec <- recipe(mode ~ ., data = train) %>%
  update_role(person_id, new_role = "id") %>%
  step_zv(all_predictors()) %>%
```

```r
  step_dummy(all_nominal_predictors())

rf_spec <- rand_forest(
  trees = 1000,
  mtry = tune(),
  min_n = tune()
) %>%
  set_engine("ranger", importance = "permutation") %>%
  set_mode("classification")

rf_wf <- workflow() %>%
  add_recipe(rec) %>%
  add_model(rf_spec)

rf_grid <- grid_regular(
  mtry(range = c(2, 20)),
  min_n(range = c(2, 40)),
  levels = 5
)

rf_res <- tune_grid(
  rf_wf,
  resamples = folds,
  grid = rf_grid,
  metrics = metric_set(accuracy, mn_log_loss)
)

best_rf <- select_best(rf_res, metric = "mn_log_loss")
rf_fit <- finalize_workflow(rf_wf, best_rf) %>% fit(train)
```

Interpretation warning. Random forest variable importance is not the same as a marginal causal effect. A variable can be important because it proxies for omitted variables, because it helps define interactions, or because it is correlated with a causal driver.

8 Gradient boosting and XGBoost

Boosting builds an ensemble sequentially. Each new weak learner is trained to improve the current model. In gradient boosting, the algorithm performs gradient descent in function space: it adds trees that point in the direction that reduces the loss.

A simplified additive form is:

$$\hat{f}_M(x) = \sum_{m=1}^{M} \nu \hat{h}_m(x), \tag{14}$$

where $\hat{h}_m$ is a weak learner and ν is the learning rate. Smaller learning rates usually require more trees but can improve generalization.

XGBoost is a scalable implementation of gradient boosted trees. Chen and Guestrin's paper introduced engineering and algorithmic improvements including sparsity-aware learning and efficient approximate split finding. In applied economics and transport, XGBoost is often a strong benchmark for tabular data.

8.1 Important boosting tuning parameters

Parameter	Meaning	Typical effect
Number of trees	How many trees are added	Too few underfit; too many can overfit without early stopping.
Learning rate	Step size of each tree	Smaller values are safer but need more trees.
Tree depth	Complexity of each tree	Deeper trees capture interactions but increase variance.
Minimum node size	Minimum observations in leaves	Larger values smooth predictions.
Subsample	Fraction of rows used per tree	Adds randomness and can reduce overfitting.
Column sample	Fraction of predictors used per tree	Adds diversity and can improve generalization.
Regularization	Penalties on tree weights	Controls complexity.

8.2 R implementation

Listing 3: XGBoost with tidymodels.

```r
# install.packages(c("tidymodels", "xgboost"))
library(tidymodels)
library(xgboost)

boost_spec <- boost_tree(
  trees = tune(),
  tree_depth = tune(),
  learn_rate = tune(),
  loss_reduction = tune(),
  sample_size = tune(),
  mtry = tune(),
  min_n = tune()
) %>%
  set_engine("xgboost") %>%
```

```
  set_mode("classification")

boost_wf <- workflow() %>%
  add_recipe(rec) %>%
  add_model(boost_spec)

set.seed(303)
boost_grid <- grid_latin_hypercube(
  trees(range = c(300, 2000)),
  tree_depth(range = c(2, 8)),
  learn_rate(range = c(-4, -1)),
  loss_reduction(range = c(-5, 1)),
  sample_prop(range = c(0.5, 1.0)),
  finalize(mtry(), train),
  min_n(range = c(5, 50)),
  size = 30
)

boost_res <- tune_grid(
  boost_wf,
  resamples = folds,
  grid = boost_grid,
  metrics = metric_set(accuracy, mn_log_loss)
)

best_boost <- select_best(boost_res, metric = "mn_log_loss")
boost_fit <- finalize_workflow(boost_wf, best_boost) %>% fit(train)
```

8.3 Choice modelling example

Dahmen, Weikl, and Bogenberger apply interpretable ML to tracking-based revealed preference mode choice data and report strong performance for XGBoost and random forest relative to a conventional multinomial logit benchmark, while using SHAP analysis for interpretation. This is a good example of where supervised ML can be valuable: predicting mode choices from passively collected data with many contextual features.

However, prediction is not the same as welfare analysis. If XGBoost predicts that a person will cycle after a bike-lane improvement, that is useful. But a welfare calculation requires stronger assumptions about utility, trade-offs, policy variables, and counterfactual validity.

9 Support vector machines

Support vector machines (SVMs) classify observations by finding a separating hyperplane with large margin. For a linear binary classifier, the decision boundary is:

$$\hat{y} = \text{sign}(w'x + b). \tag{15}$$

The hard-margin problem seeks the widest separating margin. Real data are rarely perfectly separable, so soft-margin SVMs allow some violations. The hinge loss is:

$$L(y, f(x)) = \max(0, 1 - yf(x)), \quad y \in \{-1, 1\}. \tag{16}$$

The power of SVMs comes partly from kernels. A kernel computes inner products in a transformed feature space without explicitly creating the transformed variables. Common kernels include linear, polynomial, and radial basis function kernels.

SVMs can perform well in medium-sized datasets with complex boundaries, but they require careful scaling and tuning. They are less naturally interpretable than linear models and less automatically convenient for large tabular data than tree ensembles.

9.1 R implementation

Listing 4: Support vector machine with a radial basis function kernel.

```r
# install.packages(c("tidymodels", "kernlab"))
library(tidymodels)
library(kernlab)

svm_spec <- svm_rbf(
  cost = tune(),
  rbf_sigma = tune()
) %>%
  set_engine("kernlab") %>%
  set_mode("classification")

svm_rec <- recipe(adoption ~ ., data = train) %>%
  step_zv(all_predictors()) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_normalize(all_numeric_predictors())

svm_wf <- workflow() %>%
  add_recipe(svm_rec) %>%
  add_model(svm_spec)

svm_grid <- grid_regular(
  cost(range = c(-3, 5)),
  rbf_sigma(range = c(-5, 0)),
  levels = 6
)

svm_res <- tune_grid(
```

```
    svm_wf,
    resamples = folds,
    grid = svm_grid,
    metrics = metric_set(roc_auc, mn_log_loss, accuracy)
  )
```

10 Neural networks

A feedforward neural network is a flexible function approximator. A hidden layer computes transformations of inputs, and the output layer maps those transformations to predictions.

For a single hidden layer:

$$\hat{y} = g\left(\alpha_0 + \sum_{m=1}^{M} \alpha_m \sigma(w_m' x + b_m)\right), \tag{17}$$

where $\sigma(\cdot)$ is an activation function and $g(\cdot)$ is an output function such as identity for regression or softmax for classification.

Deep learning is most useful when raw inputs have complex structure: images, text, audio, video, GPS traces, street-view images, or high-frequency sensor streams. For ordinary tabular data with moderate sample sizes, tree ensembles and regularized regression are often stronger starting points. This is especially important in stated choice experiments, where the number of respondents may be modest.

Sample-size warning for choice analysis. The uploaded ANN sample-size paper by Alwosheel, van Cranenburgh, and Chorus recommends a conservative rule of thumb for ANN-based discrete choice analysis: the sample should be large relative to the number of network weights. Neural networks can be powerful, but small DCE datasets can easily be too small for unconstrained deep models.

10.1 Neural networks for choice modelling

There are several ways neural networks enter choice modelling:

1. **Direct classifier:** predict the chosen alternative from engineered features.
2. **Utility augmentation:** add a flexible learned component to a traditional utility function.
3. **Representation learning:** learn features from images, text, GPS, or high-dimensional contexts.
4. **Latent construct modelling:** use neural networks to map psychometric indicators to latent variables or class membership.

5. **Context-aware heterogeneity:** use neural networks to map context variables to shifts in taste parameters.

The frontier is not simply replacing MNL with a neural network. The frontier is designing neural architectures that preserve some economic meaning while expanding flexibility.

10.2 Python example

R can fit neural networks, but Python is often more mature for deep learning workflows. A small scikit-learn example is shown below. For serious deep learning, PyTorch, TensorFlow/Keras, or JAX are more common.

Listing 5: A minimal Python neural-network classifier with scikit-learn.

```python
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import log_loss, accuracy_score

numeric = ["income", "age", "travel_time", "cost"]
categorical = ["urban", "car_access", "education"]

preprocess = ColumnTransformer([
    ("num", StandardScaler(), numeric),
    ("cat", OneHotEncoder(handle_unknown="ignore"), categorical)
])

model = MLPClassifier(
    hidden_layer_sizes=(64, 32),
    activation="relu",
    alpha=1e-4,
    early_stopping=True,
    random_state=123,
    max_iter=500
)

pipe = Pipeline([
    ("prep", preprocess),
    ("model", model)
])

pipe.fit(X_train, y_train)
probs = pipe.predict_proba(X_test)
preds = pipe.predict(X_test)

print(log_loss(y_test, probs))
print(accuracy_score(y_test, preds))
```

11 Metrics: what should be optimized?

A metric is not a neutral choice. It encodes what kind of error matters.

11.1 Regression metrics

Metric	Formula	Use
Mean absolute error	$\frac{1}{n}\sum_i \lvert y_i - \hat{y}_i \rvert$	Easy to interpret; robust to outliers.
Root mean squared error	$\sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}$	Penalizes large errors strongly.
R^2	$1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$	Common but can be misleading out of sample.
Mean absolute percentage error	$\frac{1}{n}\sum_i \lvert (y_i - \hat{y}_i)/y_i \rvert$	Problematic when y_i is near zero.

For environmental forecasting, RMSE may be appropriate when large errors are costly, such as missing extreme pollution episodes. MAE may be better when interpretability and robustness matter.

11.2 Classification metrics

Metric	What it measures	When useful
Accuracy	Fraction of correct predicted classes	Balanced classes and equal error costs.
Balanced accuracy	Average recall across classes	Imbalanced classes.
Precision	Share of predicted positives that are true positives	Avoiding false positives.
Recall	Share of actual positives detected	Avoiding false negatives.
F1 score	Harmonic mean of precision and recall	Imbalanced binary tasks.
ROC AUC	Ranking ability across thresholds	Binary classification with threshold choice.
Log loss	Quality of predicted probabilities	Choice prediction and probabilistic forecasts.
Brier score	Squared error of predicted probabilities	Probability calibration.

For choice modelling, log loss is often more informative than accuracy because choice models produce probabilities. A model that assigns probability 0.49 to the chosen alternative and 0.51

to another is almost as informative as one that assigns 0.51 to the chosen alternative. Accuracy treats these as failure versus success. Log loss respects probability quality.

11.3 Calibration

A classifier is calibrated if predicted probabilities correspond to observed frequencies. If a model predicts a 0.70 probability of adoption for 1000 households, about 700 should adopt among similar cases. Calibration is critical when predictions feed into policy simulations, risk maps, or expected demand forecasts.

Listing 6: Calibration curve for a binary classifier.

```
calibration_data <- pred %>%
  mutate(bin = ntile(.pred_yes, 10)) %>%
  group_by(bin) %>%
  summarize(
    mean_pred = mean(.pred_yes),
    observed = mean(adoption == "yes"),
    n = n(),
    .groups = "drop"
  )

library(ggplot2)
ggplot(calibration_data, aes(mean_pred, observed)) +
  geom_point() +
  geom_abline(linetype = "dashed") +
  labs(
    x = "Mean predicted probability",
    y = "Observed adoption rate"
  )
```

12 Feature engineering for environmental and choice data

Feature engineering is the process of creating useful predictors. In economics, this is where theory and institutional knowledge enter supervised ML.

12.1 Environmental economics features

Useful features often include:

- weather: temperature, rainfall, heating degree days, cooling degree days, drought indices;
- geography: distance to coast, elevation, floodplain status, land cover, urban density;
- exposure: pollution measures, wildfire risk, heat risk, crop stress;

- policy: treatment eligibility, subsidy availability, regulation timing;
- behaviour: past consumption, travel history, technology adoption, survey responses;
- social context: neighbourhood adoption rates, social vulnerability, local political variables.

For prediction, lagged outcomes are often powerful. For causal analysis, they must be used carefully. They can block or open causal pathways depending on the research design.

12.2 Choice modelling features

Choice data can be represented in long or wide format.

- **Long format:** one row per alternative per choice situation. This is the natural format for random utility models.
- **Wide format:** one row per choice situation, with attributes for each alternative in separate columns. This is often more convenient for generic ML classifiers.

A choice model estimates utility functions such as:

$$V_{ni} = \beta_{cost} cost_{ni} + \beta_{time} time_{ni} + \beta_{CO2} CO2_{ni} + \beta_{comfort} comfort_{ni}. \tag{18}$$

A generic ML classifier may instead receive variables such as *cost_car*, *cost_bus*, *time_car*, *time_bus*, income, age, weather, and built environment. This can improve prediction, but the model no longer automatically yields clean marginal rates of substitution.

Self-explanation prompt. Explain why converting a DCE from long format to wide format can help a random forest but hurt welfare interpretation. Write no more than five sentences.

13 Worked example: predicting participation in a conservation programme

Imagine a watershed agency offers households a conservation programme. The target is whether a household enrols. Predictors include programme cost, expected biodiversity gain, household income, education, environmental attitude, distance to the watershed, and local flood risk.

13.1 Simulate data

The following code simulates a simple dataset. In a real project you would replace this with survey or administrative data.

Listing 7: Simulating a conservation programme adoption dataset.

```r
library(tidyverse)
library(tidymodels)

set.seed(304)
N <- 2500

dat <- tibble(
  household_id = 1:N,
  income = rlnorm(N, log(45000), 0.5),
  education_years = sample(10:20, N, replace = TRUE),
  attitude = rnorm(N),
  distance_km = rexp(N, rate = 1/8),
  flood_risk = runif(N),
  cost = sample(c(20, 50, 80, 120), N, replace = TRUE),
  biodiversity_gain = sample(1:5, N, replace = TRUE),
  information_treatment = rbinom(N, 1, 0.5)
) %>%
  mutate(
    latent = -1.8 - 0.018 * cost +
      0.45 * biodiversity_gain +
      0.55 * attitude +
      0.35 * information_treatment +
      0.30 * flood_risk +
      0.000004 * income -
      0.025 * distance_km +
      0.25 * attitude * information_treatment,
    p = plogis(latent),
    enrol = factor(if_else(runif(N) < p, "yes", "no"), levels = c("no", "yes"))
  ) %>%
  select(-latent, -p)
```

13.2 Fit several models

We compare regularized logistic regression, random forest, and XGBoost. We use log loss as the primary metric because we care about predicted probabilities, not just the most likely class.

Listing 8: Comparing supervised learners.

```r
set.seed(305)
split <- initial_split(dat, prop = 0.8, strata = enrol)
train <- training(split)
test  <- testing(split)
folds <- vfold_cv(train, v = 5, strata = enrol)

base_rec <- recipe(enrol ~ ., data = train) %>%
  update_role(household_id, new_role = "id") %>%
  step_zv(all_predictors()) %>%
```

```
  step_dummy(all_nominal_predictors()) %>%
  step_normalize(all_numeric_predictors())

models <- workflow_set(
  preproc = list(base = base_rec),
  models = list(
    elastic = logistic_reg(penalty = tune(), mixture = tune()) %>%
      set_engine("glmnet") %>% set_mode("classification"),
    rf = rand_forest(trees = 800, mtry = tune(), min_n = tune()) %>%
      set_engine("ranger", importance = "permutation") %>%
      set_mode("classification"),
    xgb = boost_tree(trees = tune(), tree_depth = tune(), learn_rate = tune(),
                     loss_reduction = tune(), min_n = tune()) %>%
      set_engine("xgboost") %>% set_mode("classification")
  )
)

set.seed(306)
res <- workflow_map(
  models,
  "tune_grid",
  resamples = folds,
  grid = 20,
  metrics = metric_set(mn_log_loss, roc_auc, accuracy),
  verbose = TRUE
)

rank_results(res, rank_metric = "mn_log_loss", select_best = TRUE)
```

13.3 Use the result

Suppose XGBoost has the best log loss. What should we do?

1. Check calibration.
2. Compare predictive performance against a transparent logistic baseline.
3. Use partial dependence or SHAP to understand broad patterns.
4. Inspect whether predicted probability decreases when cost increases.
5. Avoid interpreting the information-treatment coefficient causally unless treatment was randomized or an identification strategy is valid.
6. If the goal is WTP or welfare, estimate a choice model or a structurally constrained hybrid model.

14 Supervised ML versus discrete choice models

This section gives a practical comparison. It will be deepened in later chapters.

Dimension	Supervised ML classifier	Discrete choice model
Main target	Predict labels or probabilities.	Estimate choice probabilities grounded in behavioural assumptions.
Data format	Often one row per case.	Often one row per alternative per choice situation.
Strength	Flexible prediction, interactions, high-dimensional inputs.	Interpretation, substitution patterns, WTP, welfare, counterfactual analysis.
Validation	Cross-validation and held-out test sets.	Likelihood, holdout validation, plausibility, signs, elasticities, welfare checks.
Interpretation	Post hoc or model-specific explanations.	Parameters often have direct behavioural meaning.
Policy use	Strong for local prediction within support.	Stronger for structured counterfactuals if assumptions are credible.
Risk	Data leakage, poor extrapolation, weak causal meaning.	Misspecification, restrictive substitution patterns, subjective specification search.
Best practice	Use as benchmark and discovery tool.	Use as theory-consistent model for inference and welfare.

The uploaded literature provides several lessons.

- The van Cranenburgh et al. discussion paper frames supervised ML and choice modelling as different but potentially complementary paradigms.
- The Hillel systematic and experimental reviews warn that validation design can strongly influence apparent ML superiority in mode choice.
- The Dahmen et al. tracking-data study shows that interpretable ML can be valuable when data are rich and the target is prediction.
- The Aboutaleb et al. paper argues that policy analysis requires structural assumptions and scrutiny, not only empirical fit.
- The Paredes et al. car ownership study shows that feature engineering can determine whether ML improves prediction over a discrete choice model.

15 Model selection workflow

A rigorous supervised ML workflow should be boring in the best sense. The steps are repeated across projects.

1. Define the scientific question: prediction, screening, specification discovery, policy targeting, or welfare analysis.
2. Define the prediction target and prediction horizon.
3. Define the unit of generalization: new rows, new people, new regions, new time periods, or new policies.
4. Split data in a way that matches that generalization target.
5. Build preprocessing inside the resampling workflow.
6. Fit simple baselines.
7. Fit flexible models.
8. Tune hyperparameters with resampling.
9. Evaluate once on a final test set.
10. Check calibration, subgroup performance, and plausibility.
11. Compare with theory-driven models if the application involves choice or welfare.
12. Document every choice.

Never preprocess before splitting if the preprocessing learns from the data. Normalization, PCA, imputation, target encoding, feature selection, and class balancing must be estimated using training data only inside the resampling workflow. Otherwise the validation data influence training indirectly.

16 Common mistakes

16.1 Mistake 1: choosing the metric after seeing the results

Choose the primary metric before model tuning. If the goal is probability prediction for demand forecasting, log loss or Brier score may be better than accuracy. If the goal is screening high-risk households, recall may matter more.

16.2 Mistake 2: treating ML predictions as causal effects

A model that predicts flood adaptation from risk perception does not prove that changing risk perception will change adaptation. The variable may be correlated with unobserved experience, income, local institutions, or past damage.

16.3 Mistake 3: ignoring grouped dependence

Repeated choices by the same individual, household, route, or region are not independent. If the model sees the same decision maker in training and testing, performance may be overstated for new decision makers.

16.4 Mistake 4: interpreting feature importance as utility

Feature importance measures prediction contribution, not marginal utility. Cost can have low feature importance if little cost variation exists, even though cost is behaviourally important.

16.5 Mistake 5: over-tuning

Trying hundreds of models and reporting the best validation score can overfit the validation process. Use nested resampling or a final untouched test set for credible performance claims.

16.6 Mistake 6: using a black-box classifier for WTP

WTP requires a marginal rate of substitution between a non-market attribute and money. A generic classifier does not automatically produce this object. Use a choice model, a WTP-space model, or a constrained hybrid architecture designed for welfare analysis.

17 Chapter summary

Supervised learning estimates a function from predictors to an outcome. Linear regression, logistic regression, penalized regression, random forests, gradient boosting, SVMs, and neural networks are different ways to choose the function class and control overfitting.

For environmental economics, supervised ML is valuable for forecasting, screening, targeting, imputation, high-dimensional control, and data fusion. For choice modelling, supervised ML is valuable for prediction, benchmarking, specification support, representation learning, and flexible heterogeneity discovery. However, high predictive accuracy alone does not deliver causal effects, welfare measures, or credible policy counterfactuals.

The practical rule is:

Use supervised ML when the goal is prediction, discovery, or flexible representation. Use economic structure when the goal is causal interpretation, substitution patterns, WTP, or welfare. Use both when possible: ML for disciplined prediction and discovery, choice theory for interpretation and policy use.

18 End-of-chapter exercises

Conceptual exercises

1. Explain the difference between a loss function and a performance metric. Can they be the same?
2. Why is log loss usually more informative than accuracy for choice probability prediction?
3. Give an environmental economics example where false negatives are more costly than false positives.
4. Explain why lasso can be useful for specification search but insufficient for welfare analysis.
5. Compare random forests and gradient boosting in terms of how trees are combined.
6. Why do SVMs usually require feature scaling?
7. Why might a neural network be useful for street-view image choice data but unnecessary for a small tabular DCE?
8. Explain the difference between feature importance and marginal utility.

Applied exercises in R

1. Simulate the conservation programme dataset from Section 14. Fit logistic regression, elastic net, random forest, and XGBoost. Compare log loss, ROC AUC, and calibration.
2. Change the data-generating process so that the cost effect is nonlinear. Which model improves most?
3. Create an artificial panel dataset where each household makes five adoption choices. Compare row-wise and grouped validation. How much do results change?
4. Convert a long-format DCE into wide format and fit a random forest classifier. Then estimate an MNL model in long format. Compare prediction and interpretability.
5. Use permutation importance for the random forest. Which variables are important? Which variables would you trust for behavioural interpretation?

Research design exercises

1. Design a supervised ML study to predict adoption of heat pumps. Define the target, predictors, validation split, primary metric, and baseline model.
2. Design a supervised ML benchmark for an environmental DCE. Explain how you would prevent respondent-level leakage.
3. Suppose XGBoost predicts conservation programme enrolment better than mixed logit. Write a paragraph explaining what conclusion you can and cannot draw.
4. Propose a hybrid workflow where ML is used to discover nonlinearities and a choice model is used for WTP estimation.

19 Reading guide

Start with James et al. for an accessible introduction to statistical learning. Then read Hastie, Tibshirani, and Friedman for a more advanced treatment. Read Athey and Imbens for the economist's perspective. For regularization, read Tibshirani on lasso and Zou and Hastie on elastic net. For tree ensembles, read Breiman on random forests, Friedman on gradient boosting, and Chen and Guestrin on XGBoost. For the choice-modelling connection, read van Cranenburgh et al., Hillel et al., Aboutaleb et al., Dahmen et al., and the machine-learning chapter in the second edition of the Handbook of Choice Modelling.

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Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 4

Unsupervised Learning and Representation Learning

From clustering and dimension reduction to latent classes, attitudes, text, images, and embeddings.

Generated for self-study use

July 16, 2026

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How to study this chapter

This chapter is about learning useful structure from data when no single labelled outcome is being predicted. In environmental economics and choice modelling, this means using unlabelled or weakly labelled information to discover groups, latent dimensions, representations of text and images, and lower-dimensional summaries that can later be used in valuation, policy analysis, segmentation, and discrete choice models.

Read the chapter in three passes. In the first pass, focus on the concepts: representation, distance, dimension reduction, clustering, latent classes, and embeddings. In the second pass, run the R code and change the number of clusters, number of principal components, and simulated preference segments. In the third pass, connect each method to a research question you care about: environmental attitudes, adoption of green technologies, transport mode choice, recreation demand, conservation preferences, energy demand, or stated preference valuation.

Learning objectives

By the end of this chapter, you should be able to:

1. Explain the difference between supervised and unsupervised learning.
2. Explain why representation, scaling, and distance metrics matter before clustering.
3. Use principal component analysis (PCA) for dimensionality reduction and interpret its limitations.
4. Use clustering as an exploratory tool without confusing clusters with causal types or preference classes.
5. Distinguish algorithmic clusters from latent class choice models.
6. Explain how latent variables, psychometric indicators, and learned representations can enter choice models.
7. Use topic models and embeddings to convert text into variables for environmental and choice modelling applications.
8. Design a defensible workflow for unsupervised learning in R, with Python extensions where useful.

1 Why unsupervised learning matters for economists

Supervised learning was the subject of Chapter 3. There, the problem was to learn a function that maps observed inputs to an observed target:

$$Y = f(X) + \varepsilon.$$

In unsupervised learning, we do not begin with a target variable. We observe data such as

$$X_1, X_2, \ldots, X_n,$$

and ask whether there is useful structure in the covariates themselves. That structure may be a lower-dimensional representation, a grouping of respondents, a set of topics in open-ended text, a latent attitude dimension, or an embedding of images, words, households, firms, locations, or alternatives.

Athey and Imbens describe unsupervised learning as the case where we have observations on covariates without labels and may want to partition the sample into clusters or estimate the joint distribution of the variables. They also note that mixture models in econometrics are closely related to clustering because mixture components resemble clusters, although the modelling assumptions differ (Athey and Imbens, 2019). Van Cranenburgh et al. emphasize the same high-level taxonomy for choice modelling: supervised learning maps features to outputs, while unsupervised learning searches for hidden patterns or groupings, such as clusters, in data (van Cranenburgh et al., 2022).

Core idea

Unsupervised learning is not a substitute for theory. It is a way to create, inspect, compress, and organize data before, during, or after economic modelling. In environmental economics and choice modelling, the most useful question is rarely "which cluster is real?" The better question is "does this representation help us understand heterogeneity, specify a better model, design a better experiment, or make a more credible policy analysis?"

Unsupervised learning is useful in at least five recurring settings.

First, environmental economists often work with rich data: remote sensing, weather and climate grids, pollution monitors, smart meters, administrative records, open-ended survey responses, and spatial data. These data are high-dimensional; representation learning helps turn them into variables that can be used in econometric and choice models.

Second, many environmental behaviours are heterogeneous. Households differ in environmental concern, risk perceptions, income constraints, trust in institutions, and familiarity with technologies. Clustering and latent variable models can help describe this heterogeneity.

Third, discrete choice experiments often contain psychometric indicators and attitudinal questions. These are not choices themselves, but they are often useful for modelling class membership, latent variables, or preference heterogeneity.

Fourth, choice modelling increasingly uses text, images, GPS traces, route histories, and other nontraditional data. These data require representation learning before they can be connected to utility functions.

Fifth, unsupervised learning can support scientific discovery. It can reveal patterns that were not specified in advance, which is useful for generating hypotheses. But these hypotheses must later be validated with separate data, theory, or experimental design.

2 Representation comes before clustering

A common mistake is to treat clustering as the first step. It is not. Clustering is meaningful only after deciding how the data are represented and how similarity is measured.

Suppose respondent n is described by a vector

$$x_n = (\text{income}_n, \text{age}_n, \text{environmental attitude}_n, \text{car ownership}_n, \text{distance to transit}_n).$$

A clustering algorithm needs to know whether two respondents are close. The simplest answer is Euclidean distance:

$$d(n, m) = \sqrt{\sum_{p=1}^{P} (x_{np} - x_{mp})^2}.$$

But this distance depends on scale. Income measured in euros can dominate a Likert-scale attitude variable measured from 1 to 5. Before clustering, we usually standardize continuous variables:

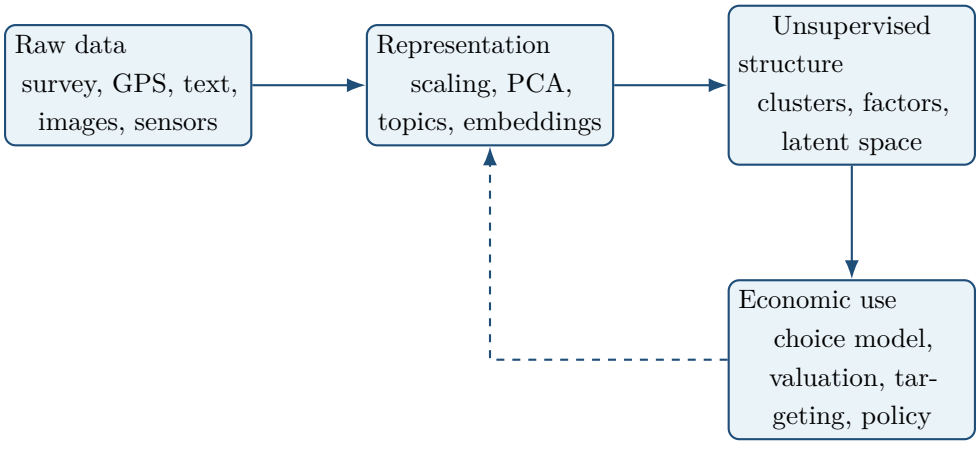
$$z_{np} = \frac{x_{np} - \bar{x}_p}{s_p}.$$

For mixed data with categorical and continuous variables, Euclidean distance may be inappropriate. Alternatives include Gower distance, matching distance for binary variables, cosine distance for text vectors, correlation distance for profiles, spatial distances for geographic data, and learned distances from embeddings.

A frequent error

Do not run k-means on a raw survey file and interpret the clusters as behavioural types. K-means assumes continuous variables, meaningful Euclidean distance, and roughly spherical clusters. These assumptions are often violated in environmental and choice data.

The representation step asks: what object should be compared? Possible objects include respondents, households, trips, choice tasks, alternatives, products, locations, firms, policy bundles, open-ended text responses, or images. The answer should follow the research question.



3 Principal component analysis

Principal component analysis (PCA) is a linear dimension-reduction method. It replaces a set of correlated variables with a smaller number of orthogonal components that explain as much variance as possible.

Let X be an $n \times p$ matrix of standardized variables. PCA finds directions $w_1, w_2, \ldots$ such that the first component

$$z_1 = Xw_1$$

has maximum variance subject to $\|w_1\| = 1$. The second component maximizes variance subject to being orthogonal to the first, and so on. Equivalently, PCA solves an eigenvalue problem for the covariance matrix:

$$S = \frac{1}{n-1} X^\top X.$$

If

$$Sw_k = \lambda_k w_k,$$

then w_k is the loading vector for component k, and λ_k is the variance explained by that component.

3.1 Interpretation

In environmental economics, PCA can be used to create indices from correlated environmental attitudes, local environmental quality measures, landscape metrics, or remote-sensing bands. In choice modelling, PCA may summarize psychometric indicators or contextual variables before entering a class membership model or utility function.

However, PCA is not a latent variable model in the behavioural sense. It is a variance-maximizing linear projection. A principal component can be useful, but it is not automatically an attitude, belief, norm, or preference.

PCA in one sentence

PCA finds low-dimensional linear summaries of high-dimensional data; it does not, by itself, prove the existence of behavioural constructs.

3.2 Choosing the number of components

Common rules include:

- Keep enough components to explain a target share of variance, such as 80 or 90 percent.
- Inspect a scree plot and look for an elbow.
- Use cross-validation if the components feed into a predictive model.

- Use substantive interpretability: do the retained components make sense?

In choice modelling, the number of components should also be judged by whether the resulting model has plausible signs, stable estimates, and meaningful behavioural interpretation.

3.3 R implementation with `recipes`

The `tidymodels` ecosystem provides a clean way to include PCA inside a reproducible preprocessing workflow. The official `recipes` documentation recommends standardization before PCA because PCA is scale-sensitive.

```r
# install.packages(c("tidyverse", "tidymodels", "factoextra"))
library(tidyverse)
library(tidymodels)
library(factoextra)

set.seed(123)

respondents <- tibble(
  id = 1:600,
  income = rlnorm(600, log(45000), 0.45),
  age = round(rnorm(600, 45, 14)),
  env_concern = rnorm(600, 0, 1),
  climate_risk = 0.7 * env_concern + rnorm(600, 0, 0.8),
  institutional_trust = rnorm(600, 0, 1),
  technology_openness = rnorm(600, 0, 1),
  car_dependency = rnorm(600, 0, 1)
)

pca_recipe <- recipe(~ income + age + env_concern + climate_risk +
                       institutional_trust + technology_openness +
                       car_dependency,
                     data = respondents) %>%
  step_normalize(all_numeric_predictors()) %>%
  step_pca(all_numeric_predictors(), num_comp = 3)

pca_prepped <- prep(pca_recipe)
pca_scores <- bake(pca_prepped, new_data = NULL)

head(pca_scores)
# Use tidy(pca_prepped, number = 2) to inspect loadings.
tidy(pca_prepped, number = 2)
```

Leakage warning

If PCA is used inside a supervised prediction workflow, estimate the PCA rotation only on the training data and apply that rotation to validation or test data. Estimating PCA

on the full dataset before splitting leaks information from the test set.

4 Nonlinear dimension reduction: t-SNE and UMAP

PCA is linear. Environmental and choice data are often nonlinear. Methods such as t-SNE and UMAP are popular for visualizing high-dimensional data in two or three dimensions.

These methods can be useful for inspecting structure in embeddings, attitude batteries, image features, GPS-derived behaviour profiles, and high-dimensional environmental indicators. However, they should be used with care.

4.1 What these methods are good for

They are useful for visualization, diagnosis, and hypothesis generation. For example, after embedding open-ended survey responses about a climate policy, UMAP may reveal clusters of comments about cost, fairness, distrust, biodiversity, or local impacts. After embedding satellite image patches, UMAP may reveal land-cover patterns. After embedding GPS trajectories, it may reveal behavioural rhythms.

4.2 What these methods are not good for

They should not be used as definitive evidence that discrete groups exist. Local neighbourhoods are often more meaningful than global distances. Results can change with hyperparameters and random seeds. Axes have no simple interpretation. Distances in the two-dimensional map should not be treated as economic distances without further validation.

```r
# install.packages("umap")
library(umap)

X <- respondents %>%
  select(income, age, env_concern, climate_risk,
         institutional_trust, technology_openness, car_dependency) %>%
  scale()

set.seed(123)
umap_fit <- umap(X)

umap_df <- as_tibble(umap_fit$layout, .name_repair = ~c("UMAP1", "UMAP2")) %>%
  bind_cols(respondents %>% select(id, env_concern, car_dependency))

ggplot(umap_df, aes(UMAP1, UMAP2, colour = env_concern)) +
  geom_point(alpha = 0.7) +
  theme_minimal()
```

5 Clustering

Clustering tries to group observations so that members of the same group are similar and members of different groups are dissimilar. Clustering is often the first unsupervised method economists encounter, but it is also one of the easiest to misuse.

5.1 K-means

K-means partitions the observations into K clusters. It solves:

$$\min_{C_1,\dots,C_K} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2,$$

where μ_k is the centroid of cluster k. The algorithm alternates between assigning observations to their nearest centroid and updating centroids.

Athey and Imbens describe the algorithm as repeatedly assigning each unit to the closest centroid and then updating the centroid as the average of the points assigned to that cluster. They also emphasize that choosing K is hard because there is no natural cross-validation benchmark in purely unsupervised clustering.

5.2 Hierarchical clustering

Hierarchical clustering creates a tree of nested clusters. Agglomerative clustering starts with each observation as its own cluster and merges clusters step by step. The analyst chooses a distance metric and a linkage rule, such as complete, average, or Ward linkage.

Hierarchical clustering is often useful in environmental economics because dendrograms can reveal nested structure: households within lifestyle groups, municipalities within vulnerability profiles, or species/conservation units within threat profiles.

5.3 K-medoids and mixed data

K-medoids is similar to k-means but represents each cluster by an actual observation rather than a centroid. It is more robust to outliers. With Gower distance, it can handle mixed continuous, binary, ordinal, and categorical data better than raw k-means.

5.4 Gaussian mixture models

A Gaussian mixture model assumes the data are generated by a mixture of distributions:

$$p(x_i) = \sum_{s=1}^{S} \pi_s \phi(x_i; \mu_s, \Sigma_s),$$

where π_s is the mixing weight of component s. This is closer to the econometric idea of latent classes because membership is probabilistic rather than deterministic. However, ordinary Gaussian mixture models are still models of the covariates, not models of choices.

5.5 DBSCAN and density-based clustering

Density-based clustering groups observations that lie in dense regions separated by sparse regions. DBSCAN can detect irregularly shaped clusters and identify noise points. It is useful when the number of clusters is unknown and outliers are meaningful. But it can be sensitive to the radius and minimum-points hyperparameters, especially in high-dimensional data.

6 How to validate clusters

Clustering validation has two sides: statistical validation and substantive validation.

6.1 Statistical validation

Common statistical checks include:

- Within-cluster sum of squares and elbow plots.
- Silhouette width.
- Cluster stability under resampling.
- Agreement across algorithms.
- Separation in holdout data, if a predictive downstream task exists.

The silhouette value for observation i compares average distance to its own cluster with average distance to the nearest other cluster. Values near 1 indicate good separation, values near 0 indicate boundary observations, and negative values suggest possible misclassification.

```
# install.packages("cluster")
library(cluster)

X_scaled <- respondents %>%
  select(income, age, env_concern, climate_risk,
         institutional_trust, technology_openness, car_dependency) %>%
  scale()

set.seed(123)
wss <- map_dbl(1:8, function(k) {
  kmeans(X_scaled, centers = k, nstart = 30)$tot.withinss
})

```

```
tibble(k = 1:8, wss = wss) %>%
  ggplot(aes(k, wss)) +
  geom_line() +
  geom_point() +
  theme_minimal()

set.seed(123)
km3 <- kmeans(X_scaled, centers = 3, nstart = 50)
sil3 <- silhouette(km3$cluster, dist(X_scaled))
mean(sil3[, 3])
```

6.2 Substantive validation

For economists, the decisive question is not whether the clusters are neat. It is whether they are meaningful for the scientific or policy problem. Ask:

- Do clusters differ in plausible ways on variables not used to construct them?
- Are clusters stable across samples, survey waves, or regions?
- Do clusters predict choices, policy support, or adoption behaviour out of sample?
- Do clusters correspond to interpretable mechanisms rather than arbitrary data geometry?
- Would a policy based on these clusters be ethically and legally defensible?

Do not overinterpret cluster labels

Labels such as "green idealists", "cost-sensitive households", or "technology sceptics" are analyst-created summaries. They are not directly observed facts. Use them as provisional interpretive tools, not as evidence of fixed psychological types.

7 Clusters versus latent classes in choice modelling

This distinction is central for choice modelling.

A clustering algorithm groups respondents using observed covariates. A latent class choice model estimates segments from the choice likelihood itself, usually with class-specific utility parameters. For individual n choosing alternative i from choice set C_n, a latent class model has probability:

$$P_{ni} = \sum_{s=1}^{S} \pi_{ns} P_{ni|s},$$

where π_{ns} is the probability that individual n belongs to class s, and $P_{ni|s}$ is the class-specific choice probability.

A class membership model may be written as:

$$\pi_{ns} = \frac{\exp(g_s(z_n))}{\sum_{r=1}^{S} \exp(g_r(z_n))},$$

where z_n contains respondent characteristics, attitudes, contextual variables, or learned representations.

Within each class, the utility may be:

$$V_{ni|s} = \beta_{s,\text{cost}}\text{cost}_{ni} + \beta_{s,\text{time}}\text{time}_{ni} + \beta_{s,\text{env}}\text{env}_{ni}.$$

Class-specific willingness to pay for an environmental attribute is:

$$WTP_{s,\text{env}} = -\frac{\beta_{s,\text{env}}}{\beta_{s,\text{cost}}}.$$

Issue	Algorithmic clustering	Latent class choice model
Object being modelled	Similarity among covariates	Choice probabilities and preference heterogeneity
Membership	Usually deterministic	Probabilistic
Uses choices directly?	Usually no, unless choices are included as features	Yes, through the likelihood
Economic interpretation	Indirect and exploratory	Direct through class-specific utility parameters
WTP and welfare	Not available by itself	Available if model is RUM-consistent and cost coefficient is identified
Validation	Cluster separation and stability	Fit, prediction, parameter signs, WTP plausibility, policy validity
Main risk	Reifying arbitrary groups	Overfitting or overinterpreting too many classes

Torres Lahoz et al. (2023) show why this distinction matters. Their latent class choice model for car-sharing uses machine learning to construct latent variables from attitudinal indicators, but the final model remains anchored in a random utility framework. Their results suggest that attitudinal information changes the class structure and helps create more realistic segments. This is an example of ML enriching a choice model rather than replacing it.

Liebe and Meyerhoff (2021) similarly argue that unsupervised ML is closely related to latent class and latent variable modelling in social-science choice research. But the relationship is conceptual, not identical: clustering can inspire a latent class specification, but it does not estimate a random utility model.

8 Latent variables, attitudes, and learned representations

Many important drivers of environmental and transport choices are not directly observed: environmental concern, trust, climate risk perception, status quo bias, place attachment, moral norms, technology anxiety, perceived control, or identity. Choice modellers often represent these with latent variables.

A simple integrated choice and latent variable (ICLV) structure is:

$$x_n^* = \gamma^\top z_n + \eta_n,$$

$$I_{np} = \alpha_p x_n^* + \nu_{np},$$

$$U_{ni} = V_{ni} + \lambda_i x_n^* + \varepsilon_{ni},$$

where x_n^* is a latent construct, I_{np} is an indicator response, and U_{ni} is utility.

This is powerful because it connects attitudes to choices while accounting for measurement error. The problem is specification. Which indicators measure which construct? Should relationships be linear? Are there interactions? Do constructs vary across classes?

Machine learning suggests a more flexible representation:

$$r_n = f_\theta(I_n, z_n),$$

where r_n is a learned latent representation of respondent n, constructed from indicators I_n and characteristics z_n. This representation can enter class membership or utility:

$$\pi_{ns} = \text{softmax}_s(\alpha_s + \delta_s^\top r_n),$$

or

$$V_{ni} = \beta^\top x_{ni} + \lambda_i^\top r_n.$$

Torres Lahoz et al. (2023) introduce artificial neural networks to formulate latent constructs in latent class choice models. Their approach is attractive because it allows nonlinear relationships between attitudinal indicators and class membership while preserving the random utility interpretation of the choice model. Wong et al. (2018) take a different route by using a restricted Boltzmann machine to infer latent behavioural representations from choice-related data, even when psychometric indicators are unavailable. Their paper is useful because it shows both the promise and the danger: richer latent representations can improve fit, but the number of parameters and interpretability burden can increase quickly.

Research design implication

If you plan to collect attitudinal indicators for a stated preference survey, decide in advance whether they will be used only descriptively, as inputs to class membership, as indicators in an ICLV model, or as inputs to a learned representation. The survey design, sample

size, and validation strategy depend on this choice.

9 Representation learning for text

Environmental economics often contains text: open-ended survey responses, public comments, news, policy documents, environmental impact statements, social media, planning documents, and stakeholder submissions. Choice modelling increasingly encounters text in descriptions of alternatives, user reviews, travel narratives, and stated preference debriefing responses.

A basic representation of text is a document-term matrix. Let C be an $N \times T$ matrix, where C_{it} counts whether word t appears in document i. Athey and Imbens describe this as a natural starting point for text analysis and discuss both unsupervised topic models and supervised text prediction. When T is large relative to N, regularization or dimension reduction is essential.

9.1 Topic models

A topic model represents each document as a mixture of latent topics. In latent Dirichlet allocation (LDA), each topic is a distribution over words, and each document has a distribution over topics. In environmental economics, topics might represent cost concerns, climate risk, biodiversity, distrust, fairness, property rights, local identity, or health impacts.

For respondent n, the estimated topic proportions can become covariates:

$$\theta_n = (\theta_{n1}, \ldots, \theta_{nK}).$$

These topic proportions can enter a class membership model, a treatment-effect model, or a choice model:

$$\pi_{ns} = \text{softmax}_s(\alpha_s + \delta_s^\top \theta_n).$$

```r
# install.packages(c("tidytext", "topicmodels", "tm"))
library(tidyverse)
library(tidytext)
library(topicmodels)

comments <- tibble(
  id = 1:8,
  text = c(
    "The policy is too expensive for low income households",
    "Biodiversity and habitat protection should be prioritized",
    "I do not trust the government to spend the tax well",
    "Flood protection is urgent because climate risk is rising",
    "The programme should protect farmers and local jobs",
    "Cleaner air is worth paying for because health matters",
    "Costs are unfair for rural households",
    "Wetlands and species recovery are important"
```

```
  )
)

# This tiny example is only pedagogical. Real topic models need many documents.
dtm <- comments %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word") %>%
  count(id, word) %>%
  cast_dtm(document = id, term = word, value = n)

set.seed(123)
lda_fit <- LDA(dtm, k = 3, control = list(seed = 123))

top_terms <- tidy(lda_fit, matrix = "beta") %>%
  group_by(topic) %>%
  slice_max(beta, n = 5) %>%
  arrange(topic, desc(beta))

top_terms

doc_topics <- tidy(lda_fit, matrix = "gamma")
doc_topics
```

9.2 Embeddings

Topic models are interpretable but limited. Modern representation learning often uses embeddings: dense vectors that place semantically similar words, sentences, documents, images, products, or alternatives close together in a vector space.

An embedding model maps an object o to a vector:

$$e(o) \in \mathbb{R}^d.$$

For text, o might be a word, sentence, or entire document. For images, it might be a street-view image or landscape photograph. For choice alternatives, it might be a product description, route description, or recreation site description.

Embeddings can be used in several ways:

- Cluster open-ended responses after a survey.
- Create respondent-level summaries of concerns or motivations.
- Build features for utility functions from product or policy descriptions.
- Compare textual descriptions of alternatives.
- Retrieve similar comments or policy cases for qualitative analysis.

```python
# Python sketch: sentence embeddings for open-ended responses
# pip install sentence-transformers umap-learn scikit-learn

from sentence_transformers import SentenceTransformer
from sklearn.cluster import KMeans
import umap

texts = [
    "The policy is too expensive for low income households",
    "Biodiversity and habitat protection should be prioritized",
    "I do not trust the government to spend the tax well",
    "Flood protection is urgent because climate risk is rising",
]

model = SentenceTransformer("all-MiniLM-L6-v2")
E = model.encode(texts, normalize_embeddings=True)

clusters = KMeans(n_clusters=2, random_state=123, n_init=20).fit_predict(E)
layout = umap.UMAP(random_state=123).fit_transform(E)
```

Embeddings are not neutral

Embeddings inherit patterns from training data and design choices. They can encode social bias, language-specific meaning, and domain mismatch. For policy use, validate embeddings against human coding, known constructs, or downstream performance.

10 Matrix factorization and recommender-style representations

Choice data sometimes have a user-item structure: consumers choose products, households choose tariffs, tourists choose sites, commuters choose routes, or participants choose policy bundles. In large sparse settings, matrix factorization can learn latent features of users and alternatives.

Let Y_{nu} be a measure of interaction between person n and item u, such as a rating, purchase, visit, or choice. Matrix factorization approximates:

$$Y_{nu} \approx a_n + b_u + p_n^\top q_u,$$

where p_n is a latent vector for person n and q_u is a latent vector for item u.

In economics, the key question is whether these latent vectors can be connected to utility and welfare. Athey and Imbens discuss matrix completion and consumer choice as part of the modern ML toolkit for economists. Work at the intersection of recommender systems and demand estimation often combines latent factors with price variation and utility maximization so that counterfactual price and welfare questions remain possible.

For choice modelling, matrix factorization is promising when the choice set is large and sparse, such as online product choice, tourism destination choice, recreation site choice, airline itinerary choice, or digital mobility services. But without economic structure, latent factors may predict well while providing limited welfare interpretation.

11 Environmental economics applications

11.1 Environmental attitude segmentation

A common environmental valuation survey includes attitudinal statements such as:

- Climate change is a serious risk.
- Biodiversity protection is a moral duty.
- I trust public agencies to manage environmental programmes.
- Environmental taxes are unfair to low-income households.
- I prefer local conservation projects over national programmes.

PCA, factor analysis, clustering, or learned representations can summarize these responses. These summaries may be used to describe the sample, explain support for policies, or model heterogeneity in willingness to pay.

A careful workflow is:

1. Pre-register or document which attitudinal constructs are expected.
2. Inspect reliability and dimensionality.
3. Estimate candidate representations on a training sample.
4. Validate against held-out responses, choices, or known groups.
5. Use representations in a choice model only after checking signs, WTP, and stability.

11.2 Remote sensing and spatial environmental data

Remote-sensing data may contain many bands, indices, textures, and time-series summaries. PCA or autoencoders can compress these features into land-cover or habitat representations. Clustering can identify landscape types, drought patterns, vegetation regimes, or pollution exposure profiles. These representations can enter environmental policy evaluation, land-use models, or choice models of recreation and residential location.

11.3 Energy and climate behaviour

Smart-meter data and energy-use profiles can be clustered into load-shape types. These types may predict responsiveness to dynamic pricing, solar adoption, heat-pump adoption, or demand-response programmes. However, if the goal is causal policy evaluation, clusters alone are not enough. They must be combined with credible identification, randomized interventions, or causal ML methods, which will be covered in Chapter 5.

11.4 Open-ended responses and deliberative valuation

Environmental valuation increasingly includes deliberative elements, debriefing questions, and open-ended comments. Topic models and embeddings can summarize these data. For example, comments in a conservation DCE may reveal latent concerns about fairness, local livelihoods, biodiversity, taxation, or trust. These can be used to explain protest responses, attribute non-attendance, or class membership.

12 Choice modelling applications

12.1 Exploratory heterogeneity analysis

Before estimating a mixed logit or latent class model, clustering can help explore whether respondents differ systematically in attitudes, experience, context, or constraints. This can guide interactions, class membership variables, or priors.

But a cluster should not simply be inserted into a utility function and treated as a revealed preference segment without caution. Because clusters are estimated, they carry uncertainty. A two-step workflow can understate standard errors and overstate confidence. When possible, estimate segmentation jointly through a latent class model.

12.2 Initialization for latent class models

Clustering can provide starting values or initial segmentation for latent class choice models. This is computationally useful, especially with many classes or complex class membership functions. The final model, however, should be evaluated through the choice likelihood and behavioural plausibility.

12.3 Attitudes in class membership

Torres Lahoz et al. show a modern version of this idea: learned latent constructs from attitudinal indicators enter the class membership model of a latent class choice model. This preserves class-specific utility interpretation while allowing nonlinear attitude structures.

12.4 Text and image alternatives

Van Cranenburgh et al. argue that ML opens choice modelling to data types such as text and images, which are difficult to use in standard theory-driven models. This is important for environmental choice tasks involving landscape images, street-view images, product descriptions, policy text, or virtual-reality experiments. A representation model can turn an image or text into features, and the choice model can then use these features as attributes.

The key design challenge is to keep the connection to economic interpretation. If a convolutional neural network predicts that respondents prefer one landscape image over another, the analyst still needs to know which visual attributes drive utility: tree cover, water visibility, biodiversity cues, development intensity, perceived safety, or accessibility.

12.5 Portfolio and participatory value evaluation experiments

Participatory Value Evaluation (PVE) experiments ask respondents to choose a portfolio of policies subject to constraints. Hernandez et al. use association rules and random forests to assist the specification of portfolio choice models. This is an example of unsupervised or weakly supervised pattern discovery helping the analyst identify interactions and specification improvements in complex choice data.

13 Worked R lab: from attitudes to clusters to choice models

This lab simulates respondents with latent environmental and technology orientations, creates an environmental choice experiment, clusters respondents using observed characteristics and attitudes, and then estimates a choice model with cluster interactions.

13.1 Step 1: simulate respondents

```r
library(tidyverse)
library(cluster)
library(mlogit)

set.seed(42)
N <- 600

respondents <- tibble(
  id = 1:N,
  income = rlnorm(N, log(45000), 0.45),
  age = pmax(18, round(rnorm(N, 45, 14))),
  rural = rbinom(N, 1, 0.35),
  env_att = rnorm(N, 0, 1),
  trust = rnorm(N, 0, 1),
  tech = rnorm(N, 0, 1),
```

```
  cost_stress = -0.000015 * income + 0.4 * rural + rnorm(N, 0, 0.5)
) %>%
  mutate(
    true_type = case_when(
      env_att > 0.6 & cost_stress < 0.2 ~ "green_supporter",
      cost_stress > 0.4 ~ "cost_constrained",
      TRUE ~ "moderate"
    )
  )
```

13.2 Step 2: cluster respondents

```
X <- respondents %>%
  select(income, age, rural, env_att, trust, tech, cost_stress) %>%
  mutate(income = log(income)) %>%
  scale()

set.seed(42)
km <- kmeans(X, centers = 3, nstart = 50)

respondents <- respondents %>%
  mutate(cluster = factor(km$cluster))

# Statistical diagnostic
sil <- silhouette(km$cluster, dist(X))
mean(sil[, 3])

# Substantive diagnostic
respondents %>%
  group_by(cluster) %>%
  summarise(
    n = n(),
    income = mean(income),
    age = mean(age),
    rural = mean(rural),
    env_att = mean(env_att),
    trust = mean(trust),
    tech = mean(tech),
    cost_stress = mean(cost_stress),
    .groups = "drop"
  )
```

13.3 Step 3: simulate a choice experiment

Each respondent faces six choice tasks with three alternatives: status quo, basic conservation programme, and ambitious conservation programme. The utility function is simulated so

that environmentally concerned respondents value biodiversity more, while cost-constrained respondents are more cost-sensitive.

```r
alts <- c("SQ", "Basic", "Ambitious")

choice_data <- expand_grid(
  id = respondents$id,
  task = 1:6,
  alt = alts
) %>%
  left_join(respondents, by = "id") %>%
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = case_when(
      alt == "SQ" ~ 0,
      alt == "Basic" ~ sample(c(10, 20, 30), n(), replace = TRUE),
      alt == "Ambitious" ~ sample(c(30, 50, 70), n(), replace = TRUE)
    ),
    biodiversity = case_when(
      alt == "SQ" ~ 0,
      alt == "Basic" ~ sample(c(1, 2), n(), replace = TRUE),
      alt == "Ambitious" ~ sample(c(2, 3), n(), replace = TRUE)
    ),
    co2 = case_when(
      alt == "SQ" ~ 0,
      alt == "Basic" ~ sample(c(1, 2), n(), replace = TRUE),
      alt == "Ambitious" ~ sample(c(2, 3), n(), replace = TRUE)
    ),
    asc = case_when(
      alt == "SQ" ~ 0,
      alt == "Basic" ~ 0.15,
      alt == "Ambitious" ~ 0.05
    ),
    beta_cost = -0.06 - 0.04 * cost_stress,
    beta_bio = 0.45 + 0.30 * env_att,
    beta_co2 = 0.25 + 0.10 * env_att,
    V = asc + beta_cost * cost + beta_bio * biodiversity + beta_co2 * co2
  ) %>%
  group_by(chid) %>%
  mutate(
    p = exp(V) / sum(exp(V)),
    chosen_alt = sample(alt, size = 1, prob = p),
    choice = as.integer(alt == chosen_alt)
  ) %>%
  ungroup()
```

13.4 Step 4: estimate a model with cluster interactions

```r
mdata <- mlogit.data(
  choice_data,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

mnl_base <- mlogit(
  choice ~ cost + biodiversity + co2 | 0,
  data = mdata
)

mnl_cluster <- mlogit(
  choice ~ cost + biodiversity + co2 +
    cost:cluster + biodiversity:cluster + co2:cluster | 0,
  data = mdata
)

summary(mnl_base)
summary(mnl_cluster)
```

13.5 Step 5: interpret cautiously

The cluster-interaction model may reveal that some clusters are more cost-sensitive or more biodiversity-sensitive. But this is a two-step model. The clusters were estimated first, then treated as observed. For publication-quality work, compare this with a latent class choice model, a mixed logit model, or a model where attitudes enter the class membership function directly.

Concept check

Why does using estimated clusters as fixed respondent types usually understate uncertainty? What would change if class membership were estimated jointly with the choice model?

14 Optional Python lab: embeddings for open-ended responses

Python is useful for modern NLP and embedding workflows. The following pattern can be used for open-ended comments in environmental valuation surveys.

```python
import pandas as pd
from sentence_transformers import SentenceTransformer
from sklearn.cluster import KMeans
```

```python
from sklearn.metrics import silhouette_score

comments = pd.DataFrame({
    "id": [1, 2, 3, 4, 5, 6],
    "text": [
        "I support the policy because it protects biodiversity",
        "The tax is too expensive for my household",
        "I do not trust the government to deliver the project",
        "Flood protection is necessary because climate risk is increasing",
        "Rural households should not pay more than urban households",
        "Species recovery and habitat restoration are worth paying for"
    ]
})

model = SentenceTransformer("all-MiniLM-L6-v2")
E = model.encode(comments["text"].tolist(), normalize_embeddings=True)

km = KMeans(n_clusters=3, random_state=123, n_init=20)
comments["cluster"] = km.fit_predict(E)

# Meaningful only with enough documents; tiny example is pedagogical.
print(comments)
```

The resulting clusters can be inspected qualitatively, compared with hand coding, and summarized at the respondent level. They can also be used as inputs to a protest-response model, a class membership model, or an analysis of attribute non-attendance.

15 Common mistakes

15.1 Mistake 1: Treating clusters as discovered truths

Clusters are outputs of an algorithm, a representation, a distance metric, and a choice of hyperparameters. Change any of these and the clusters may change.

15.2 Mistake 2: Ignoring scaling and mixed data types

If income, distance, age, and Likert-scale attitudes are all entered raw into Euclidean k-means, the resulting clusters may reflect measurement units more than behaviour.

15.3 Mistake 3: Using PCA components without interpretation

A component that explains variance may not correspond to an economic construct. Always inspect loadings and validate components against external variables or theory.

15.4 Mistake 4: Confusing visualization with evidence

UMAP and t-SNE plots can be persuasive, but they are not formal proof of groups. They are visualization tools.

15.5 Mistake 5: Using text embeddings without validation

Embeddings can be powerful but opaque. Validate them against human-coded categories, known survey constructs, or downstream performance.

15.6 Mistake 6: Turning unsupervised outputs into policy rules too quickly

Segment-specific policies can be useful, but clustering can encode protected characteristics, spatial inequality, or historical disadvantage. Ethical review is part of the modelling workflow.

16 End-of-chapter exercises

Exercise 1: PCA interpretation

Take an environmental attitudes dataset with at least six Likert-scale items. Run PCA after standardizing the variables. Report the first three components, their loadings, and a substantive interpretation. Then write one paragraph explaining why your interpretation could be wrong.

Exercise 2: Cluster stability

Using the same dataset, run k-means with $K = 2, 3, 4, 5$. For each value of K, compute silhouette width. Then bootstrap the data and examine whether cluster assignments are stable. Which K would you choose, and why?

Exercise 3: From clusters to choice models

Estimate a simple MNL model for a stated preference dataset. Then add interactions between attributes and cluster membership. Compare signs, log-likelihood, prediction, and WTP. Discuss whether the clusters provide behavioural insight or only improve fit.

Exercise 4: Latent class alternative

Write the likelihood for a two-class latent class choice model. Explain how it differs from first clustering respondents and then estimating two separate MNL models.

Exercise 5: Text representation

Collect 100 open-ended comments about an environmental policy. Fit an LDA topic model or compute sentence embeddings and cluster them. Compare the resulting categories with a manual coding of 30 comments.

Exercise 6: Research design

Design a stated preference survey on willingness to pay for river restoration. Include a plan for collecting attitudes, using open-ended comments, and incorporating unsupervised learning into a choice model. Specify which analyses are exploratory and which are confirmatory.

17 Chapter summary

Unsupervised learning helps economists organize information when labels are absent or incomplete. PCA and related methods reduce dimensionality. Clustering describes groups. Topic models and embeddings transform text into analyzable variables. Matrix factorization learns latent features of users and alternatives. In environmental economics, these tools help study attitudes, spatial patterns, energy profiles, policy narratives, and ecological data. In choice modelling, they help with heterogeneity, latent classes, class membership, text and image attributes, and model specification.

The central lesson is caution with usefulness. Unsupervised learning can discover patterns, but it does not by itself identify preferences, causal effects, or welfare. To use it well, combine it with economic theory, validation, and appropriate choice models.

18 Reading guide

Start with Athey and Imbens (2019) for the economist's overview of unsupervised learning, matrix completion, and text analysis. Read James et al. (2021, 2023) or Hastie, Tibshirani, and Friedman (2009) for statistical learning foundations. For choice modelling, read van Cranenburgh et al. (2022) and the machine-learning chapter in Hess and Daly (2024). For latent classes and attitudes, read Torres Lahoz et al. (2023) and the hybrid choice model chapters in Hess and Daly (2024). For text in R, use the `tidytext` documentation and the `topicmodels` package. For PCA and clustering workflows in R, use `recipes`, `factoextra`, `cluster`, and `umap`. For modern embeddings, Python tools such as `sentence-transformers`, `scikit-learn`, and `umap-learn` are usually more mature than native R options.

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Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 5

Causal Machine Learning

From potential outcomes and double machine learning to causal forests, policy learning, and welfare-aware choice applications.

Generated for self-study use

July 16, 2026

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How to study this chapter

This chapter explains how machine learning can be used when the target is not only prediction but a causal or policy quantity: the effect of a regulation, subsidy, information intervention, conservation program, price change, infrastructure investment, warning label, or choice architecture. Read it slowly. Causal machine learning is powerful, but it is also easy to misuse because a flexible algorithm cannot repair a weak research design.

Study the chapter in three passes. In the first pass, focus on the causal language: potential outcomes, estimands, identification, confounding, overlap, and heterogeneity. In the second pass, study double machine learning and causal forests as algorithms that combine predictive ML with causal scores. In the third pass, run the R labs and ask a design question: what assumptions would make this estimate causal in a real environmental economics or choice modelling study?

Learning objectives

By the end of this chapter, you should be able to:

1. Distinguish prediction, association, causal estimation, structural choice modelling, and welfare analysis.
2. Define average treatment effects, conditional average treatment effects, treatment-on-the-treated effects, and policy values.
3. State the key identification assumptions behind selection-on-observables causal ML.
4. Explain why naive ML plug-in estimators can be biased for causal parameters.
5. Describe the logic of double/debiased machine learning, orthogonal scores, and cross-fitting.
6. Use causal forests to estimate and summarize heterogeneous treatment effects.
7. Apply causal ML to environmental policy, conservation programs, energy demand, information treatments, and choice modelling.
8. Implement basic workflows in R using `DoubleML`, `mlr3`, and `grf`.

1 Why causal machine learning exists

Machine learning is excellent at estimating complex relationships. If the research task is to predict next month's electricity consumption, next year's deforestation risk, or the transport mode chosen in a repeated revealed preference dataset, flexible supervised learning can be very helpful. But many economic questions are not prediction questions. They are questions about interventions.

Examples:

- What is the effect of a conservation payment on landholders' adoption of biodiversity practices?
- What is the effect of real-time energy feedback on household electricity demand?
- What is the effect of a low-emission zone on vehicle kilometres travelled?
- What is the effect of providing climate-risk information on the probability of choosing flood insurance?
- What is the effect of reducing public transport travel time on mode choice?
- Which households should receive a targeted subsidy if the policy objective is maximum carbon reduction per euro?

A high-performing predictive model can answer questions such as "who currently adopts?" or "who is likely to choose the electric vehicle?" It does not automatically answer "who would adopt if treated?" or "what would happen if we changed the policy?" This is why causal machine learning has become important in economics. It uses ML to estimate high-dimensional nuisance functions, discover treatment-effect heterogeneity, or learn targeting rules, while preserving the causal logic of research design and identification.

Athey and Imbens (2019) frame this as a productive meeting point between ML and economics: predictive tools are useful, but economists often need causal, policy, and counterfactual quantities. Chernozhukov et al. (2018) provide one of the central technical bridges through double/debiased machine learning. Wager and Athey (2018) and Athey, Tibshirani, and Wager (2019) provide the forest-based bridge for heterogeneous effects.

Core message

Causal ML is not "ML plus causal language." It is causal inference that uses ML to estimate parts of the problem that are too complex for low-dimensional parametric models. Identification still comes from assumptions, experimental design, quasi-experimental design, or structural theory.

2 Causal questions in environmental economics and choice modelling

Environmental economics and choice modelling generate several kinds of causal questions.

2.1 Policy evaluation questions

A government may introduce an emissions standard, protected area, congestion charge, conservation payment, appliance rebate, or information campaign. The relevant question is often an average effect:

$$ATE = \mathbb{E}[Y(1) - Y(0)],$$

where $Y(1)$ is the outcome under the policy and $Y(0)$ is the outcome without it. Outcomes can be emissions, energy use, conservation adoption, mode share, crop yields, property values, subjective welfare, or choice probabilities.

2.2 Targeting questions

Environmental policies are often constrained by budgets and administrative capacity. A policy maker may ask:

> Which households, firms, farms, or regions should be targeted to maximize environmental gains per euro?

This requires heterogeneity. The central estimand becomes the conditional average treatment effect:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x].$$

If X includes household income, energy burden, baseline emissions, building age, political attitudes, social norms, and local infrastructure, then $\tau(x)$ describes how policy effects differ across contexts.

2.3 Choice-architecture questions

Choice modellers often study interventions inside or around choice tasks:

- providing information before a stated preference survey;
- changing default options;
- presenting environmental attributes visually rather than textually;
- changing labels, rankings, recommendations, or warning messages;
- changing travel time, price, carbon information, or reliability in a real choice environment.

Some of these interventions are randomized in stated preference designs. Others arise in revealed preference settings and require stronger assumptions. Causal ML can help estimate how such interventions affect choices, but welfare interpretation still requires caution.

2.4 Structural counterfactual questions

Discrete choice models often ask counterfactual questions such as "what happens if price changes?" or "what is the compensating variation from improved air quality?" These are causal in spirit, but the source of identification is not always the same as in treatment-effect designs. In a random utility model, counterfactuals depend on the specified utility function, the error structure, the choice set, and assumptions about stable preferences. Causal ML can complement these models but should not replace the structural assumptions needed for welfare analysis.

3 Potential outcomes and estimands

Let D_i denote treatment status for unit i: $D_i = 1$ if treated and $D_i = 0$ otherwise. Let $Y_i(1)$ and $Y_i(0)$ be the potential outcomes under treatment and control. We observe only one of them:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

This is the fundamental missing-data problem of causal inference. For every unit, one potential outcome is observed and the other is counterfactual.

3.1 Common estimands

Estimand	Formula	Example question
Average treatment effect	$\mathbb{E}[Y(1) - Y(0)]$	What is the average effect of an energy feedback program?
Average treatment effect on treated	$\mathbb{E}[Y(1) - Y(0) \mid D = 1]$	What was the effect on households that actually received the retrofit subsidy?
Conditional average treatment effect	$\mathbb{E}[Y(1) - Y(0) \mid X = x]$	For which households does the subsidy reduce energy use most?
Policy value	$\mathbb{E}[Y(\pi(X))]$	What outcome would result from treating households selected by rule π?
Choice probability effect	$\Pr(Y(1) = j) - \Pr(Y(0) = j)$	How does an information treatment change the probability of choosing the green option?

In choice modelling, Y may be a binary adoption decision, a chosen alternative, a continuous outcome after choice, or a welfare measure derived from a structural model. Be explicit about which one you are studying.

Retrieval practice 1

Without looking back, write definitions for ATE, ATT, and CATE. Then give one environmental economics example and one choice modelling example for each.

4 Identification before estimation

Causal ML is often taught through code. That is dangerous. The code estimates a statistical object. It becomes causal only under identification assumptions.

4.1 Consistency and SUTVA

Consistency means the observed outcome equals the potential outcome under the treatment actually received. If a farm receives the conservation payment, its observed outcome is $Y(1)$. If not, it is $Y(0)$.

The stable unit treatment value assumption, often abbreviated SUTVA, says that one unit's treatment does not change another unit's potential outcomes and that treatment is sufficiently well-defined. Environmental problems often challenge this assumption. Pollution, congestion, information, and conservation can spill over across space and networks.

4.2 Unconfoundedness

The selection-on-observables assumption is:

$$(Y(1), Y(0)) \perp D \mid X.$$

This says that after conditioning on observed covariates X, treatment assignment is as good as random. This assumption is plausible in some randomized or carefully designed observational settings. It is not plausible merely because X is large.

In environmental economics, unconfoundedness is questionable when policy adoption is driven by political pressure, unobserved environmental risk, expected future growth, administrative capacity, or voluntary participation. In choice modelling, unconfoundedness is questionable when travel time, price, and availability are chosen by firms or governments in response to demand.

4.3 Overlap

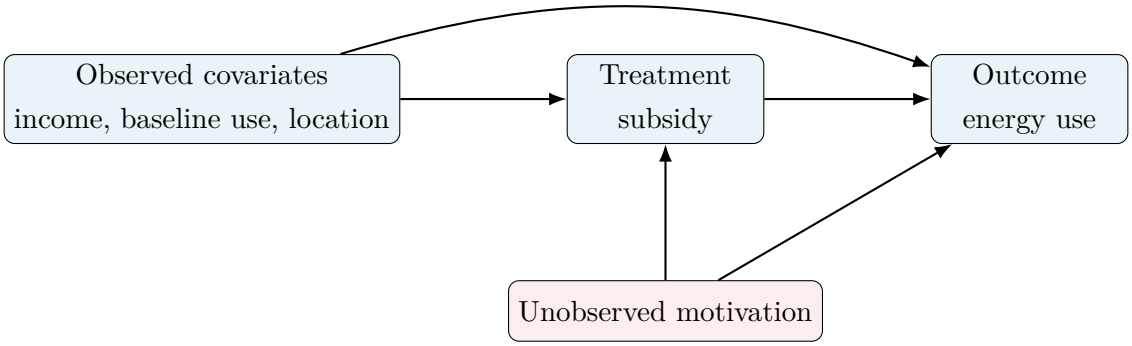
Overlap, also called positivity, requires:

$$0 < \mathbb{P}(D = 1 \mid X = x) < 1$$

for the relevant range of x. If all high-income households are treated and all low-income households are untreated, the data cannot reveal what would have happened to low-income households under treatment unless additional assumptions are imposed.

4.4 Causal diagrams

Directed acyclic graphs help express causal assumptions. They are not required for every analysis, but they are excellent tools for self-study because they force you to say what causes what.



The unobserved motivation node is a warning: if motivated households are more likely to apply for a subsidy and would have reduced energy use anyway, a naive comparison is biased. ML can use many observed covariates, but it cannot condition on unmeasured motivation unless it is proxied, instrumented, randomized, differenced out, or modelled structurally.

Design before algorithm

Before selecting a learner, write the causal estimand and the identifying assumption. A causal forest with thousands of trees is still a biased estimator if treatment assignment is confounded by unobserved factors.

5 Why naive ML is not enough

A common but flawed strategy is to fit a predictive model:

$$\hat{m}(d, x) \approx \mathbb{E}[Y \mid D = d, X = x]$$

and then compute:

$$\hat{m}(1, x) - \hat{m}(0, x).$$

This is sometimes useful, especially in randomized experiments. But in observational studies it can fail for several reasons.

5.1 Regularization bias

ML methods use regularization to improve prediction. Regularization is desirable for prediction but can bias causal parameters. For example, lasso may shrink a control variable coefficient toward zero. If that variable is a confounder, the treatment effect estimate can be biased.

Chernozhukov et al. (2018) formalize this problem: naively plugging ML estimates of nuisance functions into estimating equations can produce estimators that are not root-n consistent for low-dimensional causal parameters. The solution is not to avoid ML. The solution is to use scores that are insensitive, to first order, to small nuisance-estimation errors.

5.2 Overfitting

If the same observations are used both to learn complex nuisance functions and to estimate the treatment effect, overfitting can contaminate the causal estimate. Cross-fitting solves much of this problem by predicting each observation with nuisance models trained on other observations.

5.3 Bad target functions

Prediction algorithms minimize prediction loss. A treatment-effect estimator targets a contrast between potential outcomes. These are different targets. A model can predict Y well by learning the strong prognostic components of the outcome while learning little about treatment-effect heterogeneity.

Prediction target versus causal target

A predictor asks: "What outcome is likely for this observed unit?" A causal estimator asks: "How would the outcome change if treatment status were changed, holding the causal system fixed?" These are not the same task.

6 Double/debiased machine learning

Double/debiased machine learning, often abbreviated DML, is a general framework for using ML to estimate nuisance functions while retaining valid inference for a lower-dimensional causal parameter. The core ideas are:

1. Use ML for nuisance prediction.
2. Use an orthogonal score for the causal parameter.
3. Use sample splitting or cross-fitting so nuisance predictions are out-of-sample.

6.1 The partially linear regression model

A simple and important model is:

$$Y = \theta D + g(X) + \varepsilon,$$

where D is a treatment or policy variable, X are controls, $g(X)$ is an unknown function, and θ is the causal parameter of interest. Treatment may itself depend on X:

$$D = m(X) + v.$$

If treatment is conditionally exogenous given X, the parameter θ can be estimated by residualizing

both outcome and treatment:

$$\tilde{Y} = Y - \hat{\ell}(X), \quad \tilde{D} = D - \hat{m}(X),$$

where $\hat{\ell}(X)$ predicts Y from X, and $\hat{m}(X)$ predicts D from X. Then regress $\tilde{Y}$ on $\tilde{D}$. The point is not simply residualization; the point is that the resulting score is orthogonal to small nuisance-estimation errors.

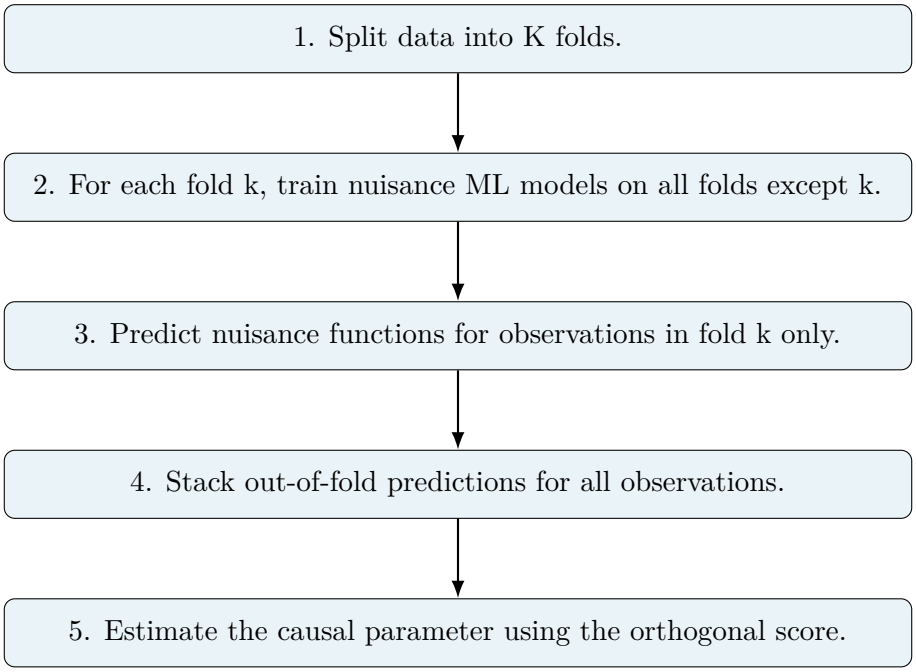
6.2 Orthogonal scores

A score $\psi(W; \theta, \eta)$ is Neyman-orthogonal if small errors in the nuisance function η have no first-order effect on the estimating equation for θ. Informally:

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta)]\big|_{\eta=\eta_0} = 0.$$

This property makes the final estimate robust to imperfect ML nuisance estimates. It does not make identification assumptions unnecessary.

6.3 Cross-fitting algorithm



6.4 When DML is a good fit

DML is useful when:

- the target parameter is low-dimensional, such as an average treatment effect or a coefficient in a partially linear model;

- confounders are many or nonlinear;
- flexible prediction of outcome and treatment assignment is needed;
- you want valid standard errors for the target parameter;
- treatment is continuous, binary, or instrumental-variable based, depending on the model class.

DML is not magic. It is not a substitute for randomized treatment, valid instruments, or credible quasi-experimental variation.

7 Meta-learners for heterogeneous effects

Sometimes the objective is not a single average effect but a function $\tau(x)$. Meta-learners turn ordinary supervised learning methods into treatment-effect estimators.

7.1 S-learner

Fit one model:

$$\hat{m}(x, d) \approx \mathbb{E}[Y \mid X = x, D = d].$$

Then estimate:

$$\hat{\tau}(x) = \hat{m}(x, 1) - \hat{m}(x, 0).$$

This is simple but can understate heterogeneity if the learner does not use treatment interactions strongly.

7.2 T-learner

Fit two models:

$$\hat{\mu}_1(x) \approx \mathbb{E}[Y \mid D = 1, X = x], \quad \hat{\mu}_0(x) \approx \mathbb{E}[Y \mid D = 0, X = x].$$

Then:

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x).$$

The T-learner is intuitive but can be unstable when one group is small.

7.3 X-learner

The X-learner, associated with Kunzel et al. (2019), imputes treatment effects for treated and untreated units and then learns these imputed effects. It can perform well when treatment and control group sizes are imbalanced.

7.4 R-learner

The R-learner, associated with Nie and Wager (2021), residualizes the outcome and treatment and learns heterogeneity from an objective that isolates the causal component of the signal. It is closely related in spirit to orthogonalization.

Learner	Main idea	Practical caution
S-learner	One model with treatment as a feature	May smooth away treatment heterogeneity
T-learner	Separate outcome models by treatment group	Can be unstable with small treated or control groups
X-learner	Impute individual treatment effects, then learn them	Useful under group imbalance, but depends on imputation quality
R-learner	Residualize treatment and outcome, then learn CATE	Needs good nuisance models and careful tuning
Causal forest	Tree ensemble designed for treatment heterogeneity	Requires overlap and careful interpretation of heterogeneity

8 Causal forests

Causal forests adapt random forests to treatment-effect estimation. A random forest for prediction partitions covariate space to reduce prediction error. A causal forest partitions covariate space to reveal heterogeneity in treatment effects.

8.1 Honesty

A central idea is honesty: use one part of the data to decide the tree splits and another part to estimate effects within leaves. Honesty reduces adaptive bias because the same outcome noise is not used both to choose the partition and to estimate the treatment effect.

8.2 Local comparison

For a target unit with covariates x, a causal forest constructs adaptive weights over training observations. Units that frequently fall in the same leaves as x receive higher weights. The treatment effect at x is estimated by a local treated-versus-control comparison adjusted for nuisance functions.

8.3 What causal forests produce

Typical outputs include:

- an average treatment effect estimate;
- individual-level or observation-level CATE estimates;
- standard errors for average effects and sometimes for CATE predictions;
- variable importance measures for heterogeneity;
- best linear projections of CATEs on covariates;
- ranked subgroups or policy rules.

Do not over-read CATEs

A causal forest CATE estimate is not an observed individual treatment effect. It is a noisy estimate of an average effect for units with similar covariates. Treat it as an estimated conditional mean, not as the known effect for a particular person or household.

9 Policy learning and targeting

Estimating heterogeneity is often only an intermediate step. A government or organization may need to allocate treatment. Let $\pi(X) \in \{0, 1\}$ be a policy rule that assigns treatment. The policy value is:

$$V(\pi) = \mathbb{E}[Y(\pi(X))].$$

A simple targeting rule treats units with $\hat{\tau}(X) > c$, where c is the cost threshold. More sophisticated policy learning methods choose a rule subject to budget, fairness, administrative simplicity, or political constraints. Athey and Wager (2021) study policy learning with observational data using doubly robust scores and regret guarantees.

In environmental economics, policy learning is relevant for weatherization subsidies, targeted conservation payments, water-saving campaigns, wildfire defensible-space programs, and transport interventions. In choice modelling, it is relevant for individualized information treatments or recommendations, but one must be careful not to optimize short-run choices at the expense of welfare, equity, or long-run environmental outcomes.

10 Extensions: instruments, panels, and difference-in-differences

This chapter emphasizes selection-on-observables because it is the entry point for DML and causal forests. Many environmental and choice applications require other designs.

10.1 Instrumental variables

If treatment is endogenous, an instrument Z can help if it affects treatment, has no direct effect on the outcome except through treatment, and is independent of unobserved outcome determinants.

DML has instrumental-variable variants for partially linear IV settings. In transport, examples might include quasi-random variation in service disruptions or policy eligibility thresholds; in environmental economics, instruments might come from administrative thresholds, weather shocks, or regulatory assignment rules. Each candidate instrument needs substantive defence.

10.2 Difference-in-differences

When policy is introduced in some units and not others, panel data can support difference-in-differences designs. ML can help model nuisance functions, heterogeneity, or counterfactual trends, but the core identifying assumption remains parallel trends or an appropriate generalized version. With staggered adoption, be cautious: ordinary two-way fixed effects can be misleading under heterogeneous dynamic effects.

10.3 Synthetic controls and matrix completion

Synthetic control and matrix completion methods estimate counterfactual outcomes for treated units using untreated units and pre-treatment information. These are especially relevant for place-based environmental policies, protected areas, regional climate policies, and infrastructure investments. They are closer to causal panel-data methods than to the cross-sectional causal forests covered here, but they share the idea of using prediction to estimate counterfactuals.

11 Applications in environmental economics

11.1 Environmental policy evaluation

Environmental policies often have heterogeneous effects. A fishery quota, protected area, carbon price, building retrofit program, or congestion charge may work differently across regions, baseline risk, compliance capacity, and market conditions. Miller (2020) is a useful environmental economics example: causal forests are adapted to estimate heterogeneous and time-varying effects of environmental policies in a panel setting.

11.2 Conservation and payments for ecosystem services

Suppose a government offers payments for riparian buffers. Adoption may depend on farm size, income, slope, soil quality, environmental attitudes, previous participation, and local norms. A naive comparison of participants and non-participants is confounded because participants self-select. Causal ML can help if the analyst has enough covariates to make selection-on-observables plausible or if treatment is randomized, phased in, or assigned by a credible rule.

Possible causal ML questions:

- What is the average effect of the payment on adoption?

- Which farms have the largest predicted conservation gain?
- Does the effect vary by baseline land quality or environmental attitudes?
- What targeting rule maximizes biodiversity gain for a fixed budget?

11.3 Energy behaviour and climate policy

Energy interventions include feedback devices, social-comparison messages, rebates, time-of-use tariffs, and default green electricity. Randomized controlled trials are common in this area, which makes causal ML especially useful for heterogeneity analysis. Causal forests can estimate which households respond most to a social norm message; DML can adjust for high-dimensional baseline consumption profiles.

11.4 Pollution exposure and environmental justice

Causal ML can help study whether policy effects differ across income, race, age, exposure, and vulnerability. However, environmental justice questions require more than maximizing average treatment effect. A policy that has high average benefits may worsen inequality if benefits accrue mainly to less vulnerable groups. Policy learning should therefore include equity constraints or subgroup reporting.

12 Applications in choice modelling

Choice modelling and causal ML meet in at least four places.

12.1 Randomized information treatments in stated preference surveys

Many stated preference studies randomize information, framing, labels, defaults, or visualizations. For example, respondents may be randomly assigned to receive information about biodiversity decline before a conservation DCE. A causal ML analysis can estimate how the information treatment changes:

- the probability of choosing the conservation alternative;
- estimated WTP;
- attribute non-attendance;
- class membership in a latent class model;
- response certainty or protest responses.

If treatment is randomized, identification is relatively strong. But the outcome must be defined carefully. A change in the probability of selecting an alternative is not automatically a welfare effect.

12.2 Revealed preference data and endogenous attributes

In revealed preference transport data, price, travel time, waiting time, and availability are often endogenous or measured with error. ML prediction of mode choice may be accurate, but causal interpretation of an attribute change requires exogenous variation or a structural model. If low travel time is correlated with high accessibility and unobserved residential sorting, a model may confuse sorting with the effect of time.

12.3 Discrete choice models as causal structural models

A random utility model can be used for counterfactuals because it specifies how choice probabilities respond to changes in attributes. But this is a model-based causal interpretation. It relies on stable utility, correct specification, appropriate substitution patterns, and exogenous attributes or valid instruments. Causal ML does not replace these requirements, but it can help diagnose heterogeneity or estimate treatment effects around the choice model.

12.4 Recent choice-modelling examples

The choice modelling literature increasingly combines causal and ML tools. The uploaded Journal of Choice Modelling paper on autonomous vehicle security concern by Sabzi et al. (2026) is a direct example: a benchmark logit model provides interpretable average effects, while generalized random forests reveal heterogeneity in how security concerns affect stated AV interest. This illustrates the complementarity emphasized throughout this textbook: discrete choice models summarize behaviour in interpretable utility terms, while causal ML can flexibly explore effect heterogeneity.

Research design template for choice studies

For any treatment embedded in a choice survey, write four objects before estimating a model:

1. Treatment: What exactly changes? Information, default, frame, attribute, price, visual display, or choice set?
2. Outcome: Choice probability, WTP, utility coefficient, class membership, response certainty, or real follow-up behaviour?
3. Identification: Random assignment, quasi-experiment, instrument, panel design, or structural assumption?
4. Estimator: Logit interaction, mixed logit, DML, causal forest, latent class treatment effect, or hybrid approach?

13 Worked R lab 1: DML for a conservation program

This lab simulates an observational conservation-payment dataset. Treatment is not randomized: farms with higher environmental motivation and better administrative access are more likely to receive the program. We observe proxies for these factors. The goal is to estimate the average effect of the program on adoption of a conservation practice.

13.1 Install packages

```r
# Install once if needed:
# install.packages(c("data.table", "mlr3", "mlr3learners", "DoubleML"))
# install.packages(c("ranger", "xgboost"))

library(data.table)
library(mlr3)
library(mlr3learners)
library(DoubleML)
library(ranger)
```

13.2 Simulate data

```r
set.seed(2026)
N <- 2500

farms <- data.table(
  farm_size = rlnorm(N, log(50), 0.7),
  slope = runif(N, 0, 20),
  baseline_biodiv = rnorm(N),
  distance_office = rexp(N, rate = 1/25),
  prior_participation = rbinom(N, 1, 0.35),
  income = rlnorm(N, log(65000), 0.4),
  env_attitude = rnorm(N),
  region = factor(sample(1:8, N, replace = TRUE))
)

# Treatment assignment: participation in a payment program
lin_p <- -0.8 +
  0.35 * scale(log(farms$farm_size))[,1] -
  0.04 * farms$distance_office +
  0.55 * farms$prior_participation +
  0.45 * farms$env_attitude +
  0.25 * farms$baseline_biodiv

p_treat <- plogis(lin_p)
farms[, program := rbinom(.N, 1, p_treat)]
```

```r

# True heterogeneous effect: bigger for high slope and high attitude farms
true_tau <- 0.08 + 0.03 * (farms$slope > 10) + 0.05 * (farms$env_attitude > 0)

# Outcome: adoption of conservation practice, binary
lin_y0 <- -1.4 +
  0.25 * scale(log(farms$farm_size))[,1] +
  0.04 * farms$slope +
  0.35 * farms$prior_participation +
  0.50 * farms$env_attitude +
  0.25 * farms$baseline_biodiv

prob_y <- plogis(lin_y0 + farms$program * qlogis(pmin(0.95, pmax(0.05, 0.5 + true_
    tau))))
farms[, adopted := rbinom(.N, 1, prob_y)]
farms[, true_tau := true_tau]

mean(farms$adopted[farms$program == 1]) - mean(farms$adopted[farms$program == 0])
mean(farms$true_tau)
```

The naive difference in means is not a credible causal estimate because program participation depends on covariates that also affect adoption. DML adjusts flexibly for these covariates under the assumption that selection is ignorable after conditioning on them.

13.3 Prepare DoubleML data

```r
# DoubleML expects outcome y, treatment d, and covariates x.
# Use one-hot encoding for factor variables.
X <- model.matrix(~ farm_size + slope + baseline_biodiv + distance_office +
                    prior_participation + income + env_attitude + region,
                  data = farms)[, -1]

dml_df <- data.table(
  adopted = farms$adopted,
  program = farms$program,
  X
)

obj_dml_data <- DoubleMLData$new(
  data = dml_df,
  y_col = "adopted",
  d_cols = "program"
)
```

13.4 Estimate a partially linear model

For a binary outcome, a more tailored model could be used, but the partially linear regression is a transparent starting point for teaching. Here the coefficient is a linear-probability-style average effect.

```r
# Learners for nuisance functions
learner_y <- lrn("regr.ranger", num.trees = 500, min.node.size = 5)
learner_d <- lrn("regr.ranger", num.trees = 500, min.node.size = 5)

set.seed(2026)
dml_plr <- DoubleMLPLR$new(
  data = obj_dml_data,
  ml_l = learner_y,  # outcome nuisance E[Y|X]
  ml_m = learner_d,  # treatment nuisance E[D|X]
  n_folds = 5
)

dml_plr$fit()
dml_plr$summary()
```

13.5 Interpretation

The estimated coefficient answers: under the identification assumptions and the partially linear approximation, what is the average effect of program participation on the probability of adoption? It does not answer which farms benefit most. For that, we need CATE estimation.

Binary outcomes

A partially linear regression with a binary outcome is often useful as a first approximation, but it can predict outside the unit interval and its estimand is a linear probability effect. For serious applied work, consider interactive regression models, doubly robust estimators for binary outcomes, or outcome-specific modelling choices.

14 Worked R lab 2: causal forests for heterogeneous effects

Now estimate heterogeneous treatment effects using `grf`. The package provides causal forests, regression forests, instrumental forests, and related tools.

14.1 Estimate the forest

```r
# install.packages("grf")
library(grf)

Y <- farms$adopted
```

```r
W <- farms$program
X_grf <- X

set.seed(2026)
cf <- causal_forest(
  X = X_grf,
  Y = Y,
  W = W,
  num.trees = 2000,
  seed = 2026
)

average_treatment_effect(cf)
```

14.2 Predict CATEs and inspect heterogeneity

```r
tau_hat <- predict(cf)$predictions
farms[, tau_hat := tau_hat]

summary(farms$tau_hat)
cor(farms$tau_hat, farms$true_tau)

# Which variables appear important for heterogeneity?
vi <- variable_importance(cf)
head(sort(vi, decreasing = TRUE), 10)

# Compare high and low predicted effect groups
farms[, cate_group := fifelse(tau_hat >= quantile(tau_hat, 0.75),
                              "top quartile", "other")]
farms[, .(
  n = .N,
  mean_tau_hat = mean(tau_hat),
  mean_true_tau = mean(true_tau),
  mean_slope = mean(slope),
  mean_env_attitude = mean(env_attitude)
), by = cate_group]
```

14.3 Best linear projection

```r
# Best linear projection of CATEs on selected covariates
blp <- best_linear_projection(
  cf,
  A = model.matrix(~ slope + env_attitude + prior_participation + baseline_biodiv,
                   data = farms)[, -1]
)
blp
```

The best linear projection is often more interpretable than a long list of individual CATE predictions. It asks: which covariates linearly explain the variation in estimated treatment effects?

How to report causal forest results

Report the average effect, the identification assumptions, the overlap diagnostics, the distribution of CATE estimates, interpretable subgroup summaries, and uncertainty. Avoid presenting a ranked list of individuals as if the CATEs were true individual effects.

15 Worked R lab 3: an information treatment in a choice experiment

Suppose a DCE asks respondents to choose between a status quo and a river restoration program. Half of respondents are randomly assigned to receive a short information module about biodiversity loss. The outcome is whether they choose the restoration alternative in each task.

This lab shows a simple logit with treatment interactions. It is not a substitute for a full panel mixed logit, but it teaches how causal treatment effects and choice modelling connect.

```r
library(data.table)
set.seed(2026)

N <- 800
Tn <- 6

respondents <- data.table(
  id = 1:N,
  info = rbinom(N, 1, 0.5),  # randomized information treatment
  income = rlnorm(N, log(45000), 0.45),
  env_attitude = rnorm(N)
)

choices <- respondents[, .(task = 1:Tn), by = .(id, info, income, env_attitude)]
choices[, cost := sample(c(10, 25, 50, 75), .N, replace = TRUE)]
choices[, biodiv := sample(c(1, 2, 3), .N, replace = TRUE)]

# Utility difference: restoration minus status quo
choices[, V := -0.045 * cost +
              0.55 * biodiv +
              0.30 * info +
              0.25 * info * (env_attitude > 0) +
              0.20 * env_attitude]
choices[, p_restore := plogis(V)]
choices[, choose_restore := rbinom(.N, 1, p_restore)]

# Simple clustered logit-style model using glm for illustration
```

```r
m <- glm(
  choose_restore ~ cost + biodiv + info + info:env_attitude + env_attitude,
  data = choices,
  family = binomial()
)
summary(m)
```

Because `info` was randomized, the coefficient on `info` has a causal interpretation for the log-odds of choosing restoration, conditional on the model. The interaction `info:env_attitude` asks whether the information effect differs by environmental attitude. In a full DCE analysis, one would account for repeated choices, alternative-specific attributes, scale, random coefficients, and WTP.

15.1 Causal ML extension

A causal forest can estimate heterogeneity in the information effect on the binary choice outcome. But the estimand is the effect on the probability of choosing restoration in the observed tasks, not automatically a welfare measure.

```r
# One row per respondent-task. Treatment = randomized information.
X_choice <- model.matrix(~ cost + biodiv + income + env_attitude, data = choices)[,
    -1]
Y_choice <- choices$choose_restore
W_choice <- choices$info

cf_choice <- causal_forest(X_choice, Y_choice, W_choice, num.trees = 1500)
average_treatment_effect(cf_choice)

choices[, tau_info := predict(cf_choice)$predictions]
summary(choices$tau_info)
```

Repeated choices

The code above is intentionally simple. In real panel choice data, repeated observations from the same respondent require grouped splitting, respondent-level clustering, or a model that handles panel structure. Random row-wise splitting can leak information across training and test folds.

16 Python alternatives

Python is often useful when causal ML is combined with deep learning, large-scale feature engineering, or production workflows. The main packages are:

- `DoubleML`: Python and R implementation of double/debiased ML.

- `econml`: Python package for heterogeneous treatment effects using ML-based causal methods.
- `dowhy`: Python package for modelling causal assumptions, identification, estimation, and refutation workflows.
- `scikit-learn`: base learners used inside causal estimators.

A minimal `econml` sketch:

```python
# Illustrative Python sketch, not run in this chapter
import numpy as np
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from econml.dml import CausalForestDML

est = CausalForestDML(
    model_y=RandomForestRegressor(n_estimators=500, min_samples_leaf=10),
    model_t=RandomForestClassifier(n_estimators=500, min_samples_leaf=10),
    discrete_treatment=True,
    n_estimators=1000,
    random_state=2026
)

est.fit(Y, T, X=X, W=None)
tau_hat = est.effect(X)
ate = np.mean(tau_hat)
```

Use Python when it serves the research design, not because it looks more advanced. For many PhD projects in environmental valuation and discrete choice, R remains more convenient because the choice modelling ecosystem is stronger.

17 Diagnostics and reporting checklist

17.1 Before estimation

1. Define the treatment, outcome, unit, and time horizon.
2. Write the estimand: ATE, ATT, CATE, policy value, or effect on choice probabilities.
3. Draw a causal diagram or write the confounding story in words.
4. Identify the source of causal variation: randomization, natural experiment, panel design, instrument, or selection-on-observables.
5. Check whether interference or spillovers are likely.

17.2 During estimation

1. Use cross-fitting or honest estimation where appropriate.

2. Tune nuisance models by prediction performance, but evaluate causal estimates by causal diagnostics.
3. Check overlap using propensity scores or treatment probabilities.
4. Use grouped splits when observations are clustered by respondent, household, firm, farm, region, or time.
5. Compare flexible estimates with transparent baselines.

17.3 After estimation

1. Report average effects with uncertainty.
2. Summarize heterogeneity with interpretable subgroups, not only individual CATEs.
3. Test robustness to learner choice, covariate set, folds, and trimming rules.
4. Discuss unobserved confounding honestly.
5. Separate behavioural interpretation from predictive or targeting usefulness.

18 Common mistakes

18.1 Mistake 1: Calling prediction causal

A model that predicts adoption of electric vehicles does not estimate the causal effect of a subsidy. Prediction answers what is likely under the observed system; causal analysis asks what would change under intervention.

18.2 Mistake 2: Thinking many controls guarantee unconfoundedness

Having hundreds of covariates can help, but unconfoundedness is still an assumption. Important unobserved confounders can remain, especially in voluntary environmental programs.

18.3 Mistake 3: Ignoring overlap

Treatment effects are hard to estimate where treated and untreated units are not comparable. Causal ML can extrapolate silently. Always check whether each subgroup contains both treated and untreated observations.

18.4 Mistake 4: Treating CATEs as individual truths

A CATE estimate is a conditional average, not a measured individual effect. This matters ethically when targeting policies or ranking households.

18.5 Mistake 5: Using causal forests for welfare without a welfare model

A causal forest can estimate the effect of an information treatment on choice probability. It cannot by itself compute compensating variation, equivalent variation, or WTP. Welfare requires a model connecting choices to utility and money-metric trade-offs.

18.6 Mistake 6: Splitting repeated choice data row by row

If the same respondent appears in training and test folds, nuisance models and predictive models may leak respondent-specific information. Use grouped cross-fitting and grouped validation whenever the unit of generalization is the person, household, farm, region, or firm.

19 End-of-chapter exercises

Exercise 1: Identify the estimand

For each question, state whether the target is ATE, ATT, CATE, policy value, or a structural welfare quantity:

1. What is the average effect of a home energy report on electricity demand?
2. Which households should receive a heat-pump rebate?
3. What is the WTP for improved river biodiversity?
4. How does a biodiversity information treatment affect the probability of choosing a restoration program?
5. What was the effect of an agri-environment payment among farms that participated?

Exercise 2: Draw a causal diagram

Draw a DAG for the effect of a congestion charge on car commuting. Include income, residential location, workplace location, public transport access, baseline car dependence, and political attitudes. Which variables should be adjusted for? Which might be mediators rather than confounders?

Exercise 3: DML replication

Run the conservation program DML lab. Replace random forests with gradient boosting using `regr.xgboost`. Compare the estimated average effect and standard error. Does a better nuisance prediction model necessarily change the causal estimate substantially?

Exercise 4: CATE interpretation

Using the causal forest lab, create four subgroups based on predicted CATE quartiles. Compare their covariate means. Write a paragraph explaining what the forest appears to have learned. Then write a second paragraph listing alternative explanations and identification concerns.

Exercise 5: Choice experiment treatment

Design a DCE on willingness to pay for urban tree planting. Randomize one of three information treatments: health benefits, biodiversity benefits, or climate adaptation benefits. Define the causal estimands and explain how you would estimate heterogeneous effects by baseline environmental attitude.

Exercise 6: Welfare caution

Suppose a causal forest shows that a green default increases selection of a renewable electricity plan by 12 percentage points for low-income households. What additional information is needed before claiming that the default improves welfare?

20 Chapter summary

Causal machine learning combines causal identification with flexible prediction. It is useful because environmental economics and choice modelling often involve high-dimensional confounding, nonlinear relationships, and heterogeneous policy effects. Double/debiased machine learning uses orthogonal scores and cross-fitting to estimate low-dimensional causal parameters while allowing flexible nuisance models. Meta-learners and causal forests estimate conditional average treatment effects. Policy learning turns treatment-effect estimates into allocation rules.

The main lesson is discipline. Causal ML improves estimation after the causal question and identifying assumptions are defined. It does not turn observational correlations into causal evidence by itself. In choice modelling, it can estimate the effects of information, defaults, and policy treatments on choices; it can reveal heterogeneity that standard logit interactions may miss; and it can support targeting. But welfare analysis still requires utility, money-metric trade-offs, and careful counterfactual interpretation.

21 Reading guide

Start with Hernan and Robins (2020) or Imbens and Rubin (2015) for causal inference foundations. Then read Athey and Imbens (2019) for the economist's overview of ML and causal economics. Read Chernozhukov et al. (2018) for double/debiased ML. Read Wager and Athey (2018) and Athey, Tibshirani, and Wager (2019) for causal forests and generalized random forests. Read Kunzel et al. (2019) and Nie and Wager (2021) for meta-learners and the R-learner.

For policy learning, read Athey and Wager (2021). For environmental economics, read Miller (2020) as a high-quality example of causal forests for environmental policy. For choice modelling, compare causal ML approaches with the broader agenda in van Cranenburgh et al. (2022), the machine-learning chapter in Hess and Daly (2024), and the AV security concern application by Sabzi et al. (2026).

For implementation, learn `DoubleML` and `grf` in R. Use `dagitty` for causal diagrams and adjustment sets, `fixest` for conventional panel and fixed-effect econometrics, and Python packages such as `econml` and `dowhy` when you need the Python causal ML ecosystem.

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How to study this chapter

This chapter teaches interpretability as a research design problem, not as a cosmetic add-on after fitting a complex model. Read it with two questions in mind. First: what must the model explain to be useful for environmental economics or choice modelling? Second: what kind of explanation is credible for that purpose?

Use three passes. In the first pass, learn the vocabulary: transparent models, post-hoc explanations, global explanations, local explanations, feature effects, feature attribution, counterfactual explanations, calibration, welfare interpretation, and behavioural validity. In the second pass, work through the R code and generate at least one permutation-importance plot, one partial-dependence or accumulated-local-effects plot, and one local explanation. In the third pass, return to your own research question and decide which outputs would be acceptable evidence: coefficients, elasticities, WTP, SHAP values, counterfactual examples, prototypes, or some combination.

Learning objectives

By the end of this chapter, you should be able to:

1. Distinguish model transparency, post-hoc explainability, behavioural interpretability, and causal interpretability.
2. Explain why interpretability requirements differ across prediction, valuation, targeting, forecasting, and policy counterfactuals.
3. Compute and interpret permutation feature importance, partial dependence, individual conditional expectation, accumulated local effects, LIME-style local surrogates, and SHAP values.
4. Translate choice-model outputs into marginal effects, elasticities, WTP measures, class-membership probabilities, and welfare changes.
5. Diagnose common interpretability failures: correlated predictors, data leakage, extrapolation, unstable explanations, implausible signs, and causal overclaiming.
6. Design an interpretable ML workflow in R for environmental economics and choice modelling.
7. Decide when to prefer an interpretable model by design over explaining a black box after the fact.

1 Why interpretability matters in environmental economics and choice modelling

Machine learning models are often evaluated by predictive accuracy, log loss, area under the ROC curve, or mean squared error. These metrics are necessary in many applications, but they are rarely sufficient in environmental economics and choice modelling. Environmental policy and choice analysis often require statements such as:

- A household is willing to pay a specified amount for a biodiversity improvement.
- A transport intervention changes mode shares by changing travel times, costs, access times, and reliability.
- A conservation policy has different effects for different groups and locations.
- A carbon price changes behaviour through marginal costs, substitution patterns, and income constraints.
- A predicted adoption probability can be used for targeting without violating fairness, plausibility, or policy validity.

These are not merely statements about prediction. They are statements about mechanisms, trade-offs, heterogeneity, counterfactuals, and sometimes welfare. A black-box model may predict well but still be unsuitable for valuation or policy analysis if it produces implausible cost responses, unstable feature attributions, or explanations that rely on artefacts of the data collection process.

Core idea

Interpretability is not one thing. An explanation that is good enough for debugging a classifier may be inadequate for estimating willingness to pay. An explanation that is good enough for describing a trained model may be inadequate for making a causal claim. A choice modeller must ask: interpretable for whom, for what decision, and under what assumptions?

The literature on machine learning and choice modelling is converging on this point. The uploaded discussion paper by van Cranenburgh, Wang, Vij, Pereira, and Walker argues that machine learning can help choice modelling with prediction, automated specification search, and nontraditional data such as text and images, while also recognizing the different goals of choice models and data-driven ML models. Dahmen, Weikl, and Bogenberger use SHAP to interpret XGBoost mode-choice predictions from tracking-based revealed preference data, showing the value of global, mode-level, and individual explanations. Alwosheel, van Cranenburgh, and Chorus propose prototypical examples to diagnose whether neural networks for discrete choice analysis have learned plausible relationships rather than opaque artefacts. These papers should be read as examples of the same general principle: prediction can be useful, but behavioural credibility needs additional diagnostics.

2 Four meanings of interpretability

The word "interpretability" is used in several ways. Confusing them is a common source of bad empirical work.

2.1 Model transparency

A model is transparent if a trained analyst can inspect its functional form and understand how inputs map to outputs. Linear regression, logistic regression, shallow decision trees, simple generalized additive models, and standard multinomial logit models are relatively transparent. A deep neural network or a large random forest is usually not transparent in this sense.

Transparency is not the same as correctness. A linear model can be transparent and wrong. A mixed logit model can be theoretically meaningful yet hard to summarize because it contains high-dimensional random coefficients and simulated likelihoods.

2.2 Post-hoc explainability

Post-hoc explanations are methods applied after fitting a model. They include permutation feature importance, partial dependence plots, accumulated local effects, SHAP values, LIME, surrogate trees, counterfactual explanations, and prototypes. These methods can be used with many model classes, including black-box models.

Post-hoc explanation methods are useful for debugging and communication, but they are not magic. They explain a fitted model, not necessarily the true data-generating process. If the model is trained on biased data or a leaky validation split, the explanation may faithfully explain a bad model.

2.3 Behavioural interpretability

Behavioural interpretability means that model outputs can be mapped to concepts in behavioural theory: marginal utility, diminishing sensitivity, loss aversion, status quo bias, taste heterogeneity, class membership, attribute non-attendance, substitution patterns, and WTP.

A discrete choice model can be behaviourally interpretable because it is built around utility comparisons. For example, in a random utility model,

$$U_{ni} = V_{ni} + \varepsilon_{ni},$$

where the systematic utility V_{ni} can contain cost, time, emissions, biodiversity, risk, or service-quality attributes. The signs, magnitudes, and ratios of coefficients have behavioural meaning if the modelling assumptions are credible.

2.4 Causal interpretability

Causal interpretability means that a quantity can be interpreted as the effect of an intervention. A SHAP value for income in a model predicting electric-vehicle adoption is not the causal effect of income. A partial dependence curve for environmental attitudes is not automatically the effect of changing attitudes. Causality requires identification assumptions, research design, and sometimes methods such as instrumental variables, experiments, differences-in-differences, causal forests, or double/debiased machine learning.

Do not confuse explanation with identification

Model explanations describe how a fitted model uses variables. They do not by themselves identify causal effects, welfare changes, or structural preference parameters. If the research question is causal or welfare-based, interpretability must be combined with identification and economic structure.

3 A taxonomy of explanation tasks

Before selecting a method, define the explanation task. Table 1 gives a practical taxonomy.

Table 1: Interpretability tasks in environmental economics and choice modelling

Task	Typical question	Useful tools
Debugging	Did the model learn something obviously wrong?	Residual plots, calibration, permutation importance, PDP/ICE/ALE, SHAP, local examples, validation audits.
Prediction explanation	Why did the model predict this choice or risk level?	SHAP, LIME, local surrogate models, counterfactual explanations, nearest prototypes.
Behavioural interpretation	What trade-offs or heterogeneity patterns does the model imply?	Coefficients, marginal effects, elasticities, WTP ratios, class-membership probabilities, posterior random coefficients, constrained feature effects.
Policy forecasting	What happens under a realistic policy scenario?	Scenario simulation, logsum measures, counterfactual prediction under support checks, sensitivity analysis.
Welfare analysis	What is the monetary value of a change in attributes?	RUM-consistent choice models, WTP-space models, logsum welfare, credible cost coefficients, uncertainty intervals.

Task	Typical question	Useful tools
Causal analysis	What is the effect of an intervention?	Experiments, quasi-experiments, DML, causal forests, orthogonal scores, sensitivity analysis.
Communication	How do we explain results to nontechnical audiences?	Simple plots, counterfactual examples, scenario tables, maps, short narratives, transparent baselines.

A useful workflow begins with the task, not the method. If the task is "understand why the random forest predicts that a trip is by car", SHAP or LIME may be appropriate. If the task is "estimate household WTP for cleaner water", SHAP is not enough: one needs a credible valuation model, usually with a cost or price coefficient and a welfare interpretation.

4 Interpretable models by design

The most direct route to interpretability is to use a model whose structure is interpretable before any post-hoc explanation. This is the point emphasized by Rudin's critique of explaining black boxes in high-stakes settings: when stakes are high and an interpretable model can perform well, it is often preferable to build interpretability into the model rather than explain a black box after the fact.

In economics and choice modelling, interpretable-by-design models include:

- Linear regression and generalized linear models.
- Penalized linear models, such as lasso and elastic net.
- Generalized additive models and shape-constrained additive models.
- Shallow trees, rule lists, and monotonic scoring rules.
- Multinomial logit, nested logit, mixed logit, latent class, and hybrid choice models.
- Behaviourally constrained neural or ML-enhanced choice models.

The point is not that these models are always simple. A mixed logit model with random coefficients can be complex. A latent class model with many classes can be difficult to communicate. A hybrid choice model can be hard to identify. But these models make explicit what quantities are meant to be interpreted.

4.1 Choice models as interpretable structures

For a standard multinomial logit model,

$$P_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in C_n} \exp(V_{nj})}.$$

If

$$V_{ni} = \beta_c c_{ni} + \beta_b b_{ni} + \beta_e e_{ni},$$

where c is cost, b is biodiversity improvement, and e is emissions reduction, then the coefficient signs and ratios can be interpreted, subject to model assumptions. A marginal WTP for biodiversity is

$$WTP_b = -\frac{\beta_b}{\beta_c}.$$

A simple logsum welfare change from baseline 0 to policy 1 is

$$\Delta CS_n = -\frac{1}{\beta_c}\left[\ln \sum_{j \in C_n} \exp(V_{nj}^1) - \ln \sum_{j \in C_n} \exp(V_{nj}^0)\right],$$

assuming a fixed marginal utility of income and a cost coefficient $\beta_c < 0$. This is a different kind of interpretability from a SHAP plot. It is welfare interpretation under a random utility model.

Choice-model interpretability is not just coefficient readability

Choice models produce behavioural objects: WTP, elasticities, substitution patterns, inclusive values, class shares, posterior distributions of tastes, and welfare changes. A black-box classifier may produce accurate probabilities but not these objects without additional structure.

4.2 Transparent ML does not have to be linear

A useful compromise between flexibility and interpretability is an additive model:

$$g(x) = \alpha + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p),$$

where each f_j is a smooth or piecewise function. If interactions are needed, one can add selected terms:

$$g(x) = \alpha + \sum_j f_j(x_j) + \sum_{(j,k) \in \mathcal{I}} f_{jk}(x_j, x_k).$$

This allows nonlinear effects while keeping interpretation tractable. In choice modelling, the analogue is a utility function with flexible but interpretable transformations, such as splines for travel time or income interactions for cost sensitivity.

For environmental valuation, shape constraints are particularly valuable. A model can be constrained so that utility decreases with cost, increases with biodiversity improvement, and increases with emissions reduction. Some flexible ML models, such as gradient boosting imple-

mentations, support monotonic constraints. Even when constraints are not used directly, they should be checked after fitting.

5 Post-hoc explanation methods

Post-hoc methods explain a fitted model $\widehat{f}$. The model may be a random forest, gradient boosting machine, neural network, support vector machine, or even a choice model. The following methods are model-agnostic unless otherwise noted.

5.1 Permutation feature importance

Permutation feature importance asks how much predictive performance worsens when a feature is randomly shuffled. Let $\mathcal{L}(y, \widehat{f}(x))$ be a loss function on a validation set. For feature j, define $X^{\pi(j)}$ as the validation data with feature j permuted. A simple importance score is

$$I_j = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}(y_i, \widehat{f}(x_i^{\pi(j)})) - \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}(y_i, \widehat{f}(x_i)).$$

Large I_j means predictive performance deteriorates when the feature is broken.

This is useful for answering: which variables does the trained model rely on for prediction? In an environmental adoption model, a high importance for previous adoption in the neighbourhood may reveal spatial diffusion. In a mode-choice classifier, a high importance for travel time or distance may be reassuring.

But there are pitfalls. If two features are strongly correlated, permuting one may create unrealistic combinations and may understate or overstate importance. For example, car travel time and trip distance are correlated. Household income, dwelling type, and neighbourhood may also be correlated. Permutation importance is a predictive diagnostic, not a causal importance measure.

5.2 Partial dependence plots

A partial dependence plot (PDP) shows the average model prediction as a feature or group of features is varied. For a set of focal features S and complement C, the PDP is

$$\widehat{PD}_S(x_S) = \frac{1}{n}\sum_{i=1}^{n} \widehat{f}(x_S, x_{iC}).$$

It answers: what does the model predict on average if we set X_S to a given value while keeping the empirical distribution of the other features?

PDPs are easy to interpret, but they can be misleading when features are correlated. If we set travel distance to 100 km while keeping observed urban walking-trip characteristics, we may ask the model to predict in regions with no empirical support. Apley and Zhu emphasize this extrapolation problem and propose accumulated local effects as an alternative.

5.3 Individual conditional expectation plots

An individual conditional expectation (ICE) curve is the individual-level version of a PDP:

$$\widehat{ICE}_S^{(i)}(x_S) = \widehat{f}(x_S, x_{iC}).$$

A PDP averages ICE curves. ICE plots reveal heterogeneity that an average curve hides. In a mode choice model, travel time may reduce public transport choice probability sharply for some individuals and barely for others. In an environmental technology adoption model, subsidies may matter differently for renters and homeowners.

ICE plots are particularly useful for detecting interactions. If ICE curves are roughly parallel, an additive representation may be adequate. If they cross or fan out, heterogeneity or interactions are important.

5.4 Accumulated local effects

Accumulated local effects (ALE) plots estimate how model predictions change locally as a feature changes, using observed data regions rather than constructing unrealistic combinations. For a continuous feature x_j, the first-order ALE function is based on local differences in intervals:

$$\widehat{ALE}_j(z) = \int_{z_0}^{z} \mathbb{E}\left[\frac{\partial \widehat{f}(X)}{\partial x_j} \mid X_j = t\right] dt - \text{constant}.$$

In practice the derivative is approximated by finite differences within bins of X_j. ALE is often preferable to PDP when predictors are correlated, a frequent situation in environmental and transport data.

5.5 LIME: local surrogate explanations

LIME explains a prediction by fitting a simple local model around the instance of interest. For observation x_0, LIME solves approximately

$$\hat{g} = \arg\min_{g \in \mathcal{G}} \sum_{z} \pi_{x_0}(z) \left[\widehat{f}(z) - g(z)\right]^2 + \Omega(g),$$

where $\pi_{x_0}(z)$ weights perturbed observations by proximity to x_0, g is an interpretable local surrogate, and $\Omega(g)$ penalizes complexity.

LIME is useful for individual cases: why did the model classify this household as likely to adopt heat pumps? Why did the model predict this route would be chosen? But the result depends on the perturbation scheme, distance measure, and local sample. For structured choice data, naive perturbations can create impossible choice tasks, such as negative costs or inconsistent alternative attributes.

5.6 SHAP values

SHAP values are based on Shapley values from cooperative game theory. They decompose a prediction into contributions from features. For feature j, a Shapley value can be written as

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left[v(S \cup \{j\}) - v(S) \right],$$

where F is the set of all features and $v(S)$ is the value of a model using feature coalition S. In applied ML, $v(S)$ is defined through conditional or marginal expectations of the fitted model.

SHAP values are popular because they provide local attributions and can be aggregated into global summaries. In mode-choice applications, SHAP can show which variables increase or decrease the predicted probability of choosing car, transit, bicycle, walking, or another mode for a specific observation. Dahmen et al. use SHAP for exactly this type of global, mode-level, and individual interpretation in tracking-based mode-choice modelling.

However, SHAP values are not causal effects. They depend on how missing features are integrated out, how correlated features are handled, and what prediction target is explained. For multi-class choice models, one must state whether SHAP values explain the probability of a specific class, the log-odds for a class, or another transformed output.

5.7 Counterfactual explanations

A counterfactual explanation asks: what small change to the input would change the prediction? For example:

> The model predicts that this household will not adopt a heat pump. The prediction would change if upfront cost were lower by EUR 1,500 and expected energy savings were higher by EUR 250 per year.

Counterfactual explanations can be intuitive, but they require feasibility constraints. It is not useful to tell a household to increase income, move location, or change building age. In choice modelling, counterfactuals must respect the choice set, budget constraints, engineering feasibility, and policy manipulability.

5.8 Prototypes and criticisms

Prototype explanations select representative examples. Criticism examples select cases the model handles poorly or unusually. For neural networks in discrete choice, Alwosheel et al. adapt the idea of prototypical examples to diagnose whether an ANN has learned intuitive choice relationships. This can be especially useful when model equations are not transparent but example-based reasoning is understandable.

6 From ML explanations to choice-model quantities

A central lesson of this textbook is that ML explanations and choice-model outputs are complementary, not interchangeable. Table 2 compares them.

Table 2: Choice-model quantities versus model-agnostic ML explanations

Output	**Meaning**	**Typical use**
Coefficient	Marginal utility or effect on systematic utility, under a specified model.	Behavioural interpretation, hypothesis testing, WTP ratios.
Elasticity	Percentage change in choice probability for a percentage change in an attribute.	Demand forecasting, policy sensitivity.
WTP ratio	Monetary trade-off between a non-market attribute and cost.	Environmental valuation, benefit-cost analysis.
Logsum welfare	Expected maximum utility converted to money.	Welfare analysis and appraisal.
Class-membership probability	Probability that a decision maker belongs to a latent preference segment.	Segmentation, heterogeneity analysis.
Permutation importance	Performance loss when a variable is disrupted.	Predictive importance and debugging.
PDP/ALE	Average feature effect on model prediction.	Model inspection, nonlinear patterns.
ICE	Individual feature effect curves.	Heterogeneity and interaction diagnosis.
SHAP	Feature attribution for one prediction, aggregable across cases.	Local explanation, global importance, debugging.
Counterfactual explanation	Minimal feasible input change that changes prediction.	Communication, targeting, policy feasibility.

The most dangerous mistake is to interpret a model-agnostic explanation as if it were a structural parameter. A SHAP value for cost in a random forest can tell us that cost contributed to a predicted probability for a case. It is not a marginal utility of income, not an elasticity, and not a WTP estimate unless the model and explanation are embedded in a structure that supports such interpretation.

6.1 Marginal effects in MNL

For a simple MNL model with alternative-specific variable x_{ni} and coefficient β, the own marginal effect is

$$\frac{\partial P_{ni}}{\partial x_{ni}} = \beta P_{ni}(1 - P_{ni}),$$

and the cross effect for alternative $j \neq i$ is

$$\frac{\partial P_{nj}}{\partial x_{ni}} = -\beta P_{nj} P_{ni}.$$

An elasticity is

$$E_{ni,x} = \frac{\partial P_{ni}}{\partial x_{ni}} \frac{x_{ni}}{P_{ni}}.$$

These quantities are not generic feature attributions; they come from a probability model with a specific behavioural structure.

6.2 Explaining choice probabilities from ML models

If a random forest or gradient boosting model predicts choice probabilities, one can explain those probabilities with SHAP, ALE, or ICE. But for choice modelling, four extra questions are needed:

1. Is the prediction target the chosen alternative, the probability of each alternative, or a transformed probability such as log-odds?
2. Does the data representation respect choice-set structure, alternative attributes, and availability?
3. Are explanations computed on held-out data split by individual or choice situation as appropriate?
4. Do the feature effects obey behavioural restrictions such as negative cost sensitivity or plausible time sensitivity?

If these conditions are not met, the explanation can be visually convincing and still be scientifically weak.

7 Interpretability workflow for environmental and choice applications

Figure 1 summarizes a defensible workflow.

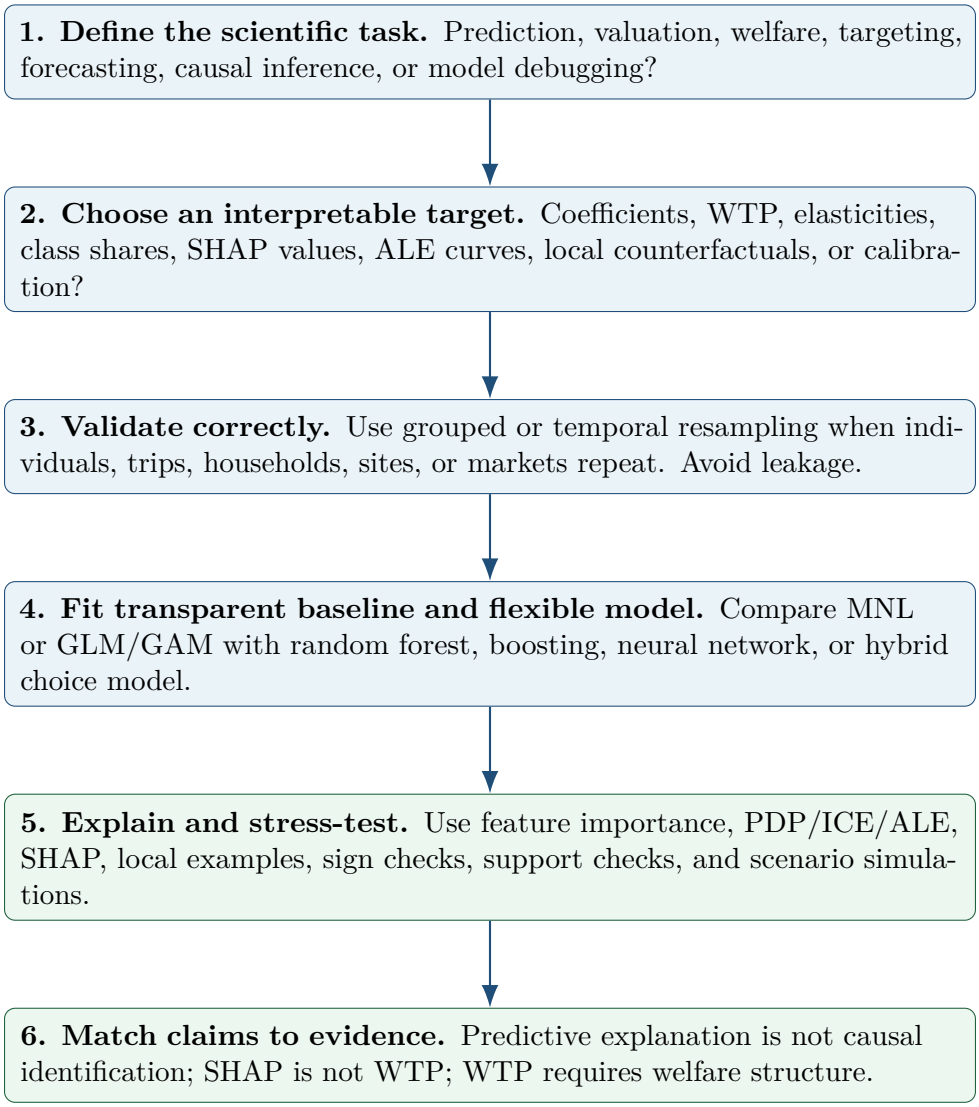


Figure 1: A defensible interpretability workflow for environmental economics and choice modelling.

8 Application 1: environmental valuation

Environmental valuation often asks how much people are willing to pay for improvements in non-market goods: cleaner water, better air quality, biodiversity, landscape protection, carbon reduction, flood protection, or ecosystem restoration. Interpretability is central because the goal is usually not only to predict which option a respondent chooses. The goal is to estimate trade-offs and welfare.

Suppose a DCE includes three alternatives in each choice task: two environmental programmes and a status quo. Attributes are annual household cost, biodiversity improvement, emissions reduction, recreation access, and policy reliability. A black-box model may predict choices well. But before using it for valuation, ask:

- Does predicted choice probability decrease as cost increases, holding other attributes fixed?

- Does predicted choice probability increase with biodiversity improvement over the range where the model has support?
- Are WTP estimates identifiable and stable?
- Are respondents ignoring some attributes? Could attribute non-attendance be present?
- Are heterogeneity patterns plausible and not artifacts of survey design?

Regularization can help detect attribute non-attendance or simplify specifications. The uploaded paper by Li and Yu applies LASSO to attribute non-attendance in a choice experiment for organic apples in Germany. Their application illustrates a general idea: shrinkage and variable selection can be used not just for prediction, but also to diagnose which attributes appear to enter the choice process.

Valuation warning

A model can rank environmental programmes accurately and still be unsuitable for benefit-cost analysis. Welfare interpretation requires a defensible monetary trade-off, normally through a cost coefficient or a WTP-space model. Flexible ML should either preserve that structure or be used as a diagnostic/specification aid.

9 Application 2: mode choice and transport behaviour

Mode choice is one of the most active areas comparing ML and discrete choice models. The results are nuanced. Several studies find that random forests, gradient boosting, and neural networks can outperform MNL or even richer logit models in prediction, especially with high-dimensional revealed preference data. But validation, interpretability, and behavioural outputs remain crucial.

Hillel, Bierlaire, Elshafie, and Jin's systematic review identifies common methodological limitations in ML mode-choice studies, including validation problems and insufficient hyperparameter tuning. Hillel's experimental review shows that trip-wise splitting in travel diary data can leak individual information across training and test sets, inflating performance estimates for flexible classifiers. This matters for interpretability: if validation is biased, explanations may explain leakage rather than travel behaviour.

Dahmen, Weikl, and Bogenberger use XGBoost and random forests on smartphone tracking-based revealed preference data and then interpret the best model with SHAP. Their analysis is a useful template: first compare predictive models, then use explanation tools at multiple levels, and finally inspect sensitivity to data source, sample size, and feature sets. This is stronger than reporting a single accuracy number or a single global importance plot.

Mode-choice lesson

For mode choice, interpretable ML should be assessed at three levels: global behaviour across the sample, mode-specific behaviour for each alternative, and local explanations for individual trips or persons. These outputs should be compared with known travel-behaviour theory: travel time, cost, distance, access, built environment, vehicle ownership, and socio-demographics.

10 Application 3: latent variables, attitudes, and hybrid choice

Environmental and transport choices are shaped by attitudes: environmental concern, risk perceptions, trust, comfort, technology anxiety, moral norms, and perceived behavioural control. Traditional hybrid choice models represent latent variables using structural and measurement equations. ML can enter this pipeline in at least three ways.

First, ML can learn nonlinear mappings from psychometric indicators to latent constructs. Torres Lahoz et al. introduce artificial neural networks to formulate latent constructs in latent class choice models, aiming to flexibly incorporate attitudinal indicators while maintaining consistency with generalized random utility assumptions.

Second, ML can model class membership or context-dependent heterogeneity. The context-aware Bayesian mixed multinomial logit model by Lukawska, Jensen, and Rodrigues uses a neural network to map contextual information into interpretable shifts in individual preference parameters. This is an example of a hybrid architecture: the neural component adds flexibility, but the final object remains preference parameters in a mixed logit structure.

Third, ML can support specification search. Ortelli et al. frame assisted specification of discrete choice models as a multi-objective combinatorial optimization problem. Hernandez et al. use association rules and random forests to support model specification for complex participatory value evaluation data. In these cases, ML is not replacing behavioural modelling; it is helping the analyst search a large specification space.

11 Application 4: text, images, and nontraditional data

The classic discrete choice dataset contains alternatives, attributes, individual characteristics, and choices. Modern environmental and transport datasets often contain text, images, maps, sensor traces, and remote sensing. Interpretability becomes more difficult because the features may not be directly meaningful.

For example, a street-view image choice experiment may ask respondents to choose cycling streets based on images. A convolutional neural network can use raw pixels, while a discrete choice model may use human-coded or segmented attributes such as tree cover, lane width, traffic, pavement, and building density. Zhu and Si compare discrete choice and ML models for street-view image choices, illustrating the trade-off between using richer inputs and preserving

interpretable semantic variables.

A practical approach is to build a two-stage representation:

1. Convert images, text, or sensor traces into semantic features: greenery, traffic density, water visibility, noise, sentiment, topics, or accessibility measures.
2. Use those features in choice models or interpretable ML, with post-hoc explanation tools to check whether the representation is plausible.

Deep embeddings can be useful, but they need additional interpretation. If an image embedding predicts park choice well, ask which visual properties drive that embedding. If a language model embedding predicts opposition to a renewable energy project, inspect representative text, topics, and nearest-neighbour documents.

12 R implementation: interpretable ML for a simulated environmental choice problem

This lab simulates respondents choosing among environmental policy options. We first create a task-level dataset for ML prediction. Then we fit a random forest, compute permutation importance, and create feature-effect explanations. Finally we connect the ML workflow to choice-model diagnostics.

12.1 Simulate data

```r
# Install once if needed:
# install.packages(c("tidyverse", "tidymodels", "ranger", "vip", "pdp", "iml", "DALEX"))

library(tidyverse)
library(tidymodels)
library(ranger)
library(vip)
library(pdp)
library(iml)
library(DALEX)

set.seed(2026)

N  <- 800   # respondents
Tn <- 6     # tasks per respondent
J  <- 3     # alternatives: A, B, status quo

respondents <- tibble(
  id = 1:N,
```

```r
  income = rlnorm(N, log(45000), 0.45),
  env_attitude = rnorm(N),
  urban = rbinom(N, 1, 0.6)
)

long <- expand_grid(
  id = 1:N,
  task = 1:Tn,
  alt = c("A", "B", "SQ")
) %>%
  left_join(respondents, by = "id") %>%
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = if_else(alt == "SQ", 0, sample(c(10, 25, 50, 80), n(), TRUE)),
    biodiversity = if_else(alt == "SQ", 0, sample(0:3, n(), TRUE)),
    co2 = if_else(alt == "SQ", 0, sample(0:3, n(), TRUE)),
    reliability = if_else(alt == "SQ", 1, sample(1:4, n(), TRUE)),
    # Data-generating utility with nonlinear attitude effect
    V = -0.055 * cost * (45000 / income)^0.25 +
         0.45 * biodiversity +
         0.28 * co2 +
         0.18 * reliability +
         0.30 * env_attitude * biodiversity +
         0.10 * urban * co2 +
         if_else(alt == "SQ", 0.25, 0)
  ) %>%
  group_by(chid) %>%
  mutate(
    p = exp(V) / sum(exp(V)),
    chosen_alt = sample(alt, 1, prob = p),
    choice = as.integer(alt == chosen_alt)
  ) %>%
  ungroup()

# Convert to one row per choice task for ML classification.
wide <- long %>%
  select(id, task, chid, alt, cost, biodiversity, co2, reliability,
         income, env_attitude, urban, chosen_alt) %>%
  pivot_wider(
    names_from = alt,
    values_from = c(cost, biodiversity, co2, reliability)
  ) %>%
  distinct(id, task, chid, .keep_all = TRUE) %>%
  mutate(chosen_alt = factor(chosen_alt))
```

The simulation is intentionally structured. Cost should reduce utility; biodiversity and emissions reduction should increase utility; environmental attitudes should interact with biodiversity. This

known structure helps us evaluate whether explanations recover plausible patterns.

12.2 Fit a random forest with grouped resampling

Because each respondent answers multiple tasks, we split by respondent. This avoids putting the same person in both training and test data.

```
set.seed(2026)
split <- group_initial_split(wide, group = id, prop = 0.8)
train <- training(split)
test  <- testing(split)

folds <- group_vfold_cv(train, group = id, v = 5)

rf_spec <- rand_forest(trees = 600, mtry = tune(), min_n = tune()) %>%
  set_engine("ranger", importance = "permutation", probability = TRUE) %>%
  set_mode("classification")

rf_wf <- workflow() %>%
  add_formula(chosen_alt ~ . - id - chid - task) %>%
  add_model(rf_spec)

rf_grid <- grid_regular(
  mtry(range = c(3, 12)),
  min_n(range = c(2, 30)),
  levels = 4
)

rf_tuned <- tune_grid(
  rf_wf,
  resamples = folds,
  grid = rf_grid,
  metrics = metric_set(accuracy, mn_log_loss)
)

best_rf <- select_best(rf_tuned, metric = "mn_log_loss")

rf_fit <- finalize_workflow(rf_wf, best_rf) %>%
  fit(train)

pred <- predict(rf_fit, test) %>%
  bind_cols(predict(rf_fit, test, type = "prob")) %>%
  bind_cols(test %>% select(chosen_alt))

metrics(pred, truth = chosen_alt, estimate = .pred_class)
mn_log_loss(pred, truth = chosen_alt, .pred_A, .pred_B, .pred_SQ)
```

12.3 Permutation feature importance

```r
# Extract fitted ranger object from workflow
rf_engine <- extract_fit_engine(rf_fit)

vip(rf_engine, num_features = 20)
```

Interpretation should be cautious. If biodiversity for alternative A and biodiversity for alternative B are important, this indicates predictive reliance. It does not directly estimate marginal utility. Also inspect whether the model relies heavily on variables that should not be available at prediction time or that encode survey design artefacts.

12.4 Partial dependence and ICE

```r
# The pdp package can be used on fitted models with custom prediction wrappers.
# Here we show the idea for the predicted probability of choosing A.

predict_A <- function(object, newdata) {
  predict(object, newdata, type = "prob")$.pred_A
}

pdp_bio_A <- partial(
  object = rf_fit,
  pred.var = "biodiversity_A",
  train = train,
  pred.fun = predict_A,
  grid.resolution = 4,
  ice = TRUE
)

autoplot(pdp_bio_A, alpha = 0.10) +
  labs(
    title = "ICE and partial dependence for biodiversity of alternative A",
    y = "Predicted probability of choosing A"
  )
```

If the PDP for biodiversity is downward sloping, that would contradict the data-generating process and should trigger debugging. If ICE curves show large heterogeneity, inspect whether it is due to environmental attitudes, income, or urban residence.

12.5 Model-agnostic explanations with `iml`

```r
# iml expects a prediction function returning a numeric vector.
# Explain the probability of choosing A.

x_train <- train %>% select(-chosen_alt, -id, -chid, -task)
```

```r
y_train <- train$chosen_alt

pred_A_iml <- function(model, newdata) {
  predict(model, newdata, type = "prob")$.pred_A
}

predictor_A <- Predictor$new(
  model = rf_fit,
  data = x_train,
  y = as.numeric(y_train == "A"),
  predict.function = pred_A_iml
)

# Permutation importance for choosing A
imp_A <- FeatureImp$new(predictor_A, loss = "ce")
plot(imp_A)

# ALE for cost of alternative A
ale_cost_A <- FeatureEffect$new(
  predictor_A,
  feature = "cost_A",
  method = "ale",
  grid.size = 20
)
plot(ale_cost_A)

# Local Shapley explanation for one choice task
one_case <- x_train[1, , drop = FALSE]
shap_one <- Shapley$new(predictor_A, x.interest = one_case)
plot(shap_one)
```

12.6 DALEX workflow

```r
# DALEX works well for comparing several models with a common interface.

explainer_rf_A <- DALEX::explain(
  model = rf_fit,
  data = x_train,
  y = as.numeric(y_train == "A"),
  predict_function = pred_A_iml,
  label = "Random forest: P(A)"
)

# Model performance
mp <- model_performance(explainer_rf_A)
plot(mp)

```

```r
# Permutation variable importance
vi <- model_parts(explainer_rf_A, type = "difference")
plot(vi)

# Partial dependence or ALE-style profiles
profile_cost <- model_profile(
  explainer_rf_A,
  variables = "cost_A",
  type = "accumulated"
)
plot(profile_cost)

# Break-down explanation for an individual prediction
bd <- predict_parts(explainer_rf_A, new_observation = one_case, type = "break_down"
    )
plot(bd)
```

13 R implementation: choice-model interpretation

Now compare the ML explanations with a choice model. The following code estimates a simple MNL using `mlogit`. The model is intentionally simpler than the data-generating process, so it may miss nonlinear heterogeneity.

```r
# install.packages("mlogit")
library(mlogit)

mdata <- mlogit.data(
  long,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

mnl <- mlogit(
  choice ~ cost + biodiversity + co2 + reliability | 0,
  data = mdata
)

summary(mnl)

b <- coef(mnl)
wtp <- tibble(
  attribute = c("biodiversity", "co2", "reliability"),
  wtp = -c(b["biodiversity"], b["co2"], b["reliability"]) / b["cost"]
```

```
)
wtp
```

The MNL model gives WTP ratios. The random forest gives flexible predictions and model explanations. The two are answering different questions. A strong applied paper would compare them:

- Does MNL recover expected signs?
- Does random forest improve log loss on grouped test data?
- Do ALE curves for cost and biodiversity match economic theory?
- Do WTP estimates change when interactions or heterogeneity are added?
- Do ML explanations reveal nonlinearities that should be added to the choice model?

Lab 6.1: Explanation comparison

Using the simulated data, estimate three models: MNL, random forest, and gradient boosting. For each model, assess: out-of-sample log loss, sign plausibility for cost, importance of biodiversity, heterogeneity by environmental attitude, and usefulness for WTP. Write one paragraph explaining which model you would use for prediction and which model you would use for welfare analysis.

14 Python implementation sketch

Python is useful when using `scikit-learn`, `xgboost`, `lightgbm`, `shap`, `alibi`, or deep-learning frameworks. The same principles apply: grouped validation, clear prediction target, and cautious interpretation.

```python
import pandas as pd
from sklearn.model_selection import GroupShuffleSplit
from sklearn.ensemble import RandomForestClassifier
from sklearn.inspection import permutation_importance, PartialDependenceDisplay
from sklearn.metrics import log_loss, accuracy_score

# X: one row per choice task; y: chosen alternative; groups: respondent id
# X = wide.drop(columns=["chosen_alt", "id", "chid", "task"])
# y = wide["chosen_alt"]
# groups = wide["id"]

gss = GroupShuffleSplit(n_splits=1, test_size=0.2, random_state=2026)
train_idx, test_idx = next(gss.split(X, y, groups=groups))

X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]
y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]
```

```

rf = RandomForestClassifier(
    n_estimators=600,
    min_samples_leaf=10,
    random_state=2026,
    n_jobs=-1
)
rf.fit(X_train, y_train)

proba = rf.predict_proba(X_test)
pred = rf.predict(X_test)
print(accuracy_score(y_test, pred))
print(log_loss(y_test, proba))

# Permutation importance on held-out data
imp = permutation_importance(
    rf, X_test, y_test,
    scoring="neg_log_loss",
    n_repeats=10,
    random_state=2026,
    n_jobs=-1
)

# Partial dependence for a selected feature
PartialDependenceDisplay.from_estimator(
    rf, X_test,
    features=["cost_A"],
    target="A"
)
```

For SHAP:

```
# pip install shap
import shap

explainer = shap.TreeExplainer(rf)
shap_values = explainer.shap_values(X_test)

# For multi-class models, inspect the class-specific output carefully.
# Example: explain class A if it is class index 0.
shap.summary_plot(shap_values[0], X_test)
shap.waterfall_plot(shap.Explanation(
    values=shap_values[0][0],
    base_values=explainer.expected_value[0],
    data=X_test.iloc[0],
    feature_names=X_test.columns
))
```

When using Python SHAP for choice data, clearly state the explained output: class probability, logit, margin, or expected utility proxy. Different outputs lead to different explanations.

15 Evaluating explanations

Explanations themselves must be evaluated. A plot is not evidence simply because it is colourful.

15.1 Fidelity

Does the explanation approximate the model accurately? For local surrogate models, report local fidelity. For surrogate trees, report how well the surrogate predicts the original model, not just the observed outcome.

15.2 Stability

Are explanations stable across resamples, tuning choices, random seeds, and plausible preprocessing choices? If the top variables change completely across folds, do not overinterpret the ranking.

15.3 Support

Do feature-effect plots evaluate the model in regions supported by data? PDPs can extrapolate into impossible combinations. ALE and conditional methods can reduce this problem but do not eliminate it.

15.4 Behavioural plausibility

Do explanations agree with basic theory? Cost should generally reduce utility. Travel time should not systematically increase the attractiveness of a slower alternative unless it is proxying for something else, such as scenic routes or comfort. Biodiversity improvement should not reduce programme attractiveness without a plausible explanation, such as access restrictions or distrust.

15.5 Calibration

A model can be accurate in class labels but poorly calibrated in probabilities. Choice modelling, welfare analysis, and policy simulation usually require credible probabilities. Use calibration plots, Brier scores, log loss, and probability integral diagnostics where appropriate.

```r
# Calibration curve for predicted probability of choosing A
calib <- pred %>%
  mutate(chose_A = as.integer(chosen_alt == "A")) %>%
  mutate(bin = ntile(.pred_A, 10)) %>%
```

```
  group_by(bin) %>%
  summarise(
    mean_pred = mean(.pred_A),
    observed = mean(chose_A),
    n = n(),
    .groups = "drop"
  )

ggplot(calib, aes(mean_pred, observed)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, linetype = 2) +
  coord_equal() +
  labs(x = "Mean predicted probability", y = "Observed frequency")
```

16 Common interpretability failures

16.1 Failure 1: Explaining a leaky model

If the validation split leaks information across individuals, trips, households, markets, or locations, flexible models may learn respondent or context fingerprints. Explanations then describe leakage. This is why grouped or temporal validation from Chapter 2 is a prerequisite for interpretability.

16.2 Failure 2: Treating feature importance as economic importance

A variable can be predictively important because it proxies for many omitted variables. A variable can be economically important even if its predictive importance is modest. For example, cost may be experimentally varied over a narrow range, leading to modest predictive importance, yet it is crucial for WTP and welfare.

16.3 Failure 3: Ignoring correlated predictors

Correlated predictors complicate permutation importance, PDPs, and SHAP values. If distance, time, and cost are correlated, attribution is ambiguous. Report correlation diagnostics and consider grouped importance, conditional importance, ALE, or theory-driven variable selection.

16.4 Failure 4: Explaining probabilities without saying which probability

In multi-class choice models, there is no single prediction to explain. Explain $P(\text{car})$, $P(\text{transit})$, $P(\text{status quo})$, or a utility-like score. Report the target explicitly.

16.5 Failure 5: Claiming WTP from arbitrary black-box attributions

WTP requires a monetary trade-off. SHAP values or feature-effect slopes do not automatically produce WTP. If WTP is the target, build a model with a cost coefficient, WTP-space structure, or constrained utility representation.

16.6 Failure 6: Equating interpretability with simplicity

Simple models are not always interpretable in the relevant sense. A simple linear model with omitted interactions can produce misleading coefficients. A more complex additive or mixed logit model may be more behaviourally interpretable if it represents heterogeneity and substitution patterns better.

17 A checklist for applied papers

Use the following checklist before submitting a paper that uses explainable AI in environmental economics or choice modelling.

1. State the primary task: prediction, explanation, valuation, causal inference, or welfare analysis.
2. Include a transparent baseline model.
3. Validate with a split that matches the deployment or inference problem.
4. Report proper probabilistic metrics, not only accuracy.
5. Explain the model on held-out data, not only training data.
6. For multi-class choice, specify which class or output is being explained.
7. Check sign and monotonicity of core economic attributes such as cost and time.
8. Inspect correlated predictors before using PDP, permutation importance, or SHAP.
9. Use multiple explanation methods when conclusions are important.
10. Separate predictive explanations from causal or welfare claims.
11. Quantify uncertainty or stability of explanations when feasible.
12. Translate results into behavioural quantities only when model structure permits it.

18 Conceptual checkpoint

Answer these questions before continuing.

Retrieval questions

1. What is the difference between a transparent model and a post-hoc explanation?
2. Why can a PDP be misleading when predictors are correlated?
3. What does a SHAP value explain, and what does it not explain?
4. Why is grouped validation necessary in many mode-choice datasets?
5. Why is a random forest feature importance score not a WTP estimate?
6. What choice-model outputs correspond most directly to welfare analysis?
7. Give one example where ML explanations can improve a choice model specification.

19 Research extensions

Extension 1: Explainability for constrained neural choice models

Fit a neural network that predicts choices but constrains the cost effect to be non-positive. Compare its predictive performance and explanation stability with an unconstrained neural network and an MNL model. Discuss whether the constraint improves behavioural credibility.

Extension 2: SHAP versus WTP in environmental DCEs

Using an environmental DCE, estimate a mixed logit model and a gradient boosting classifier. Compare WTP estimates from the mixed logit with SHAP summaries from the boosting model. Identify where the two agree, where they differ, and which claims each can support.

Extension 3: Explaining image-based choices

Use a street-view or landscape-image choice dataset. Compare three representations: raw images through a CNN, semantic segmentation variables, and human-coded attributes. Evaluate predictive performance and interpretability.

Extension 4: Explanation stability under resampling

Repeat model fitting and SHAP/ALE analysis over multiple grouped folds. Measure how often the same top variables appear and how much feature-effect curves vary. This turns explanation into a reproducible statistical object rather than a single plot.

20 Summary

Interpretability is a bridge between machine learning performance and economic usefulness. In environmental economics and choice modelling, the key issue is not whether a method produces an explanation, but whether the explanation supports the intended claim.

The chapter's main messages are:

- Interpretability has multiple meanings: transparency, post-hoc explainability, behavioural interpretability, and causal interpretability.
- Model-agnostic tools such as permutation importance, PDP, ICE, ALE, LIME, and SHAP are useful for debugging and understanding predictions.
- These tools explain fitted models, not necessarily causal mechanisms or welfare effects.
- Choice models provide behavioural quantities such as WTP, elasticities, substitution patterns, and welfare measures.
- ML can improve choice modelling by revealing nonlinearities, supporting specification search, handling rich data, and diagnosing predictive patterns.
- For policy and welfare, ML should usually be constrained, hybridized, or used alongside theory-driven choice models.

Reading guide

Read in this order:

1. Molnar's *Interpretable Machine Learning* for a broad and accessible overview of model-agnostic methods.
2. Doshi-Velez and Kim for why interpretability needs rigorous evaluation.
3. Rudin for the argument that high-stakes decisions often require interpretable models by design.
4. Friedman, Goldstein et al., Apley and Zhu, Ribeiro et al., and Lundberg and Lee for PDP, ICE, ALE, LIME, and SHAP foundations.
5. van Cranenburgh et al. for the choice-modelling and ML research agenda.
6. Dahmen et al., Zhao et al., and Hillel et al. for mode-choice applications and validation issues.
7. Alwosheel et al., Torres Lahoz et al., Lukawska et al., and Hernandez et al. for choice-specific neural diagnostics, latent variables, context-aware heterogeneity, and assisted specification.

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Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 7

Generative AI and Research Workflows

From large language models and embeddings to retrieval-augmented generation, coding assistance, survey design, synthetic respondents, and responsible workflows for environmental economics and choice modelling.

Generated for self-study use

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How to study this chapter

This chapter is different from the previous ones. It is not only about a class of models. It is about a new research instrument: a probabilistic language interface that can help you read, write, code, search, summarize, classify, and prototype. Used well, generative AI can make a PhD student more productive. Used badly, it can produce confident nonsense, privacy breaches, invalid synthetic data, hidden plagiarism, and non-reproducible research.

Study this chapter in three passes. In the first pass, learn the technical vocabulary: token, embedding, transformer, context window, prompt, retrieval, tool use, structured output, temperature, and hallucination. In the second pass, reproduce the R workflows. In the third pass, apply the validation checklists to your own research project: what can be delegated to an AI system, what must remain under human scientific judgment, and what evidence would convince a skeptical examiner?

Learning objectives

By the end of this chapter, you should be able to:

1. Explain the core mechanics of large language models (LLMs): tokenization, next-token prediction, attention, embeddings, and decoding.
2. Distinguish ordinary prompting, structured output, retrieval-augmented generation (RAG), tool use, fine-tuning, and agentic workflows.
3. Use LLMs safely for ideation, literature review support, coding, qualitative coding, survey design, and replication workflows.
4. Build a minimal R workflow for prompt templates, structured extraction, retrieval, and choice-model feature construction.
5. Explain why LLM-generated outputs are model-generated data products, not direct evidence about human preferences.
6. Design validation protocols for LLM-assisted environmental economics and choice modelling.
7. Decide when synthetic respondents may be useful for pilot testing and when they are inappropriate for inference or valuation.
8. Create a reproducibility record for LLM-assisted research: prompts, model versions, parameters, input hashes, outputs, and human review.

1 Why generative AI belongs in this textbook

Machine learning in the previous chapters was mostly about learning a mapping from inputs to outputs. A random forest maps predictors to outcomes. A causal forest maps covariates to treatment-effect estimates. A mixed logit maps alternative attributes and individual characteristics to choice probabilities. Generative AI adds a different capability: it can produce structured or unstructured content conditional on a prompt.

For economists, generative AI matters because a large part of research is language work. We read papers, design surveys, write code, translate concepts across fields, classify qualitative responses, create documentation, debug scripts, and explain findings to different audiences. Korinek's survey of generative AI for economic research organizes LLM use cases into domains such as

ideation and feedback, writing, background research, data analysis, coding, and mathematical derivations (Korinek, 2023). The message for this textbook is not that an LLM should do your economics. The message is that an LLM can become part of a reproducible workflow if its role is explicit and validated.

For environmental economics, generative AI is especially relevant because environmental evidence is heterogeneous. A single project may involve environmental valuation papers, policy documents, satellite metadata, stakeholder comments, interview transcripts, open-ended survey answers, regulatory texts, and R or Python code. For choice modelling, generative AI is relevant because the field depends on careful language: attribute descriptions, choice-task framing, labels, hypothetical scenarios, respondent instructions, latent constructs, and interpretation of behavioural outputs. The new *Handbook of Choice Modelling* explicitly places machine learning within modern model building, inference, and interpretation, while the choice modelling discussion paper by van Cranenburgh et al. argues that ML can help the field with tasks such as specification search and nontraditional data sources (Rodrigues, Krueger, and Pereira, 2024; van Cranenburgh et al., 2022).

The central warning

An LLM output is not a fact because it sounds fluent. It is a probabilistic generation conditioned on a prompt, model, system instructions, training data, and decoding settings. Treat every LLM output as you would treat an unverified research assistant's draft: useful, inspectable, but not authoritative until checked against data, theory, sources, and code.

1.1 A working definition

In this chapter, **generative AI** refers to models that generate text, code, images, tables, audio, or other content. We focus on LLMs because they are the most immediately useful tools for PhD research workflows. An LLM is a model trained on sequences of text tokens to predict or generate plausible continuations. Modern LLMs are usually based on the transformer architecture introduced by Vaswani et al. (2017), scaled up to large datasets and adapted to follow instructions, answer questions, use tools, and produce structured output.

1.2 Fit-for-purpose use

The correct question is not "Can an LLM do this?" The correct question is:

> What is the scientific status of the output, and what validation would make it acceptable evidence for the task at hand?

A grammar suggestion in a draft may require only author review. A literature claim requires source checking. A coding suggestion requires execution and tests. A qualitative classification requires a validation sample and inter-coder agreement. A synthetic stated preference response requires comparison with real respondents and should normally be treated as a pilot-test tool, not as valuation evidence.

2 Technical foundations

This section gives a minimal technical foundation. You do not need to become a language-model engineer, but you do need enough understanding to avoid common research mistakes.

2.1 Tokens and next-token prediction

LLMs do not process words exactly as humans do. Text is split into **tokens**: common character sequences, word pieces, words, punctuation, or symbols. A sentence such as "willingness to pay for biodiversity" becomes a sequence

$$x_1, x_2, \ldots, x_T.$$

An autoregressive language model learns a probability distribution over the next token:

$$p(x_{t+1} \mid x_1, \ldots, x_t).$$

Generation repeats this step many times. At each step, the model produces a distribution over possible next tokens, and a decoding rule selects or samples from that distribution. This is why a model can produce different answers to the same prompt when randomness is enabled.

2.2 Transformers and attention

The transformer architecture replaced recurrence with attention mechanisms. In simplified form, scaled dot-product attention is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

where queries Q, keys K, and values V are learned representations of tokens. Attention allows each token representation to incorporate information from other tokens in the context window. This is the technical reason why a model can use a definition from the beginning of a prompt when answering a question near the end, although long-context reliability remains an empirical issue rather than a guarantee.

2.3 Embeddings

An **embedding** maps text to a vector:

$$e(x) \in \mathbb{R}^d.$$

If two pieces of text have similar meaning, their embeddings should be close in vector space. A common similarity measure is cosine similarity:

$$\text{sim}(a, b) = \frac{a^\top b}{\|a\|\|b\|}.$$

Embeddings are useful for semantic search, clustering documents, deduplicating survey responses, finding similar policy comments, and building retrieval systems. They are not magic: they can encode bias, over-emphasize surface similarity, miss discipline-specific distinctions, and behave differently across models.

2.4 Instruction tuning and alignment

Base language models are trained to predict text. Instruction-tuned models are further adapted to follow user instructions. Many deployed systems are also trained or filtered to be safer, more helpful, or more policy-compliant. This improves usability, but it can also change how the model responds to controversial topics, survey questions, moral trade-offs, and politically sensitive environmental policies. This is important for synthetic respondents: an aligned chatbot may be more polite, cautious, or consensus-seeking than real respondents.

2.5 Decoding: temperature, top-p, and reproducibility

When a model generates text, it must choose tokens. The **temperature** parameter changes the sharpness of the token distribution. Lower temperature makes outputs more deterministic and conservative; higher temperature increases variety and risk. **Top-p** or nucleus sampling selects from a subset of tokens whose cumulative probability reaches a threshold, an approach motivated by work on neural text degeneration (Holtzman et al., 2020).

For research, lower temperature is usually preferable for extraction, coding, and reproducible analysis. Higher temperature can be useful for brainstorming, but brainstormed outputs should never be treated as evidence.

Key distinction: generation versus estimation

A discrete choice model estimates parameters from observed choices. An LLM generates text or structured outputs from a prompt. You can use LLM-generated variables inside an econometric model, but the generated variables are measurement outputs with their own error structure. They should be documented and validated.

3 The main LLM workflows

The most common research workflows can be organized into six families.

Workflow	What it does	Typical research use
Prompting	The user sends instructions and context to a model.	Brainstorming, explanation, code drafts, survey wording, quick checks.
Structured output	The model must return data in a schema such as JSON or a typed R object.	Coding open-ended responses, extracting attributes from policy documents, table creation.
RAG	The system retrieves relevant documents and asks the model to answer using only those documents.	Literature Q&A, project memory, policy-document search, reproducible source-grounded summaries.
Tool use	The model calls external functions such as a calculator, database query, or R function.	Running diagnostics, retrieving data, checking code, automating repetitive workflows.
Fine-tuning	A model is adapted using task-specific examples.	Repeated classification/extraction tasks with stable labels and enough verified training data.
Agentic workflows	A model plans, calls tools, revises, and loops through subtasks.	Complex automation, but high risk unless bounded by tests and human checkpoints.

For most PhD projects in environmental economics and choice modelling, the safest progression is:

$$\text{prompting} \rightarrow \text{structured output} \rightarrow \text{RAG} \rightarrow \text{tool use},$$

with fine-tuning only when you have validated training data and repeated tasks. Agentic workflows are powerful but should be treated as software systems requiring tests, logs, and permission boundaries.

4 Prompting as research design

A prompt is not a casual message. In a research workflow, a prompt is part of the measurement instrument. It can induce framing effects, hidden assumptions, response instability, and systematic bias. Treat prompt design like survey design or experimental design.

4.1 The anatomy of a good research prompt

A good prompt usually specifies:

1. **Role**: what expertise the model should emulate.
2. **Task**: the exact output requested.
3. **Context**: data, definitions, assumptions, sources, or excerpts.
4. **Constraints**: what not to do, required format, length, citations, or uncertainty statements.
5. **Criteria**: how the answer will be evaluated.
6. **Output format**: prose, table, JSON, R code, checklist, or equations.
7. **Verification instruction**: ask the model to identify uncertain items and separate facts from inferences.

Listing 1: A prompt template for a choice-modelling research task.

```
You are assisting with a PhD project in environmental choice modelling.

Task: review the attribute descriptions below for a discrete choice experiment.

Context: the study values urban biodiversity improvements. Respondents choose among
three policy programmes plus a status quo. The target population is urban households.

Evaluate each attribute on:
1. clarity for non-experts;
2. neutrality of wording;
3. whether the attribute is likely to be consequential;
4. risk of overlap with other attributes;
5. possible sources of hypothetical bias.

Output: return a table with columns: attribute, problem, severity, suggested revision,
and reason. Do not invent new attributes unless the existing set has a clear gap.
Flag uncertainties.

Attributes:
[PASTE ATTRIBUTE DESCRIPTIONS HERE]
```

This prompt does not ask the model to decide the final design. It asks for a structured critique that the researcher can evaluate.

4.2 Prompting patterns that are useful for PhD work

Self-explanation prompt. Ask the model to teach a concept, then ask it to create retrieval questions. This is useful when learning unfamiliar methods.

Critic prompt. Ask the model to identify weaknesses in a proposed model, survey, or code pipeline. This is useful after you have made a draft.

Adversarial prompt. Ask what would make your conclusion invalid. This is useful for pre-analysis planning.

Schema prompt. Ask for output in an explicit structure. This is useful for reproducibility.

Grounded prompt. Provide sources and require the answer to use only them. This is essential for literature and policy-document workflows.

Do not ask for hidden reasoning transcripts

For research workflows, ask for concise reasoning summaries, assumptions, checks, and intermediate quantities when needed. Do not rely on a model's private reasoning as evidence. The evidence is the data, code, sources, and externally verifiable outputs.

5 Using LLMs from R

Browser-based chat can be useful for learning, but it is difficult to reproduce. For research, prefer scripted workflows. R now has several packages for interacting with LLMs. The `ellmer` package provides an R interface to multiple LLM providers and supports chat, tool/function calling, and structured data extraction. Local-model workflows are possible through tools such as Ollama and R wrappers such as `rollama`. Provider policies, models, prices, and APIs change; always check current documentation before running a real project.

5.1 A minimal scripted chat

The following code illustrates the idea. It is written to be explicit rather than compact. You will need an API key or a local model endpoint depending on the provider you use.

Listing 2: A minimal R workflow using ellmer.

```r
# Install once if needed:
# install.packages(c("ellmer", "glue"))

library(ellmer)
library(glue)

# Store keys in environment variables, not in scripts.
# Sys.setenv(OPENAI_API_KEY = "your-key")  # do this outside the project file

chat <- chat_openai()  # or another approved provider supported by ellmer

prompt <- glue(
"You are helping with a PhD project on environmental choice modelling.
Explain the difference between prediction accuracy and welfare validity.
Use no more than 180 words and include one example involving biodiversity valuation."
)

answer <- chat$chat(prompt)
cat(answer)
```

Do not store API keys in your Quarto file, Git repository, or shared folder. Use environment variables, project secrets, or your institution's approved credential system.

5.2 Prompt templates

A prompt template is a reproducible function. It makes your LLM calls auditable.

Listing 3: A prompt template for survey wording review.

```
library(glue)

make_attribute_review_prompt <- function(study_context, attributes) {
  glue(
"You are reviewing a discrete choice experiment for environmental valuation.

Study context:
{study_context}

Attributes:
{attributes}

Return a markdown table with columns:
attribute | clarity issue | neutrality issue | behavioural risk | revision.

Rules:
- Do not add unsupported factual claims.
- Flag uncertain judgments.
- Prefer simple language suitable for a general population survey."
  )
}

study_context <- "Urban households choose among biodiversity improvement programmes."
attributes <- paste(
  "Annual cost to your household",
  "Increase in native bird and pollinator habitat",
  "Restrictions on access to restored areas",
  sep = "\n- "
)

prompt <- make_attribute_review_prompt(study_context, attributes)
cat(chat$chat(prompt))
```

For a real study, save the prompt, model, date, parameters, and output. The prompt is part of your research instrument.

6 Structured extraction from open-ended text

Environmental economics and choice surveys often include open-ended responses:

- Why did you choose the status quo?
- What did you dislike about the policy programme?
- Which attributes did you ignore?
- What does biodiversity mean to you?
- Why do you oppose or support a climate adaptation measure?

Traditional text-as-data methods use dictionaries, supervised classifiers, topic models, embeddings, or human coding (Gentzkow, Kelly, and Taddy, 2019). LLMs add a new option:

schema-guided classification or extraction. This can be useful, but it must be validated like any other measurement model.

6.1 A coding task

Suppose respondents explain their choice in a biodiversity DCE. We want to code whether each response mentions cost, distrust, fairness, biodiversity concern, or confusion. The result can be used descriptively or as a covariate in a choice model. It should not be treated as error-free.

Listing 4: Structured coding of open-ended responses.

```
# install.packages(c("ellmer", "purrr", "dplyr", "tibble"))
library(ellmer)
library(purrr)
library(dplyr)
library(tibble)

chat <- chat_openai()

responses <- tibble(
  respondent_id = 1:4,
  text = c(
    "I chose the cheaper plan because the fee was too high.",
    "More birds and insects would make the park better for my children.",
    "I do not trust the council to actually deliver this programme.",
    "I was not sure what the biodiversity score meant."
  )
)

code_type <- type_object(
  main_reason = type_enum(
    c("cost", "biodiversity", "distrust", "fairness", "confusion", "other"),
    "The main reason expressed by the respondent."
  ),
  mentions_cost = type_boolean("Does the text mention monetary cost or affordability?"
    ),
  mentions_distrust = type_boolean("Does the text express distrust in institutions or
    delivery?"),
  mentions_confusion = type_boolean("Does the text express confusion or lack of
    understanding?"),
  short_rationale = type_string("One-sentence justification for the code.")
)

code_one <- function(txt) {
  prompt <- paste(
    "Code this open-ended response from an environmental choice experiment.",
    "Use only the information in the text. If uncertain, choose 'other' and explain.",
    "Response:", txt,
    sep = "\n"
  )
  as_tibble(chat$chat_structured(prompt, type = code_type))
}

coded <- map_dfr(responses$text, code_one) %>%
  bind_cols(responses %>% select(respondent_id, text), .)

coded
```

6.2 Validation

Structured extraction is not finished when the code runs. A minimal validation workflow is:

1. Create a human-coded gold-standard subset.
2. Blind-code at least part of the subset by two humans if possible.
3. Compare LLM codes with human codes using confusion matrices, precision/recall, and agreement statistics.
4. Inspect errors by subgroup, language, education, and response length.
5. Freeze the prompt and model before coding the full sample.
6. Report the procedure and uncertainty.

Listing 5: Comparing LLM codes with a human-coded validation sample.

```
# Suppose validation contains human labels and LLM labels.
# install.packages(c("yardstick", "dplyr"))
library(yardstick)
library(dplyr)

validation <- tibble(
  human_main_reason = factor(c("cost", "biodiversity", "distrust", "confusion")),
  llm_main_reason   = factor(c("cost", "biodiversity", "other", "confusion"),
                             levels = levels(human_main_reason))
)

conf_mat(validation, truth = human_main_reason, estimate = llm_main_reason)
accuracy(validation, truth = human_main_reason, estimate = llm_main_reason)
```

For class-imbalanced labels, do not rely only on accuracy. A model that never identifies "distrust" may still look accurate if distrust is rare.

Measurement interpretation

If an LLM says a comment expresses distrust, the variable should be interpreted as "classified as distrust by this model, prompt, and validation procedure", not as direct access to the respondent's true latent attitude. This is the same logic used for any survey measure or proxy variable.

7 Embeddings and retrieval-augmented generation

LLMs are prone to factual errors when asked to answer from memory. Retrieval-augmented generation (RAG) reduces this risk by retrieving relevant documents and conditioning the answer on those documents. The original RAG paper by Lewis et al. (2020) combined a neural retriever with a sequence-to-sequence generator and showed benefits for knowledge-intensive tasks. For research workflows, the practical idea is simple:

$$\text{question} \rightarrow \text{retrieve relevant chunks} \rightarrow \text{answer using chunks} \rightarrow \text{cite chunks.}$$

RAG is not a guarantee of truth. Retrieval can miss relevant documents, retrieve the wrong chunks, or give the model misleading context. The model can still overstate. But RAG is usually safer than asking a general model to recall sources from memory.

7.1 A minimal retrieval system in R

The following R example uses lexical retrieval rather than neural embeddings. It is not a production RAG system, but it teaches the mechanics. Replace the toy documents with your own abstracts, policy excerpts, or notes.

Listing 6: A simple retrieval pipeline with TF-IDF in R.

```r
# install.packages(c("tidyverse", "tidytext"))
library(tidyverse)
library(tidytext)

chunks <- tibble(
  chunk_id = 1:4,
  source = c("DCE design notes", "Policy report", "Choice model notes", "Validation
    notes"),
  text = c(
    "A status quo alternative helps estimate welfare changes relative to current
    conditions.",
    "Urban biodiversity policy may affect pollinators, birds, access restrictions, and
     taxes.",
    "The cost coefficient must usually be negative for willingness-to-pay calculations
    .",
    "LLM-coded qualitative variables should be validated against human coding."
  )
)

query <- "How do I compute willingness to pay in a biodiversity choice experiment?"

# Tokenize documents and query.
doc_words <- chunks %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word") %>%
  count(chunk_id, source, word, name = "n") %>%
  bind_tf_idf(word, chunk_id, n)

query_words <- tibble(text = query) %>%
  unnest_tokens(word, text) %>%
  anti_join(stop_words, by = "word") %>%
  distinct(word)

scores <- doc_words %>%
  semi_join(query_words, by = "word") %>%
  group_by(chunk_id, source) %>%
  summarize(score = sum(tf_idf), .groups = "drop") %>%
  arrange(desc(score))

retrieved <- chunks %>%
  semi_join(scores %>% slice_head(n = 2), by = "chunk_id")

retrieved
```

You can then pass the retrieved text to an LLM with strict instructions.

Listing 7: Prompting an LLM with retrieved context.

```r
context <- paste(
  paste0("Source: ", retrieved$source, "\n", retrieved$text),
  collapse = "\n\n"
```

```
)

rag_prompt <- glue::glue(
"Answer the question using only the context below.
If the answer is not in the context, say that the context is insufficient.
Cite the source names in parentheses.

Question: {query}

Context:
{context}"
)

cat(chat$chat(rag_prompt))
```

7.2 Semantic embeddings with Python from R

For semantic retrieval, embeddings are often better than keyword matching. One practical route is to call Python's `sentence-transformers` package from R using `reticulate`. The exact model you choose should be justified, documented, and tested on your domain.

Listing 8: Calling sentence-transformers from R via reticulate.

```
# In a terminal or environment setup, install first:
# pip install sentence-transformers
# install.packages(c("reticulate", "Matrix"))

library(reticulate)
library(Matrix)

st <- import("sentence_transformers")
model <- st$SentenceTransformer("all-MiniLM-L6-v2")

texts <- chunks$text
emb <- model$encode(texts, normalize_embeddings = TRUE)
q_emb <- model$encode(query, normalize_embeddings = TRUE)

# Because embeddings are normalized, dot product equals cosine similarity.
sim <- as.numeric(emb %*% q_emb)
chunks$similarity <- sim
chunks %>% arrange(desc(similarity))
```

7.3 When to use RAG in this textbook's domains

RAG is useful when you have a stable source corpus and need source-grounded answers. Examples include:

- querying your annotated folder of DCE design papers;
- searching environmental policy documents for mentions of compensating variation, distributional impacts, or consultation requirements;
- retrieving previous model specifications from your lab's project archive;
- building a thesis assistant that answers only from your own notes and PDFs;

- checking whether a survey attribute wording is consistent with previous stated preference studies.

RAG is less suitable when the source corpus is missing, low quality, biased, or not legally usable. It also cannot fix a poor research question.

8 Generative AI for literature review

A literature review is not a pile of summaries. It is an argument about what is known, what is uncertain, and where your contribution fits. LLMs can help with the mechanics of reading, but they should not decide the intellectual structure of the review.

8.1 Acceptable uses

Good uses include:

- converting abstracts into preliminary themes;
- extracting research questions, methods, datasets, and outcomes into a table;
- identifying unfamiliar terms;
- generating search strings for databases;
- checking whether your summary is understandable;
- creating flashcards for key concepts;
- helping translate a method from Python to R or from R to pseudocode.

Risky uses include:

- asking for uncited literature summaries from memory;
- accepting generated references without verification;
- letting the model decide which papers are authoritative;
- using model-generated claims in a thesis without checking the original source;
- summarizing copyrighted texts in ways that violate license terms.

8.2 A source-grounded extraction template

Listing 9: A source-grounded literature extraction prompt.

```
You are extracting information from a paper for a systematic review.
Use only the excerpt provided. Do not use outside knowledge.

Return a table with columns:
- research question
- data type
- model family
- validation strategy
- main finding
- limitation stated by authors
- relevance for environmental economics or choice modelling
```

```
If a field is not present in the excerpt, write "not stated".

Excerpt:
[PASTE PAPER EXCERPT]
```

This style of prompt produces a traceable data product. You can audit it against the source.

8.3 Quality control for LLM-assisted reviews

For a publishable review, build an audit trail:

1. Search strategy: databases, keywords, inclusion/exclusion criteria.
2. Source storage: PDFs or bibliographic records with stable identifiers.
3. Extraction prompt: frozen and versioned.
4. Extraction output: saved as CSV/JSON with document IDs.
5. Human audit: random sample checked against original papers.
6. Error report: types of extraction error and correction procedure.

This is similar to the methodological discipline Hillel et al. (2021) recommend for ML mode choice comparisons: performance claims require careful validation, not casual train/test choices or unverifiable pipelines.

9 Generative AI for coding and reproducible computation

LLMs can be excellent coding assistants. They can explain error messages, generate examples, translate code, write tests, and suggest package functions. But generated code should be treated like code copied from the internet: it may run, be wrong, be inefficient, or silently implement a different model.

9.1 Coding workflow

A safe coding workflow is:

1. Ask for a minimal example, not a full thesis pipeline.
2. Run the code in a clean script.
3. Add unit tests or simulated-data checks.
4. Compare results with known formulas or package vignettes.
5. Ask the model to explain assumptions, then verify them manually.
6. Save the prompt and code output if the model contributed materially.

9.2 Example: asking for a simulated check

Suppose an LLM writes an R function to compute WTP. You should test it on known coefficients.

Listing 10: Testing an LLM-generated WTP function.

```r
wtp <- function(beta_attribute, beta_cost) {
  -beta_attribute / beta_cost
}

# Test: if beta_attribute = 0.5 and beta_cost = -0.1,
# WTP should be 5 monetary units.
stopifnot(abs(wtp(0.5, -0.1) - 5) < 1e-8)

# Test: WTP is undefined or misleading if the cost coefficient is zero.
tryCatch(
  wtp(0.5, 0),
  warning = function(w) message("warning caught"),
  error = function(e) message("error caught")
)
```

A more robust function would check for near-zero cost coefficients and report uncertainty using the delta method, simulation, or bootstrapping.

9.3 When code assistance is especially valuable

LLMs are useful for:

- converting a dataset from wide to long choice format;
- writing checks for one chosen alternative per choice set;
- translating a likelihood equation into R code;
- creating simulation data for model recovery;
- writing Quarto templates for reproducible reports;
- producing package-specific skeletons for `mlogit`, `apollo`, `gmnl`, or `logitr`.

But do not assume that a generated `apollo` or `mlogit` specification is behaviourally correct. Check coding of alternatives, signs, units, availability, panel structure, and scale.

10 Environmental economics applications

Generative AI can support environmental economics at several stages. The key is to separate **administrative productivity** from **scientific evidence**.

10.1 Policy-document analysis

Environmental economists often analyze regulations, consultation documents, environmental impact statements, and policy reports. LLMs can extract dates, policy instruments, affected sectors, stated objectives, mitigation measures, or references to ecosystem services. A robust workflow uses RAG or structured extraction and validates against human coding.

Example tasks:

- classify policy instruments into tax, subsidy, standard, information, or voluntary agreement;
- extract whether a policy mentions distributional impacts;

- identify stated target outcomes such as biodiversity, water quality, or emissions;
- summarize stakeholder objections in consultation comments.

10.2 Survey design and pretesting

LLMs can help review survey wording for clarity, neutrality, and reading level. They can generate alternative descriptions of complex environmental goods, but the researcher must ensure scientific accuracy and consequentiality. Environmental valuation guidance emphasizes careful design, implementation, and analysis of DCEs; an LLM can assist this process but does not replace the need for expert design and respondent pretesting (Johnston et al., 2021; Faccioli et al., 2025).

10.3 Open-ended response coding

In environmental valuation, open-ended responses can reveal protest responses, consequentiality concerns, distrust, attribute non-attendance, confusion, and moral objections. LLM coding may be useful, but it should be compared with human coding. The result can then be used for sensitivity analysis.

Example:

- Estimate a baseline choice model.
- Code open-ended responses for protest indicators.
- Re-estimate excluding or modelling protest responses.
- Report how WTP changes.

10.4 Connecting qualitative codes to choice models

Suppose LLM-assisted coding identifies whether respondents mention affordability. You can use this code as an interaction in a choice model.

Listing 11: Using an LLM-coded respondent variable in a choice model.

```
# choice_long: long-format choice data with id, chid, alt, choice, cost, biodiversity
# respondent_codes: one row per respondent with cost_concern coded from open text

choice_augmented <- choice_long %>%
  left_join(respondent_codes %>% select(id, cost_concern), by = "id") %>%
  mutate(cost_concern = as.integer(cost_concern))

mdata <- mlogit::mlogit.data(
  choice_augmented,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

fit <- mlogit::mlogit(
  choice ~ cost + biodiversity + cost:cost_concern | 0,
  data = mdata
)
```

```
summary(fit)
```

Interpretation must be cautious. The interaction captures association between model-coded cost concern and marginal utility of cost. It is not necessarily causal evidence that a concern causes different price sensitivity.

11 Choice modelling applications

Choice modelling is the main focus of this textbook. Generative AI affects the field in at least eight ways.

11.1 Attribute discovery and wording

LLMs can help generate candidate attribute lists from interviews, policy documents, or literature. They can also suggest simpler wording. However, attribute selection remains a theoretical and empirical task. An attribute must be relevant, understandable, policy-controllable, and not redundant with other attributes.

Do not let an LLM invent the good being valued

Environmental attributes should be grounded in ecology, policy, and respondent understanding. If a model invents a vague attribute such as "nature quality", the research team must translate it into measurable, credible, and consequential levels.

11.2 Choice-task comprehension

Before fielding a survey, ask an LLM to act as a skeptical respondent and identify confusing instructions. This is only a pretest supplement. Real cognitive interviews with humans remain necessary.

11.3 Specification search

LLMs can read a list of candidate variables and suggest plausible interactions, transformations, or random parameters. This can complement algorithmic assisted specification methods such as the multi-objective combinatorial approach proposed by Ortelli et al. (2021) or the association-rule and random-forest approach of Hernandez et al. (2023). The LLM should not replace cross-validation, likelihood-based comparison, behavioural plausibility checks, or theory.

11.4 Latent constructs and open text

Uploaded studies such as Torres Lahoz et al. (2023) show how machine learning can help incorporate attitudinal indicators into latent class choice models. LLMs provide another route for transforming open-ended text into variables related to attitudes, beliefs, distrust, fairness, or environmental identity. These variables can enter class membership models, hybrid choice models, or interaction terms, but they need measurement validation.

11.5 Code generation for choice packages

LLMs can produce skeleton code for `mlogit`, `apollo`, `gmnl`, `logitr`, or Biogeme. Use the generated code as a starting point, not as a trusted specification. Check:

- whether the data are in the correct long format;
- whether choice sets have exactly one chosen alternative;
- whether availability is coded correctly;
- whether alternative-specific variables and individual-specific variables are placed correctly;
- whether panel likelihood is specified when repeated choices are present;
- whether monetary coefficients have plausible signs;
- whether WTP calculations use consistent units.

11.6 Multimodal choice experiments

Some choice experiments use images, maps, or videos. Recent choice-modelling work compares DCMs and ML models for street-view image choices, while the broader ML-choice literature emphasizes the potential of text and image data (Zhu and Si, 2024; van Cranenburgh et al., 2022). Generative AI can help create draft stimuli, captions, or accessibility descriptions. It can also introduce uncontrolled visual confounds. If generated images are used in an experiment, the generation process, prompts, seed/settings, post-processing, and validation must be documented.

11.7 Synthetic respondents

LLMs can be prompted to answer surveys as if they were members of certain populations. Argyle et al. (2023) propose language models as proxies for specific human subpopulations in social science research; Horton, Filippas, and Manning (2023, revised 2026) discuss LLMs as simulated economic agents; recent work even tests whether LLMs can predict results of social science experiments. In choice modelling, emerging work asks whether LLMs can assist choice model specification and whether they can simulate stated preference responses in energy contexts (Sfeir et al., 2025; Zhang et al., 2025).

The right interpretation is cautious. Synthetic respondents can help with:

- debugging survey logic;
- stress-testing confusing wording;
- generating hypotheses about heterogeneity;
- checking whether alternatives are obviously dominated;
- exploring possible response patterns before expensive data collection.

They should not be used as a substitute for human respondents when estimating welfare, WTP, elasticities, or policy support.

Rule for synthetic respondents

Use synthetic respondents for pilot testing and hypothesis generation. Use human respondents, revealed behaviour, or credible experimental data for inference about human preferences and welfare.

11.8 LLM-assisted model interpretation

After estimating a choice model, an LLM can help translate coefficients into plain-language summaries. But it must be given exact model outputs and strict instructions not to invent. For example, it can draft a paragraph explaining that a negative cost coefficient implies lower utility for higher costs. You still need to verify every number, sign, and unit.

12 A full workflow: biodiversity DCE with LLM assistance

This section shows how generative AI can fit into a full PhD workflow without taking over scientific judgment.

12.1 Research question

Suppose the project asks:

> What are urban households willing to pay for biodiversity improvements, and how do distrust and perceived fairness affect support for conservation funding?

12.2 Workflow map

Stage	LLM role	Human validation
Literature search	Suggest search strings and extract methods from source excerpts.	Search databases manually; verify all citations; audit extracted tables.
Attribute development	Critique wording and identify possible overlap.	Expert review, ecological validity, focus groups, cognitive interviews.
Survey pretest	Act as a simulated respondent to identify confusion.	Real pilot respondents and timing data.
Open-ended coding	Code reasons for choices into a schema.	Human-coded validation set and error analysis.
Choice modelling	Generate code skeletons and diagnostic checklists.	Run code, test on simulated data, check signs and likelihoods.
Interpretation	Draft plain-language summaries.	Verify all estimates, uncertainty intervals, and welfare formulas.
Reporting	Create replication README and disclosure statement.	Ensure reproducibility and institutional compliance.

12.3 An integrated R skeleton

Listing 12: Integrated skeleton for an LLM-assisted DCE project.

```
# Packages
library(tidyverse)
library(ellmer)
library(mlogit)

# 1. Load choice data and open-ended responses.
```

```r
choice_long <- read_csv("data/choice_long.csv")
open_text <- read_csv("data/open_ended_responses.csv")

# 2. Set up LLM client.
chat <- chat_openai()

# 3. Define a frozen coding schema.
reason_type <- type_object(
  protest = type_boolean("True if respondent rejects the payment vehicle or survey
    premise."),
  cost_concern = type_boolean("True if respondent mentions cost or affordability."),
  distrust = type_boolean("True if respondent mentions lack of trust in institutions."
    ),
  biodiversity_motive = type_boolean("True if respondent mentions biodiversity,
    species, or habitat."),
  confidence = type_enum(c("low", "medium", "high"), "Confidence in the coding."),
  rationale = type_string("Short reason for the coding.")
)

code_response <- function(txt) {
  prompt <- paste(
    "Code this response from a biodiversity discrete choice experiment.",
    "Use only the response text. Do not infer demographics or politics.",
    "Response:", txt,
    sep = "\n"
  )
  as_tibble(chat$chat_structured(prompt, type = reason_type))
}

# 4. Code responses. In a real project, cache outputs and audit a validation subset.
# coded <- purrr::map_dfr(open_text$response, code_response)
# respondent_codes <- bind_cols(open_text %>% select(id), coded)

# 5. Merge validated codes into choice data.
# choice_augmented <- choice_long %>% left_join(respondent_codes, by = "id")

# 6. Estimate choice model.
# mdata <- mlogit.data(choice_augmented, choice = "choice", shape = "long",
#                      chid.var = "chid", alt.var = "alt", id.var = "id")
# fit <- mlogit(choice ~ cost + biodiversity + cost:cost_concern | 0, data = mdata)
# summary(fit)

# 7. Report sensitivity: baseline, validated-code model, protest-response exclusion.
```

The lines are intentionally commented where real API calls or data files are required. In your own project, cache outputs so you do not repeatedly send the same data to a provider, and so your results can be audited.

13 Synthetic respondents and choice experiments

Synthetic respondent workflows deserve a separate discussion because they are tempting and dangerous.

13.1 What the model is doing

A synthetic respondent prompt might specify age, income, location, transport habits, and environmental attitudes, then ask the LLM to choose among alternatives. The output is generated from patterns in the model and the prompt, not from a real person's preferences. Even if aggregate synthetic choices resemble a survey, the mechanism is not the same as observing human choices.

13.2 A pilot-testing example

Listing 13: A synthetic respondent prompt for pilot testing only.

```
You are simulating a respondent for survey pretesting, not producing data for
    inference.
Persona: urban renter, age 32, moderate income, uses public parks weekly, concerned
    about rent.

Choice task:
A: Annual cost 20 euros; medium biodiversity gain; no access restriction.
B: Annual cost 80 euros; high biodiversity gain; moderate access restriction.
C: Status quo; annual cost 0 euros; no biodiversity gain.

Choose A, B, or C. Then explain what information was confusing or missing.
```

The useful output is not the choice itself. The useful output is the critique: "access restriction is vague", "annual cost needs payment vehicle", or "biodiversity gain needs an example."

13.3 Validation requirements

If a researcher wants to use synthetic respondents beyond pilot testing, validation must be severe:

- compare synthetic and human marginal choice shares;
- compare estimated coefficients and WTP from human and synthetic data;
- test robustness across model providers, prompts, temperatures, and personas;
- check subgroup calibration;
- ensure no leakage from published target data;
- preregister what role synthetic data will play;
- report failures, not only successes.

Even then, synthetic data should not be merged with human data as if it were equivalent. It is better treated as a model-based simulation.

14 Responsible use: privacy, confidentiality, and ethics

Research data in environmental economics and choice modelling may include personal data, locations, income, political attitudes, health information, mobility traces, or open-ended comments. Do not paste confidential data into public or consumer AI tools unless your institution's policy and the provider's terms permit it.

Provider policies differ and change. Official OpenAI documentation states that API data is not used to train or improve OpenAI models by default unless the user opts in, and OpenAI's business privacy pages state that business offerings do not train on organization data by default. This does not mean every tool, plugin, browser extension, or third-party wrapper has the same protections. Verify the actual data path.

14.1 A practical privacy hierarchy

From safest to riskiest:

1. Use non-sensitive synthetic or simulated data.
2. Use local models on secured institutional hardware.
3. Use institution-approved enterprise/API services with appropriate data-processing terms.
4. Use de-identified excerpts after checking re-identification risk.
5. Avoid consumer chat interfaces for confidential respondent data.

14.2 Documentation and disclosure

A minimal disclosure statement should include:

- the tasks for which generative AI was used;
- the model/provider and date of use;
- whether confidential data were processed;
- how outputs were validated;
- whether generated text was edited by the author;
- where prompts and outputs are stored in the replication archive, if appropriate.

Example disclosure

Generative AI was used to draft R code skeletons, suggest survey wording alternatives, and classify open-ended responses using a preregistered prompt schema. All code was manually checked and tested. A 15 percent validation sample of qualitative classifications was human-coded and compared with the LLM outputs. No confidential respondent identifiers were sent to external AI services.

15 Reproducibility record

LLM outputs are not fully reproducible in the same way as deterministic code. Models change, APIs change, safety filters change, and decoding can be stochastic. Your job is to make the workflow as auditable as possible.

For each material LLM call, record:

- date and time;
- model/provider or local model checkpoint;
- software package versions;

- system prompt and user prompt;
- input data identifiers or hashes;
- temperature, top-p, maximum tokens, and other parameters;
- raw output;
- post-processing code;
- human edits or corrections;
- validation results.

Listing 14: A simple log for LLM-assisted steps.

```r
library(tibble)
library(digest)

log_llm_call <- function(task, model, prompt, output, temperature = NA_real_) {
  tibble(
    timestamp = as.character(Sys.time()),
    task = task,
    model = model,
    temperature = temperature,
    prompt_hash = digest(prompt, algo = "sha256"),
    output_hash = digest(output, algo = "sha256"),
    prompt = prompt,
    output = output
  )
}

# Example:
# record <- log_llm_call("attribute wording review", "provider-model-name", prompt,
    answer)
# readr::write_csv(record, "logs/llm_calls.csv", append = file.exists("logs/llm_calls.
    csv"))
```

Hashing does not replace storing the prompt and output, but it helps detect accidental changes.

16 Common mistakes

Mistake 1: Treating fluent output as expert knowledge. A fluent explanation can be wrong. Verify against sources.

Mistake 2: Asking for citations from model memory. Use RAG or manual search. Generated citations can be invalid.

Mistake 3: Uploading sensitive data casually. Check institutional policy, provider terms, and data-processing agreements.

Mistake 4: Using synthetic respondents as real respondents. Synthetic responses are simulations, not observed human choices.

Mistake 5: Ignoring prompt sensitivity. Change the wording, role, examples, or order and outputs may change.

Mistake 6: Hiding AI assistance. Material use should be documented according to institutional, journal, and funder norms.

Mistake 7: Automating without tests. LLM-generated code must be run, inspected, and tested.

Mistake 8: Confusing qualitative coding with causal measurement. LLM-coded text variables can support analysis, but they do not identify causal effects by themselves.

17 Concept checkpoint

Answer these without looking back.

1. What is the difference between an LLM and a discrete choice model?
2. Why does RAG reduce but not eliminate hallucination risk?
3. Why is a prompt a measurement instrument in qualitative coding?
4. What should you record for reproducibility when using an LLM in a research workflow?
5. Give one safe and one unsafe use of synthetic respondents in a DCE project.
6. How would you validate LLM-coded protest responses in an environmental valuation survey?
7. Why might an aligned chatbot be a poor simulator of real respondents' preferences?
8. How can LLM-coded variables be included in a choice model, and what interpretation problem remains?

18 Applied lab

Lab 7.1: Build an LLM-assisted qualitative coding workflow

Use a small set of open-ended responses from a stated preference or environmental survey. If you do not have real data, create 30 simulated comments. Your task is to:

1. Define a coding schema with five labels.
2. Write a prompt that forbids inference beyond the text.
3. Code the responses using structured output.
4. Hand-code 10 responses yourself.
5. Compare human and LLM labels.
6. Revise the schema if errors reveal ambiguity.
7. Write a 300-word methods note describing the workflow and limitations.

Lab 7.2: Build a tiny RAG assistant for your thesis

Create a folder with five abstracts or short excerpts from papers you trust. Then:

1. Split the documents into chunks.
2. Build either lexical or semantic retrieval.
3. Retrieve the top three chunks for a question.
4. Ask an LLM to answer using only those chunks.

5. Check whether the answer is fully supported.
6. Document one retrieval failure and how you would fix it.

Lab 7.3: Survey design stress test

Take one choice task from your own or a published DCE. Ask an LLM to critique the task from the perspective of three respondent personas: low education, high environmental concern, and distrustful of government. Record only issues that could plausibly affect real respondents. Then design a human pretest question to check each issue.

19 Chapter summary

Generative AI is best understood as an assistant for research workflows, not as an oracle. In environmental economics, it can help process policy texts, open-ended survey responses, and complex documentation. In choice modelling, it can assist survey wording, coding, specification brainstorming, code generation, and interpretation. The main scientific challenge is validation: the more an LLM output influences evidence, measurement, inference, or welfare analysis, the stronger the validation required.

The safest workflow is structured, source-grounded, logged, and human-reviewed. Use LLMs to accelerate low-level tasks and broaden your thinking. Do not use them to bypass data collection, identification, behavioural theory, or welfare logic.

Key terms

Term	Meaning
Token	A unit of text processed by a language model, often a word piece or character sequence.
Transformer	Neural architecture based on attention mechanisms, foundational for modern LLMs.
Embedding	Vector representation of text, image, or other content.
Prompt	Instruction and context sent to a generative model.
Structured output	Model output constrained to a schema, such as JSON or typed R objects.
RAG	Retrieval-augmented generation: retrieve relevant sources, then generate an answer conditioned on them.
Temperature	Decoding parameter controlling randomness in generation.
Hallucination	A fluent but unsupported or false model output.
Synthetic respondent	Model-generated response intended to simulate a human respondent.
Tool use	Allowing a model to call external functions or software.
Fine-tuning	Adapting a model using task-specific training examples.
AI disclosure	Statement documenting material use of AI in a research project.

Further reading

Start with Korinek (2023) for economic research workflows, Gentzkow, Kelly, and Taddy (2019) for text as data in economics, Lewis et al. (2020) for RAG, and Bommasani et al. (2021) for the broader foundation-model framing. For choice modelling, read van Cranenburgh et al. (2022) and the machine-learning chapter in the *Handbook of Choice Modelling*. For synthetic respondents, read Argyle et al. (2023), Horton, Filippas, and Manning (2023, revised 2026), and recent validation studies before using the idea in your own research.

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Machine Learning and AI for Environmental Economics and Choice Modelling

Chapter 8

Environmental Data for Machine Learning

From surveys and administrative records to satellites, sensors, text, images, and choice context

Self-study textbook for PhD students

Main implementation language: R, with selected Python workflows

Prepared for the user by OpenAI GPT-5.5 Pro
July 16, 2026

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How to study this chapter

This chapter starts Part II of the textbook. The first seven chapters built the general machine learning, causal, interpretability, and generative AI foundations. We now move to environmental economics. The main lesson is simple but easy to underestimate: in environmental applications, the hard part is often not the algorithm. The hard part is building a valid data object.

A random forest, gradient boosting model, neural network, causal forest, or mixed logit model is only as defensible as the data that enter it. Environmental data are spatial, temporal, multi-resolution, sometimes remotely sensed, often noisy, frequently linked from several institutions, and rarely collected for the exact research question at hand. Choice data add another layer: the data are structured by decision makers, alternatives, choice occasions, attributes, and choice sets.

Read this chapter in three passes. On the first pass, focus on the data taxonomy and the intuition behind spatial and temporal validation. On the second pass, work through the R labs. On the third pass, return to the design questions and ask how you would build the dataset for your own PhD project.

Learning checkpoint

After completing this chapter, you should be able to:

1. Classify the main data sources used in ML for environmental economics and choice modelling.
2. Explain why spatial, temporal, and hierarchical dependence change the way we validate ML models.
3. Link environmental context to individuals, alternatives, and choice occasions without creating data leakage.
4. Build a simple spatial ML workflow in R using vector and raster data structures.
5. Distinguish measurement error, prediction error, imputation error, and policy extrapolation error.
6. Design a defensible data pipeline for environmental valuation or environmental choice modelling.

1 Why environmental data are different

Machine learning textbooks often begin with a rectangular dataset: one row per observation, one column per feature, one target variable. Environmental economics rarely begins there. A typical project may combine household survey responses, choice experiment tasks, parcel boundaries, weather histories, satellite-derived vegetation indices, air quality monitor readings, travel diaries, smartphone tracking data, census covariates, administrative programme records, and text from policy consultations.

Economists have long used weather, climate, and geospatial data, but the scale and richness of these data have expanded. Satellite imagery and derived remote sensing products can now be used to measure economic and environmental conditions at high spatial and temporal resolution (Donaldson and Storeygard, 2016; Burke et al., 2021). Weather and climate data can provide important quasi-random variation, but the choice of weather product, aggregation rule, spatial match, and time window can change estimates substantially (Auffhammer et al., 2013). Modern ML adds tools for prediction and representation learning, but it also raises new risks: spatial leakage, temporal leakage, covariate shift, algorithmic extrapolation, and black-box treatment of measurement error.

In this chapter, *environmental data* means any data describing environmental states, exposures, amenities, risks, resources, ecological processes, or environmentally relevant behaviour. This includes both direct measurements, such as PM2.5 or forest cover, and indirect proxies, such as nighttime lights, street-view greenness, or social media text about climate risk.

1.1 The data object is part of the model

In an environmental ML project, data construction choices are not clerical. They are modelling assumptions. Consider the simple prediction target

$$Y_{igt} = f(X_{igt}) + \varepsilon_{igt},$$

where i indexes a person or household, g a location, and t time. Before estimating f, the analyst must decide:

- What is the spatial unit: point, parcel, census tract, municipality, grid cell, route segment, or protected area?
- What is the time unit: hour, day, month, year, growing season, survey wave, policy period, or choice task?
- Which environmental measure is relevant: current exposure, cumulative exposure, lagged exposure, expected future risk, average local amenity, or alternative-specific attribute?
- How are data linked: nearest monitor, inverse-distance weighting, raster extraction, buffer average, route exposure, spatial interpolation, or model-based imputation?
- What is the prediction target: observed behaviour, stated choice, welfare measure, adoption, treatment effect, ecological outcome, or latent preference segment?

These choices encode theory. A household's choice of flood insurance may depend on perceived flood risk, not only objective modeled flood risk. A commuter's mode choice may depend on weather at departure time, air pollution along the route, route comfort, and prior experience. A farmer's adoption of a conservation practice may depend on long-run rainfall variability, soil quality, market access, and programme eligibility.

1.2 Environmental ML as data fusion

A useful way to see environmental ML is as *data fusion*. Let $D^{(1)}, D^{(2)}, \ldots, D^{(K)}$ denote data layers. The analyst constructs a learning table

$$\mathcal{D} = \mathcal{J}\left(D^{(1)}, D^{(2)}, \ldots, D^{(K)}; s, t, u\right),$$

where $\mathcal{J}$ is a joining operation and (s, t, u) represent spatial, temporal, and unit-level keys. The quality of $\mathcal{D}$ depends on whether the joining operation respects the research design.

For example, suppose we study willingness to pay for urban tree planting. A defensible data object may join:

1. Stated choice experiment responses from households.
2. Household approximate residential location.
3. Current tree canopy within 300 m, 500 m, and 1 km buffers.
4. Modelled air pollution exposure.

5. Heat island intensity during recent summers.
6. Census income and demographic context.
7. Flood, drought, or heat vulnerability indices.

The key issue is not merely whether the variables are predictive. The key issue is whether they correspond to the information set and choice context that decision makers plausibly used, or that a policy counterfactual would actually change.

2 A taxonomy of environmental data for ML

Table 1 summarizes the major environmental data families used in ML projects. The categories overlap, but the table helps diagnose opportunities and risks.

Table 1: Common environmental data sources for ML in economics and choice modelling

Data family	Typical variables	ML uses	Main risks
Surveys and choice experiments	Preferences, beliefs, adoption, WTP, attitudes, stated choices	Demand estimation, valuation, segmentation, latent constructs, policy scenarios	Hypothetical bias, sampling bias, panel dependence, attribute non-attendance
Revealed preference and administrative data	Purchases, energy bills, programme participation, land transactions, permits	Prediction, causal policy evaluation, adoption models, demand estimation	Missing counterfactuals, endogenous exposure, institutional measurement changes
Remote sensing imagery and products	Land cover, NDVI, nighttime lights, forest loss, urban form, water extent	Measurement, imputation, mapping, feature extraction, monitoring	Cloud contamination, resolution mismatch, algorithmic measurement error, domain shift
Weather and climate data	Temperature, precipitation, drought, wind, humidity, extreme events	Exposure construction, impact estimation, risk prediction, scenario analysis	Aggregation choices, model-product uncertainty, spatial interpolation error
Sensors and monitoring networks	Air quality, water quality, traffic, noise, energy, soil moisture	High-frequency exposure, anomaly detection, forecasting, compliance monitoring	Non-random station placement, calibration drift, missingness, sensor changes
Spatial and built environment data	Roads, transit, parks, coastlines, protected areas, elevation, slope, amenities	Accessibility, exposure, choice-set construction, environmental justice analysis	Modifiable areal unit problem, routing assumptions, geocoding error
Text and media	Consultation comments, news, social media, policy documents, survey open text	Sentiment, topic modelling, policy salience, risk perception, qualitative coding	Selection bias, language bias, platform changes, hallucinated labels if LLMs are used poorly

Data family	Typical variables	ML uses	Main risks
Images and immersive data	Street-view images, drone images, virtual reality scenes, photographs	Perceived greenness, safety, aesthetics, image choice, visual amenities	Representation bias, privacy, unstable explanations, copyright and consent
Biodiversity and ecological data	Species occurrences, habitat quality, acoustic recordings, eDNA, ecosystem service indicators	Conservation targeting, species distribution models, biodiversity valuation	Detection bias, uneven sampling effort, taxonomic uncertainty, ecological lag structure

The literature on choice modelling and ML emphasizes that new data types such as text, images, and passively collected behavioural traces are one of the main reasons for integrating ML into choice modelling (van Cranenburgh et al., 2022). The second edition of the *Handbook of Choice Modelling* also highlights the broadening of choice modelling toward context, advanced data capture, stated choice design, virtual environments, heterogeneity, forecasting, and appraisal (Hess and Daly, 2024).

3 The environmental data cube

A simple mental model is the environmental data cube:

$$\text{unit} \times \text{space} \times \text{time} \times \text{attribute}.$$

The unit may be a household, farm, firm, trip, parcel, census tract, grid cell, protected area, or choice alternative. Space may be a point, line, polygon, raster cell, route, or network. Time may be continuous or discrete. Attributes may be observed, remotely sensed, predicted, or simulated.

For choice modelling, the cube becomes even richer. The core data object is often:

$$(n, q, i, t, s) \rightarrow (y_{nq}, X_{nqit}, Z_{nt}, C_{nq}),$$

where n is the decision maker, q is the choice occasion, i is the alternative, t is the time, s is the location, y_{nq} is the chosen alternative, X_{nqit} are alternative-specific variables, Z_{nt} are person-specific or context variables, and C_{nq} is the choice set.

This notation matters because environmental data can enter a choice model in at least four ways:

1. **Alternative attributes.** Example: beach water quality, route air pollution, forest biodiversity gain, flood risk of a house.
2. **Individual context.** Example: respondent lives in a heat-exposed neighbourhood, has prior flood experience, or is close to public transport.
3. **Choice occasion context.** Example: weather on the travel day, wildfire smoke on the survey day, energy price at the hour of consumption.
4. **Choice-set construction.** Example: recreation sites available within a driving radius, routes generated by a routing algorithm, products available in a local market.

A choice model with environmental context might be written as

$$V_{nqit} = \alpha_i + \beta_p p_{nqit} + \beta_e E_{nqit} + \delta^\top Z_{nt} + \theta^\top (E_{nqit} \times Z_{nt}),$$

where E_{nqit} is an environmental attribute and Z_{nt} is individual or contextual information. In a machine learning model, V_{nqit} could be replaced or augmented by a flexible function $g(E_{nqit}, Z_{nt}, X_{nqit})$. The challenge is to preserve the economic meaning of the variables that support welfare analysis.

4 Surveys, stated preferences, and choice experiments

Environmental economics often values non-market goods: biodiversity, air quality, water quality, landscape, recreation, ecosystem services, climate mitigation, or adaptation. Because these goods may not have direct market prices, researchers use stated preference methods, especially discrete choice experiments (DCEs).

A DCE is not just a survey. It is an experimental data-generating process. The analyst controls attributes and levels, constructs choice sets, and observes choices under hypothetical scenarios. For environmental ML, DCEs can be linked to rich context data:

- geographic context of respondents;
- environmental conditions around respondents;
- images, maps, or virtual reality stimuli;
- attitudinal and psychometric variables;
- experimental metadata such as task order and response time.

Recent work uses DCEs not only with general publics, but also with experts to inform conservation action (Erbaugh et al., 2026). In such settings, the data object may represent expert choices over interventions, attributes of conservation programmes, ecological outcomes, feasibility, cost, and uncertainty. The modelling objective can include prediction, preference estimation, and policy prioritisation.

4.1 Choice experiments as structured ML data

In ML terms, a DCE produces repeated observations with strong structure. A row-wise table may look like this:

$$\{\text{respondent}, \text{task}, \text{alternative}, \text{attributes}, \text{chosen}\}.$$

However, treating these rows as independent is wrong. Observations are clustered within respondents and choice tasks. Alternatives within the same task compete against each other. Random row-level splitting can place alternatives from the same choice task into both training and test sets, which is a severe form of leakage.

The ML mode choice literature provides an important warning. Flexible classifiers may appear to strongly outperform random utility models when validation is done incorrectly; grouped validation by person or choice context can reduce inflated performance estimates (Hillel et al., 2021; Hillel, 2021). This lesson transfers directly to environmental DCEs.

Common mistake

Never randomly split rows of long-format choice data if rows contain alternatives from the same task or repeated tasks from the same individual. Split at the respondent, household, site, trip, or spatial block level, depending on the generalization question.

4.2 Attitudes, beliefs, and attribute non-attendance

Environmental choices often depend on beliefs and values: climate concern, trust in institutions, perceived responsibility, place attachment, risk perception, and attitudes toward technology or conservation. Traditional hybrid choice models represent such latent constructs through structural and measurement equations. ML can assist by flexibly mapping survey indicators to latent constructs or class membership. For example, neural components have been introduced into latent class choice models to incorporate attitudinal indicators while retaining random utility foundations (Torres Lahoz et al., 2023).

Another recurring issue is attribute non-attendance: respondents may ignore cost, origin, ecological attributes, or technical details. Regularization methods such as LASSO can help screen attributes or interactions in choice experiments, including environmental and food valuation settings (Li and Yu, 2025). But LASSO selection should be interpreted cautiously. A coefficient shrunk to zero may reflect true non-attendance, insufficient variation, correlated attributes, or the penalty choice.

5 Revealed preference, tracking, and administrative data

Revealed preference data measure actual behaviour. In environmental economics, this may include property purchases, appliance adoption, energy use, transport choices, park visits, fishing trips, water consumption, conservation enrolment, insurance uptake, or agricultural management.

Administrative and revealed preference data have three advantages:

1. They can be large and repeated over time.
2. They reflect real incentives and constraints.
3. They can often be linked to policy variation or environmental exposure.

They also have major limitations:

1. Choice sets may be unobserved or difficult to reconstruct.
2. Prices, exposures, and policies may be endogenous.
3. The researcher may observe behaviour but not preferences, beliefs, or perceived alternatives.
4. Administrative definitions can change over time.

Smartphone tracking data illustrate the opportunity and challenge. Tracking-based revealed preference data can provide detailed mode choice, route, time, and context variables. ML models such as random forests and XGBoost have been shown to outperform conventional MNL in some tracking-based mode choice settings, and SHAP-style explanations can help interpret model predictions (Dahmen et al., 2024). Yet the analyst must still address trip segmentation, mode labelling, non-random app users, privacy, and leakage across repeated trips by the same person.

5.1 Choice-set construction in revealed preference data

Choice-set construction is central when alternatives are not directly observed. Suppose we model residential location choice and environmental amenities. We observe the chosen neighbourhood, but not all alternatives considered by the household. Or suppose we model recreation site choice. We observe the site visited, but not all sites the person would have considered.

A defensible choice-set procedure should specify:

- the universe of feasible alternatives;
- sampling rules for non-chosen alternatives;
- spatial and temporal availability constraints;
- information constraints and search costs;
- environmental attributes for chosen and non-chosen alternatives.

For ML, choice-set construction matters because a classifier trained on poorly sampled alternatives can learn availability artefacts rather than preferences. For welfare analysis, it matters because substitution patterns and compensating variation depend on the alternatives included.

6 Remote sensing and Earth observation data

Remote sensing has transformed environmental economics because it can measure environmental and economic proxies consistently across space and time. Economists have used satellite data for agriculture, urbanization, deforestation, night-time lights, air pollution, climate impacts, disaster damage, conflict, and development (Donaldson and Storeygard, 2016). ML extends this by extracting features from imagery, classifying land cover, detecting objects, imputing missing ground data, and measuring hard-to-observe socioeconomic outcomes (Jean et al., 2016; Burke et al., 2021).

6.1 What a satellite image contains

A satellite image is not a photograph in the ordinary sense. It is usually a stack of raster bands. A pixel can contain reflectance values in visible, near-infrared, short-wave infrared, thermal, radar, or other bands. Derived products transform these bands into indices or classifications, such as:

- NDVI: vegetation greenness;
- EVI: enhanced vegetation index;
- NDBI: built-up index;
- land surface temperature;
- burned area;
- impervious surface;
- water extent;
- land cover class;
- forest loss year.

Global datasets such as Hansen et al.'s forest cover change maps provide high-resolution measures of forest extent and loss derived from Landsat time-series analysis (Hansen et al., 2013). Platforms

such as NASA Earthdata and Google Earth Engine provide access to many remote sensing products and analysis-ready data collections (NASA Earthdata, 2026; Google Earth Engine, 2026).

6.2 Resolution: spatial, temporal, spectral, and thematic

Resolution is multidimensional. Four questions should be asked before using any remote sensing product:

1. **Spatial resolution.** What area does one pixel represent? 10 m, 30 m, 250 m, 1 km?
2. **Temporal resolution.** How often is the location observed? Daily, weekly, monthly, annually?
3. **Spectral resolution.** Which wavelengths are measured?
4. **Thematic resolution.** What categories or derived variables are available, and how were they classified?

A high-resolution image is not always better. If the economic unit is a municipality, daily 10 m data may create huge computation without adding meaningful signal. If the policy target is a household's walking access to shade, coarse data may hide the exposure that matters.

6.3 Remote sensing as measurement, not truth

Remote sensing products are model outputs. They may contain cloud masking errors, atmospheric correction errors, classification uncertainty, sensor changes, and temporal compositing artefacts. When an economist uses a remote-sensing-derived variable as a covariate, that variable may be a prediction with error:

$$\tilde{E}_{gt} = E_{gt} + \nu_{gt},$$

where E_{gt} is the latent environmental state and ν_{gt} is measurement error. If $\tilde{E}_{gt}$ is used as a treatment, exposure, or attribute, measurement error can attenuate estimates, bias heterogeneity analysis, or distort WTP.

Key takeaway

In environmental ML, labels and features are often predictions from earlier models. Treat them as measured with uncertainty. When possible, report the source, version, spatial resolution, temporal coverage, preprocessing steps, and known limitations of each environmental layer.

7 Weather, climate, and environmental exposure data

Weather and climate variables are central to environmental economics. They enter models of agriculture, health, labour, migration, energy demand, transport, conflict, adaptation, recreation, and risk perception. Economists often use weather as a plausibly exogenous source of variation, but the data require careful handling. Auffhammer et al. show that data product choice, interpolation, aggregation, and climate model interpretation are not neutral details (Auffhammer et al., 2013).

7.1 Weather versus climate

Weather is the realized state of the atmosphere over short timescales. Climate is the distribution of weather over longer periods. For empirical economics, the distinction matters:

- A heat wave during a survey week may affect stated preferences or travel behaviour.
- Long-run summer temperature normals may affect housing prices or agricultural land values.
- Expected future climate risk may affect adaptation investments.
- Historical shocks may identify causal effects under some assumptions.

ERA5 and ERA5-Land are widely used reanalysis datasets. ERA5 provides hourly estimates for many atmospheric, land, and oceanic variables on a global grid, while ERA5-Land provides land-focused variables at finer spatial resolution (ECMWF and Copernicus Climate Change Service, 2026). Reanalysis data combine observations and physical models. They are not raw observations, but they provide global consistency.

7.2 Exposure construction

Environmental exposure often requires aggregation:

$$\text{Exposure}_{it} = \sum_{\tau \in W_t} w_{i\tau} E_{g(i),\tau},$$

where W_t is a time window, $E_{g(i),\tau}$ is the environmental measure for location $g(i)$ and time τ, and $w_{i\tau}$ are weights. The weights may represent residence time, commuting time, route duration, land area, population, crop calendar relevance, or behavioural salience.

The same raw data can support many exposure metrics:

- daily maximum temperature;
- degree days above a threshold;
- cumulative rainfall during the growing season;
- number of smoke days in the previous month;
- PM2.5 during commuting hours;
- drought index during the previous year;
- historical flood frequency within a buffer.

Theory should guide the window. If you search across many exposure definitions and report the one with the best prediction, you risk overfitting and post-treatment interpretation mistakes.

8 Sensors, monitoring networks, and air quality data

Air quality, water quality, noise, energy use, and traffic can be measured by monitoring networks and sensors. Open platforms such as OpenAQ provide access to real-time and historical air quality data from monitoring locations through APIs (OpenAQ, 2026). Sensor data are attractive because they can be high-frequency, but they are rarely spatially random.

Monitoring stations are often placed near urban areas, roads, industrial sites, or regulatory hotspots. This creates selection. If a model learns from stations and predicts to rural regions, the prediction domain may differ sharply from the training domain.

8.1 Low-cost sensors

Low-cost sensors enable dense monitoring but introduce calibration challenges. Sensor readings can drift over time, respond to humidity, or differ by device. A typical calibration model is

$$E_{kt}^{\text{true}} = h(E_{kt}^{\text{sensor}}, H_{kt}, T_{kt}, d_k) + \eta_{kt},$$

where H is humidity, T is temperature, and d_k is device information. ML can estimate h, but the calibrated exposure should still be treated as a prediction.

For choice modelling, sensor data are especially useful when the relevant exposure is dynamic: smoke exposure during a travel episode, noise along a route, water quality at a recreation site, or real-time electricity price and grid carbon intensity.

9 Spatial, built-environment, and demographic data

Environmental economics often links environmental data to human systems. Spatial and built-environment covariates include road networks, transit stops, coastlines, parks, elevation, slope, protected areas, zoning, amenities, parcel characteristics, and settlement morphology. Population layers such as WorldPop provide open high-resolution spatial demographic data, while the Global Human Settlement Layer provides global information on built-up areas and human settlement patterns (WorldPop, 2026; European Commission Joint Research Centre, 2026).

These data allow the analyst to construct variables such as:

- distance to coast, park, road, transit, or pollution source;
- number of amenities within a buffer;
- population-weighted pollution exposure;
- green space per capita;
- road density or built-up intensity;
- flood-zone exposure of housing stock;
- travel time to conservation sites or health facilities.

9.1 The modifiable areal unit problem

Spatial aggregation can change results. A variable measured at census tract level may yield different conclusions than the same phenomenon measured at grid-cell level or municipality level. This is the modifiable areal unit problem. In ML, it appears as a feature-engineering problem; in economics, it is also an identification and interpretation problem.

For example, a model predicting adoption of home insulation may perform differently depending on whether income is measured at household, block, tract, or municipal level. If household income is unavailable, tract income may proxy both purchasing power and neighbourhood-level policy targeting.

10 Text, images, and multimodal environmental data

ML and AI are especially useful for unstructured data. Environmental economics increasingly uses open-ended survey responses, policy documents, news, social media, planning documents, satellite images, street-view images, drone imagery, and photographs.

10.1 Text

Text can measure salience, sentiment, narratives, policy attention, misinformation, or perceived risk. Examples include:

- comments in environmental impact consultations;
- open-ended survey answers about climate risk;
- news articles about drought, wildfire, or pollution;
- social media posts about air quality events;
- local government plans and zoning documents.

A text workflow may use dictionaries, topic models, supervised classifiers, embeddings, or LLM-assisted coding. The key risk is label validity. If a model classifies comments as "supportive" or "concerned", the researcher must validate against human-coded data and document uncertainty.

10.2 Images

Images can represent environmental amenities and disamenities: street greenness, litter, building quality, shade, road safety, coastal erosion, park condition, landscape beauty, and visual pollution. Choice experiments with images create a natural bridge between visual stimuli and choice modelling. Recent work compares DCMs and ML models for street-view image choice problems, including models that use semantic segmentation or raw images (Zhu and Si, 2024). Such studies highlight a fundamental tradeoff: computer vision can extract features that are hard to hand-code, while DCMs provide clearer behavioural and welfare interpretation.

10.3 Multimodal choice data

A multimodal environmental choice dataset may combine:

- tabular attributes: price, time, risk, ecosystem outcome;
- text: policy description or public comment;
- images: visual representation of landscape or street;
- spatial context: location, distance, neighbourhood exposure;
- respondent data: beliefs, demographics, prior experience.

A neural model can learn from all modalities, but a welfare model must know which elements are policy attributes, which are contextual moderators, and which are only proxies.

11 Biodiversity and ecological data

Biodiversity data are central to conservation economics and ecosystem service valuation. Common sources include species occurrence records, habitat maps, acoustic recordings, camera traps, eDNA samples, protected area boundaries, ecological monitoring, and expert elicitation. GBIF provides global access to biodiversity occurrence data through web and API services (GBIF, 2026).

Biodiversity data are powerful but difficult. Occurrence records are not usually random samples. Bird observations may cluster near roads and cities. Camera traps detect some species more easily than others. Citizen science data reflect observer effort and platform participation. Species distribution models often require pseudo-absence construction, sampling effort correction, spatial blocking, and uncertainty reporting.

For environmental choice modelling, biodiversity can enter as:

- an attribute in a DCE, such as species richness or habitat quality;
- a spatial covariate affecting recreation choices;
- an outcome to be protected by policy;
- a latent ecological state predicted from remote sensing and field observations;
- an ecosystem service supply measure linked to demand or WTP.

12 From raw data to a modelling table

The most important skill in this chapter is translating raw data into a modelling table without losing the structure of the problem. We use the term *modelling table* broadly: it may be a long-format choice table, a panel dataset, a raster stack, a graph, or a set of tensors.

12.1 A defensible data pipeline

A good environmental ML pipeline has seven stages:

1. **Define the estimand or prediction target.** Predict adoption? Estimate WTP? Map exposure? Estimate treatment effects? Forecast demand?
2. **Define the unit and support.** Individuals, trips, farms, parcels, grid cells, sites, policies, or choice alternatives.
3. **Acquire raw layers.** Surveys, administrative data, remote sensing, climate, sensors, geography, text, images.
4. **Harmonize space and time.** Coordinate reference systems, temporal windows, spatial joins, buffers, grids, calendars.
5. **Engineer theory-guided features.** Exposure windows, accessibility, environmental amenities, risks, interactions.
6. **Validate the data object.** Missingness, leakage, domain shift, measurement error, balance across groups, plausibility checks.
7. **Train and evaluate models.** Use validation designs that match the intended generalization problem.

A simple but useful documentation table is shown in Table 2.

Table 2: Minimum documentation for environmental ML data layers

Item	Example	Why it matters
Source and version	ERA5-Land hourly, accessed 2026-07	Reproducibility; later versions can differ
Spatial support	9 km grid, household point extraction	Determines interpretation of exposure
Temporal support	Daily max from hourly data, summer 2018–2025	Determines lag and exposure window
Preprocessing	Cloud mask, monthly composite, buffer average	Can create hidden assumptions

Item	Example	Why it matters
Uncertainty	Monitor interpolation RMSE, classification accuracy	Needed for credible inference
Linking rule	500 m buffer mean, population-weighted	Affects estimates and welfare interpretation
Validation split	By municipality and year	Prevents spatial and temporal leakage
Policy relevance	Variable can change under intervention?	Required for counterfactual analysis

13 Spatial and temporal validation

Random train-test splits are often wrong for environmental data. Spatial and temporal dependence means nearby observations are similar. If nearby observations from the same process appear in both training and test sets, the test error may be too optimistic. The same logic applies to repeated observations from the same person, household, station, farm, route, or site.

Roberts et al. review cross-validation strategies for data with temporal, spatial, hierarchical, and related dependence structures and recommend blocked strategies when the prediction goal involves transfer to new space, time, or predictor domains (Roberts et al., 2017). Meyer and Pebesma further argue that spatial predictions should report an area of applicability: the part of predictor space where cross-validation error is expected to be meaningful (Meyer and Pebesma, 2021).

13.1 Validation must match the generalization claim

Ask: where will the model be used?

- **New rows like the old rows.** Random split may be acceptable if observations are independent.
- **New individuals in the same places.** Split by individual or household.
- **New choice tasks for the same individuals.** Split by task carefully; estimate separately whether repeated-person effects are intended.
- **New locations.** Use spatial block cross-validation or leave-location-out validation.
- **Future periods.** Use forward-chaining or rolling-origin validation.
- **New policy regimes.** Prediction error alone is insufficient; need structural, causal, or theory-based assumptions.

13.2 Spatial leakage example

Suppose the target is PM2.5 at unmonitored locations. A random split across monitoring observations may put a station's neighbouring observations in both training and test sets. The model can exploit local spatial smoothness. It may look accurate, but fail when deployed in a region with different climate, emissions, topography, or monitoring density.

The same problem arises in choice data. If trips from the same individual are split across training and test sets, a model may learn individual habits instead of general mode choice relationships. This is why grouped validation is essential in panel and tracking data (Hillel, 2021; Dahmen et al., 2024).

14 Software ecosystem

This textbook uses R as the main implementation language. For environmental data, R is strong because it combines econometrics, geospatial data processing, ML, and choice modelling. Python is often preferred for deep learning, cloud geospatial work, and computer vision.

Table 3: Software tools for environmental ML

Task	R tools	Python or other tools
Vector spatial data	`sf`, `lwgeom`	`geopandas`, `shapely`
Raster data	`terra`, `stars`, `exactextractr`	`rasterio`, `xarray`, `rioxarray`
Spatial ML resampling	`spatialsample`, `blockCV`, `CAST`	`scikit-learn` group splits, custom blocks
ML modelling	`tidymodels`, `mlr3`, `ranger`, `xgboost`	`scikit-learn`, `xgboost`, `lightgbm`
Choice modelling	`mlogit`, `gmnl`, `apollo`, `logitr`	Biogeme, PyLogit, custom PyTorch/JAX
Causal ML	`grf`, `DoubleML`, `econml` via Python	`DoubleML`, `econml`, `causalml`
Earth observation	`rgee`, `rstac`, `gdalcubes`	Earth Engine Python API, `geemap`, `earthaccess`
Text and embeddings	`tidytext`, `text2vec`, APIs	`transformers`, `sentence-transformers`
Reproducibility	`renv`, Quarto, `targets`	`conda`, `poetry`, notebooks, Snakemake

The R packages `sf` and `terra` are the core geospatial workhorses. `sf` supports vector geometries and simple features, while `terra` handles raster and vector spatial data, including raster operations such as extraction, aggregation, focal calculations, and zonal summaries (Pebesma et al., 2026; Hijmans, 2026). For validation, `spatialsample` provides spatial resampling infrastructure in the tidymodels ecosystem, and `blockCV` provides spatially and environmentally separated folds (tidymodels, 2026; Valavi et al., 2026).

15 Coding lab 1: building a spatial environmental ML table in R

This lab simulates an environmental choice dataset and shows how spatial covariates are joined. It is deliberately small. The goal is to understand the workflow, not to create a production-quality exposure model.

Coding lab

Task. Simulate households, alternatives, and environmental raster layers. Extract local exposure and fit a simple model predicting programme support.

```
# Install once if needed:
# install.packages(c("sf", "terra", "tidyverse", "tidymodels", "ranger"))

library(sf)
library(terra)
library(tidyverse)
```

```
library(tidymodels)
library(ranger)

set.seed(8)

# 1. Simulate household points in a projected coordinate system.
# Coordinates are arbitrary metres in a 100 km by 100 km study region.
n <- 800
hh <- tibble(
  id = 1:n,
  x = runif(n, 0, 100000),
  y = runif(n, 0, 100000),
  income = rlnorm(n, log(35000), 0.45),
  env_attitude = rnorm(n)
) |>
  st_as_sf(coords = c("x", "y"), crs = 3857)

# 2. Create two artificial raster layers: heat risk and greenness.
r <- rast(nrows = 100, ncols = 100,
          xmin = 0, xmax = 100000,
          ymin = 0, ymax = 100000,
          crs = "EPSG:3857")
xy <- crds(r, df = TRUE)

heat <- setValues(r, 0.4 * xy$x / 100000 + 0.6 * xy$y / 100000 + rnorm(ncell(r), 0,
    0.08))
green <- setValues(r, 1 - 0.7 * xy$x / 100000 + rnorm(ncell(r), 0, 0.12))
names(heat) <- "heat_risk"
names(green) <- "greenness"
env_stack <- c(heat, green)

# 3. Extract environmental variables at household locations.
# In real applications, consider buffers, exposure windows, and measurement error.
hh_env <- terra::extract(env_stack, vect(hh)) |>
  as_tibble() |>
  bind_cols(st_drop_geometry(hh))

# 4. Simulate programme support using both economics and environment.
hh_env <- hh_env |>
  mutate(
    log_income = log(income),
    latent_utility = -1.2 +
      0.000015 * income +
      0.9 * env_attitude +
      1.4 * heat_risk -
      0.8 * greenness,
    support_prob = plogis(latent_utility),
    support = factor(rbinom(n(), 1, support_prob), labels = c("no", "yes"))
  )

# 5. Fit a simple random forest classifier.
set.seed(8)
split <- initial_split(hh_env, prop = 0.8, strata = support)
train <- training(split)
test <- testing(split)

rf_spec <- rand_forest(trees = 500, mtry = 3, min_n = 10) |>
  set_engine("ranger", importance = "permutation") |>
```

```
  set_mode("classification")

rf_wf <- workflow() |>
  add_formula(support ~ log_income + env_attitude + heat_risk + greenness) |>
  add_model(rf_spec)

rf_fit <- fit(rf_wf, train)

pred <- predict(rf_fit, test) |>
  bind_cols(test |> select(support))

metrics(pred, truth = support, estimate = .pred_class)
```

15.1 What this lab teaches

The code hides several decisions that would be critical in a real project:

- Point extraction assumes the household is represented by one location and one pixel.
- Heat and greenness are treated as observed without error.
- The train-test split is random; if households are spatially clustered, this may be too optimistic.
- The model predicts support, but does not estimate WTP or welfare.

In a real environmental valuation project, you would usually test multiple exposure definitions and report why one is preferred before fitting the final model.

16 Coding lab 2: spatial cross-validation

Now we alter the validation design. The goal is to estimate performance when predicting to new areas, not merely new households drawn from the same spatial distribution.

```
# Install once if needed:
# install.packages(c("spatialsample", "yardstick"))

library(spatialsample)
library(yardstick)

# Convert the household data with geometry back to an sf object
hh_model_sf <- hh |>
  bind_cols(hh_env |> select(heat_risk, greenness, log_income, support, env_attitude))

# Create spatial block folds. The exact block size should be justified by
# the scale of spatial dependence and the intended deployment setting.
set.seed(8)
sp_folds <- spatial_block_cv(
  hh_model_sf,
  v = 5
)

# A minimal tidymodels resampling workflow
rf_res <- fit_resamples(
  rf_wf,
  resamples = sp_folds,
  metrics = metric_set(accuracy, kap),
  control = control_resamples(save_pred = TRUE)
)
```

```
collect_metrics(rf_res)
```

If spatial cross-validation produces much worse performance than random cross-validation, that is not a nuisance. It is important scientific information. It means the model relies partly on spatial interpolation. This may be acceptable if the goal is interpolation within the same region, but not if the goal is transfer to new regions.

17 Coding lab 3: linking environmental context to a choice model

We now simulate a long-format choice experiment in which respondents choose among conservation programmes. Environmental context enters both as a respondent-level moderator and as an alternative-specific attribute.

Worked example

A city offers residents three options: no programme, tree planting, or wetland restoration. Each programme has a cost and expected environmental gain. Respondents living in hotter neighbourhoods may value tree planting more because it reduces urban heat.

```
# Install once if needed:
# install.packages(c("mlogit", "tidyverse"))

library(mlogit)
library(tidyverse)

set.seed(88)
N <- 500       # respondents
Tn <- 6        # choice tasks
alts <- c("statusquo", "trees", "wetlands")

respondents <- tibble(
  id = 1:N,
  heat_risk = runif(N, 0, 1),
  env_attitude = rnorm(N)
)

choice_long <- expand_grid(
  id = 1:N,
  task = 1:Tn,
  alt = alts
) |>
  left_join(respondents, by = "id") |>
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = case_when(
      alt == "statusquo" ~ 0,
      TRUE ~ sample(c(15, 30, 60, 90), n(), replace = TRUE)
    ),
    biodiversity_gain = case_when(
      alt == "statusquo" ~ 0,
      TRUE ~ sample(c(1, 2, 3), n(), replace = TRUE)
    ),
    heat_reduction = case_when(
      alt == "trees" ~ sample(c(1, 2, 3), n(), replace = TRUE),
      alt == "wetlands" ~ sample(c(0, 1), n(), replace = TRUE),
```

```
      TRUE ~ 0
    ),
    asc = case_when(
      alt == "trees" ~ 0.3,
      alt == "wetlands" ~ 0.1,
      TRUE ~ 0
    ),
    V = asc - 0.04 * cost +
      0.55 * biodiversity_gain +
      0.35 * heat_reduction +
      0.60 * heat_risk * heat_reduction +
      0.25 * env_attitude * biodiversity_gain
  ) |>
  group_by(chid) |>
  mutate(
    p = exp(V) / sum(exp(V)),
    chosen_alt = sample(alt, 1, prob = p),
    choice = as.integer(alt == chosen_alt)
  ) |>
  ungroup()

mdata <- mlogit.data(
  choice_long,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

mnl_env <- mlogit(
  choice ~ cost + biodiversity_gain + heat_reduction +
    I(heat_risk * heat_reduction) + I(env_attitude * biodiversity_gain) | 0,
  data = mdata
)

summary(mnl_env)

b <- coef(mnl_env)
wtp_heat_reduction <- -b["heat_reduction"] / b["cost"]
wtp_biodiversity <- -b["biodiversity_gain"] / b["cost"]

wtp_heat_reduction
wtp_biodiversity
```

This example shows the distinction between environmental context and environmental attributes. Heat risk is a respondent-level context variable. Heat reduction is an alternative-specific attribute. Their interaction allows the marginal utility of heat reduction to differ by local exposure.

18 Coding lab 4: remote sensing workflow templates

The next code blocks are templates. They require real files or authenticated APIs. They are included to help you recognize the standard workflow.

18.1 R raster extraction with `terra`

```r
library(sf)
library(terra)
library(exactextractr)

# Read spatial units and raster data
tracts <- st_read("data/census_tracts.gpkg")
ndvi <- rast("data/ndvi_composite_2025.tif")
pm25 <- rast("data/pm25_annual_2025.tif")

# Ensure compatible coordinate reference systems
tracts <- st_transform(tracts, crs(ndvi))

# Area-weighted extraction for polygons
tracts$mean_ndvi <- exact_extract(ndvi, tracts, "mean")
tracts$mean_pm25 <- exact_extract(pm25, tracts, "mean")

# Build modelling table
model_df <- tracts |>
  st_drop_geometry() |>
  select(tract_id, mean_ndvi, mean_pm25, income, population, outcome)
```

18.2 Python cloud access template

Python is often convenient for cloud-native geospatial data, deep learning, and computer vision.

```python
# Template only. Requires installed packages and credentials where needed.

import geopandas as gpd
import xarray as xr
import rioxarray
from sklearn.model_selection import GroupKFold
from sklearn.ensemble import RandomForestRegressor

# Read vector data
sites = gpd.read_file("data/recreation_sites.gpkg")

# Read raster or NetCDF climate data
climate = xr.open_dataset("data/era5_monthly.nc")

# Example: create a seasonal average
summer_temp = climate["t2m"].sel(month=[6, 7, 8]).mean("month")

# Convert extracted values to tabular form, then model
# In production, document CRS, time windows, aggregation, and missing values.
```

19 Measurement error, label error, and generated regressors

ML often treats features as given. Economists should not. Many environmental variables are generated regressors:

- pollution exposure predicted from monitors and satellite aerosol optical depth;
- greenness derived from satellite indices;
- flood risk from hydrological models;
- biodiversity from species distribution models;

- poverty or asset wealth predicted from imagery;
- sentiment labels predicted from text.

Let Y be a behaviour or choice, and let $\hat{E}$ be a predicted environmental variable. A naive model uses

$$Y = m(\hat{E}, X) + \varepsilon.$$

But the true target may be

$$Y = m(E, X) + \varepsilon, \quad \hat{E} = E + \nu.$$

If ν is classical measurement error, linear coefficients may be attenuated. If ν is spatially correlated, correlated with socioeconomic conditions, or generated by a biased remote sensing model, the problem is worse. When possible, use validation data, propagate uncertainty, conduct sensitivity analysis, or report estimates under multiple environmental products.

19.1 Generated features in welfare analysis

Generated environmental attributes are especially important in choice modelling. Suppose a DCE tells respondents that a policy improves biodiversity from low to high. If the analyst later links each respondent to a predicted baseline biodiversity index, WTP for biodiversity improvement may depend on the accuracy and meaning of that predicted index. A measure that is useful for ecological mapping may not match public perception or the survey attribute.

Common mistake

Do not assume that a remotely sensed or modelled environmental variable is the same as the environmental concept in the choice experiment. A respondent may value perceived water clarity, not chlorophyll-a; perceived flood safety, not a 100-year flood map; shade while walking, not average NDVI in a 1 km buffer.

20 Design patterns for environmental choice modelling

This section gives reusable templates. They are not formulas to apply mechanically; they are starting points.

20.1 Pattern 1: Context-augmented discrete choice experiment

Question. Does local exposure shape WTP for an environmental improvement?

Data. DCE responses, respondent locations, local environmental exposure, demographics, attitudes.

Model.

$$V_{nqi} = \beta_c c_{nqi} + \beta_a a_{nqi} + \theta(a_{nqi} \times E_n) + \gamma^\top Z_n.$$

ML extension. Use random forests or gradient boosting to discover nonlinear interactions, then test theory-consistent interactions in MNL, mixed logit, or latent class models. A neural context-aware mixed logit can allow preference parameters to shift with contextual variables while retaining interpretability of utility coefficients (Lukawska et al., 2024).

Risk. Environmental exposure may be endogenous to residential sorting. WTP heterogeneity by exposure is not automatically causal.

20.2 Pattern 2: Remote sensing for ecosystem service supply

Question. Which areas provide ecosystem services, and how do people value access to them?

Data. Satellite-derived land cover, vegetation indices, terrain, hydrology, recreation visits or stated choices.

ML extension. Predict ecosystem service indicators using remote sensing and ecological observations, then include predicted supply in recreation demand or DCE models.

Risk. Two-stage prediction can understate uncertainty. If the predicted service variable is biased in data-poor regions, demand or WTP estimates inherit that bias.

20.3 Pattern 3: Tracking-based revealed preference with environmental context

Question. How do weather, air pollution, heat, and infrastructure influence travel mode choice?

Data. GPS traces, trip segments, inferred mode labels, weather, pollution, route attributes, built environment.

ML extension. Use XGBoost or random forests for high-dimensional prediction and SHAP or partial dependence for model analysis; compare with MNL or mixed logit. Tracking-based mode choice studies show both the predictive promise of ML and the need for careful interpretation (Dahmen et al., 2024).

Risk. Validation by trip rather than by person can leak individual routines. Mode labels may be generated by another algorithm.

20.4 Pattern 4: Environmental justice and policy targeting

Question. Which populations face high exposure and low adaptive capacity?

Data. Pollution, heat, flood risk, green space, demographics, income, health, housing, infrastructure.

ML extension. Predict exposure and vulnerability; estimate heterogeneous treatment effects of environmental policies; build targeting tools.

Risk. Targeting can reproduce measurement bias if disadvantaged groups are undermeasured. Causal ML still requires identification assumptions (Athey and Imbens, 2019; Chernozhukov et al., 2018).

21 Data ethics, privacy, and governance

Environmental ML can be socially valuable, but it can also create risks. Geocoded survey data, tracking data, energy usage, and environmental exposure histories can reveal sensitive information. A PhD project should treat privacy and governance as part of the design, not as an administrative afterthought.

Key principles:

1. **Minimize identifiable data.** Store exact coordinates separately from analysis tables when possible.

2. **Use spatial masking carefully.** Jittering or coarsening can protect privacy but also create exposure measurement error.
3. **Document consent.** Participants should know whether data are linked to geography, sensors, images, or administrative records.
4. **Audit representativeness.** Sensor networks, smartphone apps, and online platforms rarely represent the full population.
5. **Avoid harmful targeting.** Environmental risk predictions can affect insurance, property values, or policy eligibility.
6. **Respect Indigenous and local data sovereignty.** Environmental data can encode culturally sensitive sites and knowledge.

Privacy-preserving spatial analysis often requires compromises. For example, a researcher may use neighbourhood centroids instead of exact home coordinates. This reduces disclosure risk but may bias estimates of exposure to parks, roads, or hazards. The best practice is to assess how sensitive results are to spatial masking.

22 A checklist for environmental ML datasets

Before fitting any model, answer the following checklist.

1. What is the research question: prediction, causal inference, welfare analysis, targeting, or exploration?
2. What is the unit of observation and the unit of intervention?
3. Are the spatial and temporal supports of all variables compatible?
4. Are any features generated by previous models? Is their uncertainty known?
5. Could any feature contain information from the future relative to the prediction time?
6. Could the validation split leak information across people, places, time, tasks, or alternatives?
7. Are missing values informative?
8. Does the training domain cover the deployment domain?
9. Are protected or vulnerable groups underrepresented or measured differently?
10. Which variables are policy manipulable, and which are only proxies?
11. If WTP or welfare is estimated, are price/cost variables measured and specified appropriately?
12. Are data version, code, random seeds, and preprocessing steps reproducible?

23 Worked mini-project: designing a PhD data pipeline

Imagine a PhD project on household demand for urban heat adaptation. The policy is a municipal programme that subsidizes street trees, cool roofs, and pocket parks. The researcher wants to estimate both adoption probability and WTP for programme attributes.

23.1 Step 1: Define research outputs

The project has three outputs:

1. Predict which households are likely to support each adaptation option.

2. Estimate WTP for tree canopy, heat reduction, biodiversity co-benefits, and programme cost.
3. Simulate counterfactual adoption and welfare under alternative programme designs.

Output 1 can use flexible ML. Outputs 2 and 3 require choice modelling discipline.

23.2 Step 2: Data layers

The data layers are:

- DCE survey with repeated choices over adaptation programmes.
- Respondent demographics, climate concern, heat experience, home ownership.
- Approximate geocoded residence.
- Satellite-derived tree canopy and land surface temperature.
- ERA5 or local weather summaries for recent heat waves.
- Census vulnerability indicators.
- City street tree inventory and park locations.

23.3 Step 3: Feature construction

Possible features include:

- mean canopy within 250 m, 500 m, and 1 km;
- distance to nearest park;
- number of extreme heat days in previous summer;
- census-level income and age vulnerability;
- respondent-reported heat discomfort;
- interaction between heat exposure and programme heat reduction.

The PhD student should not include every possible buffer and lag in the final welfare model. A good approach is to use exploratory ML to identify plausible nonlinearities, then pre-specify a small set of theory-driven variables for the final choice model.

23.4 Step 4: Validation

Use different validation designs for different goals:

- Prediction of new respondents in the same city: group split by respondent.
- Transfer to new neighbourhoods: spatial block validation.
- Future survey waves: temporal validation.
- Welfare estimates: sensitivity analysis across exposure definitions and model specifications.

23.5 Step 5: Model comparison

Compare:

1. MNL with context interactions.
2. Mixed logit with random cost and environmental coefficients.

3. Latent class model with exposure and attitudes in class membership.
4. Random forest or XGBoost classifier for prediction.
5. Hybrid model where ML suggests interactions or nonlinear transformations.

Report predictive performance, behavioural plausibility, WTP distributions, and counterfactual stability.

24 Retrieval practice

Try to answer these from memory before checking the brief answers.

1. What are the four dimensions of the environmental data cube?
2. Why is a remotely sensed environmental variable not automatically "ground truth"?
3. What is spatial leakage?
4. In a choice model, what is the difference between an environmental alternative attribute and an environmental context variable?
5. Why might grouped validation by respondent be necessary in stated choice data?
6. Give one example of a generated regressor in environmental economics.
7. Why can high predictive accuracy be insufficient for environmental welfare analysis?
8. What is the area of applicability in spatial prediction?

Brief answers

1. Unit, space, time, and attribute.
2. It is usually processed, corrected, classified, interpolated, or predicted from sensor measurements and models.
3. Overoptimistic validation caused by nearby or otherwise dependent observations appearing in both training and test sets.
4. An alternative attribute varies across alternatives in a choice set; a context variable describes the individual, location, or occasion and may moderate tastes.
5. Repeated tasks from the same respondent are not independent; row-wise splitting can leak respondent-specific information.
6. Predicted pollution exposure, satellite-derived greenness, flood risk maps, or ML-coded sentiment.
7. Welfare analysis requires interpretable tradeoffs, policy manipulability, and credible counterfactual structure, not just prediction.
8. The subset of predictor space where the model has training support and the validation error is expected to apply.

25 End-of-chapter exercises

Exercise 1: Data audit

Choose an environmental dataset you know. Create a documentation table with source, version, spatial support, temporal support, preprocessing, uncertainty, linking rule, validation design, and policy relevance.

Exercise 2: Validation design

You have 10,000 household observations from 50 municipalities. The goal is to predict adoption of rooftop solar in municipalities not in the training data. Propose a validation strategy and explain why random row-wise validation is not enough.

Exercise 3: Choice data structure

A DCE has 1,200 respondents, 8 choice tasks each, and 3 alternatives per task. How many long-format rows are there? At what level should you split data for a model intended to predict choices of new respondents?

Exercise 4: Remote sensing variable

Suppose you use NDVI as a measure of urban greenness. List three reasons why NDVI may not match the concept of greenness that respondents value in a DCE.

Exercise 5: Build a simple pipeline in R

Using any spatial vector layer and raster layer available to you, write R code that reads both, transforms them to a common CRS, extracts raster values to the vector units, and builds a modelling table.

Exercise 6: Welfare and ML

A random forest predicts support for a wetland restoration programme better than an MNL model. Explain what additional assumptions or modelling steps are needed before using the random forest for WTP or welfare analysis.

Exercise 7: Generated regressor sensitivity

You have two pollution exposure products: monitor interpolation and satellite-derived estimates. Fit the same demand or choice model with both. How would you report differences in coefficients, WTP, and policy simulations?

26 Common mistakes and how to avoid them

Common mistake

Mistake 1: Building the data table before defining the estimand.
Avoid this by writing the research question, prediction target, unit of intervention, and policy counterfactual before engineering features.

Common mistake

Mistake 2: Treating spatial joins as neutral.
Avoid this by documenting every buffer, distance, interpolation, aggregation, and CRS transformation.

Common mistake

Mistake 3: Using random cross-validation for spatial or panel data.
Avoid this by validating at the level where dependence occurs: respondent, household, station, site, municipality, route, or spatial block.

Common mistake

Mistake 4: Confusing policy proxies with policy levers.
A model may learn that high NDVI predicts support for tree planting. That does not mean a policy can manipulate NDVI instantly or that NDVI is the welfare-relevant attribute.

Common mistake

Mistake 5: Ignoring generated-regressor uncertainty.
Avoid this by validating environmental layers, reporting product versions, comparing alternative products, and propagating uncertainty where feasible.

27 Chapter summary

Environmental ML begins with data design. The most important issues are spatial and temporal support, data fusion, measurement error, validation design, and the distinction between prediction variables and policy-relevant attributes. Remote sensing, sensors, text, images, tracking data, and administrative records create enormous opportunities for environmental economics and choice modelling. But they also create new ways to be wrong.

For choice modelling, environmental data can define alternative attributes, individual context, choice occasion context, or the choice set itself. Machine learning can help process high-dimensional data, discover interactions, extract features from unstructured sources, and improve prediction. But welfare analysis, WTP estimation, and counterfactual policy appraisal still require theory-consistent data construction and model interpretation.

Key takeaway

A high-quality environmental ML project is not defined by the most complex algorithm. It is defined by a data pipeline whose spatial, temporal, behavioural, and policy assumptions are explicit enough that another researcher can evaluate them.

Reading guide

1. Donaldson and Storeygard (2016): the best entry point for economists using satellite data.
2. Auffhammer et al. (2013): essential reading before using weather or climate products in economics.
3. Burke et al. (2021): modern overview of satellite imagery and ML for sustainable development.

4. Roberts et al. (2017) and Meyer and Pebesma (2021): crucial for spatial validation and prediction transfer.
5. Hillel et al. (2021) and Hillel (2021): warnings on ML validation in mode choice and panel data.
6. van Cranenburgh et al. (2022): broad discussion of how ML and choice modelling can cross-fertilize.
7. Dahmen et al. (2024) and Zhu and Si (2024): examples of tracking and image data in ML-based choice modelling.
8. Hess and Daly (2024): broader choice modelling foundations, including context, stated choice design, revealed behaviour capture, and appraisal.

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Chapter 9

Prediction Problems in Environmental Economics

Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Position of this chapter. The previous chapter introduced environmental data sources: sensors, satellites, administrative records, spatial layers, text, surveys, and linked stated/revealed preference data. This chapter turns those data into prediction systems. We focus on environmental outcomes that economists often need before valuation, policy evaluation, targeting, risk assessment, or counterfactual simulation can begin.

Main idea. Prediction is not a lower-status task relative to causal inference. In environmental economics, prediction is often a necessary input into policy: pollution exposure must be predicted before health damages can be estimated; flood and fire risk must be forecast before adaptation policy can be designed; energy demand must be forecast before pricing and infrastructure can be evaluated. But prediction must be designed around the policy decision it will support.

Learning objectives

After studying this chapter, you should be able to:

- define environmental prediction problems as supervised learning tasks with explicit units, horizons, and decision uses;

- distinguish nowcasting, forecasting, spatial prediction, imputation, classification, ranking, and early warning;

- choose appropriate loss functions and validation schemes for environmental and economic decisions;

- recognize special problems created by spatial dependence, temporal dependence, rare events, measurement error, and policy feedback;

- build baseline and ML prediction workflows in R for pollution, risk, adoption, and energy-demand examples;

- connect prediction models to environmental valuation, causal ML, and choice modelling without confusing prediction with causal or welfare interpretation.

9.1 Why prediction matters in environmental economics

Environmental economics is often presented as a field of valuation, externalities, policy instruments, and welfare analysis. Those tasks remain central, but most applied projects also contain a prediction component. Before estimating the benefits of an air-quality regulation, researchers may need pollution exposure surfaces. Before studying climate adaptation, they may need predicted flood, heat, fire, or crop-yield risk. Before estimating heterogeneous programme effects, they may need baseline risk scores or vulnerability indices.

A recent review of machine learning in environmental economics organizes existing applications into environmental policy evaluation, environmental and resource markets, prediction of environmental outcomes, and media analysis. It also emphasizes the importance of new data sources such as sensors, satellites, and text. This is useful for economists because many environmental outcomes are spatially dense, temporally dynamic, or only partially observed.

Prediction is therefore not merely an engineering add-on. It changes what environmental economists can measure. High-resolution pollution estimates can replace crude monitor-distance proxies. Satellite-based land-use measures can replace infrequent surveys. Text models can transform policy documents, news, and public comments into usable variables. But all of these uses require discipline: the model must be validated for the scientific or policy question, not only for an arbitrary leaderboard metric.

A first taxonomy

Prediction task	Typical target	Economic use	Main risk
Spatial prediction	Pollution, temperature, land cover, biodiversity, flood exposure at unmonitored locations	Exposure assessment, damage estimation, spatial targeting	Overoptimistic accuracy if nearby places appear in both training and test data
Temporal forecasting	Energy demand, water demand, air pollution, fire risk, crop yields	Infrastructure planning, pricing, early warning, adaptation	Leakage from future information or poor performance under regime change
Classification	Deforestation/no deforestation, adoption/non-adoption, compliance/non-compliance	Targeting, enforcement, risk scoring	Class imbalance and threshold choices hidden behind accuracy
Ranking and prioritization	Which locations or households should receive intervention first?	Budget allocation, inspections, conservation targeting	High average accuracy may still misrank the most policy-relevant cases
Imputation and data fusion	Missing sensors, missing survey variables, harmonized exposure layers	Linking data for valuation or causal analysis	Uncertainty from imputation ignored downstream
Representation learning	Embeddings from text, images, satellite pixels, street views	Inputs to valuation, choice models, or policy evaluation	Representations may encode spurious or non-stable patterns

9.2 From economic question to prediction design

A frequent mistake is to start with an algorithm: random forest, XGBoost, neural network, or transformer. A better workflow starts with the decision problem. Define the unit, target, horizon, action, and loss before choosing the model.

The prediction design template

Element	Question to answer	Example
Unit	What object receives one prediction?	Census tract-day, household-month, farm-season, beach-day, grid cell-year
Target Y	What is being predicted?	PM2.5 concentration, electricity demand, deforestation risk, heat mortality risk, protected-area visitation

Features X	What is known at prediction time?	Weather forecast, land cover, past consumption, satellite indices, policy status, demographics
Horizon	When is the target realized?	One hour ahead, next day, next summer, next five years
Action	What decision uses the prediction?	Issue alert, inspect site, allocate subsidy, estimate exposure, design price
Loss	What kind of mistake is costly?	Missing high-risk days, over-alerting households, underpredicting low-income exposure
Validation	What future or out-of-sample situation must the model generalize to?	New years, new locations, new climate regime, new policy regime

Teaching checkpoint. Take a paper you know in environmental economics. Rewrite its empirical objective using the template above. If the paper estimates a causal effect, identify any hidden prediction problem embedded in the measurement, exposure, targeting, or counterfactual step.

9.3 Loss functions: prediction for policy, not just accuracy

Standard ML courses often introduce mean squared error for regression and accuracy for classification. These are useful starting points, but environmental policy problems usually require more careful loss functions.

Regression losses

For continuous outcomes such as pollution, electricity demand, or water consumption, common metrics include:

- Mean absolute error (MAE): easier to interpret in original units and more robust to outliers.

- Root mean squared error (RMSE): penalizes large errors strongly, useful when large misses are very costly.

- Mean absolute percentage error (MAPE): scale-free, but unstable when actual values are near zero.

- Quantile loss: useful when decision makers need prediction intervals or upper-tail forecasts.

- Weighted losses: useful when errors for vulnerable groups, extreme pollution days, or protected areas deserve greater weight.

Classification losses

For binary or multiclass outcomes such as deforestation, wildfire ignition, technology adoption, or violation detection, common metrics include accuracy, balanced accuracy, precision, recall, F1 score, ROC-AUC, PR-AUC, and log loss. Accuracy is often misleading when events are rare. A model that predicts no illegal deforestation may achieve high accuracy if illegal deforestation is uncommon, while being useless for enforcement.

Decision-aware loss

The most useful loss is often linked to the policy decision. Suppose a regulator can inspect only 5 percent of forest parcels. The relevant metric may be precision among the top 5 percent of predicted risk scores, not global accuracy. Suppose a heat-warning system is designed to avoid deaths among elderly residents. The relevant metric may penalize false negatives on high-vulnerability days more than false positives on mild days.

Common error. Do not choose a metric only because it is standard in a software package. Choose the metric that corresponds to the economic or policy consequence of prediction errors.

9.4 Validation for spatial and temporal environmental data

Environmental data violate the independent and identically distributed assumption in obvious ways. Nearby places tend to be similar. Consecutive days tend to be similar. Households, farms, beaches, or municipalities may appear repeatedly. If we randomly split rows into training and test sets, the test set can contain nearly identical observations to those in the training set. This creates leakage and inflated performance.

Four validation designs

Validation design	Use when	What it tests	Example
Random split	Observations are plausibly independent and the deployment setting is similar	Average interpolation performance	Cross-sectional household adoption data with one row per household
Grouped split	Repeated observations appear for the same unit	Generalization to new units	Train on some households, test on unseen households
Spatial block split	Nearby locations are correlated	Generalization to new places	Train on some spatial blocks, test on held-out blocks
Temporal split / rolling origin	Forecasting future periods	Generalization to future time	Train through 2022, validate on 2023, test on 2024

The choice modelling literature has a closely related warning: when panel data are split trip-wise instead of person-wise, flexible ML classifiers can exploit individual-specific leakage. Hillel and co-authors show that this problem can substantially bias apparent performance in mode-choice prediction. The same logic applies in environmental economics: if the same monitoring station, farm, or household appears in both training and test data, reported performance may measure memorization rather than generalization.

Spatial leakage example

Imagine predicting PM2.5 at 1 km grid cells using satellite aerosol optical depth, weather, road density, and land use. If the random split puts one grid cell in training and its immediate neighbour in testing, the test point is not genuinely independent. A spatial block split is more demanding: it asks whether the model predicts well in locations not near the training points. This is often the relevant question when exposure predictions are transferred to unmonitored communities.

Rule of thumb. Match the validation design to the deployment design. If the model will be used in new years, validate on held-out years. If it will be used in new locations, validate on held-out spatial blocks. If it will be used for new households, validate on held-out households.

9.5 Core prediction domains in environmental economics

9.5.1 Pollution exposure prediction

Air pollution is a natural prediction problem because monitoring stations are sparse relative to the spatial variation in exposure. ML models can combine ground monitors, satellite products, meteorology, land-use variables, road networks, industrial sources, elevation, and temporal

indicators. The economic use is usually downstream: health-damage estimation, distributional analysis, environmental justice mapping, or evaluation of regulatory exposure reductions.

The key methodological risks are spatial leakage, changes in monitor networks, measurement error, and domain shift. A model trained in dense urban monitoring networks may underperform in rural or low-income regions. A model trained before a major industrial closure, wildfire season, or regulatory change may not forecast well after the regime changes.

9.5.2 Energy and water demand forecasting

Energy and water demand are high-frequency prediction problems. Features may include lagged demand, weather, holidays, prices, building characteristics, household demographics, appliance ownership, and programme participation. Forecasts support grid planning, dynamic pricing, demand response, drought management, and infrastructure investment. For economists, forecasts can also serve as baselines in programme evaluation.

The central challenge is feedback. If a utility changes prices or issues conservation messages based on model predictions, future demand is partly caused by the prediction system itself. This does not make prediction impossible, but it means the model is embedded in a policy environment. Forecast evaluation should therefore be updated after deployment.

9.5.3 Land-use, deforestation, and ecosystem risk

Remote sensing and ML make it possible to predict forest loss, agricultural expansion, wetland conversion, habitat fragmentation, and ecosystem degradation at high spatial resolution. Inputs can include past land cover, slope, distance to roads, distance to settlements, commodity prices, protected-area status, fire history, night lights, and governance proxies.

For policy, the target is rarely just classification accuracy. A conservation agency may need to rank parcels for inspection, prioritize payments for ecosystem services, or identify locations where road expansion would create high deforestation pressure. This makes top-k performance, calibration, and spatial transfer more relevant than global accuracy.

9.5.4 Climate risk, hazards, and adaptation

Climate adaptation requires prediction of heat risk, flood risk, drought risk, wildfire risk, crop-yield losses, coastal erosion, and infrastructure vulnerability. ML can combine physical model outputs, historical weather, remote sensing, exposure data, and socioeconomic vulnerability. The economic role is to support adaptation investment, insurance pricing, disaster preparedness, and welfare analysis.

The largest difficulty is extrapolation. Climate change can move the data-generating process outside the historical range. A model trained only on past weather may fail under unprecedented heat, compound hazards, or behavioural adaptation. In this setting, hybrid approaches that combine physical models, scenario analysis, and ML are often safer than purely historical prediction.

9.5.5 Environmental technology adoption

Adoption prediction includes solar panels, electric vehicles, heat pumps, water-saving technologies, conservation practices, organic farming, climate adaptation measures, and green tariffs. Features may include prices, subsidies, local peer adoption, income, housing stock, beliefs, credit access, installer availability, and policy exposure.

These models are useful for targeting information campaigns or estimating market potential. However, prediction of adoption is not the same as causal estimation of a subsidy effect. If high-income households are both more likely to receive marketing and more likely to adopt, an ML model may predict adoption well while providing no valid estimate of what a subsidy would do.

9.5.6 Environmental values, visits, and recreation demand

Prediction problems also appear in non-market valuation. We may predict recreation-site visitation, congestion, beach crowding, wildlife-viewing demand, or response probabilities in stated preference surveys. ML can help forecast demand under observed conditions and detect nonlinear effects of weather, access, congestion, or site attributes. But welfare analysis still requires careful linkage to behavioural choice models.

9.6 A general ML workflow for environmental prediction

Step 1: define the target without post-treatment leakage

The target must be observable at the prediction horizon. If the model predicts next-day pollution alerts, it cannot use tomorrow's realized pollution or policy response as a feature. If it predicts adoption before a campaign, it cannot use variables created after the campaign.

Step 2: construct features as they would be available in practice

Feature engineering is often more important than the choice of algorithm. Environmental features may include lags, moving averages, anomalies relative to seasonal normals, spatial buffers, distances, neighbourhood summaries, vegetation indices, meteorological forecasts, and policy indicators. For causal or welfare applications, record which features are pre-treatment, post-treatment, or potentially endogenous.

Step 3: build interpretable baselines

Start with simple baselines: historical average, seasonal naive forecast, linear regression, logistic regression, or generalized additive model. If XGBoost improves RMSE by 2 percent over a strong seasonal baseline, the practical value may be small. If it improves detection of extreme pollution days by 40 percent, the value may be large.

Step 4: compare flexible models

Tree ensembles are strong default methods for tabular environmental data. Random forests are robust and easy to tune; gradient boosting often performs better but needs more careful tuning. Neural networks are powerful for images, sequences, and multimodal data, but require more data, stronger regularization, and more careful validation. For spatial raster or image tasks, convolutional neural networks or vision transformers may be appropriate, but they should be justified by the data structure.

Step 5: calibrate, interpret, and stress-test

A model with good ranking may still be poorly calibrated: predicted 80 percent risk may correspond to actual 40 percent risk. Calibration matters when probabilities enter expected benefit-cost calculations. Interpretability methods such as partial dependence, accumulated local effects, permutation importance, and SHAP can help diagnose the model, but they should be interpreted as model summaries rather than causal estimates.

Step 6: propagate prediction uncertainty

Environmental predictions often become inputs to later econometric models. If predicted pollution exposure is treated as observed truth, downstream uncertainty will be understated. Practical solutions include bootstrap prediction intervals, Bayesian predictive distributions, conformal prediction, multiple imputation, or sensitivity analysis with alternative prediction models.

9.7 Worked example in R: predicting next-day pollution

The following example uses simulated data so that it is fully reproducible without downloading external files. The structure mimics a common air-quality prediction problem: monitor-day observations, weather, satellite signal, road density, land use, spatial coordinates, and lagged pollution. The important lesson is the workflow, especially the temporal split and grouped cross-validation.

```
# Chapter 9 R lab 1: next-day pollution prediction
# install.packages(c("tidyverse", "tidymodels", "ranger", "xgboost", "vip", "finetune"))

library(tidyverse)
library(tidymodels)
library(ranger)
library(xgboost)

set.seed(901)

n_stations &lt;- 80
n_days &lt;- 365

stations &lt;- tibble(
  station_id = factor(1:n_stations),
  x = runif(n_stations, 0, 100),
  y = runif(n_stations, 0, 100),
  road_density = rgamma(n_stations, 2, 0.4),
  industrial = rbinom(n_stations, 1, 0.25),
  green_share = runif(n_stations, 0.05, 0.70)
)

dat &lt;- crossing(station_id = stations$station_id, day = 1:n_days) %&gt;%
  left_join(stations, by = "station_id") %&gt;%
  mutate(
    temp = 15 + 10 * sin(2 * pi * day / 365) + rnorm(n(), 0, 4),
    wind = pmax(0.1, rnorm(n(), 4, 1.2)),
    satellite_no2 = 20 + 0.30 * road_density + 3 * industrial -
      0.05 * green_share + rnorm(n(), 0, 3),
    spatial_hotspot = 8 * exp(-((x - 70)^2 + (y - 35)^2) / 900),
    pm25 = 8 + 0.8 * satellite_no2 - 1.4 * wind + 0.15 * temp +
      1.8 * road_density + 3.5 * industrial - 4 * green_share +
      spatial_hotspot + rnorm(n(), 0, 4)
  ) %&gt;%
  group_by(station_id) %&gt;%
  arrange(day, .by_group = TRUE) %&gt;%
  mutate(pm25_lag1 = lag(pm25),
         pm25_ma7 = slider::slide_dbl(pm25, mean, .before = 7, .complete = FALSE)) %&gt;%
  ungroup() %&gt;%
  filter(!is.na(pm25_lag1))

# Predict PM2.5 for future days. Use a temporal split, not a random split.
train &lt;- dat %&gt;% filter(day &lt;= 280)
test  &lt;- dat %&gt;% filter(day &gt; 280)

folds &lt;- group_vfold_cv(train, group = station_id, v = 5)

rec &lt;- recipe(pm25 ~ temp + wind + satellite_no2 + road_density + industrial +
                green_share + x + y + pm25_lag1 + pm25_ma7,
              data = train) %&gt;%
  step_normalize(all_numeric_predictors())

rf_spec &lt;- rand_forest(trees = 600, mtry = tune(), min_n = tune()) %&gt;%
  set_engine("ranger", importance = "permutation") %&gt;%
  set_mode("regression")

rf_wf &lt;- workflow() %&gt;% add_recipe(rec) %&gt;% add_model(rf_spec)

grid &lt;- grid_regular(mtry(range = c(3, 8)), min_n(range = c(5, 25)), levels = 4)

rf_tuned &lt;- tune_grid(
  rf_wf,
  resamples = folds,
  grid = grid,
  metrics = metric_set(rmse, mae, rsq)
)
```

```
best &lt;- select_best(rf_tuned, metric = "rmse")
rf_fit &lt;- finalize_workflow(rf_wf, best) %&gt;% fit(train)

pred &lt;- predict(rf_fit, test) %&gt;% bind_cols(test %&gt;% select(pm25, day, station_id))
metrics(pred, truth = pm25, estimate = .pred)

# Compare against a simple baseline: yesterday equals tomorrow.
baseline &lt;- test %&gt;% transmute(pm25, .pred = pm25_lag1)
metrics(baseline, truth = pm25, estimate = .pred)
```

What to learn from the lab

First, the split is temporal: the model is trained on earlier days and tested on later days. Second, cross-validation is grouped by station to reduce leakage from repeated observations. Third, a simple baseline is included. A prediction model should not be considered useful unless it improves on a baseline that a domain expert would actually use.

9.8 Worked example in R: rare-event environmental enforcement

Now consider illegal dumping, illegal deforestation, or environmental non-compliance. These are often rare-event classification problems. The policy user may care about the top-ranked cases for inspection. Global accuracy is a poor metric.

```
# Chapter 9 R lab 2: rare-event inspection targeting
# install.packages(c("tidyverse", "tidymodels", "ranger", "yardstick"))

library(tidyverse)
library(tidymodels)
library(ranger)

set.seed(902)

n &lt;- 6000
risk_dat &lt;- tibble(
  distance_road = rexp(n, rate = 1/5),
  distance_city = rexp(n, rate = 1/20),
  slope = rgamma(n, 2, 0.5),
  prior_alerts = rpois(n, 0.4),
  protected = rbinom(n, 1, 0.35),
  commodity_price = rnorm(n, 100, 12)
) %&gt;%
  mutate(
    lin = -4.2 - 0.18 * distance_road - 0.04 * distance_city -
      0.10 * slope + 0.75 * prior_alerts - 0.55 * protected +
      0.012 * commodity_price + 0.018 * commodity_price * prior_alerts,
    p_event = plogis(lin),
    violation = factor(if_else(runif(n) &lt; p_event, "yes", "no"),
                       levels = c("yes", "no"))
  ) %&gt;%
  select(-lin, -p_event)

split &lt;- initial_split(risk_dat, prop = 0.75, strata = violation)
train &lt;- training(split)
test  &lt;- testing(split)

rec &lt;- recipe(violation ~ ., data = train)

rf_spec &lt;- rand_forest(trees = 700, mtry = tune(), min_n = tune()) %&gt;%
  set_engine("ranger", importance = "permutation", probability = TRUE) %&gt;%
  set_mode("classification")

wf &lt;- workflow() %&gt;% add_recipe(rec) %&gt;% add_model(rf_spec)

folds &lt;- vfold_cv(train, v = 5, strata = violation)

tuned &lt;- tune_grid(
  wf,
  resamples = folds,
  grid = 12,
  metrics = metric_set(roc_auc, pr_auc, mn_log_loss)
)
```

```
best &lt;- select_best(tuned, metric = "pr_auc")
fit &lt;- finalize_workflow(wf, best) %&gt;% fit(train)

prob &lt;- predict(fit, test, type = "prob") %&gt;%
  bind_cols(test %&gt;% select(violation))

# Standard metrics
roc_auc(prob, truth = violation, .pred_yes)
pr_auc(prob, truth = violation, .pred_yes)

# Decision metric: precision among the top 5 percent highest-risk parcels
prob %&gt;%
  arrange(desc(.pred_yes)) %&gt;%
  mutate(rank = row_number(), top5 = rank &lt;= 0.05 * n()) %&gt;%
  filter(top5) %&gt;%
  summarise(precision_top5 = mean(violation == "yes"),
            event_rate_overall = mean(prob$violation == "yes"))
```

What to learn from the lab

This lab changes the evaluation question. For an inspection agency, the central question may be: if we inspect only the parcels with highest predicted risk, how many violations do we find? This is a ranking problem. The appropriate metric is precision or recall at the budget-constrained inspection rate, not accuracy across all parcels.

9.9 Prediction, welfare, and choice modelling

Prediction models often feed into environmental choice models. For example, a recreation-site choice model may include predicted beach water quality, predicted crowding, predicted travel time, or predicted heat stress. A household energy-tariff choice experiment may include predicted bill impacts. A conservation DCE may use predicted biodiversity gains under different policy scenarios.

This linkage is powerful but risky. If predicted attributes are treated as certain, WTP estimates may be too precise. If predicted attributes are biased for some groups or places, welfare conclusions may be biased. If the prediction model uses variables affected by the policy being evaluated, counterfactual simulations can be invalid.

Three safe-use principles

Principle 1: Preserve the timing. Features used to predict choice attributes must be available before the choice or policy decision.

Principle 2: Preserve uncertainty. When predicted environmental attributes enter a choice model, conduct sensitivity analysis over plausible prediction errors or use multiple-imputation style workflows.

Principle 3: Preserve behavioural interpretation. Flexible prediction can generate attributes; it should not automatically replace the utility model if the research goal is WTP, welfare, substitution, or counterfactual policy analysis.

9.10 Interpreting environmental prediction models

Interpretability is useful for debugging, scientific learning, and communication with policy users. But interpretation tools must be used carefully. Feature importance tells us which variables the model used for prediction, not which variables causally determine the outcome.

Useful diagnostic tools

Tool	What it answers	Caution

Permutation importance	How much does predictive performance decline when a feature is shuffled?	Correlated features can share or hide importance
Partial dependence	How does predicted outcome change as one feature varies on average?	Can evaluate unrealistic combinations of features
Accumulated local effects	How does prediction change locally across the observed feature distribution?	Less intuitive than partial dependence but often safer with correlated variables
SHAP values	How are features associated with individual predictions in a fitted model?	Model explanation, not causal decomposition
Calibration plot	Do predicted probabilities match observed frequencies?	Needs enough data in probability bins
Residual map	Where are predictions systematically wrong?	Can reveal missing spatial processes or measurement problems

9.11 A complete applied template for a PhD paper

A strong environmental prediction section in a PhD paper should contain the following elements:

- A clear definition of the prediction task: unit, target, features, horizon, and policy use.

- A diagram or table showing timing: when features are observed and when outcomes occur.

- A simple benchmark model and at least one flexible model.

- Validation that matches the deployment setting: temporal, spatial, grouped, or combinations thereof.

- Metrics aligned with the economic decision, including rare-event or top-k metrics when relevant.

- Calibration and uncertainty assessment when predictions become inputs to valuation or causal analysis.

- Interpretability diagnostics used as model checks, not causal claims.

- Sensitivity analysis for domain shift, alternative model classes, and alternative validation schemes.

- Transparent code, package versions, seeds, and data-processing documentation.

9.12 Conceptual checkpoint

Answer these questions without looking back:

1. Why can random train/test splitting overstate performance in spatial pollution prediction?

2. Why is accuracy a poor metric for rare-event environmental enforcement?

3. What is the difference between forecasting electricity demand and estimating the causal effect of dynamic pricing?

4. What can go wrong if predicted environmental attributes enter a choice model as if they were observed without error?

5. Give one example of a prediction task where RMSE is appropriate and one where top-k precision is more appropriate.

6. Explain why a feature can be highly important for prediction but not policy-controllable.

9.13 End-of-chapter exercises

Exercise 1: Reframe a paper

Choose one environmental economics paper that uses pollution, climate, biodiversity, or land-use data. Identify the hidden prediction task. Write the unit, target, features, horizon, decision use, and validation design.

Exercise 2: Redesign validation

Suppose a paper predicts farm-level drought vulnerability using 20 years of observations and randomly splits farm-year rows. Explain why this can be problematic. Propose two alternative validation designs: one for new years and one for new regions.

Exercise 3: Rare event targeting

Using the rare-event lab code, change the event rate by modifying the intercept in the data-generating equation. Compare accuracy, PR-AUC, and precision among the top 5 percent. Explain which metric you would report to an enforcement agency.

Exercise 4: Choice-model bridge

Design a stated choice experiment for coastal adaptation in which one attribute, flood-risk reduction, is generated by a prediction model. List the additional information you would report to respondents and the sensitivity analyses you would run before estimating WTP.

Exercise 5: Interpretability audit

Fit a random forest or XGBoost model to any environmental prediction dataset. Produce a permutation importance plot and one partial dependence or ALE plot. Write one paragraph explaining what you can infer and one paragraph explaining what you cannot infer.

9.14 Common mistakes

Mistake 1: Confusing interpolation with extrapolation. A model that predicts well between observed monitors may not predict well in regions with no monitors or under future climate conditions.
Mistake 2: Ignoring policy timing. Using variables that are observed after a policy decision creates leakage and can make a model unusable for real-time decision support.
Mistake 3: Reporting only one metric. Environmental decisions often require several metrics: average error, tail error, calibration, distributional performance, and decision-specific ranking metrics.
Mistake 4: Treating ML explanations as causal evidence. SHAP, partial dependence, and feature importance explain fitted predictions. They do not identify causal mechanisms by themselves.
Mistake 5: Forgetting downstream uncertainty. If predicted exposures are used in welfare, health, or choice models, uncertainty from the prediction stage must be acknowledged.

9.15 Summary

Prediction problems are central to environmental economics because environmental variables are often spatially incomplete, temporally dynamic, high-dimensional, or expensive to observe. ML expands what can be measured and forecast: pollution exposure, energy demand, ecosystem risk, climate hazards, technology adoption, and recreation pressure. However, useful environmental prediction is not algorithm-first. It is decision-first. The best model is the one whose target, horizon, validation design, loss function, and uncertainty treatment match the economic question.

The next chapter moves from prediction to policy evaluation. We will study how causal machine learning can estimate heterogeneous effects of environmental policies, when those methods are

credible, and how they connect to welfare and choice modelling.

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Chapter 10

Causal Machine Learning for Environmental Policy

Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Position of this chapter. Chapters 8 and 9 introduced environmental data and prediction. This chapter asks a harder question: how can ML help estimate the effects of environmental policies, programmes, risks, and interventions when the target is causal rather than predictive?

Main idea. Causal machine learning does not remove the need for research design. It helps estimate flexible nuisance functions, heterogeneous treatment effects, and policy rules after the analyst has made credible identification assumptions. The causal question must come first; ML is then used as an estimation engine, diagnostic device, or heterogeneity discovery tool.

Learning objectives

After studying this chapter, you should be able to:

- distinguish prediction problems from causal policy-evaluation problems;

- define potential outcomes, treatment effects, selection bias, confounding, and overlap;

- explain why naive ML plug-in estimators can be biased for causal parameters;

- describe double/debiased machine learning, orthogonalization, and cross-fitting in intuitive and mathematical terms;

- use causal forests and generalized random forests to study heterogeneous policy effects;

- connect causal ML to environmental-policy applications such as conservation, pollution regulation, energy programmes, climate adaptation, and technology adoption;

- implement basic causal ML workflows in R and recognize when they are not credible.

10.1 From prediction to causal policy evaluation

A prediction model asks: given X, what value of Y is likely? A causal model asks: what would Y have been under a different policy, treatment, or exposure? The distinction is simple but crucial. Suppose an ML model predicts that households with smart meters consume less electricity. That pattern may reflect causal effects of smart meters, but it may also reflect income, housing quality, environmental attitudes, utility targeting, or pre-existing conservation behaviour. Prediction can be excellent even when causal interpretation is invalid.

Environmental economics is full of policy questions where prediction alone is insufficient. Did low-emission zones reduce pollution exposure? Did payments for ecosystem services reduce deforestation? Did information campaigns reduce water use? Did flood-insurance reform change adaptation investments? Did protected areas improve biodiversity outcomes? Did an electric-vehicle subsidy increase adoption, or merely subsidize households that would have purchased anyway?

Causal ML is useful because environmental policy data are often high-dimensional, spatial, temporal, and heterogeneous. The treatment effect may vary across income groups, land-use types, local climate, baseline exposure, institutional quality, or ecological vulnerability. Conventional linear regressions can summarize average effects, but they may miss important heterogeneity. At the same time, flexible ML can overfit or induce regularization bias if used naively for causal parameters. The modern causal ML literature therefore combines research-design logic with ML estimation.

Policy examples

Policy question	Treatment D	Outcome Y	Confounding risk	Why ML helps
Air-quality regulation	Facility or area exposed to regulation	Pollution, health, property values	Regulated areas differ from unregulated areas	Flexible adjustment for weather, industry, geography, baseline trends
Conservation payments	Enrollment in payment scheme	Forest loss, biodiversity, income	Enrollment is voluntary and targeted	Heterogeneous effects by land pressure, slope, access, opportunity cost
Energy-efficiency rebate	Household receives rebate	Energy use, adoption, bills	Households self-select into programmes	Flexible propensity scores and demand baselines
Climate-adaptation information	Receives risk information	Insurance purchase, relocation, investment	Information may be targeted to high-risk households	Effects may vary by risk, wealth, beliefs, prior exposure
Beach access restriction	Change in crowding or access rule	Visits, welfare, ecological pressure	Policy adopted at congested or sensitive sites	Flexible controls and heterogeneous behavioural response

10.2 Potential outcomes and causal estimands

Let D_i be a treatment indicator: D_i = 1 if unit i receives a policy or exposure, and D_i = 0 otherwise. Each unit has two potential outcomes: Y_i(1), the outcome if treated, and Y_i(0), the outcome if untreated. We observe only one of them:

```
Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).
```

The individual treatment effect is tau_i = Y_i(1) - Y_i(0), but it is never directly observed because one potential outcome is missing. The average treatment effect is ATE = E[Y(1) - Y(0)]. The average treatment effect on the treated is ATT = E[Y(1) - Y(0) | D = 1]. Causal ML often focuses on the conditional average treatment effect, CATE:

```
tau(x) = E[Y(1) - Y(0) | X = x].
```

CATEs are especially relevant for environmental policy because effects are rarely homogeneous. A conservation payment may strongly reduce deforestation in high-pressure areas but have little effect where deforestation risk is already low. A heat-warning system may reduce mortality more in elderly populations than in young populations. A carbon-pricing policy may affect transport behaviour differently in urban and rural areas.

Identification assumptions

Causal ML estimates are only as credible as the identification assumptions. For observational data, the most common baseline is selection on observables: after conditioning on X, treatment assignment is as good as random. Formally:

```
{Y(1), Y(0)} independent of D conditional on X, plus overlap: 0 &lt; P(D=1 | X) &lt; 1.
```

Overlap means that comparable treated and untreated units exist at the same values of X. If all high-risk farms receive a conservation programme and no low-risk farms do, we cannot nonparametrically learn what would have happened to high-risk farms without the programme unless additional design assumptions are made.

> Important warning. Causal ML does not create identification. It can adjust flexibly for observed confounders and reveal heterogeneity, but it cannot solve omitted variables, bad instruments, post-treatment controls, spillovers, or non-overlap by itself.

10.3 Why naive ML can be biased for causal parameters

A natural idea is to fit two ML models: one for treated units and one for untreated units, then subtract predictions. This is called a T-learner. It can work in some settings, but it is not automatically valid. Flexible models may overfit, extrapolate badly where overlap is weak, or regularize away precisely the variation needed for treatment-effect estimation.

Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins explain the key problem for double/debiased ML: when high-dimensional nuisance functions are estimated by regularized ML and then naively plugged into estimating equations, regularization bias and overfitting can contaminate the low-dimensional causal parameter. Their solution combines Neyman-orthogonal scores with cross-fitting. Athey and Imbens similarly emphasize that ML methods are useful for economists but must be adapted to causal inference, policy estimation, and counterfactual questions rather than used as black-box prediction tools.

The partially linear model

A useful teaching model is the partially linear regression:

```
Y = theta D + g(X) + epsilon, E[epsilon | D, X] = 0
```

and

```
D = m(X) + v, E[v | X] = 0.
```

Here theta is the causal effect of D on Y after adjusting for X. The functions g(X) and m(X) can be complex. ML is attractive because it can estimate them flexibly. But if theta is estimated by simply controlling for many ML-selected features, regularization and overfitting can bias theta.

Residualization intuition

The double ML solution is intuitive: first remove the part of Y predictable from X, and remove the part of D predictable from X. Then regress the residualized outcome on the residualized treatment. In symbols:

```
Y_tilde = Y - E[Y | X]; D_tilde = D - E[D | X]; theta = slope from Y_tilde on D_tilde.
```

This is the Frisch-Waugh-Lovell logic generalized with flexible ML nuisance estimates. Cross-fitting strengthens the procedure: estimate the nuisance models on one fold and compute residuals on a held-out fold, then rotate across folds. This prevents the same observations from being used both to train nuisance functions and to estimate the causal parameter.

10.4 Double/debiased machine learning

Double/debiased machine learning, often abbreviated DML, is one of the most important bridges between econometrics and ML. It is designed for settings where the target parameter is low-dimensional, such as an average treatment effect or structural coefficient, while the nuisance functions are high-dimensional or nonlinear.

The algorithm in words

Step	What happens	Purpose
1. Split data into folds	Create K folds, often 5 or 10	Enable honest nuisance estimation
2. Learn nuisance functions	Estimate E[Y|X] and E[D|X] using ML on training folds	Use flexible prediction for confounding adjustment
3. Predict on held-out fold	Generate out-of-fold predictions	Avoid overfitting bias in the estimating equation
4. Residualize	Compute Y - Y_hat and D - D_hat	Remove variation explained by controls
5. Estimate target parameter	Regress residualized Y on residualized D or solve orthogonal score	Estimate treatment effect with valid inference under assumptions
6. Repeat or tune carefully	Use cross-fitting, repeated splits, and sensitivity checks	Assess stability and finite-sample reliability

Why orthogonality matters

Neyman orthogonality means that small errors in nuisance-function estimation have only second-order effects on the target parameter. In plain language: the causal estimate is made locally insensitive to imperfect ML predictions of the nuisance components. This is what allows flexible ML to be used without invalidating inference on theta, provided rates and assumptions are satisfied.

> Teaching intuition. Prediction error in nuisance models is unavoidable. Orthogonalization designs the estimating equation so that first-order nuisance errors cancel out rather than directly bias the causal estimate.

10.5 Heterogeneous treatment effects and causal forests

Many environmental policy questions are not satisfied by an average effect. A programme with zero average effect may still be valuable if it strongly benefits a subgroup that can be targeted. A harmful average effect may conceal benefits in vulnerable communities and harms elsewhere. Causal forests and generalized random forests are designed to estimate treatment-effect heterogeneity flexibly.

Generalized random forests extend random forests from pure prediction to local estimation of statistical quantities such as heterogeneous treatment effects, conditional average partial effects, quantiles, and instrumental-variable effects. In the causal-forest case, trees are built to identify splits that reveal treatment-effect heterogeneity rather than only outcome prediction. Honest estimation is often used: one subsample chooses splits, another estimates treatment effects in leaves.

Causal forest logic

Ordinary random forest	Causal forest
Goal: predict Y	Goal: estimate tau(x) = E[Y(1)-Y(0)|X=x]
Splits reduce prediction error	Splits reveal treatment-effect heterogeneity
Output is predicted outcome	Output is estimated conditional treatment effect
Validation uses prediction loss	Validation uses treatment-effect diagnostics, calibration, policy value, and robustness

Ordinary random forest	Causal forest
Interpretation: important predictors of Y	Interpretation: moderators of treatment effects, not necessarily predictors of Y

Environmental examples of heterogeneity

Causal forests are useful when an intervention may have different effects across ecological and socioeconomic contexts. A forest-conservation payment may work best where deforestation pressure is moderate: high enough for avoided loss to be possible, but not so high that payments are insufficient. A home-energy audit may work best among households with inefficient buildings but enough liquidity to invest. A heat-alert system may work best where households have access to cooling but lack risk information. A coastal retreat programme may have heterogeneous effects depending on tenure, flood history, social networks, and local amenities.

An uploaded transportation application on autonomous-vehicle adoption illustrates this logic outside environmental economics: a logit model provides an interpretable average summary of the effect of security concerns, while generalized random forests uncover heterogeneity across demographic and built-environment contexts. The methodological lesson transfers directly to environmental policy: combine interpretable average-effect models with flexible heterogeneity analysis when policy targeting matters.

10.6 Causal ML designs for environmental economics

10.6.1 Randomized experiments and adaptive interventions

Randomized experiments remain the cleanest design. In environmental economics, experiments can test energy reports, conservation messages, default green tariffs, water-saving reminders, recycling incentives, eco-labels, or climate-risk information. ML can improve experiments by increasing precision, detecting heterogeneity, and designing targeting rules. However, if ML-driven targeting changes treatment assignment over time, the design must be documented carefully so that causal interpretation remains valid.

10.6.2 Observational selection-on-observables designs

When treatment is not randomized, DML and causal forests can be useful under conditional unconfoundedness. Examples include voluntary conservation-programme enrollment, adoption of rooftop solar, receipt of energy-efficiency rebates, participation in agri-environmental schemes, or exposure to local adaptation programmes. The analyst must defend the covariate set and ensure that controls are pre-treatment.

10.6.3 Difference-in-differences with ML

Environmental policies are often adopted at different times across places. Difference-in-differences compares changes in treated and untreated units, relying on a parallel-trends assumption. ML can help flexibly adjust for covariates, estimate heterogeneous effects, or construct better comparison groups. But it cannot rescue a design if treated and control areas had systematically different counterfactual trends that are not captured by the model.

10.6.4 Instrumental variables and encouragement designs

Some environmental policies involve imperfect compliance. For example, an information mailing may encourage energy-audit participation but not force it. An instrument Z affects treatment D and, under assumptions, affects outcome Y only through D. Generalized random forests include instrumental-variable extensions for heterogeneous effects. Still, the exclusion restriction remains a substantive assumption, not a machine-learning output.

10.6.5 Regression discontinuity and thresholds

Policy thresholds are common: eligibility based on income, risk score, pollution level, protected-area boundary, or distance. ML can help choose flexible controls around the threshold or explore heterogeneity, but the core RD logic remains local comparison near the cutoff. Avoid using ML to extrapolate far from the threshold and calling it RD evidence.

10.6.6 Spatial spillovers and interference

Many environmental policies violate the stable unit treatment value assumption. Conservation in one parcel can displace deforestation to nearby parcels. Air-quality regulation in one area can shift emissions to another. Beach-access restrictions can move visitors to neighbouring beaches. Causal ML workflows should explicitly consider interference, exposure mappings, spatial buffers, and equilibrium responses.

> Environmental-policy pitfall. A local treatment effect may be misleading when ecological or market spillovers are important. If a protected area reduces deforestation inside its boundary but increases pressure just outside, a naive unit-level causal estimate may overstate net conservation benefits.

10.7 Worked R lab 1: DoubleML for an energy-efficiency programme

This lab uses simulated data. The example estimates the effect of an energy-efficiency rebate on electricity consumption. Treatment is not random: inefficient, wealthier, and environmentally motivated households are more likely to participate. We use DoubleML to adjust flexibly for observed confounders.

```
# Chapter 10 R lab 1: Double/debiased ML
# install.packages(c("DoubleML", "mlr3", "mlr3learners", "data.table", "ranger"))

library(DoubleML)
library(mlr3)
library(mlr3learners)
library(data.table)

set.seed(1001)
n &lt;- 2000

income &lt;- rnorm(n, 60, 20)
house_size &lt;- rnorm(n, 110, 35)
baseline_kwh &lt;- rnorm(n, 500, 110)
green_attitude &lt;- rnorm(n)
old_building &lt;- rbinom(n, 1, 0.45)

# Treatment propensity: participation is selective
p_treat &lt;- plogis(-3 + 0.018 * income + 0.004 * baseline_kwh +
                  0.6 * green_attitude + 0.5 * old_building)
rebate &lt;- rbinom(n, 1, p_treat)

# True effect: rebate reduces electricity use by 35 kWh on average
# There is heterogeneity: larger reductions in old buildings
true_tau &lt;- -25 - 25 * old_building

kwh_after &lt;- 120 + true_tau * rebate +
  0.65 * baseline_kwh + 0.7 * house_size - 0.8 * income -
  12 * green_attitude + 25 * old_building + rnorm(n, 0, 45)

dt &lt;- data.table(
  kwh_after, rebate, income, house_size, baseline_kwh,
  green_attitude, old_building
```

```
)

x_cols &lt;- c("income", "house_size", "baseline_kwh", "green_attitude", "old_building")

dml_data &lt;- DoubleMLData$new(
  dt,
  y_col = "kwh_after",
  d_cols = "rebate",
  x_cols = x_cols
)

learner_y &lt;- lrn("regr.ranger", num.trees = 500, min.node.size = 5)
learner_d &lt;- lrn("regr.ranger", num.trees = 500, min.node.size = 5)

# Partially linear regression model
obj &lt;- DoubleMLPLR$new(
  dml_data,
  ml_l = learner_y,
  ml_m = learner_d,
  n_folds = 5
)

obj$fit()
obj$summary()

# Interpretation:
# The coefficient on rebate estimates the average treatment effect
# under selection-on-observables and overlap assumptions.
```

Interpreting the lab

The DoubleML estimate should be close to the average of true_tau among the population if the simulated assumptions are satisfied. In real applications, the coefficient is credible only if all important confounders are measured and included as pre-treatment controls. The random forest nuisance models may predict baseline demand and participation flexibly, but they do not solve unobserved motivation, contractor availability, or utility targeting if those are omitted.

10.8 Worked R lab 2: causal forests for heterogeneous effects

The next lab estimates heterogeneous treatment effects using the grf package. The same simulated programme has larger effects in old buildings. A causal forest should recover that old-building status moderates the effect.

```
# Chapter 10 R lab 2: Causal forests with grf
# install.packages(c("grf", "tidyverse"))

library(grf)
library(tidyverse)

X &lt;- as.matrix(dt[, ..x_cols])
Y &lt;- dt$kwh_after
W &lt;- dt$rebate

cf &lt;- causal_forest(X, Y, W, num.trees = 2000)

# Average treatment effect
average_treatment_effect(cf)

# Individualized treatment-effect predictions
tau_hat &lt;- predict(cf)$predictions

results &lt;- dt %&gt;%
  mutate(tau_hat = tau_hat) %&gt;%
  group_by(old_building) %&gt;%
  summarise(
    mean_tau_hat = mean(tau_hat),
    n = n(),
    .groups = "drop"
  )

print(results)
```

```
# Variable importance for heterogeneity discovery
variable_importance(cf) %&gt;%
  as.numeric() %&gt;%
  tibble(variable = x_cols, importance = .) %&gt;%
  arrange(desc(importance))
```

Interpreting heterogeneity

If the model works well, predicted treatment effects should be more negative for old buildings. But causal-forest heterogeneity must be interpreted cautiously. A variable can be important because it predicts treatment-effect variation, because it proxies omitted moderators, or because overlap differs across the covariate space. Always combine heterogeneity results with plots, subgroup summaries, external theory, and sensitivity checks.

10.9 From heterogeneous effects to policy targeting

Causal ML often leads naturally to targeting. If the treatment has a cost, the planner may treat units for which predicted benefits exceed costs. For example, an energy-efficiency audit may be assigned to households with high expected savings; conservation payments may target parcels with high expected avoided deforestation; heat-adaptation subsidies may target households with high expected risk reduction.

> Policy rule: treat unit i if predicted benefit_i - cost_i > 0, subject to the budget constraint.

This is attractive, but it raises ethical and political questions. Targeting can improve cost-effectiveness, but it can also exclude disadvantaged groups if the outcome, treatment cost, or prediction model embeds historical inequities. Environmental policy often has distributional goals in addition to efficiency goals. For this reason, policy learning should be evaluated using both welfare criteria and equity constraints.

Evaluation of targeting rules

Criterion	Question
Expected policy value	Does the targeting rule improve expected outcomes relative to no treatment or universal treatment?
Cost-effectiveness	How much outcome improvement is achieved per euro spent?
Equity	Which groups receive treatment, and which groups are excluded?
Robustness	Does the rule still perform under alternative models, samples, and assumptions?
Transparency	Can affected stakeholders understand why treatment is assigned?
Legal and ethical acceptability	Is the use of personal, spatial, or sensitive information justified?

10.10 Connections to choice modelling

Choice modelling and causal ML meet in several ways. First, many environmental policies affect choices: travel mode, energy technology adoption, conservation participation, recreation-site selection, waste sorting, or diet. Second, stated preference experiments are often randomized by design, creating opportunities for causal interpretation of attribute effects within the experimental context. Third, causal ML can help analyze heterogeneity in responses to attributes, information treatments, or survey designs.

Choice experiments as randomized designs

In a well-designed stated choice experiment, attributes are randomized or experimentally controlled. This helps identify the effect of attributes on choices, conditional on the choice model. However, welfare interpretation still depends on random utility assumptions, cost coefficients, scale, and behavioural validity. Machine learning can model complex response surfaces, but WTP and welfare need extra structure. This is why the uploaded choice-modelling literature repeatedly emphasizes complementarity rather than replacement: ML can improve specification, prediction, and representation, while choice models preserve behavioural and welfare interpretation.

Causal questions inside choice surveys

Survey feature	Causal estimand	Possible ML role
Information treatment before DCE	Effect of information on choices or WTP	Causal forest for heterogeneous information effects
Randomized attribute levels	Effect of attribute changes on choice probability	Flexible estimation of nonlinear attribute response
Cheap-talk or consequentiality script	Effect on stated WTP or protest behaviour	DML with flexible controls and subgroup analysis
Visual or virtual-reality treatment	Effect of richer representation on choices	Image embeddings plus experimental contrasts
Survey mode or interface design	Effect on response quality or attribute non-attendance	ML detection of response patterns, causal analysis of randomized interface

10.11 Environmental-policy case templates

Template A: payments for ecosystem services

Unit: land parcel or household. Treatment: programme enrollment. Outcome: deforestation, land-cover quality, income, or biodiversity proxy. Identification: selection on observables, matched comparison, difference-in-differences, or randomized rollout. ML role: predict deforestation risk, estimate propensity scores, estimate heterogeneous treatment effects, and design targeting rules. Main dangers: voluntary enrollment, spillovers to nearby land, non-random monitoring, and equilibrium land-market responses.

Template B: air-quality regulation

Unit: facility, monitor, neighbourhood, or grid cell. Treatment: exposure to a regulation, low-emission zone, plant closure, or enforcement action. Outcome: emissions, concentrations, health outcomes, property values. Identification: difference-in-differences, event study, boundary discontinuity, wind-based exposure design, or instrumental variable. ML role: exposure prediction, flexible trend adjustment, heterogeneity by baseline exposure and vulnerability. Main dangers: regulatory targeting, spatial spillovers, non-parallel trends, and measurement error.

Template C: energy-efficiency programme

Unit: household, building, or firm. Treatment: audit, rebate, information report, default tariff. Outcome: electricity/gas use, adoption, bills, emissions. Identification: randomized encouragement, rollout, selection-on-observables, or IV. ML role: baseline demand modelling, uplift modelling, CATE estimation, policy targeting. Main dangers: self-selection, rebound effects, weather normalization, and equity concerns in targeting.

Template D: climate adaptation

Unit: household, farm, municipality, infrastructure asset. Treatment: risk information, subsidy, insurance reform, adaptation investment, warning system. Outcome: protective action, damages, migration, crop yields, mortality. Identification: experiment, natural experiment, threshold, rollout, or DiD. ML role: risk prediction, vulnerability segmentation, heterogeneous treatment effects, policy learning. Main dangers: rare events, domain shift under climate change, behavioural adaptation, and disaster-driven selection.

10.12 Diagnostics and credibility checklist

Check	Question
Design before algorithm	What identifies the causal effect? Randomization, DiD, IV, RD, selection on observables, or another design?
Treatment timing	Are all controls measured before treatment?
Overlap	Do treated and control units coexist across the relevant covariate space?
Spillovers	Can treatment of one unit affect outcomes of another?
Outcome measurement	Is Y measured similarly for treated and untreated units?
Nuisance validation	Do nuisance models predict treatment and outcome well enough without leakage?
Heterogeneity stability	Are CATE patterns stable across folds, samples, algorithms, and reasonable specifications?
Interpretability	Can the estimated effect pattern be explained substantively?
Policy relevance	Does the estimand match the policy decision?
Sensitivity	How strong would unobserved confounding need to be to change the conclusion?

10.13 Common mistakes

Mistake 1: using post-treatment controls. If a conservation programme changes road construction, and road construction is then used as a control, the model may block part of the causal effect.

Mistake 2: interpreting CATEs as discovered truth. Heterogeneity estimates are noisy and model-dependent. Treat them as hypotheses unless supported by theory, validation, or replication.

Mistake 3: evaluating causal ML with predictive metrics only. A model can predict outcomes well while estimating treatment effects poorly. Treatment-effect validation requires design diagnostics, placebo tests, calibration, and robustness checks.

Mistake 4: ignoring overlap. Flexible ML can hide extrapolation. Always inspect propensity scores and covariate distributions.

Mistake 5: ignoring spatial and market spillovers. Environmental policies often displace behaviour across space or markets. Unit-level treatment effects may not equal aggregate policy effects.

10.14 Conceptual checkpoint

1. Why can a model with excellent predictive accuracy fail to estimate a causal effect?
2. What are Y(1), Y(0), ATE, ATT, and CATE?
3. What is overlap, and why is it especially important for environmental targeting programmes?
4. Explain double ML using residualization in your own words.

5. What is the difference between a predictor of the outcome and a moderator of the treatment effect?

6. Give one environmental example where spillovers would invalidate a naive causal ML analysis.

7. Why might policy targeting based on CATEs raise equity concerns?

10.15 Suggested exercises

Exercise 1: overlap diagnosis

Simulate a treatment that is almost always assigned to high-risk units. Estimate propensity scores using random forest and plot their distribution by treatment status. Identify where overlap is weak and discuss whether the ATE is credible.

Exercise 2: policy targeting

Using the causal-forest lab, define a programme cost of 20 kWh-equivalent units. Treat households only if predicted savings exceed the cost. Compare expected policy value under universal treatment, no treatment, and targeted treatment. Then repeat with an equity constraint requiring at least 40 percent of treated households to be low-income.

Exercise 3: environmental DiD with ML

Create a simulated panel of municipalities, some treated by a pollution policy after year 3. Estimate a conventional DiD model. Then add flexible pre-treatment covariates or group-specific trends. Discuss what ML can and cannot solve if treated municipalities had different pre-trends.

Exercise 4: stated preference information treatment

Design a DCE where respondents are randomly assigned to receive information about biodiversity loss. Define a causal estimand for the effect of information on WTP. Explain how you would estimate average and heterogeneous information effects.

10.16 Reading guide

1. Chernozhukov et al. (2018) for double/debiased machine learning, orthogonal scores, and cross-fitting.

2. Athey and Imbens (2019) for a broad economist-oriented map of supervised learning, causal inference, policy learning, and counterfactual questions.

3. Wager and Athey (2018) and Athey, Tibshirani, and Wager (2019) for causal forests and generalized random forests.

4. DoubleML documentation for implementation in R and Python; grf documentation for causal forests in R; EconML documentation for Python workflows.

5. Environmental and choice-modelling applications in the uploaded papers, especially conservation DCEs, causal ML for autonomous-vehicle security concerns, and the broader choice-modelling-in-the-age-of-ML discussion.

Chapter takeaway

Causal machine learning is most useful when it is treated as a disciplined extension of econometrics rather than as a substitute for identification. In environmental economics, it helps estimate flexible confounding adjustments, heterogeneous effects, and targeting rules for policies with spatial, temporal, ecological, and behavioural complexity. But credible causal ML still begins with design: what is the treatment, what is the counterfactual, what assumptions identify the effect, and what policy decision will the estimate support?

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Chapter 11

Environmental Valuation and Discrete Choice Experiments

Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Position of this chapter. Chapters 8 to 10 covered environmental data, prediction, and causal ML. This chapter turns to one of the main bridges between environmental economics and choice modelling: valuation of non-market environmental goods using stated preference methods, especially discrete choice experiments (DCEs).

Main idea. Environmental DCEs are not ordinary prediction exercises. Their purpose is to learn trade-offs, marginal values, welfare measures, and policy counterfactuals for goods such as biodiversity, landscape quality, water quality, climate adaptation, congestion reduction, and access to recreation. ML can help with design, data quality, heterogeneity, high-dimensional covariates, adaptive surveys, and specification search, but welfare interpretation still requires economic structure.

Learning objectives

After studying this chapter, you should be able to:

- explain why environmental valuation is necessary when markets do not reveal prices for environmental goods;

- distinguish use values, non-use values, willingness to pay, willingness to accept, compensating surplus, and equivalent surplus;

- describe the logic of contingent valuation and discrete choice experiments;

- connect DCE design to random utility theory, attributes, levels, choice sets, priors, and blocking;

- estimate a basic environmental DCE in R using mlogit and understand how logitr/Apollo extend the workflow;

- compute marginal willingness to pay and simple policy welfare measures;

- identify major threats to validity: hypothetical bias, protest responses, attribute non-attendance, scale heterogeneity, learning, fatigue, and sample representativeness;

- explain how ML can support environmental valuation without replacing welfare theory.

11.1 Why environmental valuation exists

Many environmental goods are not traded in ordinary markets. There is no market price for a cleaner river that improves recreation and ecosystem health, for the existence of a species that citizens may never directly see, for reduced coastal crowding, for lower wildfire risk, for a more natural seabed, or for the option of future access to a protected landscape. Yet policy decisions require trade-offs. Public budgets are limited, environmental interventions have opportunity costs, and regulations impose costs on firms and households.

Environmental valuation provides a framework for measuring these trade-offs in monetary or welfare terms. The goal is not to claim that nature is only worth money. The goal is to make otherwise invisible benefits explicit enough to compare with costs, distributional consequences, and alternative policies. In cost-benefit analysis, non-market valuation helps avoid the implicit assumption that unpriced environmental benefits are worth zero.

Stated preference methods are especially important when revealed behaviour is unavailable or incomplete. For example, if a policy would create a future marine protected area, improve a water body beyond historical experience, or protect biodiversity with passive-use value, observed market behaviour may not contain the required counterfactual. Stated preference surveys create structured hypothetical markets that ask people to make choices or state values under carefully designed scenarios.

Value concepts

Concept	Meaning	Environmental example	Typical valuation route
Direct use value	Value from direct interaction with the resource	Swimming at a cleaner beach; hiking in a restored forest	Travel cost, recreation demand, DCE, contingent valuation
Indirect use value	Value from ecosystem functions that support welfare	Flood protection from wetlands; air filtration by trees	Production functions, avoided cost, DCE
Option value	Value of preserving future use possibilities	Keeping future access to a protected coastline	Stated preference, sometimes revealed preference
Existence value	Value from knowing a resource exists	Protection of a threatened species	Stated preference
Bequest value	Value from preserving resources for future generations	Passing biodiversity to children and grandchildren	Stated preference
Cultural or relational value	Value linked to identity, stewardship, place, or social relations	Sacred landscapes, local fishing heritage	Mixed methods, deliberative valuation, DCE with care

Interpretive caution. Monetary valuation is a policy language, not a complete moral theory. Some environmental losses may be irreversible, rights-based, culturally sacred, or distributionally unacceptable even when an average WTP estimate is positive.

11.2 Stated preference methods: CV and DCEs

Stated preference methods ask respondents to evaluate hypothetical scenarios. The two best-known families are contingent valuation (CV) and discrete choice experiments. CV usually asks for a value for a defined policy or environmental change, often through a referendum-style yes/no payment question. A DCE asks respondents to choose among alternatives described by multiple attributes and levels. The DCE decomposes value into marginal utilities and trade-offs across attributes.

Contemporary stated preference guidance emphasizes careful scenario development, consequentiality, incentive compatibility where possible, clear payment vehicles, pretesting, attention to non-use values, validity checks, and transparent reporting. These recommendations apply directly to environmental DCEs because DCE estimates are only as credible as the hypothetical market and choice tasks that produced them (Johnston et al., 2017).

CV versus DCE

Dimension	Contingent valuation	Discrete choice experiment
Typical question	Would you pay amount A for policy P?	Which alternative do you choose among A, B, and status quo?
Strength	Simple referendum framing; close to policy vote	Estimates marginal values for multiple attributes and policy bundles
Weakness	Less informative about attribute trade-offs	More complex design and cognitive burden
Welfare output	Value of the whole policy change	Marginal WTP, compensating surplus for many scenarios
Best suited for	Single well-defined policy or good	Multi-attribute environmental programmes, design trade-offs, scenario simulation

11.3 Anatomy of an environmental DCE

A DCE presents repeated choice tasks. In each task, respondents choose one alternative from a choice set. Alternatives are described by attributes. Attributes take levels. One attribute is often monetary, such as tax, fee, annual payment, entrance price, travel cost, or product price. The monetary attribute anchors welfare estimation.

A simple beach-management DCE may include: access fee, crowding level, water quality, seabed type, ecosystem quality, and travel time. A conservation DCE may include: biodiversity gain, landowner restrictions, local employment, payment, governance, and uncertainty. A climate-adaptation DCE may include: risk reduction, public cost, relocation requirement, ecological side effects, and fairness.

Design components

Component	Question for the analyst	Bad practice	Better practice
Policy scenario	What real decision is being represented?	Vague improvement in nature	Clear baseline, change, location, timing, uncertainty, institution
Attributes	Which dimensions define the alternatives?	Too many attributes; technical terms	Attributes grounded in policy relevance, qualitative work, and respondent comprehension
Levels	What values can attributes take?	Implausible or dominated levels	Plausible ranges, pilot-tested, with enough variation for identification
Payment vehicle	How is money paid?	Unrealistic voluntary donation for public good	Credible tax, fee, bill, levy, price, or contribution depending on context
Choice set	What alternatives are shown?	Only hypothetical improvements without opt-out	Include status quo or opt-out when policy-relevant
Experimental design	Which combinations are shown?	Ad hoc combinations	Efficient, balanced, or Bayesian design with blocking and pilot priors
Survey implementation	How is the task explained?	No cheap talk, no comprehension checks	Pretesting, scripts, visual aids, consequential framing, attention checks
Analysis plan	What models and validity checks?	Only one MNL and no sensitivity analysis	Baseline MNL plus heterogeneity, WTP, robustness, validity, and ML-supported diagnostics

11.4 Random utility foundation

The standard DCE is grounded in random utility theory. Respondent n chooses alternative i in choice situation t if that alternative gives the highest utility:

```
U_nit = V_nit + eps_nit, choose i if U_nit > U_njt for all j in C_nt.
```

The systematic utility V_nit is specified as a function of attributes and possibly respondent characteristics. For a simple environmental policy:

```
V_nit = beta_cost cost_nit + beta_bio bio_nit + beta_water water_nit + beta_access access_nit + ASC_i.
```

If eps is iid extreme value, the choice probability is multinomial logit:

```
P_nit = exp(V_nit) / sum_j exp(V_njt).
```

The cost coefficient should usually be negative. Environmental improvement coefficients should often be positive if higher values indicate better outcomes. The ratio of an environmental coefficient to the cost coefficient gives marginal WTP:

```
WTP_k = - beta_k / beta_cost.
```

This ratio is the amount of money that leaves utility unchanged when attribute k improves by one unit. The same logic extends to policy scenarios: compensating surplus compares expected maximum utility before and after a policy change.

Expected maximum utility and welfare

Under logit assumptions, expected maximum utility for a choice set is connected to the logsum:

```
EMU = ln(sum_j exp(V_j)).
```

For a policy change from baseline 0 to scenario 1, a simple compensating surplus expression is:

```
CS = (1 / -beta_cost) [ ln sum_j exp(V_j^1) - ln sum_j exp(V_j^0) ].
```

The formula is simple, but its credibility rests on the specification, payment vehicle, behavioural assumptions, and scenario validity. A model that predicts choices well but has an unstable or incorrectly signed cost response may be useless for welfare analysis.

11.5 Design strategy: from qualitative research to efficient design

A strong environmental DCE begins before any econometric model is estimated. The analyst must understand the policy context, the ecological change, the institutions that can deliver it, and respondents vocabulary. Qualitative interviews, focus groups, expert consultation, cognitive testing, and pilots are part of the model, not decorative add-ons. They determine whether respondents interpret attributes as intended.

A defensible DCE workflow

Stage	Main tasks	Evidence produced
1. Policy definition	Define baseline, policy change, population, timing, spatial scope, payment mechanism	Scenario memo and causal/welfare estimand
2. Qualitative development	Interviews, focus groups, expert review, cognitive testing	Attribute list, wording revisions, comprehension evidence
3. Pilot design	Initial fractional factorial or Bayesian design with weak priors	Pilot estimates, variance, dominance checks, task burden
4. Final design	Efficient design, blocking, randomization, status quo handling, visual aids	Design report and survey instrument
5. Fieldwork	Sampling, quotas, survey monitoring, attention and timing diagnostics	Response quality report
6. Baseline estimation	MNL, alternative-specific constants, interactions, scale checks	Core preference and WTP estimates
7. Heterogeneity and validity	Mixed logit, latent class, ANA, scale, protest, external validity	Robustness and interpretation
8. Policy simulation	Counterfactual scenarios, welfare, distribution, uncertainty	Policy-relevant valuation outputs

Efficient design in plain language

A DCE design determines which attribute-level combinations respondents see. A full factorial design is usually too large. Fractional factorial and efficient designs select a subset that allows estimation of the required parameters. Efficiency depends on the design matrix and, for nonlinear models, on parameter priors. A common approach is sequential: start with weak priors, run a pilot, update priors, then generate the final design.

A design can be statistically efficient but behaviourally poor. For example, it may create unrealistic alternatives, obvious dominance, implausible costs, or excessive cognitive burden. Environmental DCE design is therefore a joint problem of statistical identification, policy realism, and respondent comprehension.

11.6 Threats to validity

Environmental DCEs are powerful but fragile. The main threats are not only econometric; many arise from survey design and respondent cognition.

Common threats

Threat	What it means	Diagnostic or mitigation
Hypothetical bias	Stated choices differ from real choices because payment is hypothetical	Consequential framing, referendum payment vehicle, cheap talk, certainty scales, validation where possible
Protest response	Respondent rejects the scenario, payment vehicle, or institution rather than the environmental good	Debriefing questions, sensitivity analysis, transparent coding
Attribute non-attendance	Respondent ignores one or more attributes	Follow-up ANA questions, latent-class ANA, LASSO or regularization diagnostics, model comparisons
Scale heterogeneity	Different respondents have different error variance or choice consistency	Scale-adjusted models, generalized MNL, careful interpretation of coefficient magnitudes
Learning and fatigue	Responses change across tasks because respondents learn or tire	Task-order interactions, shorter blocks, randomized order, response-time analysis
Starting point and anchoring	Cost levels or first tasks anchor valuations	Design sensitivity, randomized blocks, pilot testing
Status quo bias	Respondents prefer current conditions independent of attributes	Status quo constants, reference-dependent models, careful baseline description
Sample selection	Survey sample differs from policy population	Sampling frame, quotas, weights, non-response analysis
Embedding and scope insensitivity	Values do not respond properly to the size of the good	Scope tests, split-sample designs, clear magnitude descriptions

Attribute non-attendance and ML

Attribute non-attendance is especially important because ignoring the cost attribute or an environmental attribute can bias WTP. Traditional approaches include stated ANA follow-up questions, equality restrictions, latent class models, and random parameters. ML can add another layer. For example, regularization methods such as LASSO can screen parameters and interactions, helping identify attributes that appear to contribute little to predictive or likelihood performance. The uploaded paper on organic apples applies LASSO to attribute non-attendance in a choice experiment and interprets selected or screened-out attributes as diagnostic evidence rather than as an automatic final behavioural truth (Li and Yu, 2025).

> Do not overinterpret ML selection. If LASSO drops an attribute, this may indicate non-attendance, weak design variation, collinearity, insufficient sample size, scale issues, or model misspecification. Use ML as a diagnostic, then return to behavioural interpretation and robustness checks.

11.7 How ML can support environmental valuation

Machine learning enters environmental valuation in at least eight useful ways. The key is to preserve the distinction between predictive support and welfare interpretation.

ML role	What it can do	What it cannot do alone
Survey design support	Adaptive designs, pilot-prior updating, detecting dominated or uninformative tasks	Guarantee incentive compatibility or policy realism
Text and open-ended responses	Classify protest reasons, concerns, perceived realism, policy themes	Replace careful coding validation
Data-quality diagnostics	Flag speeders, straight-liners, inconsistent patterns, unusual task behaviour	Decide normative exclusion rules automatically
Specification search	Screen interactions, nonlinear transformations, class-membership predictors	Provide final theory-consistent utility specification by itself
Heterogeneity discovery	Suggest segments or moderators for WTP	Turn predictive segmentation into welfare-valid heterogeneity automatically
Attribute non-attendance	Assist with regularized selection and latent patterns	Prove psychological non-attendance
Flexible prediction benchmark	Compare DCM predictions with random forest, boosting, neural nets	Replace WTP or compensating surplus formulas

ML role	What it can do	What it cannot do alone
Images and spatial data	Extract environmental features from street view, satellite, maps, photos	Ensure respondents perceive image-derived features as intended

Recent choice-modelling literature treats this as complementarity. Machine learning can help choice modellers with feature learning, specification search, nontraditional data, and prediction, while discrete choice models contribute behavioural structure, welfare formulas, and counterfactual discipline. This complementarity is consistent with the broader research agenda in the uploaded discussion paper on choice modelling in the age of ML, and with the 2024 Handbook of Choice Modelling chapter on model building and interpretation in the age of ML.

Adaptive DCEs

Adaptive DCEs choose later tasks based on earlier responses. The uploaded gradient-based survey paper proposes a machine-learning-guided adaptive design for multiattribute product design, emphasizing scalability and robustness to parametric misspecification. The idea is attractive for environmental valuation when there are many attributes or when individual-level preference learning is important. However, adaptive designs require careful documentation because each respondent may face a different design, and standard errors, priors, and welfare interpretation need to respect the adaptive data-generating process.

Expert DCEs for conservation

Not all DCEs survey the general public. Expert DCEs can inform conservation prioritization when choices require ecological expertise. The uploaded Environmental Research Letters paper on conservation action uses DCEs with experts to elicit preferences over conservation decisions. This is a reminder that stated choice methods are not limited to household WTP; they can also structure expert judgment, institutional priorities, or policy trade-offs. For welfare analysis, however, expert preferences should not be confused with citizen WTP unless the normative target is explicitly expert decision making.

11.8 Worked R lab 1: simulate and estimate a simple environmental DCE

This lab simulates a DCE with three alternatives: two improvement programmes and a status quo. Each improvement has a cost, biodiversity gain, water-quality gain, and access restriction. The respondent chooses one alternative in each task.

```
# Chapter 11 R lab 1: Simulate and estimate a simple environmental DCE
# install.packages(c("tidyverse", "mlogit"))

library(tidyverse)
library(mlogit)

set.seed(1101)
N  &lt;- 600   # respondents
Tn &lt;- 8     # tasks per respondent
alts &lt;- c("A", "B", "SQ")

# True preference parameters
beta_cost   &lt;- -0.055
beta_bio    &lt;-  0.55
beta_water  &lt;-  0.40
beta_access &lt;- -0.25
asc_A &lt;- 0.10
asc_B &lt;- 0.05

long &lt;- expand_grid(id = 1:N, task = 1:Tn, alt = alts) %&gt;%
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = if_else(alt == "SQ", 0, sample(c(10, 25, 50, 80), n(), replace = TRUE)),
    biodiversity = if_else(alt == "SQ", 0, sample(c(0, 1, 2, 3), n(), replace = TRUE)),
    water_quality = if_else(alt == "SQ", 0, sample(c(0, 1, 2, 3), n(), replace = TRUE)),
    access_restriction = if_else(alt == "SQ", 0, sample(c(0, 1), n(), replace = TRUE)),
    asc = case_when(alt == "A" ~ asc_A, alt == "B" ~ asc_B, TRUE ~ 0),
    V = asc + beta_cost * cost + beta_bio * biodiversity +
        beta_water * water_quality + beta_access * access_restriction
  ) %&gt;%
  group_by(chid) %&gt;%
  mutate(prob = exp(V) / sum(exp(V)),
```

```
        chosen_alt = sample(alt, size = 1, prob = prob),
        choice = as.integer(alt == chosen_alt)) %&gt;%
  ungroup()

mdata &lt;- mlogit.data(long, choice = "choice", shape = "long",
                     chid.var = "chid", alt.var = "alt", id.var = "id")

m1 &lt;- mlogit(choice ~ cost + biodiversity + water_quality + access_restriction | 0,
             data = mdata)
summary(m1)

b &lt;- coef(m1)
wtp &lt;- tibble(
  attribute = c("biodiversity", "water_quality", "access_restriction"),
  wtp = c(-b["biodiversity"] / b["cost"],
          -b["water_quality"] / b["cost"],
          -b["access_restriction"] / b["cost"])
)
wtp
```

Interpreting the output

The cost coefficient should be negative. Biodiversity and water quality should be positive. Access restriction should be negative if respondents dislike restrictions. The WTP values translate the environmental and access attributes into monetary equivalents. For example, if WTP for biodiversity is 10, respondents are estimated to be willing to pay about 10 monetary units for a one-level biodiversity improvement, holding other attributes constant.

11.9 Worked R lab 2: WTP space with logitr

Preference-space models estimate beta coefficients and then compute WTP ratios. WTP-space models directly parameterize utility in terms of WTP. This can be useful because ratios of random coefficients can be unstable in mixed logit models. The logitr package is designed to estimate MNL and mixed logit models in both preference space and WTP space.

```
# Chapter 11 R lab 2: WTP-space estimation with logitr
# install.packages("logitr")

library(logitr)

# logitr expects one row per alternative. We reuse the simulated long data.
dat &lt;- long %&gt;%
  mutate(obsID = chid,
         altID = alt,
         choice = as.logical(choice))

# Preference-space MNL
m_pref &lt;- logitr(
  data = dat,
  outcome = "choice",
  obsID = "obsID",
  pars = c("cost", "biodiversity", "water_quality", "access_restriction")
)

summary(m_pref)

# WTP-space MNL: cost is the scale/price variable
m_wtp &lt;- logitr(
  data = dat,
  outcome = "choice",
  obsID = "obsID",
  pars = c("biodiversity", "water_quality", "access_restriction"),
  scalePar = "cost",
  numMultiStarts = 5
)

summary(m_wtp)
```

In applied work, compare preference-space and WTP-space estimates, confidence intervals, convergence behaviour, and sensitivity to starting values. WTP-space can be helpful, but it is not a magic solution. The monetary attribute still needs to be credible, sufficiently variable, and attended to by respondents.

11.10 Worked R lab 3: ML as a diagnostic for choice data

The next example uses a random forest to predict chosen alternatives from the DCE design. This does not produce welfare estimates. Its purpose is diagnostic: identify nonlinearities, interactions, and variables that may deserve attention in the choice model.

```
# Chapter 11 R lab 3: ML diagnostic benchmark for a DCE
# install.packages(c("tidymodels", "ranger"))

library(tidymodels)
library(ranger)

wide &lt;- long %&gt;%
  select(id, chid, task, alt, cost, biodiversity, water_quality,
         access_restriction, chosen_alt) %&gt;%
  pivot_wider(names_from = alt,
              values_from = c(cost, biodiversity, water_quality, access_restriction)) %&gt;%
  distinct(id, chid, .keep_all = TRUE) %&gt;%
  mutate(chosen_alt = factor(chosen_alt))

# Split by respondent, not by task, to avoid leakage.
set.seed(1102)
split &lt;- group_initial_split(wide, group = id, prop = 0.8)
train &lt;- training(split)
test  &lt;- testing(split)
folds &lt;- group_vfold_cv(train, group = id, v = 5)

rf_spec &lt;- rand_forest(trees = 500, mtry = tune(), min_n = tune()) %&gt;%
  set_engine("ranger", importance = "permutation") %&gt;%
  set_mode("classification")

wf &lt;- workflow() %&gt;%
  add_formula(chosen_alt ~ . - id - chid - task) %&gt;%
  add_model(rf_spec)

grid &lt;- grid_regular(mtry(range = c(2, 8)), min_n(range = c(2, 25)), levels = 4)

res &lt;- tune_grid(wf, resamples = folds, grid = grid,
                 metrics = metric_set(accuracy, mn_log_loss))

best &lt;- select_best(res, metric = "mn_log_loss")
fit &lt;- finalize_workflow(wf, best) %&gt;% fit(train)

pred &lt;- predict(fit, test) %&gt;% bind_cols(test %&gt;% select(chosen_alt))
metrics(pred, truth = chosen_alt, estimate = .pred_class)

# Extract fitted ranger object and inspect variable importance
rf_engine &lt;- extract_fit_engine(fit)
sort(rf_engine$variable.importance, decreasing = TRUE)
```

Use this benchmark carefully. If the random forest predicts much better than the MNL, ask why. Perhaps utility is nonlinear, interactions matter, the status quo is reference-dependent, or respondents use simplifying heuristics. Then translate the insight into a theory-consistent choice model: interactions, splines, latent classes, random parameters, or reference-dependent terms. Do not report random-forest variable importance as WTP.

11.11 Advanced environmental valuation topics

11.11.1 Preference heterogeneity

Environmental preferences are heterogeneous. Urban and rural respondents may value access restrictions differently. Frequent visitors may value water quality more than non-users. High-income households may be less cost-sensitive. Mixed logit allows continuous heterogeneity by treating parameters as random. Latent class models allow discrete segments. Hybrid choice models incorporate latent attitudes and perceptions.

ML can help find candidate moderators, but the final welfare story needs coherent heterogeneity. For example, an XGBoost model may suggest strong interactions between income, visit frequency, and environmental attitudes. A choice modeller can then estimate a mixed logit with income interactions or a latent class model whose class membership depends on these variables.

11.11.2 Scale heterogeneity

Respondents differ not only in preferences but also in choice consistency. Some answer carefully with low error variance; others answer noisily. In logit models, coefficient scale is inversely related to error variance, so differences in scale can be confused with differences in taste. Generalized MNL, scale-adjusted latent class models, and repeated-task diagnostics can help.

11.11.3 Attribute processing and decision rules

Respondents may not maximize a fully compensatory utility function. They may use lexicographic rules, thresholds, elimination-by-aspects, majority-of-confirming-dimensions, or status quo heuristics. Environmental choices involving moral or rights-based issues may be especially prone to non-compensatory behaviour. This does not invalidate DCEs, but it requires careful testing and transparent interpretation.

11.11.4 Combining revealed and stated preference

Revealed preference data, such as recreation trips or actual purchases, can be combined with stated preference data to improve external validity and estimate scenarios not observed in the market. Classical joint RP-SP models account for scale differences across data sources. Newer ML and neural multitask approaches can share representation across RP and SP tasks, but the welfare interpretation of joint models still depends on utility structure and monetary variation.

11.11.5 Distribution and equity

Mean WTP is not enough. Environmental policies often have distributional consequences. A coastal access fee may generate high aggregate benefits but exclude low-income users. A biodiversity programme financed through a flat tax may be regressive. Report WTP distributions, subgroup welfare, affordability, and incidence. Consider whether willingness to pay should be complemented with deliberative, rights-based, or multi-criteria analysis.

11.12 Reporting standards for an environmental DCE

A DCE report should be reproducible and policy-readable. It should allow another researcher to understand the scenario, design, sample, data exclusions, models, and welfare calculations.

Report section	Minimum contents
Policy context	Decision problem, environmental good, baseline, affected population, spatial and temporal scale
Scenario and payment vehicle	Institution, payment frequency, payment duration, realism, consequentiality wording
Attribute development	Source of attributes, qualitative work, expert input, pretest evidence
Experimental design	Attributes/levels, design method, priors, blocking, number of tasks, status quo handling
Survey and sample	Sampling frame, recruitment, response rate, quotas/weights, timing, survey mode
Data cleaning	Exclusions, speeders, protests, failed attention checks, missing data
Model specifications	MNL baseline, heterogeneity models, interactions, scale, ANA, ML diagnostics where used
Welfare estimates	WTP, confidence intervals, Krinsky-Robb or delta method, scenario compensating surplus
Validity evidence	Scope sensitivity, signs, monotonicity, predicted shares, external comparison, robustness
Code and data	R scripts, seed, packages, design files, anonymized data where possible

11.13 Software map

Task	R tools	Python/other tools	Notes
Design construction	idefix, support.CEs, AlgDesign, apollo examples	Ngene, JMP, custom Python	Ngene is common in practice; R tools are useful for reproducible open workflows
MNL estimation	mlogit, logitr, apollo	Biogeme, statsmodels custom likelihood	Start with MNL as baseline
Mixed logit	gmnl, logitr, apollo, mixl	Biogeme, PyLogit alternatives, custom PyTorch/JAX	Use simulation draws and check convergence
Latent class	apollo, gmnl, flexmix for related tasks	Biogeme, custom likelihood	Useful for segments and ANA

Task	R tools	Python/other tools	Notes
Hybrid choice	apollo	Biogeme, custom Bayesian tools	Requires measurement model and identification care
ML benchmarks	tidymodels, mlr3, ranger, xgboost, glmnet	scikit-learn, xgboost, lightgbm	Prediction and diagnostics, not automatic WTP
Deep learning	torch, keras	PyTorch, TensorFlow, JAX	Use when data are large or inputs include images/text
Text coding	tidytext, quanteda, text, embeddings via APIs	transformers, sentence-transformers	Useful for protests and open-ended comments
Reporting	Quarto, modelsummary, gt, marginaleffects	Jupyter, quarto	Keep workflow reproducible

11.14 Mini-project: value a coastal restoration programme

Use this mini-project to consolidate the chapter. Suppose a coastal region is considering a restoration programme that improves seabed naturalness, water clarity, biodiversity, and crowding management. Households would pay an annual coastal protection fee for five years.

Project tasks

- Define the policy baseline and two improvement scenarios. Be precise about location, duration, institution, uncertainty, and payment vehicle.

- Choose four non-monetary attributes and one monetary attribute. Justify each attribute using policy relevance and respondent comprehension.

- Propose three or four levels for each attribute. Explain why the levels are plausible and not obviously dominated.

- Sketch three example choice cards, including a status quo alternative.

- Write a short pretest protocol: what would you ask respondents to check comprehension, realism, and burden?

- Simulate data under plausible coefficients and estimate an MNL model in R.

- Compute marginal WTP for each attribute and compensating surplus for one restoration scenario.

- Estimate a random forest prediction benchmark and identify one interaction that might be added to a choice model.

- Write a one-page policy memo explaining what the WTP estimates mean and what they do not mean.

11.15 Concept checkpoint

- Why is a monetary attribute essential for WTP estimation in a DCE?

- What is the difference between direct use value and existence value?

- Why might a random forest be useful in a DCE study even though it cannot directly produce welfare estimates?

- How can attribute non-attendance bias WTP estimates?

- Why can a statistically efficient design still be a poor environmental valuation design?

- What is the difference between preference-space and WTP-space estimation?

- Why should row-wise splitting be avoided when benchmarking ML prediction on repeated DCE tasks from the same respondent?

- How would you test whether respondents are sensitive to the scope of an environmental good?

11.16 Common mistakes

Mistake 1: Treating the DCE as just a classification problem. Classification accuracy is useful, but the goal of environmental valuation is often welfare analysis. A classifier does not automatically provide marginal utility of income, WTP, or compensating surplus.

Mistake 2: Using unrealistic payment vehicles. If respondents do not believe the payment mechanism, the monetary trade-off is weak. This directly threatens WTP estimates.

Mistake 3: Too many attributes. More attributes can make the policy richer but also increase cognitive burden, attribute non-attendance, and fatigue.

Mistake 4: Reporting WTP without uncertainty. WTP is a ratio or transformed parameter and can be imprecise. Use delta method, simulation, bootstrap, or Bayesian credible intervals.

Mistake 5: Ignoring the status quo. The status quo is often central in environmental policy. Its description, realism, and coding affect substitution and welfare estimates.

Mistake 6: Letting ML choose the final model without theory. ML can reveal patterns, but environmental welfare analysis requires credible behavioural structure and policy interpretation.

11.17 Reading guide

For stated preference best practice, begin with Johnston et al. (2017), which gives contemporary guidance for stated preference studies used in decision making. For environmental DCE implementation in R, use Mariel et al. (2025), an open-access book that covers design, implementation, and analysis with R scripts. For random utility and mixed logit foundations, read Train (2009), Louviere, Hensher and Swait (2000), and the relevant chapters of the Handbook of Choice Modelling, second edition. For ML-choice integration, read van Cranenburgh et al. (2022), Hillel et al. (2021), Ortelli et al. (2021), and Hernandez et al. (2023). For environmental and agricultural ANA with regularization, read Li and Yu (2025).

Chapter takeaway

Environmental DCEs estimate policy-relevant trade-offs for goods that markets often fail to price. ML can improve survey design, data quality, heterogeneity discovery, text/image processing, and specification search. But the welfare outputs that make environmental valuation valuable - WTP, compensating surplus, distributional analysis, and policy counterfactuals - still depend on random utility structure, credible payment vehicles, valid scenarios, and transparent behavioural interpretation.

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Chapter 12

Random Utility Theory and the Multinomial Logit Model

Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Position of this chapter.
Chapters 8 to 11 moved from data and prediction to environmental valuation and discrete choice experiments. This chapter begins the core choice-modelling sequence. It formalizes choice as utility maximization, derives the multinomial logit model, and shows why this apparently simple model remains the baseline against which both advanced econometric choice models and machine-learning classifiers should be judged.

Main idea.
A multinomial logit model is not merely a classifier. It is a probabilistic behavioural model based on alternatives, choice sets, utility differences, and identifying restrictions. Its coefficients can support marginal effects, elasticities, willingness to pay, and welfare calculations, provided that the specification and data-generating assumptions are credible.

Learning objectives

- explain the random utility model and distinguish systematic and unobserved utility;

- derive the multinomial logit probability from utility maximization and Extreme Value Type I errors;

- understand identification, normalization, scale, alternative-specific constants, and the role of the choice set;

- estimate a basic MNL model in R using long-format choice data;

- compute marginal effects, elasticities, willingness to pay, and simple welfare measures;

- explain the independence of irrelevant alternatives property and when it is problematic;

- connect MNL to modern machine learning through softmax classification and cross-entropy loss;

- recognize what MNL can and cannot do in environmental economics and choice modelling.

12.1 Why start with random utility?

Many machine-learning textbooks begin with prediction functions. Choice modelling begins with a decision problem. A person, household, firm, or policy maker faces a set of mutually exclusive alternatives and chooses one. In environmental economics this may be a beach destination, a conservation programme, an electricity tariff, a climate-adaptation policy, or a transport mode. The modeller does not only want to classify the observed choice. The modeller wants to learn how attributes enter utility, how people trade money against environmental improvements, how substitution works, and what a counterfactual policy would imply for welfare.

The random utility model, usually abbreviated RUM, provides the common language for this task. It says that choices are systematic enough to be modelled, but not perfectly observable. The researcher observes some attributes and characteristics, but not all motives, perceptions, constraints, measurement errors, or momentary states of mind. Random utility theory therefore combines economic structure with probabilistic modelling.

This chapter focuses on the multinomial logit model because it is the benchmark model. It is simple enough to understand fully, but rich enough to reveal almost all core issues that later chapters will revisit: identification, scale, substitution, heterogeneity, model fit, prediction, elasticities, welfare, and the boundary between behavioural modelling and machine learning.

12.2 Choice data: alternatives, attributes, decision makers, and tasks

A discrete choice observation is not just a row with a class label. It is a structured comparison. The basic indices are: n for decision maker, t for choice occasion or task, i for alternative, and C_nt for the choice set faced by decision maker n in task t.

```
n = 1,...,N decision makers t = 1,...,T_n choice tasks or occasions i in C_nt
alternatives available in task t
```

Choice datasets are usually stored in long format for econometric estimation. Each row is an alternative within a choice task. A binary indicator records whether that alternative was chosen. Alternative-specific attributes vary across alternatives, while individual characteristics are repeated across alternatives or interacted with alternative indicators.

Element	Example in environmental economics	Why it matters
Decision maker n	Respondent in a stated preference survey	Preferences may vary across people
Choice task t	One DCE card comparing policy A, policy B, and status quo	Repeated choices create panel structure
Alternative i	Beach site, policy package, transport mode, energy contract	Alternatives have attributes and are compared within sets
Attributes x_nit	Cost, travel time, water quality, crowding, biodiversity	Attributes enter systematic utility
Choice y_nt	The selected alternative	Outcome used to form likelihood

12.3 The random utility model

For each decision maker n, task t, and alternative i, total utility is written as the sum of an observed systematic component and an unobserved component:

```
U_nit = V_nit + eps_nit
```

The systematic utility V_nit is a function of observed attributes and parameters. The unobserved component eps_nit collects everything that affects utility but is not included in V_nit. The model does not assume that people make mistakes in a psychological sense. It assumes that the analyst sees only part of the utility-relevant information.

```
V_nit = beta_cost * cost_nit + beta_quality * quality_nit + beta_crowding * crowding_nit
+ ...
```

The behavioural rule is deterministic in total utility: the chosen alternative is the one with the highest utility. The probability enters because the analyst does not observe eps_nit.

```
Choose i if U_nit >= U_njt for every alternative j in C_nt
```

Teaching checkpoint.
The random part is in the analyst's model, not necessarily in the person's mind. This distinction is important when comparing RUMs with machine-learning classifiers. A classifier may learn P(choice | features); a RUM also specifies how those probabilities arise from comparisons of latent utilities.

12.4 From random utility to multinomial logit

The multinomial logit model follows if the unobserved utility terms are independent and identically distributed Extreme Value Type I. Under this assumption, the probability that alternative i is chosen has the familiar closed form:

```
P_nit = exp(V_nit) / sum_{j in C_nt} exp(V_njt)
```

The denominator is the sum of exponentiated systematic utilities over all available alternatives in the choice set. This is why a choice model must know which alternatives were available, not only which alternative was chosen.

The logit probability has an important machine-learning connection. It is mathematically the same softmax transformation used in multinomial classification. If the systematic utilities are linear scores, maximum likelihood estimation of MNL is equivalent to minimizing cross-entropy loss for a linear softmax classifier. The difference is not the algebra of softmax; the difference is the behavioural interpretation, the data structure, the restrictions imposed on utilities, and the post-estimation quantities of interest.

12.5 Identification, normalization, and scale

Choice models are identified from utility differences, not utility levels. If the same constant is added to every alternative in a choice set, the chosen alternative does not change. Therefore at least one alternative-specific constant must be normalized. Usually the status quo or one reference alternative receives an alternative-specific constant of zero.

```
If U_nit is replaced by U_nit + a_nt for all i, choices are unchanged.
```

Scale is also central. In logit models, the variance of the unobserved utility term is not separately identified from the magnitude of the coefficients. More noise in utility produces flatter choice probabilities; larger coefficients produce sharper choice probabilities. In a single MNL model, scale is normalized. When combining datasets, such as revealed preference and stated preference data, scale differences can matter and must be modelled explicitly.

Identification also requires variation. A cost coefficient cannot be estimated if cost does not vary across alternatives or tasks. An individual-specific variable such as income cannot enter a generic utility function by itself in a standard MNL with all alternatives available, because it is constant across alternatives within a choice task. It must enter through alternative-specific effects or interactions with attributes.

Common identification rule.
Alternative-specific attributes can enter utility directly. Individual-specific characteristics must usually be interacted with alternatives or with alternative-varying attributes. For example, income can interact with the status-quo constant, with the cost coefficient, or with a high-quality environmental attribute.

12.6 Likelihood estimation

Let y_nit equal 1 if decision maker n chooses alternative i in task t and 0 otherwise. The likelihood contribution of one choice task is the probability assigned to the chosen alternative. For a sample of independent choice tasks, the log-likelihood is:

```
LL(beta) = sum_n sum_t sum_{i in C_nt} y_nit * log(P_nit(beta))
```

The maximum likelihood estimator chooses the parameters that maximize this log-likelihood. In practice, panel data with repeated choices by respondents require careful treatment of standard errors and validation. For simple MNL with independent tasks, the likelihood is straightforward. For mixed logit and latent class models, later chapters will add integration, simulation, or class membership probabilities.

A machine-learning reader should notice that negative log-likelihood is cross-entropy loss. A choice modeller should notice that maximizing this likelihood also estimates interpretable marginal utilities under the assumptions of the RUM. Both views are useful.

12.7 Interpreting coefficients

The sign of a coefficient indicates how the attribute shifts utility, all else equal. A negative cost coefficient means that higher cost reduces utility. A positive biodiversity coefficient means that higher biodiversity improvement increases utility. Coefficient magnitudes are in utility units, not directly in euros or probabilities. Their ratios and implied changes in probabilities are often more interpretable.

```
V_nit = beta_cost * cost_nit + beta_bio * biodiversity_nit + beta_access * access_nit
```

For continuous attributes, the marginal utility is beta_k. For categorical attributes, dummy-coded coefficients represent utility differences relative to a reference category. For interactions, interpretation is conditional. For example, beta_bio_income indicates whether higher-income respondents value biodiversity improvements differently.

Alternative-specific constants capture average utility for alternatives after controlling for observed attributes. In a DCE, a positive alternative-specific constant for a policy alternative relative to the status quo can reflect general support for change, left-out policy attributes, presentation effects, or other systematic preference components. It should not be over-interpreted without design knowledge.

12.8 Marginal effects and elasticities

MNL coefficients are not probability effects. The same coefficient can imply different probability changes depending on the baseline probabilities and the attributes of all alternatives in the choice set. For a generic continuous attribute x_k that enters only alternative i, the own marginal effect is:

```
dP_i / dx_ik = P_i * (1 - P_i) * beta_k
```

The cross marginal effect on another alternative j is:

```
dP_j / dx_ik = - P_j * P_i * beta_k
```

Elasticities express percentage changes. For example, the own direct elasticity of probability P_i with respect to attribute x_ik is:

```
E_i,x = (dP_i / dx_ik) * (x_ik / P_i) = (1 - P_i) * beta_k * x_ik
```

Elasticities are especially useful in transport, energy, and recreation demand applications. They allow researchers to summarize how sensitive predicted choices are to changes in price, time, distance, or quality. But they are local quantities: they depend on the baseline scenario.

12.9 Willingness to pay and welfare

In environmental economics, one major reason to estimate choice models is to translate non-market environmental improvements into money-metric values. If cost enters utility linearly and beta_cost is negative, the marginal willingness to pay for attribute k is:

```
WTP_k = - beta_k / beta_cost
```

For example, if beta_biodiversity = 0.50 and beta_cost = -0.05, then the marginal WTP for a one-unit biodiversity improvement is 10 monetary units. This ratio interpretation is simple but fragile: it depends on the model specification, attribute coding, treatment of cost, scale, and heterogeneity.

For a policy change involving multiple attributes, the compensating surplus under MNL can be computed using the logsum formula:

```
CS_n = (1 / -beta_cost) * [ ln sum_j exp(V_nj^1) - ln sum_j exp(V_nj^0) ]
```

Here V^0 is the baseline scenario and V^1 is the policy scenario. The logsum measures the expected maximum utility from the available choice set. This is one of the key reasons that RUM-based choice models remain central in policy appraisal. A black-box classifier may predict adoption, but it does not automatically yield a theoretically valid logsum welfare measure.

12.10 The independence of irrelevant alternatives property

The MNL model implies independence of irrelevant alternatives, or IIA. The odds of choosing alternative i over alternative j depend only on the systematic utilities of i and j, not on the attributes or existence of a third alternative k.

```
P_i / P_j = exp(V_i) / exp(V_j) = exp(V_i - V_j)
```

IIA is a direct consequence of independent and identically distributed logit errors. Sometimes it is reasonable. If two environmental programmes are clearly distinct and no unobserved factor links them, IIA may be acceptable. Sometimes it is implausible. If two alternatives are close substitutes, such as two similar beaches, two electric tariffs from the same supplier, or two cycling routes with overlapping geography, MNL may impose unrealistic substitution patterns.

The classic red-bus/blue-bus intuition applies: adding a nearly identical alternative should mostly draw demand from its close substitute, not proportionally from all alternatives. Nested logit, cross-nested logit, mixed logit, error-components models, and latent class models relax this limitation in different ways.

12.11 Environmental applications

The MNL model can be used in many environmental economics settings as a first model, diagnostic baseline, or teaching model. In a beach choice study, alternatives may differ by travel cost, travel time, crowding, water quality, seabed type, and access restrictions. In a conservation DCE, alternatives may differ by species protected, habitat quality, management intensity, and household contribution. In an energy tariff DCE, alternatives may differ by price, renewable share, contract flexibility, peak pricing, and provider type.

For PhD research, MNL should almost never be the final word, but it should almost always be estimated. It reveals the basic signs, scale, choice-set coding, status-quo effect, and predictive baseline. It also helps detect mistakes before moving to more flexible models. If MNL produces a positive cost coefficient, implausible WTP, or unstable alternative-specific constants, a more complex model may hide rather than solve the problem.

> Practical advice.
> Before estimating mixed logit, latent class models, hybrid choice models, or neural choice models, estimate a clean MNL model. Check signs, standard errors, likelihood, predicted shares, WTP magnitudes, and sensitivity to coding decisions. The MNL model is a diagnostic instrument as much as a substantive model.

12.12 MNL and machine learning: softmax is not enough

The connection between MNL and machine learning is both deep and easy to misunderstand. The logit probability is a softmax. The MNL likelihood is cross-entropy. A neural network with a softmax output can predict choices. But replacing V_nit with a flexible neural network changes the status of the model. The scores may no longer be interpretable utilities unless the architecture, inputs, and constraints are designed to preserve behavioural meaning.

This distinction explains why the uploaded choice-modelling literature does not simply ask whether ML beats logit in accuracy. It asks how ML can support specification search, representation learning, context-dependent heterogeneity, latent variables, and prediction while retaining the behavioural outputs needed for policy analysis. The second edition of the Handbook of Choice Modelling explicitly includes a chapter on model building in the age of machine learning, and recent discussion papers argue that ML and choice modelling should cross-fertilize rather than replace one another.

For empirical work, compare models using more than one criterion: out-of-sample log loss, prediction accuracy, calibration, sign plausibility, marginal effects, WTP, elasticities, and counterfactual stability. A model that wins on accuracy but implies positive utility of cost or unreasonable welfare changes is not suitable for many environmental policy tasks.

12.13 Worked example in R: estimating an MNL model

The following lab simulates a small environmental DCE. Respondents choose among two policy alternatives and a status quo. The goal is to understand data structure, estimation, WTP, and prediction. In your own research, replace the simulated data with your survey data, but keep the same logic: one row per alternative per choice task.

```
# Packages
# install.packages(c("tidyverse", "mlogit"))
library(tidyverse)
library(mlogit)

set.seed(2026)

N  <- 500   # respondents
Tn <- 6     # choice tasks per respondent
alts <- c("policy_A", "policy_B", "status_quo")

# True parameters used only for simulation
b_cost  <- -0.055
b_bio   <-  0.55
b_water <-  0.35
asc     <- c(policy_A = 0.15, policy_B = 0.05, status_quo = 0.00)

long <- expand_grid(
  id = 1:N,
  task = 1:Tn,
  alt = alts
) |>
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = if_else(alt == "status_quo", 0,
                   sample(c(10, 25, 50, 75), n(), replace = TRUE)),
    biodiversity = if_else(alt == "status_quo", 0,
                           sample(0:3, n(), replace = TRUE)),
    water_quality = if_else(alt == "status_quo", 0,
                            sample(0:3, n(), replace = TRUE)),
    V = asc[alt] + b_cost * cost + b_bio * biodiversity +
        b_water * water_quality
  ) |>
  group_by(chid) |>
  mutate(
    prob = exp(V) / sum(exp(V)),
    chosen_alt = sample(alt, size = 1, prob = prob),
    choice = as.integer(alt == chosen_alt)
  ) |>
  ungroup()

# Convert to mlogit format
mdata <- mlogit.data(
  long,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

# Estimate MNL with alternative-specific constants
mnl <- mlogit(
  choice ~ cost + biodiversity + water_quality,
  data = mdata
)

summary(mnl)
```

Interpreting the output

The estimated cost coefficient should be negative, while biodiversity and water quality should be positive. Alternative-specific constants are reported relative to a base alternative. Their interpretation depends on the reference alternative and the design.

```
# Willingness to pay estimates
```

```
b <- coef(mnl)

wtp_bio   <- - b["biodiversity"] / b["cost"]
wtp_water <- - b["water_quality"] / b["cost"]

wtp_bio
wtp_water
```

The result gives the marginal willingness to pay for a one-unit improvement in biodiversity or water quality, in the same monetary units as the cost variable. If cost is annual household cost in euros, WTP is euros per household per year.

12.14 Predicting choice probabilities and policy shares

Choice models are also predictive models. After estimating coefficients, one can construct policy scenarios and compute predicted probabilities. The important point is that predictions are scenario-based: probabilities change when the attributes of any alternative in the choice set change.

```
# A simple policy scenario for one hypothetical respondent/task
scenario <- tibble(
  alt = c("policy_A", "policy_B", "status_quo"),
  cost = c(25, 50, 0),
  biodiversity = c(2, 3, 0),
  water_quality = c(2, 3, 0)
)

# Use estimated coefficients. Here names of ASCs may differ depending on base alt.
# For teaching, we focus on attribute utility.
scenario <- scenario |>
  mutate(
    V_attr = b["cost"] * cost +
             b["biodiversity"] * biodiversity +
             b["water_quality"] * water_quality,
    pred_prob = exp(V_attr) / sum(exp(V_attr))
  )

scenario
```

This simplified example ignores alternative-specific constants to keep the code transparent. In applied work, include all estimated terms, including constants, interactions, nonlinear transformations, and individual-specific effects.

12.15 Using logitr and Apollo

The R package mlogit is excellent for teaching and standard RUM workflows. For environmental valuation, logitr is convenient because it directly supports preference-space and willingness-to-pay-space logit models. Apollo is broader and more flexible, especially for advanced models such as mixed logit, latent class, hybrid choice, and custom likelihoods.

```
# logitr sketch (not run here)
# install.packages("logitr")
# library(logitr)
# fit <- logitr(
#   data = long,
#   outcome = "choice",
#   obsID = "chid",
#   pars = c("cost", "biodiversity", "water_quality")
# )
# summary(fit)

# Apollo is more verbose but much more flexible. A typical Apollo script has:
# 1. apollo_initialise()
# 2. apollo_control list
# 3. database and apollo_beta
# 4. apollo_probabilities() function
# 5. apollo_validateInputs()
# 6. apollo_estimate()
```

12.16 Python option: Biogeme

Python users often use Biogeme for discrete choice models. Biogeme is designed for maximum likelihood estimation of parametric models in general and discrete choice models in particular. A typical workflow defines variables, parameters, utilities, availability conditions, a logit or more advanced probability expression, and a likelihood. Python is also useful when the workflow combines choice models with PyTorch, TensorFlow, JAX, scikit-learn, or geospatial/image pipelines.

```
# Biogeme sketch only
# import biogeme.database as db
# import biogeme.biogeme as bio
# from biogeme.expressions import Beta
# from biogeme.models import logit
#
# beta_cost = Beta('beta_cost', 0, None, None, 0)
# beta_bio  = Beta('beta_bio', 0, None, None, 0)
#
# V1 = beta_cost * cost_A + beta_bio * biodiversity_A
# V2 = beta_cost * cost_B + beta_bio * biodiversity_B
# V3 = 0
# V = {1: V1, 2: V2, 3: V3}
# av = {1: 1, 2: 1, 3: 1}
# prob = logit(V, av, choice)
```

12.17 Concept checks

- Why is a row-wise class-label formulation insufficient for most choice-modelling tasks?

- What changes in predicted probabilities if a new alternative is added to the choice set?

- Why are utility levels not identified, while utility differences are?

- Why can income not usually enter a generic MNL utility directly unless interacted?

- What does IIA imply for substitution between similar alternatives?

- Why is WTP a ratio of coefficients rather than a coefficient by itself?

- In what sense is MNL a softmax classifier, and in what sense is it more than that?

- Why should a PhD researcher estimate MNL before moving to mixed logit or neural choice models?

12.18 Applied exercises

Exercise 1: data structure

Take any DCE dataset and draw its structure: respondents, tasks, alternatives, attributes, and chosen alternative. Convert it into long format and verify that each choice task has exactly one chosen alternative.

Exercise 2: baseline MNL

Estimate a basic MNL model with cost and all main environmental attributes. Report signs, standard errors, log-likelihood, AIC/BIC, predicted shares, and WTP. Explain whether the results are plausible.

Exercise 3: coding sensitivity

Re-estimate the model with alternative coding choices: dummy coding versus effects coding; linear cost versus log cost; generic versus alternative-specific coefficients. Compare WTP and predicted policy shares.

Exercise 4: IIA diagnosis

Identify two alternatives in your data that are close substitutes. Explain why IIA may or may not be problematic. If possible, compare MNL with nested logit or mixed logit in a later chapter.

Exercise 5: ML comparison

Train a random forest or gradient boosting classifier on a wide version of the same data. Compare predictive performance with MNL. Then ask: which model gives interpretable WTP and welfare measures? Which model is more useful for short-term prediction?

12.19 Common mistakes

- Ignoring the choice set. A choice probability is conditional on available alternatives. Missing availability information can invalidate predictions.

- Interpreting coefficients as probability effects. Coefficients are utility-scale effects; probability effects depend on baseline probabilities and the full choice set.

- Forgetting normalization. One alternative-specific constant must be normalized, and utility levels are arbitrary.

- Treating cost as just another predictor. In environmental economics, the cost coefficient is central because it anchors WTP and welfare.

- Using MNL despite implausible substitution patterns. If alternatives are close substitutes or share unobserved factors, nested or mixed models may be needed.

- Moving to complex models too soon. A mixed logit, latent class model, or neural model cannot compensate for badly coded data or an implausible basic specification.

- Confusing softmax prediction with welfare analysis. A softmax output does not guarantee economically meaningful utility, WTP, or counterfactual validity.

12.20 Reading guide

For foundations, read McFadden on conditional logit and qualitative choice behaviour, then Train for the modern simulation-based treatment of logit, GEV, probit, and mixed logit. The second edition of the Handbook of Choice Modelling is useful for a broad map of the field and includes foundations, stated choice design, heterogeneity, hybrid choice models, estimation, appraisal, and forecasting. For software, consult the mlogit, Apollo, logitr, and Biogeme documentation. For the ML interface, read the choice-modelling discussion paper by van Cranenburgh and co-authors and the Handbook chapter on model building in the age of machine learning.

Chapter summary

The multinomial logit model is the essential baseline for choice modelling. It arises from a random utility model with independent Extreme Value Type I errors, producing softmax choice probabilities. It is simple, interpretable, and computationally convenient. It also has limitations: especially IIA, homogeneous taste parameters, linear-in-parameters utility unless extended, and strong assumptions about unobserved utility. For environmental economics, MNL is valuable because it connects choices to trade-offs, WTP, welfare, elasticities, and policy scenarios. For machine learning, MNL provides the conceptual bridge between softmax prediction and behavioural modelling. The next chapter extends this baseline to models that relax IIA and represent richer substitution patterns and heterogeneity.

References and further reading

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Official documentation: CRAN mlogit package; CRAN/Apollo choice modelling package; logitr package documentation; Biogeme Python documentation.

Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 13

Beyond MNL: Nested, Cross-Nested, Mixed Logit, Probit, and GEV Models

The multinomial logit model is the natural starting point for discrete choice analysis, but its simplicity comes with restrictions. This chapter explains how advanced random utility models relax these restrictions while preserving behavioural interpretation, welfare analysis, and counterfactual policy use. The chapter also prepares the bridge toward machine-learning-assisted specification and neural choice models in later chapters.

How to study this chapter
First read Sections 13.1-13.3 conceptually. Then run the R code in Sections 13.8-13.10. Finally, return to the model comparison checklist in Section 13.12 and apply it to a dataset from your own field.

Learning objectives

- Explain why the independence of irrelevant alternatives assumption is often too restrictive.
- Distinguish similarity-driven substitution from taste heterogeneity and serial correlation.
- Write the intuition behind nested logit, cross-nested logit, mixed logit, and multinomial probit.
- Choose an appropriate model for environmental DCEs, recreation demand, transport choices, and adoption studies.
- Estimate basic nested and mixed logit models in R, and understand when specialised software such as Apollo, Biogeme, or Python-based Bayesian tools is useful.
- Evaluate advanced choice models using behavioural plausibility, predictive performance, welfare outputs, and computational feasibility.

13.1 Why go beyond the multinomial logit model?

The multinomial logit (MNL) model is attractive because it has a closed-form probability, a globally concave log-likelihood under standard linear-in-parameters utility, and easily interpretable coefficients. In many stated preference and discrete choice experiment applications, MNL is the first model to estimate and often the first model to report. It is also an important benchmark for machine learning because it gives a transparent lower-complexity baseline.

However, MNL relies on strong assumptions about unobserved utility. With independent and identically distributed type-I extreme value errors, the ratio of probabilities for two alternatives is independent of the existence or attributes of other alternatives. This is the independence of irrelevant alternatives (IIA) property. In practice, IIA can be implausible whenever some alternatives are closer substitutes than others.

```
P_ni / P_nj = exp(V_ni) / exp(V_nj) for any two alternatives i and j
```

In environmental economics and choice modelling, IIA can be problematic in at least four recurrent cases. First, alternatives may be grouped by common unobserved features: two beach sites may be both sandy, two forest recreation sites may be close to the same city, or two transport modes may share unobserved comfort attributes. Second, repeated choices by the same respondent create panel dependence. Third, tastes vary across people: the same monetary cost, travel time, biodiversity improvement, or access restriction may be interpreted differently by different groups. Fourth, some attributes may be processed non-compensatorily or ignored by some respondents.

Advanced random utility models are responses to these issues. They do not all solve the same problem. Nested and GEV models primarily handle correlation and substitution structure across alternatives. Mixed logit handles random taste heterogeneity and repeated choice correlation. Probit handles flexible normal error correlation. Latent class models, covered in the next chapter, handle discrete heterogeneity. A key research skill is diagnosing which problem you actually need to solve.

13.2 The taxonomy: what each model fixes

Model family	Main restriction relaxed	Best suited for	Typical cost
Nested logit	IIA across all alternatives; allows stronger substitution within nests	Mode choice, recreation site choice, grouped policy alternatives	Requires defensible nesting structure
Cross-nested logit	Alternatives can belong partly to several nests	Choices with overlapping similarity, e.g. multimodal transport or multi-attribute environmental programmes	More complex specification and identification

Model family	Main restriction relaxed	Best suited for	Typical cost
Mixed logit	Fixed tastes; iid errors conditional on observed variables	Repeated DCEs, WTP heterogeneity, random coefficients, panel data	Simulation, distributional assumptions, local optima
Multinomial probit	Extreme-value errors and simple correlation patterns	Flexible covariance among alternatives, small to medium choice sets	High-dimensional integration
Generalized extreme value	Specific substitution structures under EV errors	Nested, cross-nested, paired combinatorial and other structured models	Specialised derivations and software
Latent class	Single population-level preference vector	Segmented markets, behavioural classes, attribute non-attendance	Number and interpretation of classes

Table 13.1. Advanced choice models are not interchangeable. They are answers to different behavioural and statistical problems.

13.3 Nested logit: modelling similarity among alternatives

Nested logit is the most common extension of MNL for relaxing the global IIA property. The idea is simple: alternatives are partitioned into nests. Unobserved utility may be correlated within a nest, so alternatives in the same nest are closer substitutes than alternatives in different nests. For example, in a recreation site choice model, nearby beaches may form a 'coastal' nest and inland natural parks may form another nest. In transport, car and ride-hailing may share unobserved private-mobility features, while bus and rail may share public-transport features.

```
U_ni = V_ni + ε_ni, with corr(ε_ni, ε_nj) > 0 if i and j are in the same nest
```

The model can be understood as a two-level choice process. Conditional on choosing a nest, the individual chooses an alternative inside the nest. At the upper level, the probability of choosing the nest depends on the expected utility available inside it, captured by the inclusive value or logsum.

```
IV_nm = log( sum_{j in m} exp(V_nj / λ_m) )
P_ni = P_n(m) * P_n(i | m)
```

The nesting parameter λ_m controls the degree of independence within nest m. In the common normalisation, values closer to one imply less within-nest correlation and values below one imply stronger similarity among alternatives in the nest. Values outside the theoretically admissible range signal misspecification, weak identification, poor nesting design, or data that do not support the imposed structure.

> Behavioural interpretation
> Nested logit is not just a statistical correction. The nest should correspond to a credible behavioural similarity. A model that fits better but has an arbitrary nest structure can be hard to defend in environmental policy evaluation.

Example: recreation demand

Suppose tourists choose among four day-trip options: protected beach, open-access beach, coastal trail, and inland forest. A nested specification might place protected beach, open-access beach, and coastal trail in a coastal recreation nest, while inland forest forms an inland recreation nest. The policy implication is important: an access fee on the protected beach may shift demand mostly to other coastal alternatives, rather than proportionally to all alternatives.

13.4 Cross-nested logit: when similarity overlaps

The nested logit model requires each alternative to belong to one nest. That is often too rigid. Cross-nested logit allows an alternative to belong partly to multiple nests. This is useful when similarity is multidimensional. A bicycle-share option might belong partly to an 'active travel' nest and partly to a 'shared mobility' nest. A marine restoration project might belong partly to a 'biodiversity' nest and partly to a 'recreation improvement' nest. An electric car might belong partly to a 'private vehicle' nest and partly to a 'low-emission technology' nest.

```
Alternative i has allocation weights α_im across nests m, with α_im >= 0 and sum_m α_im =
1
```

Cross-nested models are powerful but demanding. The analyst must specify or estimate allocation parameters, ensure identification, and explain what the nests mean. In many PhD applications, cross-nested logit is not the first model to try. It becomes attractive when there is a strong conceptual reason to expect overlapping similarity and enough data to identify the extra structure.

13.5 Mixed logit: random taste heterogeneity and panel dependence

Mixed logit is the workhorse extension for environmental DCEs and many modern choice applications. The core idea is that coefficients vary across decision makers, and sometimes across choice occasions. Instead of assuming that all respondents have the same marginal utility of cost, travel time, biodiversity, risk, or crowding, mixed logit treats one or more coefficients as random.

```
U_nit = β_n' x_nit + ε_nit
β_n = b + η_n, η_n ~ f(η | θ)
```

The probability of respondent n's observed sequence of choices is obtained by integrating over the distribution of random coefficients. Because this integral generally has no closed form, it is approximated by simulation.

```
P_n(sequence) =  [ Π_t P_nit(β) ] f(β | θ) dβ
```

Mixed logit can approximate a wide range of random utility models. Its flexibility comes from the distribution of β. Common choices are normal, lognormal, triangular, uniform, and constrained distributions. In WTP-space models, the monetary trade-offs themselves are parameterised directly. The choice of distribution is not a technical detail: it shapes welfare estimates, tail behaviour, and the implied share of respondents with counterintuitive signs.

Fixed parameters, random parameters, and random scale

A random parameter captures taste heterogeneity. A random scale captures variation in the variance of the unobserved utility term. These are conceptually different but empirically hard to separate. In DCEs, respondents may differ because they truly value biodiversity differently, because some answer more noisily, or because survey tasks have different salience. Models that combine random coefficients, scale heterogeneity, and latent classes can become behaviourally rich but also fragile.

> Practical warning
> Do not randomise every coefficient simply because software allows it. Each random parameter needs a behavioural rationale, enough variation in the data, and diagnostic checks on signs, standard deviations, WTP distributions, and convergence.

Panel data

In stated choice surveys, each respondent often answers several choice tasks. Treating these tasks as independent observations understates uncertainty and ignores within-person consistency. Mixed logit naturally handles panel structure by drawing one vector of random coefficients per respondent and using it across that respondent's sequence of choices. This is one reason mixed logit is central in

environmental valuation.

13.6 Multinomial probit: flexible normal errors

The multinomial probit model assumes that the unobserved utility terms follow a multivariate normal distribution. This permits flexible covariance across alternatives and does not impose the extreme-value structure of logit models. In principle, this makes probit attractive for modelling rich substitution patterns.

```
ε_n = (ε_n1, ..., ε_nJ) ~ MVN(0, Σ)
```

The cost is computation. Choice probabilities require evaluating multivariate normal integrals. For small choice sets, this can be feasible. For large choice sets, repeated panel data, or many random coefficients, probit can become cumbersome. In practice, many applied researchers choose mixed logit because it offers comparable flexibility with simulation methods and software that are more widely used in transport, marketing, health, and environmental economics.

13.7 GEV models: the broader family

Nested and cross-nested logit are members of the generalized extreme value (GEV) family. The GEV framework constructs choice probabilities from a generating function that allows flexible substitution patterns while preserving closed-form probabilities for many structures. The main advantage is that some correlation patterns can be represented without high-dimensional simulation. The main limitation is that each model requires a valid generating function and a defensible structure.

For environmental economists, GEV models are most useful when alternatives naturally have a hierarchical or overlapping structure. Examples include recreation destinations grouped by region and ecosystem type, energy contracts grouped by supplier and renewable content, and transport alternatives grouped by technology and access mode. The key is not mathematical elegance alone; the nesting or generating structure should correspond to how respondents plausibly compare alternatives.

13.8 R lab I: estimating MNL and nested logit with mlogit

The following example simulates a choice dataset with four alternatives grouped into two nests. It then estimates an MNL benchmark and a nested logit model. In real research, replace the simulated data with your DCE or revealed preference dataset.

```
# install.packages(c("tidyverse", "mlogit"))
library(tidyverse)
library(mlogit)

set.seed(1301)
N  <- 600       # individuals
Tn <- 6         # tasks per individual
alts <- c("protected_beach", "open_beach", "coastal_trail", "inland_forest")

beta_cost <- -0.06
beta_quality <- 0.55
beta_crowd <- -0.40
asc <- c(protected_beach = 0.40, open_beach = 0.15,
         coastal_trail = 0.05, inland_forest = 0.00)

long <- expand_grid(id = 1:N, task = 1:Tn, alt = alts) |>
  mutate(
    chid = paste(id, task, sep = "_"),
    nest = if_else(alt == "inland_forest", "inland", "coastal"),
    cost = if_else(alt == "inland_forest", sample(c(5,10,15), n(), TRUE),
                   sample(c(10,20,40,60), n(), TRUE)),
    quality = sample(1:3, n(), TRUE),
    crowding = sample(1:3, n(), TRUE),
    V = asc[alt] + beta_cost * cost + beta_quality * quality + beta_crowd * crowding
  ) |>
  group_by(chid) |>
  mutate(p = exp(V) / sum(exp(V)),
         chosen = sample(alt, 1, prob = p),
         choice = as.integer(alt == chosen)) |>
  ungroup()

mdata <- mlogit.data(long, choice = "choice", shape = "long",
                     chid.var = "chid", alt.var = "alt", id.var = "id")

mnl <- mlogit(choice ~ cost + quality + crowding | 0, data = mdata)
summary(mnl)

nl <- mlogit(choice ~ cost + quality + crowding | 0,
             data = mdata,
             nests = list(coastal = c("protected_beach", "open_beach", "coastal_trail"),
                          inland = c("inland_forest")),
             un.nest.el = TRUE)
summary(nl)
```

After estimating the models, inspect the sign and magnitude of coefficients, the nesting parameter, the log-likelihood, and information criteria. A better likelihood is not enough: the nest parameter must be in a theoretically admissible range and the nest structure must be behaviourally meaningful.

13.9 R lab II: mixed logit with random coefficients

The next example estimates a mixed logit model with a random quality coefficient. In real environmental DCEs, cost is often kept fixed and environmental attributes are random, or the model is estimated in WTP space. Some analysts randomise cost, but this can produce unstable WTP tails unless distributional choices are carefully constrained.

```
# Continuing from the mlogit.data object above
# Random coefficient on quality, normally distributed

mxl <- mlogit(choice ~ cost + quality + crowding | 0,
              data = mdata,
              rpar = c(quality = "n"),
              panel = TRUE,
              R = 200,
              halton = NA)
summary(mxl)

# Marginal WTP using the mean quality coefficient
b <- coef(mxl)
wtp_quality <- -b["quality"] / b["cost"]
wtp_quality

# In applied work, increase R substantially and test stability.
```

For publication-quality work, 200 draws are usually only a teaching value. You should increase simulation draws, compare pseudo-random and quasi-random draws, check multiple starting values, and report

whether substantive conclusions are stable. In larger models, Apollo, Biogeme, or custom Bayesian estimation may be more convenient than a compact teaching package.

13.10 Apollo sketch: why specialised software matters

Apollo is widely used in applied choice modelling because it handles many model structures in a coherent R workflow, including MNL, nested logit, mixed logit, latent class, hybrid choice, panel likelihoods, and simulation-based estimation. The following sketch is not a full script, but it shows the logic. The analyst defines parameters, validates inputs, specifies utilities, builds probabilities, and calls the estimation routine.

```
# install.packages("apollo")
library(apollo)
apollo_initialise()

apollo_control <- list(
  modelName  = "mxl_environmental_dce",
  modelDescr = "Mixed logit for environmental DCE",
  indivID    = "id",
  mixing     = TRUE
)

apollo_beta <- c(
  asc_A = 0, asc_B = 0,
  b_cost = -0.05,
  mu_b_quality = 0.4,
  sigma_b_quality = 0.2,
  b_crowding = -0.3
)

apollo_fixed <- c()

apollo_draws <- list(
  interDrawsType = "halton",
  interNDraws = 500,
  interNormDraws = c("draw_quality")
)

apollo_randCoeff <- function(apollo_beta, apollo_inputs){
  randcoeff <- list()
  randcoeff[["b_quality"]] <- mu_b_quality + sigma_b_quality * draw_quality
  return(randcoeff)
}

# The probability function then defines utilities V for each alternative,
# calls apollo_mnl, and multiplies probabilities across tasks for each person.
```

The reason for learning specialised choice software is not merely convenience. Advanced models require careful handling of panel likelihoods, random draws, constraints, gradients, simulation, and post-estimation welfare quantities. General-purpose ML software is excellent for prediction, but it usually does not provide these choice-specific structures out of the box.

13.11 Model comparison: beyond fit

A common mistake is to rank advanced choice models only by log-likelihood, AIC, BIC, or predictive accuracy. These are useful, but they are not sufficient. In environmental economics, a model can predict well and still produce implausible WTP, wrong signs, unstable counterfactuals, or misleading substitution patterns.

Criterion	Question to ask	Warning sign
Behavioural plausibility	Do signs, elasticities, nests, and heterogeneity match theory?	Positive cost coefficient, implausible nesting parameter, extreme WTP tails
Prediction	Does the model generalise to held-out individuals or markets?	High in-sample fit but weak grouped cross-validation performance

Criterion	Question to ask	Warning sign
Welfare	Are compensating variation and WTP stable across specifications?	Policy ranking changes solely because of distributional tail assumptions
Identification	Are extra parameters supported by the data?	Large standard errors, boundary estimates, unstable convergence
Computation	Can the model be reliably estimated and replicated?	Strong dependence on starting values or number of draws
Communication	Can policy users understand what the model assumes?	Black-box structure without interpretable behavioural restrictions

Table 13.2. Advanced model comparison should combine statistical, behavioural, welfare, and computational criteria.

13.12 Links with machine learning

The advanced models in this chapter are not old-fashioned alternatives to ML. They are structured statistical learning models with economic content. They can be combined with ML in at least four ways.

- Use ML to discover nonlinear transformations, interactions, or segmentation patterns, then test them in a choice model.
- Use random forests, gradient boosting, or association rules to support specification search while preserving interpretable utility functions.
- Use neural networks to model context-dependent shifts in random coefficients, as in context-aware mixed logit architectures.
- Use deep learning to process images, text, GPS traces, or remote-sensing features and feed economically meaningful representations into a choice model.

The uploaded choice-modelling literature repeatedly stresses this complementarity. Work on choice modelling in the age of ML argues that ML can help overcome labour-intensive specification search and work with text or image data, while choice models retain strengths in theory, interpretation, and policy analysis. The recent Handbook of Choice Modelling also includes machine learning as a foundational topic, signalling that the field increasingly treats ML as an opportunity for cross-fertilisation rather than as a replacement for random utility modelling.

13.13 Environmental economics examples

Marine recreation and access fees

A coastal manager wants to evaluate access fees for protected beaches, open-access beaches, and inland substitutes. MNL may overstate substitution from a protected beach to inland forest because of IIA. Nested logit can group coastal alternatives and produce more plausible diversion patterns. Mixed logit can additionally account for heterogeneous sensitivity to crowding, seabed quality, and price.

Biodiversity conservation programmes

Households choose among conservation programmes differing in cost, species protection, landscape effects, and restrictions on land use. Mixed logit is useful because WTP for species protection is likely heterogeneous. Latent class models may reveal segments such as biodiversity-focused respondents, cost-sensitive respondents, and status-quo-oriented respondents. Cross-nested models may be considered if alternatives overlap across ecosystem type and policy instrument.

Energy contracts and green technology adoption

Consumers choose electricity contracts or heating technologies. Alternatives may cluster by supplier, renewable content, technology risk, and contract flexibility. Nested or cross-nested logit can represent

similarity among technologies. Mixed logit captures heterogeneous price sensitivity and risk attitudes. ML can support feature construction from text descriptions of contracts or smart-meter histories.

13.14 Concept checks

- Why does MNL imply proportional substitution, and when is that implausible?
- What behavioural story would justify a coastal recreation nest? What evidence would make you reject it?
- Why is a random cost coefficient potentially dangerous for WTP estimation?
- How does a mixed logit model use repeated choices from the same respondent?
- When would cross-nested logit be preferable to nested logit?
- Why can a model with better predictive accuracy still be less useful for welfare analysis?

13.15 Coding exercises

Exercise 1: test IIA

Use your own DCE or the simulated dataset in this chapter. Estimate an MNL model. Remove one alternative and re-estimate. Compare whether the ratios of probabilities among the remaining alternatives behave plausibly. Discuss whether a formal test is informative in your setting, and whether behavioural reasoning suggests a nested model.

Exercise 2: compare MNL, nested logit, and mixed logit

Estimate three models on the same data. Compare log-likelihood, AIC/BIC, grouped out-of-sample log loss, signs, WTP, and substitution patterns. Write a one-page recommendation for a policy audience.

Exercise 3: sensitivity of WTP

Estimate a mixed logit model under at least two distributional assumptions for one environmental attribute. Compare mean, median, and upper-tail WTP. Explain whether the policy conclusion is robust.

Exercise 4: ML-assisted specification

Fit a random forest or gradient boosting classifier to the same choice data in wide format. Use variable importance, partial dependence, or accumulated local effects to identify candidate interactions. Then include one theoretically defensible interaction in a choice model and evaluate whether it improves both fit and interpretation.

13.16 Common mistakes

- Treating nested logit as an automatic fix for IIA without a behavioural nesting story.
- Estimating a mixed logit with too many random coefficients relative to the information in the data.
- Using too few simulation draws and reporting unstable estimates.
- Ignoring panel structure in repeated stated choice data.
- Reporting only mean WTP when the WTP distribution is skewed or has implausible tails.
- Comparing ML and choice models only by accuracy, without checking welfare and counterfactual implications.

13.17 Chapter takeaway

The multinomial logit model is the baseline, not the destination. Nested and GEV models relax substitution restrictions by structuring similarity among alternatives. Mixed logit relaxes fixed tastes and

handles panel dependence through random coefficients. Probit allows flexible normal error correlation but is often computationally demanding. These models remain essential because environmental and transport policy often require welfare analysis, substitution patterns, and counterfactual simulations, not only prediction. Machine learning enriches this toolkit by assisting specification, feature construction, heterogeneity discovery, and computation, but the economic structure remains the anchor for interpretation.

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Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 14

Latent Class and Hybrid Choice Models: Segmentation, Attitudes, and Unobserved Heterogeneity

Many environmental and transport decisions are not made by one homogeneous population. Some people are cost-sensitive, some are environmentally committed, some ignore particular attributes, and some react strongly to attitudes such as perceived risk, trust, or moral obligation. This chapter explains latent class choice models and hybrid choice models, two model families that make such hidden heterogeneity explicit while preserving the random utility logic needed for welfare analysis and counterfactual policy evaluation.

How to study this chapter
Read Sections 14.1-14.4 to understand latent class models. Then read Sections 14.5-14.7 for hybrid choice models and machine-learning extensions. Finally, run the R skeletons in Sections 14.9-14.10 with one of your own stated preference datasets.

Learning objectives

- Distinguish continuous taste heterogeneity, discrete taste heterogeneity, and latent psychological constructs.
- Write the probability structure of a latent class choice model and interpret class membership probabilities.
- Estimate and diagnose a latent class model with class-specific utilities and class membership covariates.
- Explain what hybrid choice models add through structural equations, measurement equations, and latent variables.
- Recognise identification, scale, local optimum, and label-switching problems in latent class and hybrid models.
- Use machine learning as support for segmentation, latent construct modelling, and specification search without surrendering behavioural interpretation.

14.1 From continuous heterogeneity to classes and latent constructs

The mixed logit model in Chapter 13 represents preference heterogeneity as a continuous distribution of coefficients. Latent class models take a different view: the population is a mixture of a finite number of unobserved groups. Within each group, individuals share a class-specific preference vector. Across groups, preferences may differ sharply. This is useful when the data suggest distinct behavioural segments rather than a smooth distribution of tastes.

Hybrid choice models answer a related but different question. They ask whether attitudes, perceptions, beliefs, norms, or other latent psychological constructs help explain choices. For example, a respondent's support for a marine protected area may depend not only on cost and access restrictions, but also on trust in public institutions, perceived ecological responsibility, or perceived effectiveness of conservation policy.

Model idea	Hidden object	Typical use	Main risk
Mixed logit	Continuous random coefficients	WTP heterogeneity, repeated choices, random taste variation	Distributional assumptions and extreme WTP tails
Latent class choice model	Discrete classes of respondents	Market segmentation, policy targeting, attribute non-attendance, different behavioural rules	Over-interpreting classes and local optima
Hybrid choice / ICLV	Latent attitudes or perceptions	Linking psychometric indicators to choices and welfare	Weak identification and indicators that do not add choice-relevant information
ML-supported hybrid model	Flexible nonlinear latent representation	High-dimensional indicators, images, text, complex attitudes	Black-box constructs that are hard to validate

Table 14.1. Three ways of representing hidden behavioural structure. They can be combined, but each adds assumptions and computational burden.

14.2 Latent class choice models: the core structure

A latent class choice model assumes that each decision maker belongs to one of Q unobserved classes. Conditional on class q, the individual's choices are generated by a standard choice model, usually multinomial logit. The analyst does not observe class membership; the model estimates the probability that each respondent belongs to each class.

```
U_nitq = beta_q' x_nit + epsilon_nitq
P_nit|q = exp(V_nitq) / sum_j exp(V_njtq)
```

For panel data, the probability of respondent n's sequence of observed choices conditional on class q is the product of conditional probabilities over tasks.

```
L_n|q = product_t P_n i(n,t) t | q
```

The unconditional likelihood contribution is a weighted average of class-specific likelihoods, where the weights are class membership probabilities.

```
L_n = sum_q pi_nq * L_n|q
```

The class membership probability can be constant across respondents or modelled as a multinomial logit function of respondent characteristics. In an environmental DCE, class membership may depend on income, education, age, environmental concern, local residence, previous site use, climate-risk exposure, or experience with the policy problem.

```
pi_nq = exp(gamma_q' z_n) / sum_r exp(gamma_r' z_n), with gamma_1 normalised to zero
```

Interpretation rule
The classes are not directly observed social groups. They are statistical summaries of preference patterns. A class can be called 'cost-sensitive' or 'environmentally oriented' only after inspecting coefficients, WTP, class shares, posterior probabilities, and external validation variables.

14.3 What classes mean in environmental economics

Latent classes are useful because many environmental policies create heterogeneous trade-offs. A single average WTP can hide policy-relevant disagreement. One class may support biodiversity improvements even at high cost; another may accept environmental improvements only if access is not restricted; a third may choose the status quo frequently because it distrusts the implementing agency.

In environmental valuation, latent classes can represent at least six recurring behavioural patterns.

- Preference intensity: high-WTP and low-WTP groups for biodiversity, landscape, water quality, climate mitigation, or recreation access.
- Cost sensitivity: groups that react strongly or weakly to monetary cost, travel cost, time, taxes, or entrance fees.
- Attribute non-attendance: groups that appear to ignore cost, quality, risk, crowding, or policy restrictions.
- Status quo orientation: groups with high inertia, scepticism, loss aversion, or distrust of hypothetical policy change.
- Ethical or pro-environmental segments: groups whose choices reflect responsibility, identity, norms, or stewardship motives.
- Use-related segments: residents versus tourists, frequent versus occasional users, experienced versus inexperienced site visitors.

Example: beach management
A latent class DCE for coastal access could reveal one class that pays to avoid crowding, one class that prioritises ecological quality, one class that strongly dislikes access restrictions, and one class that almost always chooses the cheapest option. These segments imply different tariff, information, and rationing policies.

14.4 Choosing the number of classes

There is no mechanical answer to the question of how many classes to estimate. A two-class model may be too crude; a seven-class model may fit the sample but fail to generalise. The number of classes should be chosen by combining statistical fit, out-of-sample validation, behavioural interpretability, stability, and usefulness for the research question.

Criterion	What it asks	Warning
Log-likelihood	Does the model fit the estimation data better?	Always improves or stays similar as classes increase.
AIC/BIC/CAIC	Does fit improve enough to justify more parameters?	BIC often favours fewer classes, but criteria may disagree.
Holdout log loss	Does the model predict new respondents or tasks better?	Split by respondent, not by row, in panel choice data.
Posterior entropy	Are respondents clearly assigned to classes?	High entropy is not required, but diffuse classes are hard to interpret.
Class size	Are classes large enough to be credible?	Tiny classes may be real minorities or numerical artefacts.
Behavioural meaning	Can coefficients, WTP, and class covariates be explained?	Plausible names do not prove causal mechanisms.
Stability	Do repeated random starts give similar solutions?	Local maxima are common in mixture models.

Table 14.2. Model selection in latent class modelling is multi-criteria. Do not select classes by fit statistics alone.

A good reporting practice is to estimate models with Q = 1, 2, 3, ... classes; use many random starts; report information criteria; compare validation performance; and present a short justification for the selected model. When the selected number of classes is sensitive to starting values or data splits, say so explicitly.

14.5 Posterior class probabilities and policy targeting

After estimation, each respondent receives posterior probabilities of belonging to each class. These are obtained by Bayes' rule: prior class probability times class-specific likelihood, divided by the total likelihood.

```
Pr(class q | choices_n, z_n) = pi_nq * L_n|q / sum_r pi_nr * L_n|r
```

Posterior probabilities are useful for description and targeting. For example, managers may want to know whether frequent beach users are more likely to belong to a crowding-sensitive class, or whether low-income households are more likely to belong to a cost-sensitive class. However, posterior assignment is probabilistic. Classifying each respondent into the highest-probability class and then treating that class as observed creates additional uncertainty that is often ignored.

> Do not reify classes
> A respondent with posterior probabilities 0.52, 0.31, and 0.17 is not unambiguously a member of class 1. Use posterior probabilities directly when possible and avoid strong individual-level claims.

14.6 Hybrid choice models and latent variables

Hybrid choice models, often called integrated choice and latent variable models, add latent psychological constructs to the random utility model. The latent variable is not observed directly. Instead, it is inferred from observed indicators such as Likert-scale responses, psychometric items, risk perceptions, trust questions, or attitudinal statements.

A simple hybrid model has three parts: a structural equation for the latent variable, a measurement equation for the indicators, and a choice equation in which the latent variable enters utility.

```
Structural equation: LV_n = alpha' s_n + sigma_LV * eta_n
Measurement equation: I_nm = tau_m + lambda_m * LV_n + nu_nm
Choice equation: U_nit = beta' x_nit + theta * LV_n * attribute_nit + epsilon_nit
```

The structural equation explains the latent variable using socio-demographics or experiences. The measurement equation links the latent variable to indicators. The choice equation tests whether the latent

variable helps explain choices. For example, environmental concern may increase the marginal utility of biodiversity improvement, while distrust may increase status quo preference.

Example: marine protected areas
Survey indicators such as 'I trust environmental agencies', 'marine ecosystems are fragile', and 'access limits are unfair' can be used to infer latent trust, ecological concern, or perceived fairness. These latent variables can then interact with cost, access restrictions, or biodiversity improvements in the choice model.

14.7 Identification and interpretation problems in hybrid models

Hybrid models are attractive because they appear to connect economics, psychology, and survey measurement. They are also easy to overclaim. A latent variable is only identified up to scale and sign unless normalisations are imposed. Multiple latent-variable structures can produce similar choice probabilities. Indicators may explain attitudes well but add little to the choice model. Conversely, a model may fit choices well but assign a weak or unstable interpretation to the latent variable.

The identification literature emphasises that the presence of indicators does not automatically make the latent variable behaviourally meaningful. Researchers should show why the latent construct is theoretically motivated, how it is measured, whether the measurement model behaves sensibly, whether the latent variable improves prediction or interpretation, and whether the results are robust to alternative specifications.

Problem	Symptom	Diagnostic response
Scale normalisation	Changing units or fixed loadings changes parameter magnitudes	Fix one loading or latent variance; report the chosen normalisation.
Sign indeterminacy	A latent variable can be multiplied by -1 if loadings also change sign	Anchor the sign using a meaningful indicator loading.
Weak indicators	Low or unstable loadings; poor measurement fit	Improve survey design; remove weak indicators only with theory and diagnostics.
Latent variable not useful for choice	Good psychometric model but weak theta in utility	Report that attitudes are measured but not choice-relevant in this specification.
Endogeneity of attitudes	Attitudes may be rationalisations after behaviour	Interpret carefully; use timing, instruments, experiments, or panel data where possible.
Overfitting	Many indicators, interactions, and latent variables in a small sample	Use theory, validation, regularisation, and simpler benchmarks.

Table 14.3. Hybrid choice models require explicit identification and diagnostic checks.

14.8 Machine learning connections

Machine learning enters latent class and hybrid choice modelling in several ways. The safest role is often supportive: use ML to discover patterns, reduce dimensionality, represent complex indicators, or suggest interactions, then evaluate the resulting model with choice-theoretic diagnostics. More ambitious approaches embed neural networks inside the choice model.

The uploaded Torres Lahoz et al. study is a clear example: artificial neural networks are used to formulate latent constructs inside latent class choice models, allowing more flexible relationships between psychometric indicators and class membership or choice behaviour while maintaining the random utility foundation. The context-aware Bayesian mixed logit model discussed earlier in the book follows a similar philosophy: a neural network maps context into shifts in interpretable preference parameters rather than replacing the choice model entirely.

ML tool	Useful role	Choice-modelling safeguard
Clustering	Exploratory segmentation before LCCM	Do not treat clusters as classes unless validated in the choice model.
LASSO / elastic net	Select class membership covariates or interactions	Respect panel structure and avoid post-selection overclaiming.
Random forests / boosting	Identify nonlinear predictors of posterior class probabilities	Use as diagnostics; do not infer WTP from unconstrained trees.
Neural networks	Flexible latent constructs, embeddings, nonlinear class allocation	Check monotonicity, signs, WTP, sensitivity, and sample size.
Embeddings	Represent text, images, or high-dimensional attitudinal items	Link representations back to interpretable constructs or policy attributes.
Mixture-of-experts	ML analogue of class-specific models with learned gating network	Constrain utilities and report behavioural implications.

Table 14.4. ML can improve segmentation and latent representation, but welfare analysis still requires economic structure.

Sample-size warning
Neural latent-variable structures can contain many more weights than a conventional latent class model. In small environmental DCEs, a simple LCCM or ICLV model may be more credible than a high-dimensional neural architecture.

14.9 R lab I: latent class model with gmnl

The following template shows the workflow for estimating a latent class model in R with gmnl. The exact formula may need adaptation to your dataset, but the logic is stable: convert to long choice format, fit several class counts, use respondent-level panel structure, and compare fit and interpretability. The fifth part of the gmnl formula is used for class membership covariates; at minimum it should include a constant for latent class models.

```
# install.packages(c("tidyverse", "mlogit", "gmnl"))
library(tidyverse)
library(mlogit)
library(gmnl)

# Your long data should contain:
# id        respondent identifier
# chid      choice-task identifier
# alt       alternative name
# choice    1 if chosen, 0 otherwise
# cost, bio, crowd  alternative-specific attributes
# income, env_att    respondent-level covariates repeated on each row

mdata <- mlogit.data(
  df_long,
  choice   = "choice",
  shape    = "long",
  chid.var = "chid",
  alt.var  = "alt",
  id.var   = "id"
)

# MNL benchmark
mnl <- gmnl(choice ~ cost + bio + crowd | 0,
            data = mdata, model = "mnl", panel = TRUE)

# Two-class latent class model. The fifth formula part is class allocation.
lc2 <- gmnl(choice ~ cost + bio + crowd | 0 | 0 | 0 | 1,
            data = mdata, model = "lc", Q = 2, panel = TRUE)

# Class allocation with respondent covariates
lc2_cov <- gmnl(choice ~ cost + bio + crowd | 0 | 0 | 0 | income + env_att,
                data = mdata, model = "lc", Q = 2, panel = TRUE)

# Repeat with multiple class counts and multiple starting values in real work
models <- list(mnl = mnl, lc2 = lc2, lc2_cov = lc2_cov)
sapply(models, AIC)
sapply(models, BIC)
summary(lc2_cov)
```

This is not the end of the analysis. After estimating the model, inspect class-specific cost coefficients, WTP values, class shares, posterior probabilities, convergence, and sensitivity to starting values. If a class has a positive cost coefficient, an implausibly tiny share, or unstable parameters, the model needs revision.

14.10 R lab II: latent class model skeleton in Apollo

Apollo is especially useful when you need flexible latent class, mixed logit, hybrid choice, or custom likelihood structures. The following skeleton illustrates the logic of a two-class model. It is deliberately simplified: real Apollo code requires data preparation, control settings, availability variables, and careful starting values.

```
# install.packages("apollo")
library(apollo)
apollo_initialise()

apollo_beta <- c(
  asc_sq_a = 0, b_cost_a = -0.05, b_bio_a = 0.5, b_crowd_a = -0.3,
  asc_sq_b = 0, b_cost_b = -0.12, b_bio_b = 0.2, b_crowd_b = -0.8,
  delta_b = 0, gamma_income_b = 0, gamma_env_b = 0
)
apollo_fixed <- c("asc_sq_a", "asc_sq_b")

# Class allocation utility: class A is the reference class
# pi_b depends on income and environmental attitude
apollo_probabilities <- function(apollo_beta, apollo_inputs, functionality = "estimate"){
  apollo_attach(apollo_beta, apollo_inputs)
  on.exit(apollo_detach(apollo_beta, apollo_inputs))

  P <- list()
  classAlloc_settings <- list(
    classes   = c(class_a = 1, class_b = 2),
    utilities = list(
      class_a = 0,
      class_b = delta_b + gamma_income_b * income + gamma_env_b * env_att
    )
  )
  pi_values <- apollo_classAlloc(classAlloc_settings)

  V_a <- list(
    alt1 = b_cost_a * cost_alt1 + b_bio_a * bio_alt1 + b_crowd_a * crowd_alt1,
    alt2 = b_cost_a * cost_alt2 + b_bio_a * bio_alt2 + b_crowd_a * crowd_alt2,
    sq   = asc_sq_a
  )
  V_b <- list(
    alt1 = b_cost_b * cost_alt1 + b_bio_b * bio_alt1 + b_crowd_b * crowd_alt1,
    alt2 = b_cost_b * cost_alt2 + b_bio_b * bio_alt2 + b_crowd_b * crowd_alt2,
    sq   = asc_sq_b
  )

  mnl_settings_a <- list(alternatives = c(alt1=1, alt2=2, sq=3),
                         avail = list(alt1=1, alt2=1, sq=1),
                         choiceVar = choice, utilities = V_a)
  mnl_settings_b <- list(alternatives = c(alt1=1, alt2=2, sq=3),
                         avail = list(alt1=1, alt2=1, sq=1),
                         choiceVar = choice, utilities = V_b)

  P[["class_a"]] <- apollo_mnl(mnl_settings_a, functionality)
  P[["class_b"]] <- apollo_mnl(mnl_settings_b, functionality)

  lc_settings <- list(inClassProb = P, classProb = pi_values)
  P[["model"]] <- apollo_lc(lc_settings, functionality)
  P <- apollo_panelProd(P, apollo_inputs, functionality)
  P <- apollo_prepareProb(P, apollo_inputs, functionality)
  return(P)
}
```

14.11 Hybrid choice model skeleton

Hybrid choice models are more complex because the likelihood combines choices and indicators. In Apollo, the model usually combines an MNL component for choices and ordered-logit or normal-density components for indicators. The exact implementation depends on whether indicators are continuous, binary, ordered categorical, or count variables.

```
# Conceptual pseudo-code for a one-latent-variable HCM
# LV = environmental concern

LV <- zeta + gamma_age * age + gamma_member * env_org_member + sigma_lv * draw_lv

# Choice utility: environmental concern changes sensitivity to biodiversity
V_alt1 <- b_cost * cost_alt1 + (b_bio + theta_lv_bio * LV) * bio_alt1
V_alt2 <- b_cost * cost_alt2 + (b_bio + theta_lv_bio * LV) * bio_alt2
V_sq   <- asc_sq

# Measurement equations for indicators, e.g. Likert items
# indicator_1: "Protecting biodiversity is a personal responsibility"
# indicator_2: "I trust public environmental agencies"
# indicator_3: "Access restrictions are acceptable if ecosystems benefit"

I1_star <- tau1 + lambda1 * LV + error1
I2_star <- tau2 + lambda2 * LV + error2
I3_star <- tau3 + lambda3 * LV + error3

# Joint likelihood for respondent n:
# L_n = integral [ P(choices_n | LV_n) * P(indicators_n | LV_n) * f(LV_n) ] dLV_n
```

For a PhD project, begin with a simple specification: one latent variable, a small number of high-quality indicators, and one or two theoretically motivated interactions in utility. Add complexity only after checking that the measurement model, choice model, and combined likelihood are stable.

14.12 Software choices

Software	Strengths	Best use in this chapter
R: gmnl	MNL, mixed logit, generalized MNL, latent class and mixed-mixed logit; convenient formula interface	Latent class models and comparison with mixed logit.
R: Apollo	Highly flexible user-defined likelihoods, latent class, mixed logit, hybrid choice, panel data, mixtures	Serious applied LCCM and HCM work, custom indicators, complex DCEs.
R: mlogit/logitr	Fast baselines and clear data handling	MNL or mixed logit benchmarks before estimating LCCM/HCM.
Python: Biogeme	Powerful choice model estimation and simulation, flexible specifications	Advanced DCM estimation, teaching, and reproducible Python workflows.
Python: PyTorch/TensorFlow/JAX	Neural networks, embeddings, differentiable programming	Neural latent constructs or hybrid architectures; requires custom validation.
Commercial tools	Legacy support in some applied fields	Replicating older studies or institutional workflows; open tools are usually preferable for reproducibility.

Table 14.5. Use simple tools for simple baselines and flexible tools only when the research question requires them.

14.13 Model reporting checklist

- Start with MNL and mixed logit benchmarks before reporting a latent class or hybrid model.
- Report the number of random starts, convergence criteria, and whether multiple starts reached the same optimum.
- For LCCM, report class shares, class-specific coefficients, WTP by class, class membership effects, and posterior probability diagnostics.
- For HCM, report structural equations, measurement equations, loadings, normalisations, and how latent variables enter utility.
- Use respondent-level validation splits for panel DCEs and report predictive log loss or holdout likelihood, not only in-sample fit.
- Check whether cost coefficients have plausible signs in every class and whether WTP distributions are economically credible.

- Avoid causal language for classes or attitudes unless the design supports causal identification.
- Translate model results into policy simulations: class-specific adoption, welfare, revenue, distributional impacts, and equity concerns.

14.14 Worked interpretation example

Imagine a three-class model for a DCE on paid access to restored coastal areas. Class 1, representing 45 percent of the sample, has a strongly negative cost coefficient and moderate positive coefficients for biodiversity and water clarity. Class 2, 35 percent, has a weaker cost coefficient and a very high coefficient for crowding reduction. Class 3, 20 percent, has a large positive status quo constant and weak responses to ecological attributes.

The policy implications differ from the average model. A uniform entrance fee may be acceptable to class 2 if it effectively reduces crowding, but it may exclude class 1 unless discounts or resident passes are introduced. Ecological communication may persuade class 1 and class 2 differently. Class 3 may require trust-building or proof of effectiveness before it supports the intervention. The latent class model therefore supports distributional policy design, not just average welfare estimation.

Link to machine learning
A random forest or gradient boosting model could help predict posterior class probabilities from rich respondent profiles. But the WTP and welfare calculations should still come from the class-specific utility coefficients, not from the unconstrained classifier.

14.15 Common mistakes

Naming classes too confidently

Labels such as 'green citizens' or 'sceptics' are shorthand. They must be supported by coefficients, class covariates, posterior probabilities, and external variables.

Choosing classes by AIC alone

Information criteria are useful but insufficient. Validation, stability, and behavioural plausibility matter.

Ignoring local maxima

Mixture likelihoods often have multiple local optima. Use many starting values and report the search procedure.

Treating attitudes as exogenous

Attitudes may be jointly formed with behaviour or affected by previous experiences. HCMs are not automatically causal.

Adding too many latent variables

More latent constructs can make the model look psychologically rich while weakening identification and interpretability.

Using ML as a black-box class generator

Unsupervised clusters are not substitutes for latent class choice probabilities unless linked back to observed choices and utility theory.

14.16 Concept checkpoint

- Why is a latent class model called a finite mixture model?
- What is the difference between prior class probability and posterior class probability?
- Why can a class with a positive cost coefficient be a warning sign?
- In a hybrid choice model, what is the difference between a structural equation and a measurement equation?
- Why do latent variables require scale and sign normalisations?
- Give one way ML can help latent class or hybrid choice modelling, and one way it can mislead.

14.17 Chapter takeaway

Latent class and hybrid choice models are powerful tools for modelling hidden heterogeneity. Latent class models reveal discrete preference segments; hybrid choice models connect choices to latent attitudes, perceptions, and beliefs. In environmental economics, these models are especially useful because policy acceptance, WTP, and behavioural response are often distributed unevenly across the population. Machine learning can enrich these models through flexible segmentation and representation learning, but the welfare and counterfactual value of the analysis still depends on careful identification, validation, and economic interpretation.

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R package documentation: gmnl, mlogit, logitr, and Apollo manuals, consulted for implementation guidance.

Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 15

Machine Learning versus Discrete Choice Models: Prediction, Explanation, Welfare, and Policy Use

This chapter addresses the central methodological question of the book: when should an applied researcher use machine learning, when should they use discrete choice models, and when should they combine them? The answer is not a slogan. It depends on the target of the analysis: prediction, behavioural explanation, welfare measurement, causal policy evaluation, or model discovery.

How to study this chapter
Read the chapter twice. The first reading should focus on the distinction between tasks. The second should focus on the protocol for fair model comparison. Then run the R lab and deliberately create one invalid validation split to see how leakage changes the conclusion.

Learning objectives

- Explain why the debate should not be framed as machine learning versus choice modelling in the abstract.
- Distinguish prediction metrics, likelihood-based metrics, behavioural outputs, welfare outputs, and causal policy outputs.
- Recognise why a multinomial logit model and a multi-class classifier can be mathematically close but scientifically different.
- Design a fair comparison between ML and discrete choice models using grouped resampling, tuning, calibration, and uncertainty.
- Identify when flexible ML is likely to improve performance and when economic structure is more important.
- Use ML as a complement to choice modelling through specification search, representation learning, residual diagnostics, and constrained hybrid models.

Chapter map

15.1 The false dichotomy
15.2 What counts as performance?
15.3 The common mathematical core: utility, softmax, and cross-entropy
15.4 Predictive performance: when ML tends to win
15.5 Validation and leakage in choice data
15.6 Interpretation: coefficients, elasticities, SHAP, and behavioural discipline
15.7 Welfare and policy counterfactuals
15.8 Extrapolation, support, and invariance
15.9 A protocol for fair comparison
15.10 Complementary workflows
15.11 Environmental economics examples
15.12 R and Python labs
15.13 Exercises and checklist

15.1 The false dichotomy

The expression 'machine learning versus discrete choice models' is useful as a teaching device, but misleading as a research philosophy. Machine learning is a broad family of methods for learning functions from data. Discrete choice modelling is a theory-informed framework for modelling choices among alternatives, usually grounded in random utility maximisation or related behavioural theories. The two do not compete on one dimension only.

A random forest, gradient boosting model, or neural network may be excellent at predicting which alternative a person chooses from observed data. A multinomial logit, nested logit, mixed logit, or latent class model may be better at answering how utility changes with cost, how alternatives substitute for each other, how welfare changes under a policy, and what happens under counterfactual attribute changes. A researcher who asks only which method is more accurate is often answering a narrower question than the policy problem requires.

The uploaded discussion paper by van Cranenburgh, Wang, Vij, Pereira, and Walker argues for cross-pollination rather than replacement. Machine learning can help choice modellers with model selection, high-dimensional data, images, text, flexible interactions, and validation practices. Choice modelling can help machine learning with behavioural structure, interpretability, welfare logic, and counterfactual discipline. This chapter translates that position into a practical workflow for environmental economics and choice modelling.

Core idea
Do not ask: 'Which family is best?' Ask: 'Best for what target, with what data, under what validation design, and with what behavioural or welfare requirements?'

15.2 What counts as performance?

The first mistake in ML-DCM comparisons is to treat performance as a single number. Choice models and ML classifiers can be compared on out-of-sample prediction, but environmental economics usually requires more than prediction. A coastal access model, for example, may need to predict visitation, estimate willingness to pay for water quality, simulate crowding regulations, and support distributive policy analysis. A model can perform well on one target and poorly on another.

Target	Typical question	Useful criteria	Risk if ignored
Prediction	Which alternative will be chosen?	Accuracy, top-k accuracy, log loss, Brier score, calibration	A model that predicts labels well may produce poor probabilities.
Behavioural explanation	How do cost, time, crowding, and quality affect choices?	Signs, elasticities, marginal effects, heterogeneity, substitution patterns	A black box may learn implausible or unstable relationships.
Welfare	What is the WTP or compensating variation for an environmental improvement?	Cost coefficient, WTP ratios, uncertainty intervals, welfare formulae	High accuracy does not imply meaningful monetary valuations.
Policy counterfactual	What happens if a new regulation or tariff is introduced?	Structural assumptions, invariance, support, scenario diagnostics	The model may extrapolate beyond the data in an uncontrolled way.
Causal inference	What is the effect of a policy intervention?	Identification, orthogonalisation, cross-fitting, sensitivity checks	Predictive associations can be mistaken for causal effects.
Design and discovery	Which interactions or segments should be considered?	Regularisation, stability, holdout diagnostics, expert review	Specification search can become data mining.

Table 15.1. Model performance is multi-dimensional. The evaluation metric must match the scientific or policy target.

In predictive ML, cross-entropy or log loss is often more informative than accuracy because it evaluates predicted probabilities, not only the most likely class. In choice modelling, the log-likelihood plays a similar role. But even a well-calibrated predictive probability is not automatically a behavioural or welfare object. The scientific interpretation depends on what generated the probability and which assumptions allow a change in attributes to be interpreted as a change in utility, utility trade-offs, or welfare.

15.3 The common mathematical core: utility, softmax, and cross-entropy

There is more overlap between ML and discrete choice than their cultures suggest. A multinomial logit model can be seen as a softmax classifier with a structured utility index. For individual n, alternative j, and attributes x_nj, the systematic utility is:

```
V_nj = beta' x_nj
```

The MNL probability is:

```
P_nj = exp(V_nj) / sum_k exp(V_nk)
```

A neural classifier often computes a score s_nj for each class and sends the scores through the same softmax function:

```
P_nj = exp(s_nj) / sum_k exp(s_nk)
```

The difference is in the score. In a linear MNL, the score is a utility index with an economic interpretation. In a generic neural network, the score may be a flexible nonlinear function of inputs:

```
s_nj = f_theta(x_nj, z_n, context_nj)
```

Training by cross-entropy minimisation is numerically close to maximum likelihood estimation for discrete outcomes. The difference is not that one is probabilistic and the other is not. The difference is the level of structure imposed on the probability model and the meaning attached to the score. If f_theta is unconstrained, the model may predict well but lack cost monotonicity, plausible substitution patterns, or interpretable WTP.

MNL as a disciplined softmax
The softmax is not the dividing line. The dividing line is whether the score is treated as an economically interpretable utility function and whether the data, assumptions, and model structure support welfare and counterfactual use.

A minimal taxonomy

Model type	Score or utility	Strength	Weakness
MNL	Linear-in-parameters utility	Transparent, fast, welfare-ready under assumptions	IIA and restrictive functional form
Nested/GEV	Structured utility plus correlated errors	Substitution patterns and nests	Requires theory-guided nesting
Mixed logit	Random coefficients utility	Continuous heterogeneity and flexible substitution	Simulation cost and distributional assumptions
Latent class	Class-specific utilities and membership model	Discrete segments and interpretable heterogeneity	Local maxima and class interpretation risk
Random forest/boosting	Ensemble of trees	Strong nonlinear prediction with tabular data	WTP and counterfactual interpretation not automatic
Deep neural network	Learned nonlinear representation	High flexibility, images/text/large data	Data-hungry and difficult to diagnose
Hybrid ML-DCM	ML component inside or beside utility	Flexible yet partly interpretable	Design, estimation, and validation become more complex

Table 15.2. Different models impose different structure on the score that becomes a choice probability.

15.4 Predictive performance: when ML tends to win

The empirical ML-choice literature often finds that ensemble methods and neural networks can outperform simple logit models in out-of-sample prediction. This is not surprising. Many real choice processes contain nonlinearities, high-order interactions, context dependence, unobserved segmentation, and high-dimensional inputs. A hand-specified linear utility may miss these patterns.

The strongest evidence is not a single case study but a pattern. Hillel, Bierlaire, Elshafie, and Jin review ML classification methods for passenger mode choice and show that the literature is methodologically diverse but often affected by validation and reporting problems. Wang, Mo, Zheng, Hess, and Zhao benchmark many ML and DCM families across thousands of experiments and find that ensemble methods and deep learning often have strong predictive performance, while also stressing that context, dataset, input structure, sample size, and choice categories explain much of the variation in results. Dahmen, Weikl, and Bogenberger show that XGBoost and random forest can outperform a conventional MNL on smartphone tracking-based revealed preference data, with SHAP used for interpretation.

These results should be read carefully. They show that flexible prediction can be valuable, especially in rich data settings. They do not show that ML automatically dominates for welfare analysis, policy appraisal, or causal inference. They also do not show that every environmental DCE with a few hundred respondents should use a deep neural network.

ML advantage	Why it matters for choices	Typical example
Nonlinearity	Effects of cost, time, distance, crowding, or risk may be nonlinear	Mode choice with thresholds in travel time or parking cost
Interactions	Preferences may depend on income, habit, context, or previous choices	Beach users react differently to access restrictions when crowding is high
High-dimensional data	Images, text, sensor data, and GPS traces cannot be easily hand-coded	Street-view image choices or smartphone-based travel diaries
Automatic feature learning	Useful when relevant variables are not known in advance	Remote sensing features for environmental behaviour prediction
Robust prediction	Ensembles reduce variance relative to single trees	Adoption prediction for energy-saving technologies
Large sample scalability	Stochastic optimisation can exploit very large datasets	Platform, smart-meter, or passive mobility data

Table 15.3. Situations where flexible ML is often useful.

When choice models may still win

Choice models may outperform ML in small samples, in sparse designs, under strong theory, or when the test scenario differs from the training data. This is not because linearity is true. It is because structure can reduce variance and prevent the model from learning spurious patterns. In stated preference data, experimental design gives the analyst substantial control over attributes and variation. A well-specified mixed logit may use that structure more efficiently than a high-capacity black box.

In environmental economics, sample sizes are often modest, and policy questions often require WTP, compensating variation, or distributional welfare. A generic classifier may provide an attractive accuracy table but no defensible monetary valuation. The right comparison is not 'MNL versus the most flexible ML model'. It is 'the simplest defensible economic model versus flexible alternatives, evaluated against the target outputs required by the study'.

15.5 Validation and leakage in choice data

Validation is where many ML-DCM comparisons become invalid. A random row-wise split is often wrong for panel choice data. If the same respondent contributes multiple tasks or trips, then row-wise splitting places information about that respondent in both training and testing sets. Flexible ML models can exploit this leakage and appear to generalise better than they really do.

The Hillel validation work is especially important for this book. It shows that trip-wise sampling in travel diary data can create substantial data leakage and bias performance estimates, especially for flexible nonlinear classifiers. The lesson generalises beyond transport. In a beach DCE, energy tariff DCE, conservation survey, or stated adoption experiment, repeated tasks from the same respondent should normally be split by respondent when testing generalisation to new people.

The leakage rule
If the same decision maker appears more than once, do not use naive row-wise train/test splits unless your target is explicitly prediction of additional choices from already-seen individuals. For population-level generalisation, split by individual or another relevant group.

Data structure	Bad split	Better split	Reason
Repeated DCE tasks	Random choice-task rows	Respondent-level split	Avoids learning respondent-specific patterns in both train and test
Travel diary trips	Trip-wise split	Person-level split	Prevents leakage from the same traveller
Route choice with repeated OD pairs	Random route alternatives	OD-pair or traveller split, depending on target	Tests the intended form of generalisation
Spatial environmental data	Random pixels or parcels	Spatial blocks or region holdout	Avoids overoptimistic performance from spatial autocorrelation
Time series policy data	Random time points	Forward-chaining or temporal holdout	Respects forecasting and policy timing
Choice experiments by country	Pooled random split	Country holdout for transferability	Tests external validity across policy contexts

Table 15.4. Resampling must match the desired generalisation target.

Hyperparameter tuning must be nested

Another common error is using the test set for model selection. For ML, tune hyperparameters inside the training data using cross-validation, then evaluate once on the holdout set. For choice models, the equivalent problem is searching many specifications and reporting the best holdout performance without accounting for search. The evaluation protocol should treat all model families symmetrically: if one model gets tuning, all relevant competitors should receive reasonable specification effort.

```
Training data -> model tuning and specification search
Holdout data -> final unbiased comparison
External data -> transferability and policy robustness, if available
```

15.6 Interpretation: coefficients, elasticities, SHAP, and behavioural discipline

Choice modellers often say that DCMs are interpretable and ML is black-box. This is partly true and partly too simple. A linear MNL coefficient is easy to read, but it may be misleading if the model is misspecified. A gradient boosting model is harder to read, but tools such as partial dependence, accumulated local effects, individual conditional expectation, and SHAP values can reveal useful patterns. The relevant question is not whether an output looks interpretable, but whether it is valid for the intended interpretation.

Output	Choice-model version	ML version	Caution
Direction of effect	Sign of beta	PDP/ALE/SHAP slope or sensitivity	ML effects are often conditional on data distribution and correlated features
Magnitude	Marginal effect or elasticity	Numerical perturbation of predicted probabilities	Not structural unless assumptions justify intervention
Trade-off	WTP = - beta_attribute / beta_cost	May be approximated from constrained utility networks	Unconstrained classifiers do not identify WTP
Heterogeneity	Random coefficients, classes, interactions	ICE curves, SHAP distributions, subgroup predictions	Patterns may reflect correlation, not preference heterogeneity
Substitution	Cross-elasticities and nest structure	Counterfactual probability changes	Need coherent choice-set treatment

Table 15.5. Interpretation tools exist in both traditions, but their scientific status differs.

The safest way to use ML interpretation in applied choice work is as diagnostic evidence rather than final economic proof. For example, if a random forest suggests a sharp nonlinear effect of crowding, the researcher can test a spline or piecewise crowding term in a mixed logit model. If SHAP values suggest interactions between income and cost, the researcher can estimate a model with income-cost interactions or WTP-space heterogeneity. ML interpretation becomes a bridge to better specification.

Interpretability principle
A model-agnostic explanation explains the fitted model. It does not automatically explain the human decision process, the causal effect of an attribute, or the welfare consequence of a policy.

15.7 Welfare and policy counterfactuals

The strongest reason to keep choice modelling at the centre of environmental economics is welfare analysis. Many environmental goods are not traded in markets. DCEs and related choice models are used precisely because they can convert utility trade-offs into monetary measures such as marginal WTP and compensating variation.

```
Marginal WTP for attribute k = - beta_k / beta_cost

Compensating variation for policy change 0 -> 1
CV_n = (1 / -beta_cost_n) * [ log sum_j exp(V_nj^1) - log sum_j exp(V_nj^0) ]
```

These formulae depend on strong assumptions: utility is specified correctly enough, the monetary coefficient is interpretable, the error distribution and scale are handled consistently, and the policy scenario is within a credible range. But at least the assumptions are visible. A generic random forest or neural classifier does not usually produce a cost coefficient, a utility scale, or a welfare measure. One can perturb cost and attributes and observe predicted probabilities, but that is not the same as measuring welfare.

Welfare warning
Do not report WTP from an unconstrained black-box classifier unless you can state exactly what utility structure, scale normalisation, cost monotonicity, and behavioural assumptions make the calculation meaningful.

What can ML contribute to welfare analysis?

ML can still help welfare analysis. It can discover interactions, represent complex context variables, estimate nuisance functions in causal settings, improve imputation, identify latent segments, and test whether traditional specifications leave systematic residual structure. More recent neural choice models try to preserve the random utility logic while replacing selected parts of the model with flexible functions. The key design principle is to make the part used for welfare analysis explicit and constrained.

ML contribution	Welfare-compatible design
Flexible utility	Use sign constraints, monotonicity constraints, or interpretable additive components for cost and policy attributes.
Context-dependent tastes	Let ML map context to shifts in preference parameters, while keeping parameters interpretable.
Image or text attributes	Use embeddings as covariates in a choice model and examine how they affect utility.
Specification search	Use ML to suggest candidate terms; estimate final welfare model with choice-theoretic structure.
Causal nuisance estimation	Use double/debiased ML or causal forests with a credible identification strategy.

Table 15.6. ML can support welfare analysis when the welfare-relevant structure remains explicit.

15.8 Extrapolation, support, and invariance

A model trained on observed data may fail when a policy changes the choice environment. This is not only an ML problem, but flexible ML can make the problem less visible. Consider a coastal policy where access prices in the survey range from 0 to 10, but the policy scenario considers a 30 fee. A random forest cannot extrapolate linearly beyond split points in the way a parametric utility function can. Gradient boosting and neural networks may extrapolate in opaque ways. A logit model extrapolates through its functional form, which may also be wrong, but the extrapolation is at least governed by an explicit assumption.

Before using any model for policy simulation, ask whether the counterfactual lies within the support of the training data. For DCEs, this means attribute levels, combinations, choice-set composition, and respondent populations. For revealed preference data, it also means market conditions, availability, prices, and constraints.

Counterfactual problem	Diagnostic	Response
New attribute levels	Compare policy values with experimental or observed ranges	Avoid unsupported extrapolation or design a follow-up experiment
New alternatives	Check whether similar alternatives were observed	Use structured utility and carefully specified alternative-specific constants
Changed choice set	Check substitution assumptions	Use nested/mixed logit or RUM-consistent neural model if appropriate
New population	Compare covariate distribution and preference indicators	Use transportability checks, reweighting, or external validation
Changed incentives	Check whether hypothetical or revealed context changes behaviour	Use RP-SP scale models, incentive-compatible design, or sensitivity analysis

Table 15.7. Counterfactual simulation requires support and invariance diagnostics.

15.9 A protocol for fair ML-DCM comparison

Fair comparison is a research design problem. Many published comparisons are asymmetric. A hand-specified MNL is compared to many tuned ML models; or a default random forest is compared to a carefully crafted mixed logit; or row-wise validation inflates ML performance; or accuracy is reported but log loss and calibration are ignored. A fair protocol should specify the prediction target, resampling unit, tuning budget, feature set, model families, metrics, behavioural diagnostics, and reporting rules before looking at results.

Step	Question	Recommended practice
1. Define target	Prediction, explanation, welfare, or policy?	Write the target before choosing models.
2. Define unit	New choices, new people, new places, or new time periods?	Choose grouped, spatial, or temporal resampling accordingly.
3. Harmonise inputs	Do models see comparable information?	Document all features and transformations.
4. Tune fairly	Did each family receive reasonable effort?	Use nested CV or training-only tuning.
5. Use multiple metrics	Labels, probabilities, calibration, and welfare?	Report accuracy plus log loss or Brier score at minimum.
6. Diagnose behaviour	Are signs, elasticities, and substitutions plausible?	Compare DCM coefficients with ML interpretation and perturbation tests.

Step	Question	Recommended practice
7. Assess uncertainty	Are differences stable?	Use repeated resampling, confidence intervals, or paired tests.
8. Test transfer	Does performance generalise across contexts?	Use external or blocked holdouts when possible.
9. Report limits	What cannot be inferred?	Separate prediction claims from welfare and causal claims.

Table 15.8. A minimum protocol for comparing ML and discrete choice models.

Suggested benchmark set

For a typical environmental DCE, a strong benchmark set would include: MNL, MNL with theory-driven nonlinearities, mixed logit or latent class model, penalised multinomial model, random forest, gradient boosting, and possibly a shallow neural network if the sample size is sufficient. For large revealed preference or sensor-linked datasets, add XGBoost, neural networks, and context-aware or RUM-constrained neural choice models. For small stated preference datasets, do not add a neural network unless the research question requires it and the sample size supports it.

Do not over-benchmark
Running 200 models without a clear target can create the illusion of scientific rigour while increasing researcher degrees of freedom. Benchmark broadly enough to learn, but narrowly enough to interpret.

15.10 Complementary workflows

The most productive frontier is not replacing DCMs with ML. It is designing complementary workflows. The following workflows are practical for PhD research and publishable applied work.

Workflow	What ML does	What DCM does	Use case
ML as benchmark	Provides a strong predictive upper benchmark	Provides behavioural and welfare model	Show whether a simple DCM leaves large predictive gains untapped
ML-assisted specification	Finds nonlinearities, interactions, or segmentation candidates	Estimates interpretable final utility	Complex environmental DCE with many attributes
Residual diagnostics	Models errors or unexplained patterns	Acts as baseline behavioural model	Detect hidden misspecification
Representation learning	Transforms images, text, or GPS traces into features	Uses features as utility covariates	Street-view, beach images, or remote-sensing attributes
Constrained neural utility	Learns flexible utility components	Maintains RUM, sign, monotonicity, or welfare constraints	Large data with welfare interest
Causal ML plus DCM	Estimates heterogeneous treatment effects or nuisance functions	Models behavioural substitution and WTP	Policy evaluation with rich covariates

Table 15.9. Productive hybrid workflows.

The uploaded literature gives examples of these workflows. Ortelli and coauthors frame assisted specification as a multi-objective combinatorial search problem. Hernandez and coauthors use association rules and random forests to support specification in participatory value evaluation. Torres Lahoz and coauthors use neural networks for latent constructs within latent class choice models. Lukawska, Jensen, and Rodrigues use a neural network to map context into interpretable shifts in a Bayesian mixed logit model. Each example treats ML as a tool for improving choice modelling rather than as a total substitute.

15.11 Environmental economics examples

Example 1: coastal recreation and access pricing

Suppose an agency wants to price access to a restored beach while controlling crowding. A random forest may predict which beach a tourist chooses using weather, crowding, distance, price, age, household composition, and past behaviour. This is useful for operational forecasting. But if the agency needs WTP for lower crowding or the welfare cost of a fee, a choice model with a cost coefficient and crowding utility is still needed. ML can help identify whether crowding effects are nonlinear or interact with trip purpose, but the welfare calculation should be based on a transparent utility specification.

Example 2: conservation programme choice

In a conservation DCE, experts or citizens choose among programmes that differ in cost, biodiversity gains, enforcement, uncertainty, and equity. ML may help predict which profiles are attractive and discover interactions between uncertainty and programme type. But if the objective is to value biodiversity improvement or compare compensating variation across programme bundles, the final model must preserve monetary trade-offs and support policy counterfactuals.

Example 3: energy technology adoption

For rooftop solar, heat pumps, or dynamic tariffs, ML can use high-dimensional household, weather, grid, and building data to predict adoption or demand response. Choice models can represent the adoption decision under alternative subsidies, payback periods, comfort restrictions, and environmental attributes. A combined workflow may use ML to estimate baseline adoption risk and a DCE to estimate how policy attributes change utility.

Example 4: image-based environmental choices

Street-view images, virtual reality beach scenes, or landscape images create attributes that are not easily reducible to a few survey variables. Convolutional neural networks or embeddings can process images. The choice modeller can then use the image representation as an attribute or interact it with cost, access, or environmental quality. This is one of the clearest cases where ML expands what can be included in a choice model.

15.12 R lab: comparing MNL and random forest on a simulated DCE

This lab is intentionally simple. The goal is not to prove that one model is better, but to show the workflow for a fair comparison. We simulate a three-alternative environmental DCE, estimate an MNL, estimate a random forest, and compare prediction on a respondent-level holdout split. In a real project, you would add mixed logit, latent class, XGBoost, and repeated resampling.

```r
# Install once if needed:
# install.packages(c("tidyverse", "mlogit", "tidymodels", "ranger"))

library(tidyverse)
library(mlogit)
library(tidymodels)

set.seed(15)

# 1. Simulate a three-alternative environmental DCE
N  <- 700       # respondents
Tn <- 8         # choice tasks per respondent
alts <- c("A", "B", "SQ")

beta_cost <- -0.07
beta_bio  <-  0.55
beta_co2  <-  0.35
beta_crowd <- -0.30

long <- expand_grid(id = 1:N, task = 1:Tn, alt = alts) %>%
  mutate(
    chid = paste(id, task, sep = "_"),
    income = rep(rlnorm(N, log(2500), .35), each = Tn * length(alts)),
    cost = if_else(alt == "SQ", 0, sample(c(10, 25, 50, 75), n(), TRUE)),
    bio  = if_else(alt == "SQ", 0, sample(0:3, n(), TRUE)),
    co2  = if_else(alt == "SQ", 0, sample(0:3, n(), TRUE)),
    crowd = if_else(alt == "SQ", 2, sample(0:3, n(), TRUE)),
    # nonlinear income-cost sensitivity in the data-generating process
    cost_sens = beta_cost * (1.2 - 0.00008 * pmin(income, 6000)),
    V = cost_sens * cost + beta_bio * bio + beta_co2 * co2 + beta_crowd * crowd
  ) %>%
  group_by(chid) %>%
  mutate(
    p = exp(V) / sum(exp(V)),
    chosen_alt = sample(alt, 1, prob = p),
    choice = as.integer(alt == chosen_alt)
  ) %>%
  ungroup()

# 2. Respondent-level split
set.seed(15)
id_split <- initial_split(tibble(id = unique(long$id)), prop = 0.8)
train_ids <- training(id_split)$id
test_ids  <- testing(id_split)$id

train_long <- long %>% filter(id %in% train_ids)
test_long  <- long %>% filter(id %in% test_ids)

# 3. Estimate MNL
mdata_train <- mlogit.data(
  train_long, choice = "choice", shape = "long",
  chid.var = "chid", alt.var = "alt", id.var = "id"
)

mnl <- mlogit(choice ~ cost + bio + co2 + crowd | 0, data = mdata_train)
summary(mnl)

# 4. Predict with MNL on the test set
# For real projects, use package-specific prediction carefully and verify ordering.
# Below is a transparent manual probability calculation for this simple model.
coef_mnl <- coef(mnl)

test_pred_mnl <- test_long %>%
  mutate(Vhat = coef_mnl["cost"] * cost + coef_mnl["bio"] * bio +
                coef_mnl["co2"] * co2 + coef_mnl["crowd"] * crowd) %>%
  group_by(chid) %>%
  mutate(phat = exp(Vhat) / sum(exp(Vhat))) %>%
  ungroup()

mnl_choice_pred <- test_pred_mnl %>%
  group_by(chid) %>%
  slice_max(phat, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  select(chid, pred_mnl = alt)

truth <- test_long %>%
  filter(choice == 1) %>%
  select(chid, truth = alt)

mean(left_join(truth, mnl_choice_pred, by = "chid")$truth ==
     left_join(truth, mnl_choice_pred, by = "chid")$pred_mnl)

# 5. Build wide data for ML classifier
wide <- long %>%
  select(id, chid, task, alt, cost, bio, co2, crowd, income, chosen_alt) %>%
```

```
  pivot_wider(names_from = alt, values_from = c(cost, bio, co2, crowd)) %>%
  distinct(id, chid, .keep_all = TRUE) %>%
  mutate(chosen_alt = factor(chosen_alt))

train_wide <- wide %>% filter(id %in% train_ids)
test_wide  <- wide %>% filter(id %in% test_ids)

# 6. Tune random forest inside training data using respondent groups
folds <- group_vfold_cv(train_wide, group = id, v = 5)

rf_spec <- rand_forest(trees = 500, mtry = tune(), min_n = tune()) %>%
  set_engine("ranger", probability = TRUE, importance = "permutation") %>%
  set_mode("classification")

wf <- workflow() %>%
  add_formula(chosen_alt ~ . - id - chid - task) %>%
  add_model(rf_spec)

grid <- grid_regular(mtry(range = c(3, 10)), min_n(range = c(2, 30)), levels = 4)

rf_tuned <- tune_grid(
  wf, resamples = folds, grid = grid,
  metrics = metric_set(accuracy, mn_log_loss)
)

best_rf <- select_best(rf_tuned, metric = "mn_log_loss")
rf_fit <- finalize_workflow(wf, best_rf) %>% fit(train_wide)

rf_pred <- predict(rf_fit, test_wide, type = "prob") %>%
  bind_cols(predict(rf_fit, test_wide, type = "class")) %>%
  bind_cols(test_wide %>% select(chosen_alt))

metrics(rf_pred, truth = chosen_alt, estimate = .pred_class)
mn_log_loss(rf_pred, truth = chosen_alt, .pred_A, .pred_B, .pred_SQ)
```

What to look for

- The MNL coefficients should have meaningful signs. If cost is positive, the model is not behaviourally credible even if accuracy is acceptable.
- The random forest may predict better because the simulated data include nonlinear cost sensitivity through income.
- The random forest does not provide a clean WTP ratio. It can suggest that income-cost interactions matter, which you can then test in a choice model.
- If you replace the respondent-level split with a row-wise split, compare the results. If performance improves suspiciously, you have demonstrated leakage.

15.13 Python lab outline

Python is useful when you need modern ML pipelines, neural networks, image processing, or large-scale prediction. For classical choice modelling, Biogeme provides a Python-based environment for discrete choice models. For ML baselines, scikit-learn and XGBoost are common choices. The same validation rules apply.

```
# Python outline: not a complete runnable script

import pandas as pd
from sklearn.model_selection import GroupShuffleSplit, GroupKFold, GridSearchCV
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, log_loss

# wide data: one row per choice task, columns include id, chosen_alt, and features
X = wide.drop(columns=["chosen_alt", "chid"])
y = wide["chosen_alt"]
groups = wide["id"]

splitter = GroupShuffleSplit(n_splits=1, test_size=0.2, random_state=15)
train_idx, test_idx = next(splitter.split(X, y, groups=groups))

X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]
y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]
groups_train = groups.iloc[train_idx]

param_grid = {"max_features": ["sqrt", 0.5], "min_samples_leaf": [1, 5, 20]}
cv = GroupKFold(n_splits=5)

rf = RandomForestClassifier(n_estimators=500, random_state=15)
search = GridSearchCV(rf, param_grid, scoring="neg_log_loss", cv=cv)
search.fit(X_train.drop(columns=["id"]), y_train, groups=groups_train)

pred_class = search.predict(X_test.drop(columns=["id"]))
pred_prob = search.predict_proba(X_test.drop(columns=["id"]))

print(accuracy_score(y_test, pred_class))
print(log_loss(y_test, pred_prob, labels=search.classes_))
```

For neural choice models, PyTorch or TensorFlow can implement utility networks with a softmax likelihood. For applied PhD work, start with tabular ML and classical choice models before moving to deep learning. Deep learning becomes more credible when the data are large, inputs are high-dimensional, or the architecture is deliberately constrained to preserve choice-theoretic meaning.

15.14 Practical decision rules

Research situation	Recommended starting point	Add ML when…	Keep DCM central when…
Small environmental DCE	MNL, mixed logit, latent class	You need specification diagnostics or regularisation	The main output is WTP or welfare
Large RP mobility dataset	MNL plus flexible ML benchmark	Prediction is central and covariates are rich	Policy simulations require substitution and welfare
Image or text choice data	Representation learning plus choice model	Raw inputs cannot be hand-coded	You must value interpretable policy attributes
Policy targeting	Predictive ML plus causal/choice analysis	The task is identifying likely adopters	The task is estimating welfare from changing incentives
Model discovery	Transparent baseline choice model	There are many potential interactions or covariates	Final claims require economic interpretation

Table 15.10. Practical model choice rules.

15.15 Common mistakes

Comparing a weak DCM to a highly tuned ML model
A default MNL is not the best representative of choice modelling. Include meaningful nonlinearities, heterogeneity, or latent classes when relevant.

Using row-wise splits with panel data
This often creates leakage. Split by respondent, household, route, location, or time depending on the target.

Reporting only accuracy
Probability quality, calibration, log loss, and behavioural plausibility often matter more than label accuracy.

Treating SHAP as causal
SHAP explains the fitted prediction function. It does not identify causal effects or WTP.

Inferring WTP from an unconstrained classifier
WTP requires utility scale and a meaningful marginal utility of money.

Ignoring support
A counterfactual outside the observed or experimental range is an extrapolation problem, not a routine prediction.

Using too complex a model for too small a DCE
High-capacity neural networks can overfit modest survey samples. Alwosheel, van Cranenburgh, and Chorus recommend a conservative sample-size perspective for ANN-based discrete choice analysis.

15.16 End-of-chapter exercises

Conceptual exercises

- Explain why a random forest can outperform MNL in accuracy but still be unsuitable for estimating WTP.
- Give an example of a policy counterfactual that is inside the support of a DCE and one that is outside the support.
- Why does grouped cross-validation matter in a stated preference survey with eight choice tasks per respondent?
- Compare a coefficient, an elasticity, a partial dependence curve, and a SHAP value. What does each explain?
- Describe a hybrid workflow in which ML improves a choice model without replacing the welfare calculation.

Applied exercises

- Using your own DCE data, estimate an MNL and a random forest. Split by respondent. Report accuracy, log loss, and one behavioural diagnostic.
- Repeat the comparison with a row-wise split. Quantify how much the performance estimate changes. Interpret the difference as a leakage diagnostic.
- Use a random forest or XGBoost model to identify the most important interactions. Re-estimate a choice model with one theoretically plausible interaction and compare WTP.
- Choose one policy scenario and write a support diagnostic: Are all attributes and combinations within the design or observed data range?
- Write a one-page methods appendix that transparently separates prediction claims, behavioural claims, welfare claims, and causal claims.

15.17 Checklist for your own paper

- Have you written the target of the analysis before comparing models?
- Is the resampling unit correct for the intended generalisation?
- Are all model families tuned or specified with comparable effort?

- Do you report probability metrics, not only accuracy?
- Do you check calibration or probability quality?
- Do you report signs, elasticities, WTP, or other behavioural outputs when relevant?
- Do you avoid WTP claims from models that do not identify utility scale?
- Do you check support before policy counterfactuals?
- Do you present uncertainty around model comparisons?
- Do you clearly state what the model cannot answer?

Chapter summary

Machine learning and discrete choice models should be evaluated in relation to the research target. ML is often strong for prediction, nonlinear pattern discovery, high-dimensional representation, and large-scale data. Discrete choice models remain central for behavioural interpretation, substitution patterns, welfare, and policy counterfactuals. The most useful applied work combines the two traditions: ML for prediction and discovery, choice theory for utility, welfare, and policy discipline.

> One-sentence takeaway
> Use ML to learn flexibly from data; use choice modelling to discipline what the learned patterns mean for behaviour, welfare, and policy.

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Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 16

Machine-Learning-Assisted Choice Model Specification

This chapter is about using machine learning to help specify discrete choice models without surrendering the behavioural, welfare, and policy logic that makes choice modelling valuable. The central object is not the most accurate black box. The central object is a defensible utility specification: one that predicts reasonably well, is behaviourally plausible, is not the product of hidden trial-and-error, and can be used for elasticities, willingness to pay, and policy counterfactuals.

How to study this chapter
Read Sections 16.1-16.5 before touching code. Then run Lab 16A and Lab 16B. After the labs, return to the checklist in Section 16.14 and ask whether your candidate model would still be defensible if a reviewer asked for every model you tried.

Learning objectives

- Explain why model specification is a scientific decision, not merely a technical search problem.
- Build a candidate feature library for a discrete choice model, including transformations and interactions.
- Use penalised likelihood, lasso, elastic net, group penalties, and Bayesian automatic relevance determination as screening tools.
- Use random forests, gradient boosting, association rules, and interpretable ML to discover candidate nonlinearities and interactions.
- Translate ML-selected patterns into theory-consistent utility functions.
- Design a transparent, reproducible, and human-in-the-loop specification workflow.
- Recognise post-selection inference problems, leakage, overfitting, and behaviourally implausible automated specifications.
- Apply ML-assisted specification to environmental DCEs, recreation demand, energy choice, transport mode choice, and participatory value evaluation.

Chapter map

16.1 Why specification matters
16.2 The specification search problem
16.3 The candidate feature library
16.4 Penalised likelihood for specification search
16.5 Grouped, hierarchical, and constrained selection
16.6 Bayesian automatic relevance determination
16.7 Tree-based ML as a specification diagnostic
16.8 Association rules and pattern mining
16.9 Assisted specification as multi-objective optimisation
16.10 Attribute non-attendance and sparse utility weights
16.11 A reproducible human-in-the-loop workflow
16.12 Environmental economics applications
16.13 R and Python labs
16.14 Reporting checklist and exercises

16.1 Why specification matters

In a discrete choice model, the researcher does not merely choose a prediction algorithm. The researcher specifies how observed attributes, individual characteristics, context, and unobserved factors enter utility. This specification determines signs, elasticities, marginal rates of substitution, willingness to pay, substitution patterns, and welfare measures. A weak specification can make a model look statistically acceptable while producing misleading policy conclusions.

Consider a stated preference experiment for coastal recreation. Respondents choose among beaches described by travel cost, travel time, crowding, water quality, seabed type, and access restrictions. A simple utility specification might include only main effects. But the real behaviour may involve interactions: high water quality may matter more when natural reefs are present; crowding may matter more for older visitors; access restrictions may be acceptable only when ecological quality is high. If the analyst includes too few terms, the model is biased by omission. If the analyst includes too many terms, estimation becomes unstable and interpretation collapses.

```
U_ni = V_ni + eps_ni
V_ni = beta_cost cost_ni + beta_quality quality_ni + beta_crowd crowd_ni + ...
```

The dots hide the real research problem: which terms should be included, in what functional form, with what interactions, and with what heterogeneity? Traditional practice often relies on theory, analyst judgement, sequential testing, information criteria, and trial-and-error. ML-assisted specification keeps theory and judgement, but makes the search more systematic, more reproducible, and better suited to high-dimensional data.

The specification principle
Machine learning should help identify candidate structure. It should not automatically decide the final behavioural model. The final model must still pass economic, statistical, and policy-use checks.

16.2 The specification search problem

Specification search is difficult because the number of possible utility functions grows quickly. Suppose a model has 12 potential main effects, 20 possible transformations, 30 two-way interactions, 8 socio-demographic interactions, alternative-specific constants, scale terms, and candidate nests or random coefficients. Even a modest DCE can generate hundreds or thousands of plausible terms. Exhaustive search is usually impossible, and informal manual search risks producing a model that is overfit to the analyst's decisions.

The problem is not only computational. It is also epistemic. A model selected after many unreported trials has weaker evidential value than the same model selected by a pre-declared or reproducible protocol. Machine learning tools can reduce the burden of manual search, but they can also increase researcher degrees of freedom unless they are embedded in a disciplined workflow.

Challenge	Why it matters	ML-assisted response	Remaining judgement
Many candidate variables	Manual search becomes slow and incomplete	Use penalisation, feature screening, and candidate libraries	Which variables are theoretically admissible?
Many interactions	Important heterogeneity may be missed	Use tree models, SHAP interactions, association rules	Which interactions can be interpreted and defended?
Nonlinear effects	Linear utility may misrepresent response curves	Use splines, bins, trees, GAMs, partial dependence	Which nonlinearities imply plausible elasticities?
Correlated predictors	Selection may be unstable	Use elastic net, group penalties, stability selection	How should correlated attributes be grouped?
Behavioural constraints	Automated search may select wrong signs	Impose sign, monotonicity, or hierarchy constraints	Which constraints are theory-based rather than convenient?
Post-selection inference	Standard errors after selection may be misleading	Use sample splitting, bootstrap, sensitivity reporting	What inference is needed for the research question?

Table 16.1. Specification search has both computational and scientific components.

16.3 The candidate feature library

The first operational step is to create a candidate feature library. This is a structured list of terms that may enter utility or class membership, together with rules about which terms are allowed, forbidden, grouped, or required. The library prevents the analysis from becoming a fishing expedition. It also makes the modelling workflow reproducible.

16.3.1 Feature classes

- Core alternative attributes: cost, time, distance, environmental quality, emissions, crowding, reliability, access, risk, service frequency.
- Individual characteristics: income, age, education, household composition, location, prior experience, environmental attitudes, political trust.
- Alternative-specific interactions: cost by income, time by employment status, quality by environmental concern, crowding by trip purpose.
- Context terms: weather, season, congestion, information treatment, survey wave, neighbourhood, pandemic period, electricity price regime.
- Nonlinear transformations: log cost, piecewise cost, splines, quadratic effects, threshold indicators, crowding bins, diminishing returns to quality.
- Availability and feasibility: car availability, beach access, distance thresholds, technology ownership, site closures.
- Heterogeneity structures: random coefficients, latent classes, class membership covariates, scale heterogeneity, error components, nests.

In environmental valuation, the cost coefficient is special because it anchors WTP. Automated methods that remove cost, allow a positive cost coefficient, or select interactions that make cost non-monotone can destroy the welfare interpretation. A defensible candidate library usually treats cost as mandatory and constrains its sign or checks sign stability very carefully.

> Welfare warning
> For DCEs used to estimate WTP, do not let a purely predictive screening procedure remove the monetary attribute. If the cost coefficient is weak, unstable, or positive, that is a substantive problem to diagnose, not a variable-selection victory.

16.3.2 Required, optional, and forbidden terms

Term type	Meaning	Examples
Required	Always included because theory, design, or welfare requires it	Cost, alternative-specific constants, status quo indicator, experimental treatment indicator
Optional	Candidate terms selected or rejected by ML-assisted search	Quality by income, time squared, crowding by visit purpose, region-specific access effect
Grouped	Terms selected together to preserve interpretation	All levels of a categorical attribute; all dummy variables for a policy package
Hierarchical	Lower-order terms included if higher-order interaction is included	Include quality and income if quality x income is included
Forbidden	Terms excluded because they leak information or violate design logic	Chosen alternative label, post-choice satisfaction, variables observed after the decision

Table 16.2. A candidate library should encode modelling rules before the automated search begins.

16.4 Penalised likelihood for specification search

Penalised likelihood is the most direct way to connect machine learning and econometric specification. Instead of choosing a model by adding or removing variables one at a time, we estimate a model with many candidate terms and penalise complexity. In a linear or generalised linear model, a typical penalised objective is:

```
min_beta Loss(beta) + lambda * Penalty(beta)
```

The penalty controls the amount of shrinkage. When lambda is large, the model is simple. When lambda is small, the model approaches the unpenalised fit. The most common penalties are ridge, lasso, and elastic net.

Penalty	Formula	Effect	Use in choice specification
Ridge	lambda sum beta_k^2	Shrinks coefficients but rarely sets them exactly to zero	Stabilise estimates with collinearity; not a pure selector
Lasso	lambda sum |beta_k|	Can set coefficients exactly to zero	Screen large candidate sets; create sparse utility specifications
Elastic net	lambda[(1-alpha)sum beta_k^2/2 + alpha sum |beta_k|]	Combines ridge stability and lasso sparsity	Better when candidate terms are correlated
Group lasso	lambda sum_g w_g ||beta_g||_2	Selects or removes groups	Keep levels of categorical attributes or alternative-specific blocks together
Hierarchical penalties	Penalty respects interaction hierarchy	Avoids isolated high-order terms	Include main effects when interactions are selected

Table 16.3. Common regularisation tools for specification search.

The lasso, introduced by Tibshirani, is attractive because it combines shrinkage and selection: some coefficients are reduced exactly to zero. The elastic net, developed by Zou and Hastie, is often more stable when variables are correlated. The general lesson is simple: penalisation turns specification search from a sequence of informal yes-or-no tests into a regularised optimisation problem.

16.4.1 Penalised choice likelihood

For a multinomial logit-style model with choice probability P_ni(beta), the negative log-likelihood is:

```
Loss(beta) = - sum_n sum_i y_ni log P_ni(beta)
```

A lasso-penalised choice model solves:

```
min_beta - sum_n sum_i y_ni log P_ni(beta) + lambda sum_k |beta_k|
```

This formulation is conceptually straightforward, but implementation details matter. In many DCE datasets, the data are alternative-specific and panel-structured. A standard multinomial classifier on a wide dataset can be useful as a screening tool, but it is not the same as estimating a conditional logit with full choice-set structure. When using off-the-shelf software such as glmnet, be explicit about whether the objective is preliminary feature screening, prediction, or final choice-model estimation.

> Screening is not estimation
> A lasso classifier estimated on a wide table can suggest useful terms, but the final welfare model should usually be re-estimated as a proper choice model with the correct choice-set, panel, availability, and likelihood structure.

16.4.2 Choosing lambda

The penalty parameter lambda is usually chosen by cross-validation. For panel choice data, cross-validation must respect respondent grouping: all choice tasks from the same respondent should be in the same fold. Otherwise the validation error may be optimistic because the model learns respondent-specific patterns in training and is tested on the same respondents in the test fold.

Two common choices are lambda.min and lambda.1se. The first minimises the cross-validated error. The second chooses the simplest model whose error is within one standard error of the minimum. For specification search in economics, lambda.1se is often attractive because it favours parsimony and interpretability.

```
lambda.1se = largest lambda with CV error <= min(CV error) + 1 standard error
```

16.5 Grouped, hierarchical, and constrained selection

Standard lasso selects individual coefficients. Choice modelling often needs selection at a more structured level. Suppose seabed type is represented by four dummies. Selecting one level and dropping the others may be hard to interpret. Suppose a two-way interaction between water quality and income is selected while both main effects are omitted. This violates the hierarchy that many applied researchers expect. Suppose a cost coefficient becomes positive in a subgroup. This violates a core behavioural constraint.

16.5.1 Group selection

Group penalties select blocks of coefficients together. This is useful for categorical attributes, alternative-specific effects, splines, and multi-level policy variables. For example, all dummy variables representing beach habitat type can be treated as one group. The model then decides whether habitat type matters, rather than selecting an arbitrary subset of its labels.

16.5.2 Hierarchical selection

A strong hierarchy rule says that an interaction can enter only if both associated main effects enter. A weak hierarchy rule says that at least one associated main effect must enter. Hierarchical rules are valuable in environmental choice models because they preserve interpretability. If water quality x income is included, the model should also contain water quality and income-related terms needed to interpret the interaction.

16.5.3 Sign and monotonicity constraints

Environmental valuation models often require monotonicity in cost and sometimes in environmental quality. A predictive algorithm may learn local non-monotonicities because of sampling noise, omitted variables, or correlations in the design. Before translating ML-discovered nonlinearities into a DCE specification, ask whether the implied marginal utilities and WTP values are plausible over the policy-relevant range.

Constraint	Reason	Example diagnostic
Cost negative	Needed for meaningful WTP	Check sign in full sample and subgroups; inspect WTP distribution
Quality non-negative	Environmental improvement should not reduce utility unless trade-offs or context explain it	Plot marginal utility of quality by context
Interaction hierarchy	Avoid uninterpretable isolated terms	If quality x income selected, include quality and income terms
Grouped categorical terms	Avoid label-specific artefacts	Select all levels of habitat type together
Availability consistency	Unavailable alternatives should not be chosen	Audit availability coding before ML screening

Table 16.4. Behavioural constraints are not a nuisance; they define what a useful choice model means.

16.6 Bayesian automatic relevance determination

Automatic relevance determination (ARD) is a Bayesian approach to feature relevance. Instead of placing the same prior variance on every coefficient, ARD assigns each coefficient or group of coefficients its own shrinkage parameter. Irrelevant terms receive strong shrinkage toward zero; relevant terms retain larger prior variance. In the choice-modelling literature, DCM-ARD uses this idea to search utility specifications among many candidate terms.

```
beta_k ~ Normal(0, tau_k^2)
small tau_k -> coefficient k shrunk toward zero
large tau_k -> coefficient k allowed to matter
```

Compared with lasso, ARD has a probabilistic interpretation and can be naturally extended to Bayesian choice modelling. Compared with manual testing, it can explore very large candidate libraries. But ARD

still requires a candidate set, priors, convergence checks, and human interpretation. An ARD-selected term is not automatically a causal mechanism; it is evidence that the term helped explain choices under the model and data used.

ARD in one sentence
ARD asks: which candidate utility terms need non-negligible prior variance to explain the observed choices?

16.7 Tree-based ML as a specification diagnostic

Random forests and gradient boosting are often used as predictive benchmarks. In this chapter, we use them differently: as diagnostic devices for discovering candidate nonlinearities, thresholds, interactions, and neglected heterogeneity. A tree naturally partitions the data: for example, crowding may matter mostly for weekend visits, or travel time may matter only beyond 45 minutes. An ensemble averages many such trees and can reveal robust patterns.

16.7.1 What to extract from a tree model

- Permutation importance: which variables reduce predictive performance when shuffled?
- Partial dependence or accumulated local effects: how does predicted choice probability vary with one attribute?
- SHAP values: which variables drive predictions globally and locally?
- Interaction strength: which pairs of variables repeatedly interact in the trees?
- Residual diagnostics: where does a simple MNL fail compared with a flexible ML model?

The point is not to copy the tree into the utility function. The point is to translate repeated predictive patterns into a small number of behaviourally plausible terms. If XGBoost finds that crowding matters nonlinearly, you might test piecewise crowding, log crowding, or threshold indicators in a mixed logit. If random forests show that environmental quality interacts with prior experience, you might include quality x experience or estimate separate heterogeneity structures.

Tree importance warning
Variable importance is not causal importance. A variable can be important because it proxies for another variable, because of design imbalance, or because it is correlated with unobserved context.

16.7.2 From ML diagnostic to utility term

ML finding	Possible utility specification	Check before accepting
Cost effect is steeper at low costs	log(cost + c), piecewise cost, or cost spline	Does WTP remain plausible?
Crowding threshold around 0.5 people per 10 m2	indicator(crowding > 0.5) or piecewise crowding	Is threshold design-supported or sample artefact?
Quality matters mostly for experienced visitors	quality x experience interaction	Is experience pre-choice and measured reliably?
Status quo chosen by older respondents	SQ ASC x age or latent class membership	Does it reflect inertia, risk aversion, or survey fatigue?
Policy package combinations recur	interaction between package attributes	Is interaction theoretically meaningful or driven by design correlation?

Table 16.5. ML diagnostics should be translated into small, interpretable candidate terms.

16.8 Association rules and pattern mining

Association rules identify frequent co-occurrences. In retail, the classic example is that people who buy item A also buy item B. In complex choice experiments, especially portfolio choices or participatory value evaluation, association rules can reveal combinations of alternatives or attributes that respondents frequently select together. This is useful when the choice object is a bundle rather than a single alternative.

```
Rule: A -> B
Support = Pr(A and B)
Confidence = Pr(B | A)
Lift = Pr(A and B) / [Pr(A) Pr(B)]
```

In participatory value evaluation, respondents may choose a set of policies subject to a budget constraint. Association rules can suggest complementarity or substitution between policy items. For example, respondents selecting wetland restoration may also select flood-risk communication. The analyst can then test whether a portfolio choice model should include interaction terms between these policy alternatives.

The uploaded Hernandez et al. study is directly relevant here: it uses association rules and random forests to support the specification of a portfolio choice model for complex PVE data. The important lesson is not that association rules replace behavioural modelling. They provide structured clues for improving the behavioural model.

16.9 Assisted specification as multi-objective optimisation

The Ortelli, Hillel, Pereira, de Lapparent, and Bierlaire approach frames assisted specification as a multi-objective combinatorial optimisation problem. The idea is close to how expert modellers work: search over candidate specifications, balance statistical fit and complexity, and retain a set of promising models rather than blindly selecting one winner.

This perspective is useful because specification is not a single-objective problem. Maximum fit alone favours overly complex models. Minimum complexity alone favours overly simple models. A practical choice model needs a Pareto compromise between fit, parsimony, interpretability, behavioural plausibility, and policy usefulness.

```
Objective 1: improve model fit
Objective 2: reduce complexity
Constraints: required terms, sign rules, hierarchy, interpretability, estimation
feasibility
```

The output should be a Pareto frontier: a set of models where no model is strictly better on all objectives. The human modeller then inspects the frontier, checks signs and behavioural outputs, compares predictive and likelihood performance, and selects a specification that is scientifically defensible.

Stage	Automated component	Human component
Candidate generation	Enumerate transformations, interactions, and parameter structures	Define what is meaningful and admissible
Model search	Metaheuristic or optimisation algorithm explores specifications	Impose constraints and stop rules
Pareto filtering	Keep non-dominated models by fit and complexity	Reject behaviourally implausible models
Final estimation	Estimate short-listed models	Check signs, WTP, elasticities, substitution patterns
Reporting	Archive search settings and candidate set	Explain why the final model was chosen

Table 16.6. Assisted specification should be human-in-the-loop, not human-free.

16.10 Attribute non-attendance and sparse utility weights

Attribute non-attendance occurs when respondents ignore one or more attributes in a stated choice task. It is important in environmental valuation because ignoring cost or environmental quality can bias WTP. Sparse modelling is tempting here: if a coefficient is shrunk to zero, perhaps the attribute was ignored. But the interpretation is delicate. A near-zero coefficient can reflect true indifference, design weakness, measurement error, collinearity, heterogeneity, or non-attendance.

A lasso-based approach can be useful as a diagnostic for potential non-attendance, especially when combined with respondent-level or class-level modelling. For example, if an organic label or country-of-origin attribute is repeatedly screened out conditional on other attributes, that pattern may indicate non-attendance or weak relevance. But it should be triangulated with stated ANA questions, response times, certainty measures, latent class models, or follow-up tasks.

Evidence	What it may indicate	What else to check
Coefficient shrunk to zero	Low predictive contribution	Design variation, collinearity, scale, subgroup effects
Respondent self-reports ignoring attribute	Stated non-attendance	Consistency with choices and response time
Class with near-zero attribute coefficient	Segment-level non-attendance or indifference	Class size, interpretability, stability
Fast repeated choices	Heuristic processing	Straight-lining, task complexity, learning/fatigue
Attribute ignored only under high complexity	Cognitive burden	Experimental design and task order

Table 16.7. Sparse coefficients are diagnostic evidence, not definitive proof of non-attendance.

16.11 A reproducible human-in-the-loop workflow

The best ML-assisted specification workflow is transparent, auditable, and conservative. It uses ML to reduce the search space and identify patterns, but it reserves final interpretation for a theory-consistent choice model. The workflow below is designed for a PhD student analysing an environmental DCE or transport mode choice dataset.

Step	Action	Output
1. Define target	Prediction, WTP, welfare, segmentation, policy simulation, or model discovery	Primary and secondary evaluation criteria
2. Audit data	Check choice sets, availability, missingness, panel structure, experimental design, leakage variables	Clean modelling dataset
3. Build library	List required, optional, grouped, hierarchical, and forbidden terms	Candidate feature dictionary
4. Split properly	Use respondent-level or market-level splits; keep panel units intact	Training, validation, and test partitions
5. Screen	Run lasso/elastic net/ARD/tree diagnostics on training folds	Short list of candidate effects
6. Translate	Convert ML patterns to utility terms	Interpretable candidate specifications
7. Estimate DCMs	Fit MNL, mixed logit, latent class, or hybrid models	Behavioural model outputs
8. Compare	Evaluate log likelihood, prediction, calibration, signs, WTP, elasticities	Model comparison table
9. Stress test	Bootstrap, stability selection, subgroup checks, policy-range diagnostics	Robustness evidence
10. Report	Document search space, constraints, software, tuning, rejected models	Reproducible appendix

Table 16.8. A full workflow for ML-assisted choice model specification.

Practical rule
Use the test set only once for final comparison. Use cross-validation inside the training data for selection and tuning. If you use the test set to choose the specification, it is no longer a test set.

16.12 Environmental economics applications

16.12.1 Coastal recreation and crowding

A coastal DCE may contain beach quality, seabed type, access restrictions, distance, cost, and crowding. ML can help identify nonlinear crowding thresholds, interactions between crowding and activity type, or different responses among experienced and inexperienced visitors. The final choice model might include a piecewise crowding variable, an interaction between natural reef and water quality, and random taste heterogeneity in crowding sensitivity.

16.12.2 Energy tariffs and green technology adoption

Energy tariff choices often involve complex interactions among price, reliability, contract duration, renewable share, information treatments, and household characteristics. Gradient boosting may reveal that the renewable share matters only above a threshold, or that price sensitivity differs sharply by income and energy literacy. These patterns can be translated into spline terms, income interactions, or latent classes.

16.12.3 Conservation portfolios and participatory value evaluation

In conservation portfolio problems, respondents choose bundles of policies under a budget constraint. Association rules can reveal policy complementarities, while random forests can identify which policy attributes predict inclusion in the bundle. The behavioural model can then test selected interactions while preserving the budget and choice-set structure.

16.12.4 Food labels and attribute non-attendance

In food and agricultural valuation, lasso can screen label, origin, price, and sustainability terms. If a label is consistently screened out when origin and size are included, the analyst might investigate whether consumers ignore the label, confuse it with another attribute, or respond only in certain segments. This mirrors the logic of recent work using LASSO to study attribute non-attendance in choice experiments.

16.13 R and Python labs

Lab 16A. Lasso screening for a simulated environmental DCE

This lab uses a wide-format multinomial classifier for preliminary screening. It is not a substitute for the final choice model. Its purpose is to identify useful main effects and interactions that can then be translated into a proper discrete choice specification.

```
# Lab 16A: lasso screening in R
# Install once if needed:
# install.packages(c("tidyverse", "glmnet", "rsample", "mlogit"))

library(tidyverse)
library(glmnet)
library(rsample)
library(mlogit)

set.seed(1601)
N  <- 600
Tn <- 8
alts <- c("A", "B", "SQ")
```

```
long <- expand_grid(id = 1:N, task = 1:Tn, alt = alts) %>%
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = if_else(alt == "SQ", 0, sample(c(5, 15, 35, 60), n(), TRUE)),
    quality = if_else(alt == "SQ", 0, sample(0:3, n(), TRUE)),
    crowding = if_else(alt == "SQ", 0, runif(n(), 0.1, 1.5)),
    reef = if_else(alt == "SQ", 0, rbinom(n(), 1, .45)),
    access = if_else(alt == "SQ", 0, rbinom(n(), 1, .35))
  ) %>%
  group_by(id) %>%
  mutate(income = sample(c("low", "mid", "high"), 1),
         experienced = rbinom(1, 1, .50)) %>%
  ungroup() %>%
  mutate(
    V = -0.065 * cost + 0.45 * quality - 0.80 * crowding +
        0.35 * reef - 0.25 * access +
        0.35 * quality * reef - 0.45 * crowding * experienced +
        if_else(alt == "SQ", 0.20, 0)
  ) %>%
  group_by(chid) %>%
  mutate(prob = exp(V) / sum(exp(V)),
         chosen_alt = sample(alt, 1, prob = prob),
         choice = as.integer(alt == chosen_alt)) %>%
  ungroup()

wide <- long %>%
  select(id, chid, task, alt, cost, quality, crowding, reef, access,
         income, experienced, chosen_alt) %>%
  pivot_wider(names_from = alt,
              values_from = c(cost, quality, crowding, reef, access)) %>%
  distinct(chid, .keep_all = TRUE) %>%
  mutate(chosen_alt = factor(chosen_alt))

# Respondent-level split to avoid leakage.
split <- group_initial_split(wide, group = id, prop = .8)
train <- training(split)
test  <- testing(split)

x_train <- model.matrix(
  chosen_alt ~ . - id - chid - task,
  data = train
)[, -1]
y_train <- train$chosen_alt

cvfit <- cv.glmnet(
  x = x_train,
  y = y_train,
  family = "multinomial",
  alpha = 1,
  type.measure = "deviance",
  nfolds = 5
)

plot(cvfit)
coef(cvfit, s = "lambda.1se")

# Interpret the non-zero terms as screening evidence only.
# Translate selected interactions into a proper long-format DCM.
```

Lab 16B. Re-estimate a theory-consistent choice model

After screening, re-estimate a parsimonious discrete choice model in long format. Here the selected terms are quality x reef and crowding x experienced. Cost is kept mandatory because the model is intended for WTP analysis.

```
mdata <- mlogit.data(
  long,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

mnl_screened <- mlogit(
  choice ~ cost + quality + crowding + reef + access +
    I(quality * reef) + I(crowding * experienced) | 0,
  data = mdata
)

summary(mnl_screened)

b <- coef(mnl_screened)
wtp_quality <- -b["quality"] / b["cost"]
wtp_reef_interaction <- -b["I(quality * reef)"] / b["cost"]
```

```
wtp_quality
wtp_reef_interaction
```

Lab 16C. Tree diagnostics for nonlinearities

```
# Install once if needed:
# install.packages(c("tidymodels", "ranger", "vip"))

library(tidymodels)
library(ranger)
library(vip)

set.seed(1602)
folds <- group_vfold_cv(train, group = id, v = 5)

rf_spec <- rand_forest(trees = 600, mtry = tune(), min_n = tune()) %>%
  set_engine("ranger", importance = "permutation") %>%
  set_mode("classification")

rf_wf <- workflow() %>%
  add_formula(chosen_alt ~ . - id - chid - task) %>%
  add_model(rf_spec)

rf_grid <- grid_regular(mtry(range = c(2, 10)), min_n(range = c(2, 25)), levels = 4)

rf_tuned <- tune_grid(
  rf_wf,
  resamples = folds,
  grid = rf_grid,
  metrics = metric_set(mn_log_loss, accuracy)
)

best <- select_best(rf_tuned, metric = "mn_log_loss")
rf_fit <- finalize_workflow(rf_wf, best) %>% fit(train)

# Variable importance as diagnostic evidence.
rf_fit %>% extract_fit_parsnip() %>% vip(num_features = 20)
```

Lab 16D. Python sketch with scikit-learn and Biogeme

Python is useful for flexible ML screening and for Biogeme-based choice estimation. A typical workflow is to use scikit-learn for regularised screening or tree diagnostics, then use Biogeme to estimate a theory-consistent choice model.

```
# Python sketch: screening with scikit-learn
import pandas as pd
from sklearn.model_selection import GroupKFold, cross_val_score
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression

features = ["cost_A", "quality_A", "crowding_A", "reef_A",
            "cost_B", "quality_B", "crowding_B", "reef_B",
            "income", "experienced"]
X = wide[features]
y = wide["chosen_alt"]
groups = wide["id"]

preprocess = ColumnTransformer([
    ("cat", OneHotEncoder(drop="first"), ["income"]),
    ("num", StandardScaler(), [c for c in features if c != "income"])
])

clf = Pipeline([
    ("prep", preprocess),
    ("model", LogisticRegression(
        penalty="l1", solver="saga", multi_class="multinomial", C=0.2,
        max_iter=5000))
])

cv = GroupKFold(n_splits=5)
scores = cross_val_score(clf, X, y, cv=cv, groups=groups,
                         scoring="neg_log_loss")
print(scores.mean())

# Then translate selected terms into a Biogeme utility specification.
```

Lab interpretation
The lasso and random forest labs are screening tools. The final model should be estimated and interpreted as a choice model, not as a generic classifier, when the target is welfare, WTP, substitution, or policy simulation.

16.14 Reporting checklist and exercises

Reporting checklist

- State the primary target: prediction, WTP, welfare, segmentation, policy counterfactual, or specification discovery.
- Report the candidate library: required terms, optional terms, transformations, interactions, groups, hierarchy rules, and forbidden variables.
- Describe the resampling design and confirm that panel units or markets were not split incorrectly.
- Report tuning choices: penalty type, lambda rule, alpha, number of folds, random seeds, and software versions.
- Distinguish screening results from final estimated choice models.
- Report rejected specifications when they are substantively important, especially if they had good fit but implausible signs.
- Check cost sign, WTP distributions, elasticities, substitution patterns, calibration, and predictive performance.
- Use stability checks: bootstrap selection frequencies, alternative folds, subgroup estimation, and policy-range diagnostics.
- Provide code and a reproducible appendix where possible.

Conceptual exercises

- A lasso screen drops the cost variable from a DCE model. What should you do before reporting WTP?
- A random forest strongly predicts beach choice and identifies crowding as important. Explain why this does not automatically identify the welfare loss from crowding.
- Construct a candidate feature library for a DCE on renewable electricity contracts. Mark required, optional, grouped, hierarchical, and forbidden terms.
- Explain the difference between lambda.min and lambda.1se. Which would you prefer for a chapter in a PhD thesis and why?
- A tree model suggests that water quality matters only for high-income respondents. Propose two theory-consistent ways to test this inside a choice model.
- Why is group-level splitting essential in panel DCEs? Create an example of leakage from row-wise splitting.
- Describe a Pareto frontier for assisted specification. What objectives would you include besides likelihood?
- Explain why post-selection p-values from a final refitted model should be interpreted cautiously.

Applied mini-project

Choose a stated preference or revealed preference dataset from your own research. Build a candidate feature library and run two screening methods: one sparse method, such as lasso or elastic net, and one diagnostic method, such as random forest or gradient boosting. Translate no more than five ML-discovered patterns into candidate utility terms. Estimate a baseline MNL, a screened MNL, and one heterogeneity model. Compare them on prediction, signs, WTP, and policy simulation. In the appendix, report the candidate library and the selected terms.

Chapter takeaway

ML-assisted specification is most valuable when it makes the modelling process more systematic, transparent, and empirically informed. It is most dangerous when it hides researcher discretion behind an algorithm. The goal is not to automate away economic judgement. The goal is to use regularisation, tree diagnostics, association rules, Bayesian relevance determination, and multi-objective search to create better candidate models, then judge those models using the standards of choice modelling: behavioural plausibility, welfare validity, counterfactual discipline, and reproducibility.

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Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 17
Neural Choice Models and Hybrid Architectures

Neural choice models use neural networks to learn part, or all, of a choice model. Some are ordinary classifiers applied to choice data. Others are explicitly designed to respect random utility maximisation, alternative-specific attributes, availability sets, substitution patterns, taste heterogeneity, and welfare logic. This distinction is crucial. A neural network can be excellent at predicting which option is chosen and still be unsuitable for willingness-to-pay estimation or policy counterfactuals. A hybrid neural choice model tries to keep the predictive flexibility of deep learning while preserving the behavioural structure that makes discrete choice modelling useful for environmental economics.

How to study this chapter. Read Sections 17.1-17.6 for the conceptual framework. Then run Lab 17A to see the difference between a softmax classifier and a neural utility model. Read Sections 17.7-17.9 before using neural models for welfare analysis. Finally, use the reporting checklist in Section 17.13 as if you were preparing an empirical paper for review.

Learning objectives

After studying this chapter, you should be able to:

- Distinguish ordinary neural classifiers from neural models that are compatible with random utility theory.
- Explain the role of embeddings, residual utilities, taste networks, context networks, and multitask learning in choice modelling.
- Write the likelihood for a neural utility model and understand how it is estimated by gradient-based optimisation.
- Recognise when neural flexibility threatens behavioural interpretation, welfare analysis, and external validity.
- Use sign constraints, monotonicity constraints, separability restrictions, and money-fungibility restrictions to discipline neural choice models.
- Compute marginal utilities, elasticities, and willingness-to-pay measures from differentiable neural utility functions.
- Implement basic neural choice workflows in R and Python and understand when specialised software is preferable.

Chapter map

17.1	Why neural choice models?
17.2	From logit to neural networks
17.3	Four families of neural choice architectures
17.4	Random utility consistency and architectural restrictions
17.5	Hybrid architectures in recent choice-modelling research
17.6	Training, validation, sample size, and regularisation
17.7	Interpretation: gradients, elasticities, WTP, and diagnostics
17.8	Environmental economics applications
17.9	R lab: from softmax prediction to a neural utility workflow
17.10	Python lab: a minimal PyTorch neural utility model
17.11	Software landscape
17.12	Common mistakes
17.13	Reporting checklist and exercises

17.1 Why neural choice models?

Discrete choice models and neural networks both transform covariates into choice probabilities. In a multinomial logit model, the systematic utility is often specified as a linear index,

$$V_{ni} = x'_{ni}\beta,$$

and the probability of alternative i for decision maker n is

$$P_{ni} = \frac{\exp(V_{ni})}{\sum_{j \in C_n} \exp(V_{nj})}.$$

A neural classifier also often ends with a softmax layer,

$$\hat{p}_{ni} = \frac{\exp(z_{ni})}{\sum_{j \in C_n} \exp(z_{nj})},$$

where the logits z_{ni} are produced by a neural network. This mathematical similarity is the source of both the opportunity and the danger. The opportunity is that neural networks can learn nonlinearities, high-order interactions, embeddings, and context-dependent heterogeneity. The danger is that the scores z_{ni} may not have a behavioural meaning. They may mix alternative attributes, decision-maker characteristics, labels, and context in ways that predict well but violate the logic required for welfare analysis.

In environmental economics, this distinction is central. If the task is to forecast which households are likely to adopt a green electricity contract, a black-box neural classifier may be useful. If the task is to estimate willingness to pay for biodiversity, the cost coefficient, the environmental attribute derivatives, and the counterfactual welfare measure must be interpretable and stable. Neural choice modelling is therefore not about replacing econometrics with deep learning. It is about deciding which parts of the model can be learned flexibly and which parts must be constrained by economic theory.

Key idea. A neural choice model is useful only if its architecture matches the research question. Prediction tasks can tolerate more flexible black boxes. Welfare, elasticity, and counterfactual policy tasks require utility structure, sign discipline, scale awareness, and careful validation.

17.1.1 The practical motivation

Neural architectures are attractive in choice modelling for at least six reasons.

1. **Nonlinear utility.** The marginal utility of travel time, income, crowding, pollution, or biodiversity may not be constant.
2. **High-dimensional interactions.** Preferences may depend on combinations of attributes and individual characteristics, such as income by cost, age by access restrictions, or prior experience by environmental quality.
3. **Latent representations.** Categorical variables, text, images, GPS traces, and survey items can be mapped into low-dimensional representations.
4. **Large revealed-preference data.** Smartphone, smart-meter, mobility, and platform data may be too large or complex for conventional manual specification.
5. **Context-dependent heterogeneity.** Preferences may vary across weather, congestion, day of week, season, information exposure, and choice occasion.

6. **Joint learning across tasks.** Revealed-preference and stated-preference data, or multiple related choice tasks, can be modelled jointly with shared layers.

However, all six advantages have corresponding risks. Nonlinear models can learn implausible effects. High-dimensional interactions can overfit. Latent representations can be difficult to interpret. Large data can contain leakage. Context variables can proxy for unobserved confounding. Multitask learning can borrow strength across tasks but also transfer bias.

17.2 From logit to neural networks

This section builds the neural choice model from the multinomial logit model.

17.2.1 The multinomial logit as a shallow network

A linear MNL can be viewed as a very small neural network with no hidden layers. The inputs are alternative-specific attributes x_{ni}, the weights are β, the score is $V_{ni} = x'_{ni}\beta$, and the final layer is softmax. Estimation by maximum likelihood minimises the negative log-likelihood,

$$\mathcal{L}(\beta) = -\sum_{n}\sum_{i \in C_n} y_{ni} \log P_{ni},$$

where $y_{ni} = 1$ if alternative i is chosen and zero otherwise. In machine learning language, this is the cross-entropy loss.

A neural utility model replaces the linear utility with a function learned from data:

$$V_{ni} = f_\theta(x_{ni}, s_n, c_{nt}),$$

where s_n are decision-maker characteristics and c_{nt} are contextual variables. The probability remains logit-like:

$$P_{ni}(\theta) = \frac{\exp(f_\theta(x_{ni}, s_n, c_{nt}))}{\sum_{j \in C_n} \exp(f_\theta(x_{nj}, s_n, c_{nt}))}.$$

The model is trained by minimising

$$\mathcal{L}(\theta) = -\sum_{n}\sum_{i \in C_n} y_{ni} \log P_{ni}(\theta) + \lambda\Omega(\theta),$$

where $\Omega(\theta)$ is a regularisation penalty. This makes the connection with MNL direct: neural choice models often retain a choice likelihood but learn a more flexible utility function.

17.2.2 What changes when utility is nonlinear?

In a linear MNL, the marginal utility of attribute x_k is simply β_k. In a neural utility model,

$$\frac{\partial V_{ni}}{\partial x_{nik}} = \frac{\partial f_\theta(x_{ni}, s_n, c_{nt})}{\partial x_{nik}},$$

which may vary across alternatives, individuals, and contexts. The implied willingness to pay for attribute k is then a local derivative ratio,

$$WTP_{nik} = -\frac{\partial V_{ni}/\partial x_{nik}}{\partial V_{ni}/\partial cost_{ni}}.$$

This is attractive because WTP can vary flexibly. It is also dangerous because the denominator may be near zero, positive, noisy, or learned from extrapolation. For welfare work, the model must be trained and constrained so that the cost derivative is negative and economically meaningful over the relevant support.

17.3 Four families of neural choice architectures

Neural choice models can be organised into four broad families. The distinction is pedagogical; real models often combine elements from more than one family.

17.3.1 Family 1: ordinary neural classifiers

The simplest approach reshapes the choice task into a classification problem. A row contains variables for all alternatives, and the neural network predicts the chosen label. This approach is easy to implement with standard packages, but it is the least structurally disciplined.

$$\hat{y}_n = g_\theta(x_{n1}, x_{n2}, \ldots, x_{nJ}, s_n, c_n).$$

This can work well for prediction, especially when the choice set is fixed and alternatives are naturally labelled. But it can fail when choice sets vary, availability changes, new alternatives are introduced, or welfare outputs are required. It can also violate invariance to alternative ordering unless the architecture is carefully designed.

17.3.2 Family 2: neural utility models

A neural utility model computes a utility score for each alternative, then applies a softmax over the available alternatives. This preserves the within-choice-set comparison logic.

$$V_{ni} = f_\theta(x_{ni}, s_n, c_n), \qquad P_{ni} = \text{softmax}_i(V_{n1}, \ldots, V_{nJ}).$$

If the same function f_θ is shared across alternatives, the model respects a kind of alternative exchangeability. If each alternative has its own function f_{θ_i}, the model can capture alternative-specific effects but requires more data and stronger diagnostics.

17.3.3 Family 3: hybrid neural-discrete choice models

Hybrid models keep part of the utility explicitly econometric and let a neural network learn another part. A common residual form is

$$V_{ni} = x'_{ni}\beta + h_\theta(x_{ni}, s_n, c_n),$$

where $x'_{ni}\beta$ is the interpretable base utility and h_θ is a flexible residual. The residual can be constrained so that it only captures specific nonlinearities or interactions. This family is useful when the analyst wants to preserve known effects, such as a negative cost effect, while letting the data learn complex patterns elsewhere.

A taste-network version writes

$$\beta_n = b_\theta(s_n), \qquad V_{ni} = x'_{ni}\beta_n,$$

where a neural network maps respondent characteristics into taste parameters. This is a flexible version of observed taste heterogeneity. If the network output is interpretable, the analyst can study how tastes vary across the population.

17.3.4 Family 4: representation-learning choice models

Some choice data contain high-dimensional inputs such as images, text, route traces, street-view scenes, or long categorical histories. Representation learning maps these inputs into lower-dimensional features:

$$z_n = e_\theta(\text{text, image, GPS, sequence}), \qquad V_{ni} = x'_{ni}\beta + z'_n\gamma_i.$$

Examples include convolutional networks for images, recurrent or transformer models for sequences, and embedding layers for categorical variables. In environmental economics, representation learning can help bring satellite imagery, landscape photographs, open-ended survey responses, or social-media text into choice models. The challenge is to connect these representations to interpretable economic trade-offs rather than treating them as opaque prediction features.

Table 17.1. Neural choice architecture families

Family	Typical use	Strength	Main risk
Neural classifier	Fixed-label prediction	Easy and predictive	Weak behavioural structure
Neural utility model	Choice probabilities over alternatives	Preserves softmax choice logic	Utility may be hard to interpret
Hybrid neural-DCM	Prediction plus behavioural outputs	Combines structure and flexibility	Residual may absorb core effects
Representation-learning choice model	Text, images, GPS, large categorical data	Uses nontraditional data	Latent features may be opaque

17.4 Random utility consistency and architectural restrictions

A neural model is not automatically a random utility model. The relevant question is not whether the final layer is a softmax. The question is whether the model's scores can be interpreted as systematic utilities from a behavioural model, and whether the resulting probabilities are stable under the operations needed for policy analysis.

17.4.1 Choice-set structure

A choice model must condition on the set of available alternatives C_n. If an alternative is unavailable, it should not enter the denominator. In software, this means masking unavailable alternatives before the softmax:

$$P_{ni} = \frac{\exp(V_{ni})\mathbf{1}(i \in C_n)}{\sum_{j \in C_n} \exp(V_{nj})}.$$

Ignoring availability can severely bias predictions and welfare calculations. In transport, not every person has access to a car. In environmental recreation, not every site is available in every season. In energy contracts, not every tariff is offered in every region.

17.4.2 Alternative-specific separability

A choice model used for welfare analysis often requires that the utility of alternative i depends on its own attributes and relevant decision-maker or context variables, not directly on irrelevant coding artefacts from other alternatives. This does not mean cross-alternative attributes can never matter. Relative attributes, menus, and context can matter. But the model should make the role of cross-alternative information explicit.

A dangerous architecture is one that feeds all attributes of all alternatives into a generic dense network and outputs all utilities. Such a model can learn arbitrary relationships such as the cost of alternative B entering the utility of alternative A in uninterpretable ways. A safer architecture uses a shared utility module applied to each alternative, plus explicitly defined comparison features if needed.

17.4.3 Monotonicity and sign discipline

For many economic variables, the expected sign is known. Utility should generally decrease with monetary cost, travel time, risk, pollution, and crowding, holding other attributes constant. Utility should generally increase with environmental quality, biodiversity, reliability, or access to desirable amenities, although context may matter.

Neural networks do not respect signs unless told to. Sign discipline can be introduced through:

- constrained linear components, such as $\beta_{cost} = -\exp(\alpha)$;
- monotonic neural networks, where weights connected to a variable are constrained to be nonnegative or nonpositive;
- post-estimation derivative checks over the empirical support;
- penalty terms for sign violations;
- partial dependence or accumulated local effect diagnostics.

17.4.4 Money fungibility

In a standard WTP calculation, one euro should have the same marginal utility regardless of which non-cost attribute is being evaluated. If a neural architecture uses cost in many hidden interactions, the implied marginal utility of money can vary in ways that may or may not be economically acceptable. For environmental valuation, this is especially sensitive because welfare estimates are ratios involving the cost derivative. Recent economically consistent neural choice architectures explicitly restrict how cost enters the network so that willingness-to-pay measures remain meaningful.

17.4.5 Scale and error structure

In random utility models, the scale of utility is tied to the variance of the unobserved error. Neural networks can produce arbitrarily large scores unless regularised. A model with excellent classification accuracy may be overconfident, producing poorly calibrated probabilities. Calibration matters for market shares, policy simulations, and expected welfare. Neural choice models should therefore report not only accuracy but also log loss, calibration, probability distributions, and sensitivity to scale.

17.5 Hybrid architectures in recent choice-modelling research

This section reviews several influential architecture ideas. The purpose is not to memorise model names, but to learn how researchers combine neural flexibility with choice-theoretic structure.

17.5.1 Neural networks for latent variables and latent classes

Latent class choice models segment the population into unobserved classes. Hybrid choice models use latent constructs such as attitudes, norms, or perceptions. A conventional hybrid choice model often specifies structural equations for latent variables. Neural networks can replace or supplement these structural equations when the relationship between indicators and latent constructs is nonlinear.

Torres Lahoz et al. propose incorporating attitudinal indicators into latent class choice models by using artificial neural networks to formulate latent variable constructs. The important design idea is that the neural network is not used as a free-standing classifier. It is inserted into

a latent class choice structure so that the final model still has class membership, class-specific utilities, and interpretable choice parameters.

For environmental economics, this is promising when attitudes toward climate change, trust in institutions, place attachment, perceived ecological responsibility, or scepticism about policy claims are measured with multiple survey items. A neural latent construct can summarise these indicators flexibly while the choice component estimates trade-offs over cost, environmental quality, and policy design.

17.5.2 Context-aware mixed multinomial logit

Mixed logit allows random taste heterogeneity, but many specifications assume that an individual's preference parameters are stable across choice occasions. The context-aware Bayesian mixed multinomial logit model relaxes this by using a neural network to map contextual information into shifts in preference parameters. In simplified form,

$$\beta_{nt} = \bar{\beta}_n + \Delta_\theta(c_{nt}), \qquad V_{nit} = x'_{nit}\beta_{nt}.$$

The neural network $\Delta_\theta(\cdot)$ learns how context shifts tastes. The key is that the output remains an interpretable parameter shift. This is different from a black-box classifier because the neural network does not directly output a choice label; it modifies a random utility model.

Environmental applications are numerous. The value of beach crowding may depend on season and weather. The value of urban green space may depend on heatwaves. The disutility of travel time to a recreation site may depend on whether the trip is made during holidays. The value of energy reliability may depend on recent outage experience.

17.5.3 Multitask deep learning for RP and SP data

Combining revealed-preference and stated-preference data is a classic choice-modelling problem. RP data are behaviourally realistic but may have limited variation. SP data provide experimental variation but may suffer from hypothetical bias. Multitask learning provides a neural framework in which related tasks share some representation layers and have task-specific output heads.

A simplified architecture is

$$h_n = r_\theta(x_n), \qquad P_n^{RP} = \text{softmax}(W_{RP}h_n), \qquad P_n^{SP} = \text{softmax}(W_{SP}h_n).$$

The shared representation h_n captures common behavioural information, while the task-specific heads allow RP and SP choice processes to differ. Wang, Wang and Zhao use multitask deep neural networks to combine RP and SP data in the context of autonomous vehicle adoption. Their work shows the appeal of automatic feature learning, but also illustrates the need to inspect behavioural outputs such as elasticities and substitution patterns.

17.5.4 Embeddings for categorical variables

Choice datasets often contain high-cardinality categorical variables: neighbourhoods, origin-destination pairs, brands, vehicle models, recreation sites, schools, tariffs, providers, or product labels. Dummy coding can create thousands of sparse variables. Embeddings map each category to a dense vector.

$$\text{site ID} \rightarrow e_{site} \in \mathbb{R}^d.$$

The embedding can then enter utility. For example,

$$V_{ni} = x'_{ni}\beta + e'_{site(i)}\gamma.$$

Embeddings can capture similarities among alternatives, but their dimensions are not automatically interpretable. A careful architecture can align embedding dimensions with alternatives or behavioural constructs, improving interpretability.

In environmental valuation, site embeddings could summarise unobserved amenities of lakes, forests, beaches, protected areas, or urban parks. But the analyst should not use embeddings to hide omitted environmental variables when the policy goal is to value those variables explicitly.

17.5.5 Image and street-view choice models

When alternatives are represented by images, such as cycling streets, landscapes, coastal scenes, or park designs, a convolutional neural network can extract visual features. Zhu and Si compare discrete choice models and machine learning models for street-view image choices. This line of work is important for environmental economics because many environmental attributes are visual, spatial, and difficult to summarise with simple numeric variables.

A useful hybrid workflow is:

1. Use computer vision to extract interpretable visual features, such as greenery, visible water, shade, traffic, litter, or crowding.
2. Estimate a choice model using those features as attributes.
3. Compare with a CNN using raw images.
4. Use explainability tools to verify whether the CNN is relying on meaningful environmental features or spurious visual artefacts.

17.5.6 Reference-dependent neural choice models

Reference-dependent choice models recognise that utility depends on gains and losses relative to a reference point. The difficulty is that the reference point is often assumed rather than estimated. Recent reference-dependent ANN work uses a neural network to learn individual- and alternative-specific reference points, while embedding the behavioural structure of reference dependence into the architecture. This is an example of a broader principle: let the neural network learn the difficult latent object, but keep the behavioural theory explicit.

For environmental applications, reference dependence is relevant when respondents compare policies to the status quo, past beach conditions, normal electricity bills, expected travel times, or current conservation access rules. Neural reference-point learning may be useful, but it must be handled carefully because welfare conclusions can be highly sensitive to the reference point.

17.5.7 RUM-compatible neural estimators

Recent RUM-compatible neural frameworks attempt to connect neural estimation with the random utility principle more directly. RUMnets approximate random utility choice models using neural networks, while RUM-NN proposes a neural estimation framework compatible with random utility maximisation and flexible error distributions. These models are important because they move beyond the shallow similarity that both MNL and neural classifiers use softmax. They ask how to design neural architectures that are consistent with the stochastic utility model itself.

17.6 Training, validation, sample size, and regularisation

Deep learning success depends as much on workflow as on architecture. Neural choice models are especially vulnerable to misleading validation because choice data often have panel structure, repeated tasks, availability restrictions, and correlated alternatives.

17.6.1 Loss functions

For a neural utility model, the natural loss is negative log-likelihood:

$$\mathcal{L}(\theta) = -\sum_{n,t} \log P_{ni^*_{nt}t}(\theta),$$

where i^*_{nt} is the chosen alternative. This is equivalent to cross-entropy. For welfare and policy analysis, log loss is often more informative than accuracy because it evaluates the probability assigned to the observed choice.

Other useful metrics include:

- top-1 accuracy, for classification performance;
- log loss or negative log-likelihood, for probabilistic prediction;
- Brier score, for probability calibration;
- calibration plots, for overconfidence;
- out-of-sample likelihood ratio index against a null model;
- implied elasticities and WTP distributions, for behavioural plausibility.

17.6.2 Grouped validation

If each respondent answers multiple choice tasks, random row-wise splitting leaks information. The model can learn respondent-specific patterns from the training rows and appear to perform well on the test rows from the same respondent. This is especially harmful for flexible neural networks. Use respondent-level splitting, household-level splitting, route-level splitting, or spatial/temporal splitting depending on the prediction target.

17.6.3 Sample size

Neural networks can have far more weights than conventional choice models. Alwosheel, van Cranenburgh and Chorus propose a conservative rule of thumb for artificial neural networks in discrete choice analysis: use a sample size at least fifty times the number of network weights. The rule is not a theorem, and requirements depend on signal strength, regularisation, task complexity, and architecture. But it is a useful warning against training large networks on small stated-preference samples.

A neural network with two hidden layers of 64 units can easily have thousands of weights. A typical environmental DCE with 600 respondents and 8 choice tasks has only 4800 choice observations. That may be enough for a compact neural utility model, but not for a deep black-box architecture unless strong regularisation, pretraining, or external data are available.

17.6.4 Regularisation

Neural choice models should use regularisation by default. Common tools include:

- weight decay, equivalent to an L2 penalty;
- dropout, which randomly drops hidden units during training;
- early stopping using a validation set;
- small architectures before large architectures;
- monotonicity or sign-violation penalties;
- Bayesian priors or hierarchical shrinkage;

- data augmentation for images or text, when appropriate.

For choice modelling, regularisation should not be only statistical. It should also be behavioural: cost signs, availability masks, alternative-specific separability, known monotonicities, and plausible substitution patterns.

17.6.5 Hyperparameter tuning

Hyperparameters include learning rate, batch size, number of layers, number of hidden units, activation function, dropout rate, weight decay, patience for early stopping, and architecture-specific constraints. Tuning must be nested inside the training data. The test set should be used once, after model selection. In panel data, folds should be grouped by respondent or other relevant unit.

17.7 Interpretation: gradients, elasticities, WTP, and diagnostics

A neural choice model for economics should not end with a prediction score. It should be interrogated as an economic model.

17.7.1 Local marginal utilities

For a differentiable utility function $V_{ni} = f_\theta(x_{ni})$, the local marginal utility of attribute k is

$$MU_{nik} = \frac{\partial f_\theta(x_{ni})}{\partial x_{nik}}.$$

In software, this derivative can be obtained by automatic differentiation. Marginal utilities should be summarised across respondents, alternatives, and contexts. A good report shows means, medians, quantiles, and sign-violation rates.

17.7.2 Elasticities

The direct elasticity of the choice probability with respect to attribute x_{nik} is

$$E_{nik} = \frac{\partial P_{ni}}{\partial x_{nik}} \frac{x_{nik}}{P_{ni}}.$$

For neural models, the derivative $\partial P_{ni}/\partial x_{nik}$ can be computed by automatic differentiation or finite differences. Elasticities are essential because they translate model flexibility into behavioural response. For example, an environmental transport model may predict mode choice accurately but imply near-zero response to fuel prices, which would be problematic for policy simulation.

17.7.3 Willingness to pay

For environmental valuation, the local WTP for attribute k is

$$WTP_{nik} = -\frac{\partial V_{ni}/\partial x_{nik}}{\partial V_{ni}/\partial cost_{ni}}.$$

For a sample average, one may report

$$\overline{WTP}_k = \frac{1}{N} \sum_n WTP_{nik},$$

but this can be unstable if the cost derivative is close to zero for some observations. Recommended diagnostics include:

- distribution of $\partial V/\partial cost$;
- share of observations with positive cost derivatives;
- WTP quantiles and trimmed means;
- sensitivity of WTP to architecture and random seed;
- comparison with MNL, mixed logit, and WTP-space models;
- checks that counterfactual attribute changes remain within the training support.

17.7.4 Prototypical examples

Alwosheel, van Cranenburgh and Chorus propose using prototypical examples to diagnose neural networks for discrete choice analysis. The idea is to synthesise inputs that strongly activate a predicted class or choice pattern, revealing what the network has learned. This is useful because a neural model may achieve strong predictive performance by relying on spurious combinations of variables. Prototypes provide a qualitative diagnostic: do the learned relationships look like plausible choice behaviour?

17.7.5 Partial dependence, ALE, and SHAP

Partial dependence plots vary one variable while averaging over the empirical distribution of other variables. Accumulated local effects are often safer when variables are correlated. SHAP values attribute predictions to features, although their behavioural interpretation depends on model and data assumptions. For choice models, these tools should be used at the level of utilities, probabilities, and behavioural outputs, not only raw class labels.

17.8 Environmental economics applications

Neural choice models are not yet standard in environmental economics, but they are highly relevant in several areas.

17.8.1 Environmental discrete choice experiments

Environmental DCEs often involve cost, environmental quality, biodiversity, access restrictions, risk, and policy design. Neural utility models can discover nonlinearities such as threshold effects in water quality or diminishing returns to biodiversity. Hybrid models can preserve a negative cost effect while learning flexible non-cost utility.

A defensible workflow is:

1. Estimate MNL and mixed logit baselines.
2. Train a compact neural utility model with cost sign constraints.
3. Compare out-of-sample log loss and calibration.
4. Compute derivative-based WTP and compare with conventional estimates.
5. Check sign violations, WTP stability, and counterfactual validity.
6. Use neural results to refine a transparent econometric specification if the welfare task requires it.

17.8.2 Recreation demand and site choice

Site choice models often have many alternatives and rich spatial data. Neural embeddings can represent sites, while environmental attributes remain explicit. A hybrid site-choice utility might be

$$V_{ni} = \beta_c cost_{ni} + \beta_q quality_i + \beta_a access_i + e_i'\gamma + h_\theta(s_n, c_{ni}),$$

where e_i is a site embedding and h_θ captures observed heterogeneity. The analyst must decide whether the embedding is a nuisance control or a policy-relevant object. If the policy changes water quality, water quality should be explicit, not hidden inside the embedding.

17.8.3 Energy and climate policy choices

Household choices over tariffs, solar adoption, heat pumps, electric vehicles, insulation, and dynamic pricing can involve high-dimensional household histories and context. Neural networks can learn from smart-meter data or adoption histories, while choice theory can structure policy simulations. For example, a model may combine a neural representation of electricity load profiles with an explicit cost and emissions component in utility.

17.8.4 Visual environmental preferences

Environmental amenities are often visual: landscapes, coastal crowding, green infrastructure, tree canopy, litter, beach morphology, or urban design. CNNs can process images, but the most policy-relevant model often decomposes images into interpretable features. A CNN-only model may predict preferences for a park image, but a policy maker needs to know whether people value shade, water visibility, biodiversity, safety, cleanliness, or absence of crowding.

17.9 R lab: from softmax prediction to a neural utility workflow

This lab is designed as a conceptual implementation. It uses simulated data so that the workflow can be understood without downloading a specialised dataset.

Lab 17A: simulate a three-alternative environmental DCE

```
library(tidyverse)

set.seed(17)
N <- 600   # respondents
Tn <- 8    # choice tasks per respondent
alts <- c("A", "B", "SQ")

true_beta_cost <- -0.06
true_beta_bio  <-  0.55
true_beta_risk <- -0.35

long <- expand_grid(id = 1:N, task = 1:Tn, alt = alts) |>
  mutate(
    chid = paste(id, task, sep = "_"),
    income = rep(rnorm(N, 0, 1), each = Tn * length(alts)),
    cost = if_else(alt == "SQ", 0, sample(c(10, 25, 50, 75), n(), TRUE)),
    biodiversity = if_else(alt == "SQ", 0, sample(0:3, n(), TRUE)),
    flood_risk = if_else(alt == "SQ", 2, sample(0:3, n(), TRUE)),
    V = true_beta_cost * cost * exp(-0.15 * income) +
        true_beta_bio  * sqrt(biodiversity + 1) +
        true_beta_risk * flood_risk
  ) |>
```

```
  group_by(chid) |>
  mutate(
    prob = exp(V) / sum(exp(V)),
    chosen = sample(alt, 1, prob = prob),
    choice = as.integer(alt == chosen)
  ) |>
  ungroup()
```

The data-generating process contains nonlinear biodiversity utility and income-varying cost sensitivity. A linear MNL can approximate this but cannot recover the structure exactly unless the analyst specifies the transformations.

Lab 17B: conventional MNL baseline

```
library(mlogit)

mdata <- mlogit.data(
  long,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

mnl <- mlogit(choice ~ cost + biodiversity + flood_risk | 0, data = mdata)
summary(mnl)

b <- coef(mnl)
wtp_bio <- -b["biodiversity"] / b["cost"]
wtp_risk_reduction <- b["flood_risk"] / b["cost"]
```

This baseline is deliberately simple. In an empirical paper, you would also estimate richer specifications, mixed logit, latent class, and possibly WTP-space models before comparing with neural models.

Lab 17C: neural classifier as a prediction benchmark

A standard neural classifier requires a wide format: one row per choice task, with attributes for each alternative. This is useful for prediction, but it is less natural for welfare analysis.

```
library(tidymodels)

wide <- long |>
  select(id, task, chid, alt, cost, biodiversity, flood_risk, chosen) |>
  pivot_wider(
    names_from = alt,
    values_from = c(cost, biodiversity, flood_risk)
  ) |>
  distinct(chid, .keep_all = TRUE) |>
  mutate(chosen = factor(chosen))

set.seed(17)
split <- group_initial_split(wide, group = id, prop = 0.8)
train <- training(split)
test  <- testing(split)

# A lightweight neural benchmark using nnet through parsnip.
mlp_spec <- mlp(hidden_units = 8, penalty = 0.01, epochs = 300) |>
  set_mode("classification") |>
  set_engine("nnet", trace = FALSE)
```

```
mlp_wf <- workflow() |>
  add_formula(chosen ~ . - id - task - chid) |>
  add_model(mlp_spec)

mlp_fit <- fit(mlp_wf, data = train)

pred <- predict(mlp_fit, test) |>
  bind_cols(test |> select(chosen))

metrics(pred, truth = chosen, estimate = .pred_class)
```

This model can be a useful benchmark, but it does not automatically provide marginal utilities or WTP. It also assumes a fixed set of labelled alternatives.

Lab 17D: what a neural utility workflow adds

A neural utility model should be trained in long format. The utility function is applied alternative by alternative, and the softmax is computed within each choice set. In R, the most flexible implementation uses `torch` or `keras`. The following pseudocode shows the structure rather than a complete package-specific script.

```
# Pseudocode for a neural utility model in R torch/keras style
# 1. Inputs: one tensor for alternative-specific attributes.
# 2. Apply the same neural utility network to each alternative.
# 3. Mask unavailable alternatives if needed.
# 4. Apply softmax within each choice set.
# 5. Minimise negative log-likelihood.

utility_network <- function(x_alt, s_person) {
  # Example structure:
  # explicit cost term + neural non-cost term
  # beta_cost is constrained to be negative: -exp(alpha_cost)
  # h_theta learns nonlinear effects of biodiversity, risk, and interactions
  V <- -exp(alpha_cost) * x_alt$cost + h_theta(x_alt, s_person)
  return(V)
}
```

The conceptual difference from the classifier is the explicit utility layer. The cost effect can be constrained, the softmax is calculated over the available alternatives, and derivatives of utility can be used for WTP diagnostics.

17.10 Python lab: a minimal PyTorch neural utility model

Python is currently more natural for custom neural architectures. The following example shows the essential structure of a neural utility model. It assumes a tensor `X` of shape `[N, J, K]` where `N` is the number of choice tasks, `J` the number of alternatives, and `K` the number of attributes. The target `y` is the chosen alternative index.

```
import torch
import torch.nn as nn
import torch.nn.functional as F

class NeuralUtilityMNL(nn.Module):
    def __init__(self, k_features, hidden=16):
        super().__init__()
        # alpha_cost is transformed to guarantee negative cost utility.
        self.alpha_cost = nn.Parameter(torch.tensor(0.0))
        self.net = nn.Sequential(
            nn.Linear(k_features - 1, hidden),
```

```
            nn.Tanh(),
            nn.Linear(hidden, 1)
        )

    def forward(self, X, availability=None):
        # X shape: [batch, alternatives, features]
        # feature 0 is cost; remaining features are non-cost attributes.
        cost = X[:, :, 0]
        noncost = X[:, :, 1:]
        batch, J, Kminus1 = noncost.shape

        h = self.net(noncost.reshape(batch * J, Kminus1)).reshape(batch, J)
        beta_cost = -torch.exp(self.alpha_cost)
        V = beta_cost * cost + h

        if availability is not None:
            V = V.masked_fill(availability == 0, -1e9)
        return V

model = NeuralUtilityMNL(k_features=3, hidden=16)
optimizer = torch.optim.Adam(model.parameters(), lr=1e-3, weight_decay=1e-4)

for epoch in range(500):
    optimizer.zero_grad()
    V = model(X_train, availability=A_train)
    loss = F.cross_entropy(V, y_train)
    loss.backward()
    optimizer.step()
```

This model is still simple, but it already encodes several good practices: utility is alternative-specific, the softmax is within the choice set, unavailable alternatives can be masked, and the cost coefficient is constrained to be negative. The next step is to compute derivatives.

```
# Local marginal utilities and WTP by automatic differentiation
X_eval = X_test.clone().detach().requires_grad_(True)
V = model(X_eval, availability=A_test)

# Example: derivative of utility of chosen alternative with respect to inputs
chosen_V = V[torch.arange(V.shape[0]), y_test]
grads = torch.autograd.grad(chosen_V.sum(), X_eval)[0]

mu_cost = grads[torch.arange(V.shape[0]), y_test, 0]
mu_bio  = grads[torch.arange(V.shape[0]), y_test, 1]
wtp_bio = -mu_bio / mu_cost

print(torch.quantile(wtp_bio.detach(), torch.tensor([0.1, 0.5, 0.9])))
print((mu_cost.detach() > 0).float().mean())  # sign violation share
```

In a real application, the derivative diagnostics should be reported by alternative, subgroup, context, and support region. A single mean WTP is not enough.

17.11 Software landscape

R remains excellent for conventional choice modelling and ML workflows. Python currently has the advantage for custom deep learning. In practice, many researchers use both.

Table 17.2. Software options for neural and hybrid choice workflows

Tool	Best use	Notes	Language
`mlogit`, `gmnl`, `apollo`, `logitr`	Baseline DCMs	Essential for MNL, mixed logit, latent class, WTP-space comparisons	R
`tidymodels`	Predictive benchmarks	Good workflow for splitting, resampling, tuning, and comparison	R
`torch` / `keras`	Custom neural utility models	Requires more programming; flexible for constraints and derivatives	R/Python
`torch-choice`	Large-scale choice modelling with PyTorch	Provides choice datasets and logit/nested logit estimation with GPU support	Python
`choice-learn`	Scalable choice modelling and neural choice examples	Useful for large-scale choice data and custom models	Python
`xlogit`	GPU-accelerated mixed logit	Not primarily neural, but useful for large-scale DCM baselines	Python

The most important software advice is methodological: always estimate strong conventional baselines. A neural model that beats a weak MNL baseline is not convincing. Compare against a well-specified MNL, mixed logit, latent class, and, where relevant, a gradient boosting or random forest benchmark.

17.12 Common mistakes

Mistake 1: treating choice alternatives as ordinary class labels

A mode, site, tariff, or policy alternative is not just a label. It has attributes, availability, and substitution relationships. A fixed-label classifier may fail when alternatives change or when new policies are simulated.

Mistake 2: reporting accuracy only

Accuracy ignores probability calibration, likelihood, welfare plausibility, elasticities, and minority alternatives. A model can improve accuracy while producing worse behavioural outputs.

Mistake 3: allowing cost to enter an unconstrained black box

For environmental valuation, the cost derivative is the denominator of WTP. If it is positive, unstable, or near zero, WTP becomes meaningless. Cost should usually be explicit and sign-constrained.

Mistake 4: using row-wise splitting in panel choice data

Repeated choices from the same respondent must usually be split by respondent, not by row. Otherwise, validation estimates can be optimistically biased.

Mistake 5: extrapolating outside the support

Neural networks can behave unpredictably outside the training region. Policy simulations should identify whether counterfactual attribute combinations are within empirical support.

Mistake 6: interpreting embeddings as if they were measured attributes

Embeddings are learned representations. They can be useful controls, but they are not automatically policy-relevant variables. Do not replace explicit environmental attributes with opaque embeddings if the goal is valuation.

17.13 Reporting checklist and exercises

Reporting checklist

A paper using neural choice models should report:

1. The exact prediction or policy target.
2. The choice-set structure and availability handling.
3. Whether the architecture is a classifier, neural utility model, or hybrid model.
4. The utility equation or computational graph.
5. Which parameters or derivatives are behaviourally interpreted.
6. Training, validation, and test splits, including grouped splitting decisions.
7. Hyperparameter search protocol and random seeds.
8. Baseline models, including strong DCM baselines.
9. Out-of-sample log loss, calibration, and accuracy.
10. Sign, monotonicity, and derivative diagnostics.
11. Elasticity and WTP distributions, not only means.
12. Sensitivity to architecture, seed, and sample definition.
13. Evidence that counterfactuals are inside or near the training support.
14. Code and data processing details sufficient for replication.

Conceptual exercises

1. Explain why a softmax output layer is not sufficient to make a neural classifier a random utility model.
2. In an environmental DCE, why might a neural utility model learn a positive cost effect? List at least three possible causes.
3. Compare a latent class model with a neural taste-network model. What is discrete in one and continuous in the other?
4. Suppose a neural model predicts beach choice very accurately using beach ID embeddings. Can it estimate WTP for water quality? Under what conditions?
5. Explain how you would validate a neural mode-choice model estimated from smartphone tracking data.

Applied exercises

1. Simulate a DCE where the true utility of biodiversity is concave. Estimate a linear MNL and a neural utility model. Compare WTP estimates.
2. Modify the Python lab so that the non-cost network is monotonic in biodiversity. Compare sign violations before and after the constraint.
3. Train a neural classifier and a neural utility model on the same simulated data. Compare accuracy, log loss, and derivative-based WTP.
4. Design a hybrid architecture for valuing urban tree canopy from street-view images. Which variables should be explicit and which can be learned?
5. Write a one-page reviewer report for a paper that claims neural networks outperform discrete choice models but reports only accuracy.

Chapter takeaway

Neural choice models are most useful when they are designed as choice models, not merely as classifiers. Their promise is flexibility: nonlinear utilities, learned representations, context-dependent tastes, and large-scale data. Their risk is that flexibility can destroy the behavioural structure needed for welfare analysis. The frontier of the field is therefore hybrid: neural networks learn the difficult parts of the model, while random utility theory, economic restrictions, and transparent validation discipline the outputs.

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Machine Learning and AI for Environmental Economics and Choice Modelling

A self-study textbook for PhD students

Chapter 18

Choice Modelling with Text, Images, GPS, and Passively Collected Data

Many choices are no longer observed only through survey tables. Researchers now observe reviews, open-ended survey answers, product descriptions, policy texts, images, videos, GPS trajectories, mobile-phone records, smart-card transactions, app logs, street-view scenes, and sensor streams. These data can make choice models richer and more realistic. They can also make them less interpretable, less representative, and more vulnerable to leakage. This chapter explains how to transform text, images, GPS traces, and passively collected records into inputs for machine learning and discrete choice models without losing the economic meaning of the choice problem.

How to study this chapter. First, read Sections 18.1-18.4 to understand the logic of unstructured data and representation learning. Then study Sections 18.5-18.8 as the applied core of the chapter. Run the R labs in Sections 18.10 and 18.11. Finally, use the validation checklist in Section 18.13 before applying these methods to real data.

Learning objectives

After studying this chapter, you should be able to:

- Distinguish structured variables, unstructured raw objects, engineered features, and learned representations.
- Explain how text, images, GPS traces, and passive records can enter random utility models and ML classifiers.
- Design a defensible pipeline from raw traces to choice observations, alternatives, choice sets, labels, and covariates.
- Use text features, image features, and spatial features in R and Python workflows.
- Recognise privacy, selection, measurement-error, and data-leakage risks in passively collected data.
- Decide when a feature derived from text, images, or GPS can support welfare analysis and when it should be treated as predictive only.

Chapter map

18.1	Why nontraditional data matter for environmental economics and choice modelling
18.2	A taxonomy: raw objects, features, embeddings, and economic variables
18.3	From traces to choices: observation units, labels, alternatives, and choice sets
18.4	Text data in choice modelling
18.5	Image and video data in choice modelling
18.6	GPS, smartphone, smart-card, and passive revealed-preference data
18.7	Multimodal data fusion and hybrid utility functions
18.8	Validation, leakage, privacy, and representativeness
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18.10	R lab: text features for environmental choice data
18.11	R lab: GPS-style spatial feature engineering
18.12	Python lab: image and text embeddings for choice models
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18.1 Why nontraditional data matter

Classical discrete choice datasets are usually rectangular. Each row is a decision maker, a choice situation, or an alternative within a choice situation. Each column is a variable: cost, travel time, access restriction, biodiversity gain, income, age, or attitude score. This format remains essential because it maps naturally into random utility models.

However, many relevant determinants of choices are not naturally rectangular. A beach may be represented by a photograph, a crowding image, a satellite-derived water-quality measure, and thousands of online reviews. A transport mode may be represented by a GPS trace, a sequence of stops, the weather at departure, the availability of cycle lanes along the route, and a smartphone-app label. A conservation policy may be described by a long text, not only by a few attributes. A landscape-improvement programme may affect visual quality, sense of naturalness, perceived safety, and emotional response.

Machine learning helps because it can transform these raw objects into features. The transformation may be simple, such as counting words or computing trip length from GPS points. It may be complex, such as using a convolutional neural network to represent a street-view image or a language model to embed a policy description. The key question for choice modelling is not only whether the features predict choices. It is whether the features have a clear role in the choice process.

> **Key idea.** Nontraditional data are useful when they help define the choice situation, describe the alternatives, approximate latent perceptions, construct realistic choice sets, or measure context. They are dangerous when they create apparently high predictive accuracy without a defensible behavioural or causal interpretation.

18.1.1 Examples in environmental economics

Environmental economics offers many natural applications:

- **Recreation-site choice.** GPS traces and mobile-phone data can identify visits, travel times, substitute sites, crowding, and access constraints.
- **Landscape and biodiversity valuation.** Images, videos, virtual reality, and semantic labels can represent visual quality, habitat restoration, reef type, beach width, and crowding.
- **Climate adaptation.** Text descriptions and images can communicate adaptation packages, flood defences, relocation schemes, forest management, or coastal protection.
- **Energy and transport.** Smart-meter data, charging logs, public transport smart cards, vehicle telematics, and app logs can reveal repeated decisions over time.
- **Environmental perceptions.** Open-ended survey answers, reviews, complaints, and social media text can measure perceived pollution, safety, congestion, noise, and trust.

The uploaded choice-modelling literature also points in this direction. Van Cranenburgh et al. argue that ML can help choice modelling work with text, images, and continuous data streams. Zhu and Si study a stated-preference image choice problem in which respondents choose cycling streets based on street-view images, comparing MNL, mixed logit, DNN, and CNN models. Dahmen et al. apply interpretable ML to tracking-based revealed-preference mode-choice data. Axhausen's chapter in the *Handbook of Choice Modelling* gives a broad taxonomy of movement traces, ranging from travel diaries and GPS to mobile-phone records, payment cards, vehicle GPS, and social media.

18.2 Raw objects, features, embeddings, and economic variables

A central distinction is between *raw objects* and *economic variables*.

- A **raw object** is an unprocessed item such as an image, text, video, GPS trace, or clickstream.
- An **engineered feature** is a variable constructed from the raw object using rules, domain knowledge, or algorithms.
- A **learned representation** or **embedding** is a vector constructed by a model, often trained to preserve semantic or predictive information.
- An **economic variable** is a variable with a clear interpretation in the utility function, such as cost, time, quality, risk, access, crowding, or income.

Let R_{nit} denote a raw object associated with decision maker n, alternative i, and choice occasion t. For example, R_{nit} may be a street-view image of a cycling route, a textual description of a policy, or a GPS-derived route geometry. A representation model maps the raw object into a vector:

$$e_{nit} = g_\theta(R_{nit}),$$

where g_θ may be a hand-coded feature extractor, a topic model, a CNN, a language model, or a geospatial algorithm. The utility function may then use both structured attributes and representations:

$$V_{nit} = x'_{nit}\beta + e'_{nit}\gamma + z'_n\delta_i + h_\phi(c_{nt}, x_{nit}).$$

Here x_{nit} contains conventional alternative attributes, e_{nit} contains representation features, z_n contains individual characteristics, and h_ϕ is an optional flexible function for context or interactions.

18.2.1 Three levels of interpretability

A representation can be interpreted at three levels.

1. **Transparent features.** Examples: number of trees in an image, share of road covered by cycle lane, travel distance, number of pollution-related words, percentage of beach occupied by people. These are easiest to use in welfare analysis.
2. **Semi-transparent features.** Examples: topic proportions, sentiment scores, semantic-segmentation shares, visual greenness indices. These are interpretable but depend on a modelling step.
3. **Opaque embeddings.** Examples: BERT text embeddings, CLIP image embeddings, CNN activations. These are powerful for prediction but difficult to translate into marginal willingness to pay unless connected to policy-controllable attributes.

> **Warning.** Do not compute WTP for an opaque embedding unless you can explain what policy-relevant change in the embedding means. A coefficient on "image embedding dimension 17" may predict choices, but it is not automatically an environmental value.

18.3 From traces to choices

Before modelling, nontraditional data must be transformed into a choice dataset. This is often the hardest part of the research design.

18.3.1 The unit of observation

A passive trace is not automatically a choice. A smartphone may record a sequence of points. A smart card may record boarding events. A search log may record impressions and clicks. A review site may contain text about a destination. The researcher must define the choice event.

Table 1: From raw data to choice observations

Raw data	Possible choice event	Key modelling problem
GPS trajectory	Mode, route, destination, activity location	Trip segmentation, map matching, unchosen alternatives
Smart-card tap records	Transit route, stop, departure time, fare product	Missing alighting, transfer inference, captive demand
Search and booking logs	Product, itinerary, hotel, energy tariff	Choice-set visibility, ranking endogeneity, stock-outs
Street-view images	Street, route, neighbourhood, recreational site	Visual attributes, image comparability, scene-level validation
Open-ended survey text	Acceptance of policy package or programme	Constructing latent perceptions without post-treatment bias
Social media or reviews	Destination demand or perceived environmental quality	Representativeness, bots, time-varying sentiment

A good rule is to write the choice event in a sentence before doing any ML:

> Decision maker n chose alternative i from choice set C_{nt} at occasion t, where all alternatives in C_{nt} were available and known enough to be considered.

If this sentence cannot be defended, the resulting model may be a useful prediction model but not a defensible choice model.

18.3.2 Labels and ground truth

Choice modelling needs a dependent variable. In passive data, labels often come from inference rather than direct observation. For example, mode may be inferred from GPS speed profiles, accelerometer data, stop patterns, and proximity to transit lines. Purpose may be imputed from land use or time of day. Destination may be inferred from stays. These labels contain measurement error.

Dahmen et al. distinguish fully passive and semi-passive travel diaries. In semi-passive diaries, the app generates draft diaries and users validate or correct them. This improves behavioural labels but requires user involvement. Their Munich study illustrates the scale of these data: 1,192 app users, average participation of 209 days, and 882,790 trips after processing. This type of scale is attractive for ML, but it does not remove the need for careful cleaning, validation, and interpretation.

18.3.3 Unchosen alternatives and choice sets

In stated preference experiments, the analyst usually controls the choice set. In passive revealed-preference data, the unchosen alternatives must be constructed. For a route-choice problem, this may involve generating plausible alternative routes. For a mode-choice problem, this may involve computing travel time, cost, and availability for car, public transport, walking, cycling, ride-hailing, or micro-mobility. For a recreation-site problem, this may involve defining substitute beaches, parks, lakes, trails, or protected areas.

Choice-set generation is not a technical detail. It changes the model. If the alternatives are unrealistic, the estimated substitution patterns and welfare measures will be wrong. If the generated alternatives are too similar to the chosen alternative, the model may overstate taste precision. If relevant alternatives are missing, the model may confuse choice-set availability with preferences.

18.4 Text data in choice modelling

Text can enter environmental economics and choice modelling in several ways.

- **Alternative descriptions:** policy texts, product descriptions, energy-tariff descriptions, conservation programme descriptions.
- **Perceptions:** open-ended survey answers, complaints, reviews, discussion forums, participatory-planning comments.
- **Context:** news coverage, social media sentiment, regulatory texts, weather warnings, risk communication.
- **Research workflow:** coding qualitative survey responses, constructing experimental attributes, screening literature, and generating survey variations for pilot testing.

18.4.1 Representing text

The simplest text representation is a document-term matrix. Each row is a text and each column is a token. Entries may be counts, binary indicators, or TF-IDF weights. More flexible representations include topic models, word embeddings, sentence embeddings, and large language model embeddings.

Table 2: Common text representations

Representation	What it captures	Use in choice modelling
Keyword indicators	Presence of predefined concepts	Transparent interaction with attributes
Bag-of-words / TF-IDF	Salient terms in documents	Prediction, screening, exploratory modelling
Topic proportions	Mixtures of latent themes	Semi-interpretable perceptions or narratives
Sentiment scores	Positive/negative affect	Destination image, trust, perceived quality
Embeddings	Semantic similarity in dense vectors	Prediction, clustering, matching, retrieval
LLM-coded labels	Structured labels extracted from text	Useful if audited and validated by humans

18.4.2 Text in utility functions

Suppose a coastal-adaptation DCE includes a short textual description of each policy option. A language model maps the description R_{nit}^{text} to an embedding e_{nit}^{text}. A utility specification could be:

$$V_{nit} = \beta_c cost_{nit} + \beta_q quality_{nit} + \beta_r restriction_{nit} + \gamma' e_{nit}^{text} + \alpha_i.$$

This may improve prediction, but the interpretation of γ is difficult. A better research design may use text embeddings to discover recurring themes, then convert themes into interpretable attributes: fairness, local employment, biodiversity, managed retreat, flooding risk, or uncertainty. The final welfare model can then use transparent attributes.

18.4.3 Good uses of text

Text is especially useful when it captures perceptions that are otherwise missing. For example, reviews of protected areas may reveal concerns about crowding, litter, water quality, safety, parking, or trail maintenance. Open-ended survey responses may reveal why respondents reject a conservation policy even when the monetary cost is low. In participatory processes, comments may identify attributes that were missing from the original DCE.

The strongest workflow is often sequential:

1. Use NLP to explore and classify text.
2. Validate text-derived categories with human coding on a subset.
3. Convert validated categories into interpretable variables or experimental attributes.
4. Estimate a choice model using those variables.
5. Test whether text-derived variables improve out-of-sample prediction and behavioural interpretation.

18.5 Image and video data in choice modelling

Images are increasingly relevant because many environmental goods are visual. Landscapes, beaches, rivers, forests, streets, reefs, crowding, litter, erosion, and built-environment quality are often better communicated visually than verbally. Stated-preference experiments may present images or virtual reality environments. Revealed-preference studies may use street-view images, satellite images, or user-posted photographs to describe alternatives.

Zhu and Si's street-view image choice study is a useful example. Respondents chose cycling streets based on real-world street-view images. The study compared MNL, mixed logit, DNN, and CNN models and tested different image-related data formats, including semantic interpretation, semantic segmentation, raw images, semantic maps, and enriched images. Their results show that discrete choice models had nearly equal predictability to DNN models and outperformed CNN models in that setting, while models using semantic interpretation data predicted better than models using segmentation or raw imagery. This is an important lesson: more complex raw-data models are not always better than carefully engineered, interpretable features.

18.5.1 Image representation strategies

Table 3: Ways to bring images into choice models

Strategy	Example	Strengths and weaknesses
Manual coding	Human labels: trees, people, litter, reef type	Interpretable but costly and subjective
Semantic segmentation	Pixel shares: sky, road, vegetation, water	Semi-interpretable; depends on segmentation model
Computer-vision scores	Greenness, enclosure, walkability, visual disorder	Useful if validated; may be context-specific
CNN or ViT embeddings	Dense image representation	Powerful but opaque
CLIP-style multimodal embeddings	Image and text in shared space	Useful for retrieval and semantic matching; welfare interpretation difficult
End-to-end image choice model	CNN predicts choice directly	Strong prediction possible; high data need and low transparency

18.5.2 Images as attributes, not only decoration

Images in experiments are often treated as visual aids. For choice modelling, they are more than aids: they may define the alternative. This creates design challenges.

First, images can contain uncontrolled attributes. A beach image may vary in weather, light, tide, people, camera angle, buildings, water colour, and image quality. If the experimental design manipulates only crowding, but the photos also differ in naturalness and weather, the crowding coefficient may be confounded.

Second, images create scale and attention effects. Respondents may react to salient visual features that are not listed as attributes. This can be realistic, but it makes interpretation harder.

Third, images make counterfactuals difficult. A policy counterfactual should modify a defined attribute, such as increasing vegetation cover or reducing crowding. Replacing one image with another may change many things at once.

Key idea. For welfare analysis, prefer image-derived features that can be connected to policy-controllable attributes: vegetation share, beach width, litter presence, crowding density, water clarity, habitat type, or visibility of infrastructure. Use opaque embeddings mainly for prediction, matching, or exploratory analysis.

18.6 GPS, smartphone, smart-card, and passive RP data

Passive data can dramatically increase the scale and resolution of revealed-preference choice modelling. A GPS trace may record the time and route of a trip. A smart card may reveal a transit boarding. A mobile-phone record may show broad movement patterns. A vehicle telematics device may reveal route, speed, and charging behaviour. An app log may reveal search, exposure, and purchase.

Axhausen's chapter in the *Handbook of Choice Modelling* emphasises that movement traces differ in social content. A self-reported diary can include purpose, mode, companions, and meaning. A GPS trace has precise coordinates but may lack purpose and subjective context. A mobile-phone record may reveal a movement field but not the socially meaningful trip. This distinction is crucial for choice modelling: high spatial resolution does not automatically imply high behavioural validity.

18.6.1 The passive-data pipeline

A defensible GPS or app-based choice pipeline usually includes:

1. **Consent and sampling.** Who agreed to be tracked? For how long? Under what incentives?
2. **Raw-data cleaning.** Remove impossible speeds, duplicate points, gaps, device artefacts, and low-quality users.
3. **Trip segmentation.** Convert point streams into stays, trips, stages, or activities.
4. **Mode and purpose inference.** Infer or validate mode and purpose; quantify uncertainty.
5. **Map matching.** Match observed points to networks when route or network attributes matter.
6. **Alternative generation.** Construct unchosen modes, routes, destinations, or activity locations.
7. **Context enrichment.** Add weather, land use, slope, infrastructure, congestion, fares, accessibility, and environmental exposure.
8. **Choice model estimation.** Estimate DCM, ML, or hybrid model with grouped validation.
9. **Sensitivity analysis.** Test label uncertainty, alternative-generation assumptions, and feature availability.

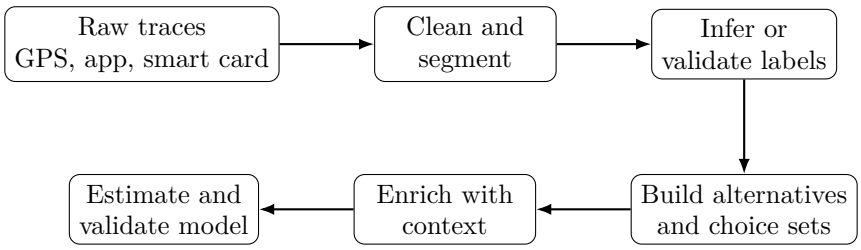


Figure 1: A passive-data workflow for choice modelling. The modelling step is only as credible as the cleaning, labelling, choice-set construction, and validation steps.

18.6.2 Mode-choice example

A tracking-based mode-choice dataset may contain one row per trip and features such as distance, departure time, rain, temperature, parking availability, public transport travel time, car travel time, walking distance to transit, car ownership, bike ownership, income, and neighbourhood density. A DCM may estimate utility for each mode. An ML classifier may predict the observed mode. A hybrid model may let context shift preference parameters, as in the context-aware mixed logit approach of Lukawska et al., where a neural network maps contextual information into interpretable shifts in preference parameters. Their real-world case uses a large bicycle route-choice dataset with 119,448 trips made by 8,555 cyclists.

18.6.3 Route-choice and recreation-site choice

Route and site choice require careful alternative construction. For route choice, analysts often generate paths using shortest-time, shortest-distance, safest, scenic, or low-slope criteria. For recreation-site choice, analysts define feasible sites based on distance, availability, information, and access. In environmental economics, site choice is central to travel-cost models and recreation-demand models. Passive data can help observe trips, but it may not identify the purpose of the trip or the set of sites actually considered.

18.7 Multimodal data fusion and hybrid utility functions

Many modern applications combine multiple data types. A cycling-street choice model may use tabular attributes, street-view images, route geometry, weather, and individual characteristics. A beach choice model may use survey choices, images, satellite-derived water quality, crowding from mobile-phone data, and open-ended text responses.

18.7.1 Fusion strategies

There are four common strategies.

1. **Sequential feature extraction.** Use ML to convert raw data into features, then estimate a conventional choice model.
2. **Parallel predictive modelling.** Train ML models on each data source and combine predictions.
3. **Joint neural architecture.** Feed text, image, spatial, and tabular data into one neural model.
4. **Hybrid utility model.** Use a structured utility function with flexible learned components.

   A hybrid utility specification could be:

$$V_{nit} = \beta_c cost_{nit} + \beta_t time_{nit} + \beta_q q_{nit} + \gamma' e_{nit}^{image} + \eta' e_{nit}^{text} + \alpha_i,$$

   or, more cautiously,

$$V_{nit} = \beta_c cost_{nit} + \beta_t time_{nit} + \beta_q q_{nit} + \lambda_1 vegetation_{nit} + \lambda_2 crowding_{nit} + \lambda_3 litter_{nit} + \alpha_i,$$

   where vegetation, crowding, and litter are validated image-derived features. The second model is usually preferable for WTP because each coefficient has a clearer environmental interpretation.

18.7.2 End-to-end versus two-stage modelling

End-to-end modelling means the model learns directly from raw data to choice probabilities. For example, a CNN takes images and outputs choice probabilities. This can work with large data and clear labels, but it often needs many observations and may be difficult to interpret.

Two-stage modelling means the raw data are first transformed into features or embeddings, then used in a choice model. This is often better for PhD research in environmental economics because it separates measurement from preference modelling. The researcher can validate the measurement model before estimating WTP.

Table 4: Choosing a modelling strategy

Research goal	Recommended strategy	Main diagnostic
Prediction of choices within observed domain	ML classifier or hybrid utility model	Out-of-sample log loss and calibration
WTP for environmental attributes	Structured choice model with transparent features	Sign, scale, confidence intervals, WTP stability
Discover missing attributes	NLP/CV exploration followed by human validation	Coding validity and robustness
Large-scale RP forecasting	ML or DCM with passive data and grouped validation	External and temporal validation
Policy counterfactuals	Theory-consistent choice model or constrained hybrid	Plausibility under interventions

18.8 Validation, leakage, privacy, and representativeness

Nontraditional data make validation harder. They also make bad validation easier to miss.

18.8.1 Leakage in text, images, and GPS

Leakage occurs when the model has access to information at training time that would not be available in a genuine prediction or policy setting.

- **Individual leakage.** Trips from the same person appear in both training and test sets.
- **Spatial leakage.** Nearly identical locations, images, or routes appear in both sets.
- **Temporal leakage.** Future context is used to predict past choices.
- **Label leakage.** Variables created using the chosen outcome enter as predictors.
- **Design leakage.** Repeated versions of the same image or scenario appear across splits.
- **Post-treatment leakage.** Text or behaviour measured after the choice is used to explain the choice as if it were prior information.

For panel choice data, grouped splitting by individual is usually essential. For image choice data, split by image or scene, not only by row. For GPS route data, consider splitting by individual and time, and test transfer to new areas if spatial generalization matters. Hillel's experimental review of ML classifiers for mode choice shows why trip-wise splitting can overstate performance when observations from the same individual leak across train and test sets.

18.8.2 Privacy and ethical design

Passive data can reveal home, workplace, religion, health visits, social networks, income proxies, and routines. Good practice includes data minimization, consent, transparent purpose limitation, aggregation where possible, secure storage, anonymization or pseudonymization, and careful handling of maps and case-study visuals. Even when data are legally usable, researchers should ask whether the inference would surprise the participant.

18.8.3 Representativeness

Smartphone tracking data are often collected from people willing and able to install an app. Review data are written by people motivated to review. Social-media posts reflect platform users, not the population. Street-view coverage differs across places. Satellite imagery differs

by cloud cover, resolution, and season. These selection processes can affect both predictive performance and welfare estimates.

Warning. Large passive datasets are not automatically representative. A million trips from app users may be less representative than a smaller, carefully sampled travel survey. Scale reduces variance, not necessarily bias.

18.9 Welfare and policy interpretation

The welfare logic of a logit-type model is usually based on changes in systematic utility. For individual n, a compensating variation under a logit model can be written as:

$$CV_n = -\frac{1}{\beta_{cost}} \left[\ln \left(\sum_j \exp(V_{nj}^1) \right) - \ln \left(\sum_j \exp(V_{nj}^0) \right) \right],$$

where superscripts 0 and 1 denote before and after a policy change. This expression is meaningful only if the change from V^0 to V^1 corresponds to a clear policy intervention and if β_{cost} has a stable monetary interpretation.

Text, images, and GPS-derived features can support welfare analysis when they are mapped to manipulable attributes. Examples:

- reducing crowding from 2.0 to 0.5 people per 10 m^2,
- increasing visible vegetation share by 15 percentage points,
- improving water clarity from medium to high,
- adding a protected cycle lane to a route,
- reducing travel time by 10 minutes,
- changing a policy description from uncertain to guaranteed funding.

Welfare analysis is much weaker when the counterfactual is "change the image embedding" or "increase topic 4". These quantities may not have a direct policy meaning. They can still be useful for screening, clustering, or prediction, but the welfare chapter of the paper should be based on interpretable transformations.

18.9.1 A useful decomposition

For applied work, separate measurement and preference:

$$R_{nit} \xrightarrow{measurement} m_{nit} \xrightarrow{preference} V_{nit} \xrightarrow{choice} y_{nt}.$$

Here R_{nit} is the raw text, image, or trace; m_{nit} is a validated measurement such as visible crowding or travel time; V_{nit} is utility; and y_{nt} is the observed choice. This decomposition makes it easier to report uncertainty and to identify which part of the pipeline is substantive economics and which part is measurement.

18.10 R lab: text features for environmental choice data

This lab simulates a small environmental policy-choice dataset where each alternative has a short text description. We extract simple text features and use them in a choice model. The goal is not to build a state-of-the-art NLP model, but to understand the workflow.

Step 1: simulate choice data with text

```
library(tidyverse)
library(mlogit)
library(text2vec)

set.seed(18)

N <- 300
Tn <- 6
alts <- c("A", "B", "SQ")

policy_text <- tibble(
  text_id = 1:6,
  description = c(
    "restores dunes and improves biodiversity",
    "builds grey flood barriers near the coast",
    "improves water quality and beach cleanliness",
    "restricts access to protect nesting birds",
    "expands parking and visitor facilities",
    "keeps current coastal management unchanged"
  ),
  biodiversity_words = c(1,0,0,1,0,0),
  restriction_words  = c(0,0,0,1,0,0)
)

d <- expand_grid(
  id = 1:N,
  task = 1:Tn,
  alt = alts
) %>%
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = if_else(alt == "SQ", 0, sample(c(10, 25, 50, 75), n(), TRUE)),
    text_id = if_else(alt == "SQ", 6L, sample(1:5, n(), TRUE))
  ) %>%
  left_join(policy_text, by = "text_id") %>%
  mutate(
    V = -0.05 * cost +
         0.80 * biodiversity_words -
         0.35 * restriction_words +
         if_else(alt == "SQ", 0.15, 0)
  ) %>%
  group_by(chid) %>%
  mutate(
    p = exp(V) / sum(exp(V)),
    chosen_alt = sample(alt, size = 1, prob = p),
    choice = as.integer(alt == chosen_alt)
  ) %>%
  ungroup()
```

Step 2: create simple NLP features

```
# Tokenize the unique descriptions
it <- itoken(policy_text$description,
             preprocessor = tolower,
             tokenizer = word_tokenizer,
             progressbar = FALSE)

vocab <- create_vocabulary(it, stopwords = c("and", "the", "to"))
vectorizer <- vocab_vectorizer(vocab)
```

```
dtm <- create_dtm(it, vectorizer)

# Convert document-term matrix to a data frame
text_features <- as.matrix(dtm) %>%
  as_tibble(.name_repair = "unique") %>%
  mutate(text_id = policy_text$text_id)

# Join to choice data
model_data <- d %>%
  left_join(text_features, by = "text_id")
```

Step 3: estimate an MNL with transparent text-derived variables

```
mdata <- mlogit.data(
  model_data,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

fit_text <- mlogit(
  choice ~ cost + biodiversity_words + restriction_words | 0,
  data = mdata
)

summary(fit_text)

b <- coef(fit_text)
wtp_biodiversity <- -b["biodiversity_words"] / b["cost"]
wtp_restriction  <- -b["restriction_words"] / b["cost"]

wtp_biodiversity
wtp_restriction
```

What to learn

The text itself is not magic. We made it useful by extracting interpretable variables. If the text feature is `biodiversity_words`, WTP has a possible interpretation, though still imperfect. If the feature is an opaque embedding dimension, WTP is hard to interpret.

18.11 R lab: GPS-style spatial feature engineering

This lab uses simulated coordinates to illustrate how spatial data are turned into variables for choice modelling. In real work, you would use a projected coordinate system, real networks, map matching, and external sources such as OpenStreetMap, transit feeds, weather records, elevation, land use, or protected-area boundaries.

```
library(tidyverse)
library(sf)

set.seed(1811)

# Simulated origins and destinations around an arbitrary centre
trips <- tibble(
  trip_id = 1:100,
  mode = sample(c("car", "bike", "pt", "walk"), 100, TRUE),
```

```
  ox = rnorm(100, 11.35, 0.03),
  oy = rnorm(100, 46.50, 0.03),
  dx = ox + rnorm(100, 0.02, 0.02),
  dy = oy + rnorm(100, 0.02, 0.02),
  rain_mm = rgamma(100, shape = 1, rate = 1),
  temp_c = rnorm(100, 18, 5)
)

origins <- st_as_sf(trips, coords = c("ox", "oy"), crs = 4326)
dests   <- st_as_sf(trips, coords = c("dx", "dy"), crs = 4326)

# Work in a metric projection for distances
origins_m <- st_transform(origins, 3857)
dests_m   <- st_transform(dests, 3857)

# Straight-line distance as a simple placeholder
trips$dist_m <- as.numeric(st_distance(origins_m, dests_m, by_element = TRUE))

# Example: construct mode-specific features
choice_like <- trips %>%
  mutate(
    peak = as.integer(sample(0:1, n(), TRUE)),
    bike_disutility_context = dist_m / 1000 * (1 + 0.10 * rain_mm),
    walk_disutility_context = dist_m / 1000 * (1 + 0.05 * rain_mm),
    car_time_min = dist_m / 1000 / 35 * 60 * (1 + 0.25 * peak),
    bike_time_min = dist_m / 1000 / 14 * 60,
    walk_time_min = dist_m / 1000 / 5 * 60
  )

summary(choice_like[, c("dist_m", "rain_mm", "car_time_min", "bike_time_min")])
```

What to learn

This code does not yet estimate a real choice model. It shows the measurement step. In a real mode-choice or route-choice dataset, you would create one row per alternative per choice occasion, with alternative-specific travel times, costs, weather exposure, infrastructure quality, slope, and availability. The difficult part is constructing the attributes of unchosen alternatives.

18.12 Python lab: embeddings for choice models

This lab is pseudocode for a modern workflow. It is included because image and transformer-based text models are often easier in Python than in R. The idea is to compute embeddings first, save them, then use them in R or Python choice models.

```
# Text embeddings with sentence-transformers
# pip install sentence-transformers pandas

import pandas as pd
from sentence_transformers import SentenceTransformer

texts = pd.DataFrame({
    "alt_id": [1, 2, 3],
    "description": [
        "restores dunes and improves biodiversity",
        "builds flood barriers near the coast",
        "keeps current coastal management unchanged"
    ]
})

model = SentenceTransformer("all-MiniLM-L6-v2")
```

```
E = model.encode(texts["description"].tolist(), normalize_embeddings=True)

emb = pd.DataFrame(E)
emb["alt_id"] = texts["alt_id"]
emb.to_csv("text_embeddings.csv", index=False)
```

```
# Image embeddings with CLIP-style models
# This is a schematic example. Exact model names and APIs can change.

from PIL import Image
import pandas as pd
import torch
from transformers import CLIPProcessor, CLIPModel

image_files = ["street_a.jpg", "street_b.jpg", "street_c.jpg"]
model = CLIPModel.from_pretrained("openai/clip-vit-base-patch32")
processor = CLIPProcessor.from_pretrained("openai/clip-vit-base-patch32")

images = [Image.open(f).convert("RGB") for f in image_files]
inputs = processor(images=images, return_tensors="pt", padding=True)

with torch.no_grad():
    image_features = model.get_image_features(**inputs)
    image_features = image_features / image_features.norm(dim=1, keepdim=True)

emb = pd.DataFrame(image_features.numpy())
emb["image_file"] = image_files
emb.to_csv("image_embeddings.csv", index=False)
```

How to use these embeddings

There are three common options.

1. Use embeddings in a prediction model, such as random forest, XGBoost, or neural network.
2. Reduce the embeddings with PCA and include a few components in a choice model.
3. Use the embeddings to cluster alternatives or create interpretable labels, then estimate the welfare model with validated labels.

For environmental valuation, option 3 is often safest. Options 1 and 2 are useful for prediction and exploration, but they are harder to defend for WTP.

18.13 Common mistakes, checklist, and exercises

18.13.1 Common mistakes

Mistake 1: confusing trace data with choice data. A GPS point is not a trip. A trip is not necessarily a choice. A choice requires alternatives and availability.

Mistake 2: treating predicted labels as certain. Inferred mode, purpose, or sentiment may contain error. Sensitivity analysis should vary label quality assumptions.

Mistake 3: using row-wise splits in panel trace data. Trips from the same individual should not leak into both training and test sets when the goal is generalization to new individuals.

Mistake 4: computing WTP for black-box features. WTP requires a meaningful attribute change and a stable monetary scale.

Mistake 5: ignoring choice-set construction. Passive data usually show the chosen path, destination, or mode. The unchosen alternatives are constructed, not observed.

Mistake 6: forgetting privacy. Exact trajectories, images, and text can re-identify people or reveal sensitive routines.

18.13.2 Reporting checklist

For a paper using text, images, GPS, or passive data in choice modelling, report:

1. What raw objects were collected, from whom, when, and under what consent conditions.
2. How the raw objects were transformed into observations, alternatives, labels, and covariates.
3. How unchosen alternatives and choice sets were constructed.
4. Which features are transparent, semi-transparent, or opaque.
5. Whether splits are grouped by individual, location, image, route, or time as appropriate.
6. How label uncertainty, missingness, and measurement error were handled.
7. Whether the model is intended for prediction, behavioural interpretation, welfare analysis, or all three.
8. Which policy counterfactuals are within the support of the data.
9. What privacy protections and data-minimisation steps were used.
10. Which code and feature-construction steps are reproducible.

18.13.3 Exercises

Exercise 1. You have 10,000 reviews of coastal recreation sites and a DCE about beach management. Design a workflow to use the reviews to improve the DCE without using post-treatment information in the utility model.

Exercise 2. A city provides GPS trajectories from a cycling app. Define a route-choice model. What is the choice event? How will you generate unchosen routes? Which features are policy-controllable?

Exercise 3. A stated-preference survey shows respondents images of restored and unrestored wetlands. List five visual confounders that could bias the estimated restoration coefficient.

Exercise 4. Suppose a CNN predicts street-image choices better than an MNL model with semantic labels. Would you use the CNN for welfare analysis? Explain your answer.

Exercise 5. Propose a grouped validation strategy for a dataset with repeated trips by individuals across many neighbourhoods and seasons.

18.14 Reading guide

Start with Axhausen's chapter on self-tracing and reporting in the *Handbook of Choice Modelling* for the conceptual difference between socially meaningful travel diaries and movement traces. Then read Dahmen et al. for an applied tracking-based revealed-preference ML workflow with SHAP interpretation. Read Zhu and Si for image choice and the comparison between semantic image features and deep image models. Read van Cranenburgh et al. for a broader agenda on choice modelling in the age of ML, especially the discussion of text, images, and data streams. For context-dependent GPS-rich choice modelling, read Lukawska et al. on context-aware Bayesian mixed logit. For validation pitfalls in panel choice data, read Hillel's work on ML classifier validation in mode choice.

Chapter takeaway

Text, images, GPS traces, and passive records can make choice models more realistic, but they do not remove the need for economic structure. The best applied work separates measurement

from preference, validates derived features, constructs plausible choice sets, prevents leakage, and reserves welfare claims for attributes that have a clear behavioural and policy interpretation. The frontier is not simply to feed more raw data into larger models. It is to use richer data to measure the choice situation better while preserving the logic that makes choice modelling useful for environmental economics.

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Chapter 19: Welfare, WTP, Elasticities, and Counterfactuals under Machine Learning

Machine Learning and AI for Environmental Economics and Choice Modelling

Self-study textbook for PhD students

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Learning objectives

By the end of this chapter, you should be able to:

1. Explain why **prediction**, **welfare analysis**, and **counterfactual policy analysis** are different tasks.
2. Derive marginal willingness to pay (WTP) from a random utility model.
3. Compute direct and cross elasticities from multinomial logit and from more general prediction models.
4. Use the logsum formula to compute compensating variation or equivalent variation in standard logit-type models.
5. Understand why off-the-shelf machine learning models do not automatically deliver welfare-consistent WTP.
6. Use R to compare a discrete choice model and a machine learning classifier for prediction, elasticities, and counterfactual simulation.
7. Decide when ML outputs can be used for descriptive sensitivity analysis, when they can support counterfactual prediction, and when they should **not** be used for welfare analysis.

19.1 The core problem: prediction is not welfare

A trained machine learning model can often answer a question such as:

> Which alternative is this person most likely to choose?

or:

> What is the predicted market share after we change the attributes of the alternatives?

These are useful questions. In transport, energy, environmental valuation, land-use, climate adaptation, agriculture, and recreation-demand studies, good predictive models help forecast behaviour under observed or near-observed conditions.

But welfare analysis asks a deeper question:

> How much better or worse off are people after the policy change?

This requires a **money-metric** interpretation of the change in utility. In environmental economics, this is central because many goods are not traded in markets: clean water, biodiversity, reduced crowding, landscape quality, wildfire risk reduction, flood protection, ecosystem restoration, and protected-area access.

A random forest, gradient boosting model, or neural network can predict that a respondent is more likely to choose a high-biodiversity beach. This alone does not tell us how much that respondent is willing to pay for the biodiversity improvement. To get WTP, we need assumptions linking predicted choices to preferences, preferences to utility, and utility to money.

This chapter is therefore about a boundary:

ML can help us predict and explore behavioural patterns. Welfare analysis requires additional economic structure.

The important task is not to reject ML. The important task is to identify **which outputs are valid under which assumptions**.

19.2 What choice models add: utility, money, and welfare

A standard random utility model writes the utility that individual n obtains from alternative i as:

$$U_{ni} = V_{ni} + \varepsilon_{ni},$$

where V_{ni} is observed systematic utility and ε_{ni} is unobserved utility.

For a simple environmental choice experiment:

$$V_{ni} = \beta_c c_{ni} + \beta_b b_{ni} + \beta_q q_{ni} + \beta_r r_{ni},$$

where:

- c_{ni} is cost;
- b_{ni} is biodiversity quality;
- q_{ni} is water quality;
- r_{ni} is crowding or access restriction;
- β_c is expected to be negative;
- positive environmental attributes should usually have positive coefficients.

The model has an economic interpretation because cost enters the utility function. This allows non-market attributes to be converted into money units.

For attribute x, the marginal WTP is:

$$WTP_x = -\frac{\beta_x}{\beta_c}.$$

The negative sign appears because $\beta_c < 0$. If $\beta_x > 0$, then WTP_x is positive: respondents are willing to pay to obtain more of attribute x.

This ratio is not just a predictive summary. It is a marginal rate of substitution between money and the non-market attribute. This is why discrete choice models are so important in environmental economics.

19.3 Welfare measures in discrete choice

19.3.1 Consumer surplus and the logsum

For logit-type random utility models, the expected maximum utility from a choice set C_n is proportional to the logsum:

$$E\left[\max_{j \in C_n} U_{nj}\right] = \gamma + \ln\left(\sum_{j \in C_n} \exp(V_{nj})\right),$$

where γ is Euler's constant. The constant cancels when we compare two policy scenarios.

Let scenario 0 be the baseline and scenario 1 be the policy. The change in expected consumer surplus for individual n is often written as:

$$\Delta CS_n = -\frac{1}{\beta_c}\left[\ln \sum_{j \in C_n^1} \exp(V_{nj}^1) - \ln \sum_{j \in C_n^0} \exp(V_{nj}^0)\right].$$

Because β_c is negative, a policy that increases expected utility produces a positive money-metric welfare gain.

This formula is widely used because it can handle:

- changes in attributes of existing alternatives;
- changes in prices or costs;
- introduction of a new alternative;
- removal of an alternative;
- changes in choice probabilities across the full choice set.

19.3.2 Equivalent and compensating variation

In welfare economics, two related measures are common.

Compensating variation (CV) asks: after the change occurs, how much money would need to be taken away or given to the individual to restore the original utility?

Equivalent variation (EV) asks: before the change occurs, how much money would the individual be willing to pay or accept to be as well off as under the changed situation?

In many applied discrete choice settings with small changes and a constant marginal utility of income, the same logsum-based formula is used as an operational approximation. For environmental valuation, the difference between CV and EV can matter conceptually, but applied DCEs often report WTP-style measures based on the marginal utility of cost or income.

19.3.3 Welfare in mixed logit

In mixed logit, preferences vary across individuals:

$$\beta_n \sim f(\beta \mid \theta).$$

The individual-specific welfare change is:

$$\Delta CS_n(\beta_n) = -\frac{1}{\beta_{c,n}} \left[\ln \sum_j \exp(V^1_{nj}(\beta_n)) - \ln \sum_j \exp(V^0_{nj}(\beta_n)) \right].$$

Because β_n is usually unobserved, we approximate the welfare distribution by simulation. Draw R parameter vectors β_{nr} from the estimated distribution or from the posterior conditional on individual choices. Then compute:

$$\widehat{\Delta CS}_n = \frac{1}{R} \sum_{r=1}^{R} \Delta CS_n(\beta_{nr}).$$

The researcher can then report the mean, median, quantiles, share of negative welfare changes, and distributional effects across income, geography, exposure, or experience groups.

19.4 WTP: ratios, WTP-space models, and nonlinear utility

19.4.1 Preference-space WTP

In preference space, we estimate utility coefficients and then compute WTP as a ratio:

$$WTP_x = -\frac{\beta_x}{\beta_c}.$$

This is simple but can be unstable if β_c is close to zero or randomly distributed with support near zero. It can also produce heavy-tailed WTP distributions.

19.4.2 WTP-space models

In WTP-space models, parameters are estimated directly in money units. A simple utility can be written as:

$$V_{ni} = \lambda_n \left(w_{b,n} b_{ni} + w_{q,n} q_{ni} - c_{ni} \right),$$

where:

- $w_{b,n}$ is WTP for biodiversity;
- $w_{q,n}$ is WTP for water quality;
- λ_n is a scale or marginal utility of income term.

This parameterization is often useful when WTP is the main target. It can also make it easier to impose distributional restrictions on WTP rather than on utility coefficients.

19.4.3 Nonlinear utility and marginal WTP

If utility is nonlinear, WTP is local. Suppose:

$$V_{ni} = f(c_{ni}, x_{ni}, z_n).$$

The marginal WTP for x is:

$$MWTP_x = -\frac{\partial V_{ni}/\partial x_{ni}}{\partial V_{ni}/\partial c_{ni}}.$$

This is the expression that matters for flexible ML-assisted utility functions. If a neural network is used to represent V_{ni}, the WTP can be computed only if the model has a utility interpretation and if the derivative with respect to cost behaves like a marginal utility of money.

19.4.4 Discrete changes

For a discrete attribute change, for example moving from low to high biodiversity, a finite-difference WTP may be preferable:

$$WTP_{x:a\to b} = -\frac{V(c, x = b, z) - V(c, x = a, z)}{\partial V/\partial c}.$$

If cost effects are nonlinear, we may need to solve for the compensating cost change δ such that:

$$V(c + \delta, x = b, z) = V(c, x = a, z).$$

This root-finding formulation is useful for flexible models.

19.5 Elasticities: behavioural sensitivity without full welfare

Elasticities measure sensitivity of demand or choice probabilities to changes in attributes. They are often easier to compute than welfare measures and can be computed for both choice models and ML prediction models.

For alternative i and variable x_{ni}, the point elasticity is:

$$E_{i,x} = \frac{\partial P_{ni}}{\partial x_{ni}} \cdot \frac{x_{ni}}{P_{ni}}.$$

For a multinomial logit model with $V_{ni} = \beta_x x_{ni} + \cdots$, the direct elasticity is:

$$E_{i,x_i} = \beta_x x_{ni}(1 - P_{ni}),$$

and the cross elasticity for $j \neq i$ is:

$$E_{i,x_j} = -\beta_x x_{nj} P_{nj}.$$

Elasticities can also be computed by finite differences:

$$E_{i,x} \approx \frac{P_i(x(1+h)) - P_i(x)}{hP_i(x)}.$$

This finite-difference approach works for random forests, gradient boosting, neural networks, and any model that returns probabilities. However, the resulting elasticity is a **model sensitivity measure**, not automatically a welfare measure.

Box 19.1: The hierarchy of outputs

Outputs differ in how much economic structure they require.

Output	Can ordinary ML provide it?	Economic structure needed?	Safe interpretation
Classification accuracy	Yes	Low	Predictive performance
Probability prediction	Yes	Low to medium	Forecasting under similar support
Market share simulation	Yes	Medium	Counterfactual prediction if intervention is credible
Marginal effects	Yes	Medium	Sensitivity of predicted probabilities
Elasticities	Yes	Medium	Sensitivity, not necessarily welfare
WTP	Not automatically	High	Requires utility and money metric
Consumer surplus	Not automatically	High	Requires welfare-consistent choice structure
Policy welfare comparison	Not automatically	Very high	Requires identification, support, and welfare theory

19.6 Counterfactual prediction: what changes, and why?

A counterfactual is a statement about what would happen under a scenario that did not occur in the observed data.

In choice modelling, common counterfactuals include:

- increasing an entry fee for a national park;
- reducing beach crowding through access restrictions;
- improving water quality;
- introducing a congestion charge;
- adding a new public transport route;
- removing a fossil-fuel subsidy;
- adding an eco-labelled product;
- introducing a carbon-neutral shipping option.

The counterfactual workflow is:

1. Define the baseline choice set and attributes.
2. Define the policy choice set and attributes.
3. Predict probabilities under both scenarios.
4. Aggregate probabilities to market shares, expected visits, emissions, revenues, or participation.
5. If the model is welfare-consistent, compute WTP or consumer surplus.
6. Report uncertainty and sensitivity to assumptions.

The crucial issue is **support**. A model is more credible when the counterfactual lies near the observed data. If all observed park fees are between 0 and 20 EUR, predicting behaviour at 200 EUR is extrapolation. Theory can sometimes discipline extrapolation; flexible ML may simply extend patterns in unreliable ways.

19.7 What ML can do in welfare and counterfactual work

ML contributes in at least six ways.

19.7.1 Better prediction within the support of the data

When the policy change is close to observed variation, ML may improve probability forecasts. For example, a gradient boosting model can capture nonlinear effects of income, travel distance, household composition, environmental attitudes, and past recreation experience.

19.7.2 Flexible nuisance functions for causal estimation

In causal ML, flexible models estimate nuisance functions such as propensity scores and outcome regressions. Double/debiased ML and causal forests use ML while targeting causal estimands with orthogonalization, sample splitting, or honesty. This is valuable for environmental policy evaluation when the target is heterogeneous treatment effects rather than WTP.

19.7.3 Discovery of heterogeneity

ML can reveal which groups are most sensitive to a policy. For instance, a causal forest might show that a conservation message increases programme enrolment mainly among households with previous outdoor recreation experience. This is a distributional insight, not a welfare estimate by itself.

19.7.4 ML-assisted specification search

Choice model specification can be labour-intensive. ML can help find interactions, nonlinearities, segmentation variables, and candidate random parameters. The final welfare model can remain a theory-consistent DCM.

19.7.5 Flexible but structured utility functions

Neural networks can be used inside utility functions, provided the architecture respects the requirements needed for welfare interpretation. Examples include neural utility components, learning MNL models, RUM-compatible neural networks, and models that impose sign or monotonicity constraints on cost.

19.7.6 Surrogate modelling for expensive structural models

Some structural or simulation-based models are slow to estimate or simulate. ML can approximate model outputs, accelerate scenario analysis, or emulate posterior predictive distributions. In this role, ML is not replacing welfare theory; it is approximating a welfare-consistent simulator.

19.8 When ML outputs are not welfare measures

An ML classifier maps covariates to a predicted label or probability:

$$\hat{P}(Y = i \mid X = x).$$

This is not the same as a random utility model unless additional restrictions hold. In a DCM, probabilities arise from utility comparisons and a stochastic structure. In a generic classifier, probabilities arise from a loss function and an algorithmic representation.

A generic ML probability can be useful for counterfactual prediction if:

- the counterfactual change is similar to variation in the training data;
- the model includes the relevant policy variable;
- the training data variation is plausibly exogenous or appropriately controlled;
- validation is grouped correctly for panel or repeated-choice data;
- uncertainty and extrapolation risk are reported.

But WTP and consumer surplus require more:

- a utility interpretation;

- a monetary attribute with a valid marginal utility of income;
- consistency between probability changes and utility changes;
- no uncontrolled endogeneity in the monetary attribute;
- meaningful substitution patterns;
- plausible behaviour outside the training support;
- welfare-relevant preferences, not merely correlations.

Box 19.2: A quick diagnostic question

Before using an ML model for welfare, ask:

> If I increase cost while holding everything else constant, does the model guarantee that utility weakly decreases and that WTP remains economically meaningful?

If the answer is no, do not report WTP from that model as an economic welfare measure. You may still report predicted probabilities, sensitivity curves, or elasticities as descriptive model outputs.

19.9 Economically constrained ML

A growing literature tries to combine ML flexibility with economic structure. The aim is to obtain some of the predictive power of ML without losing the interpretation required for policy and welfare.

19.9.1 Structured utility networks

A structured utility network writes:

$$V_{ni} = g_\theta(x_{ni}, z_n),$$

where g_θ is a neural network. To make this welfare-relevant, the network may impose:

- alternative-specific utility representation;
- a common monetary scale;
- monotonicity in cost;
- separability of cost from non-cost attributes;
- RUM-consistent probability generation;
- regularization to avoid implausible curvature.

Then marginal WTP is computed from derivatives:

$$MWTP_x(n,i) = -\frac{\partial g_\theta / \partial x_{ni}}{\partial g_\theta / \partial c_{ni}}.$$

The derivative can be obtained by automatic differentiation in Python or by finite differences in R.

19.9.2 Learning MNL and residual utility models

A useful compromise decomposes utility into an interpretable and a flexible component:

$$V_{ni} = V_{ni}^{econ}(\beta) + h_\theta(x_{ni}, z_n).$$

Here V_{ni}^{econ} contains key policy variables such as cost, travel time, biodiversity, or emissions. The ML component captures residual nonlinearities and interactions. This can improve prediction while preserving interpretability for selected variables.

But caution is needed. If the ML component also absorbs variation in cost or environmental attributes, the economic coefficients may no longer represent the full marginal utility of those variables. One strategy is to restrict h_θ to variables that are not the target of welfare analysis, or to impose orthogonality/regularization conditions.

19.9.3 Monotonic ML

Some algorithms can impose monotonicity. For example, gradient boosting implementations can constrain predictions to decrease when cost increases. This is useful but not sufficient. Monotonic probability predictions do not automatically create a valid money-metric utility function.

Monotonicity is a necessary diagnostic, not a complete welfare theory.

19.9.4 Differentiable welfare models

When a model is differentiable and RUM-consistent, automatic differentiation can compute:

- marginal utilities;
- WTP distributions;
- elasticities;
- policy gradients;
- uncertainty through bootstrapping or Bayesian posterior draws.

This is one reason Python frameworks such as PyTorch, TensorFlow, JAX, and Bayesian probabilistic programming are becoming relevant to advanced choice modelling. R remains excellent for conventional DCE analysis and reproducible empirical workflows, while Python is often convenient for custom differentiable architectures.

19.10 Environmental economics examples

19.10.1 Beach management and crowding

A coastal manager considers restricting daily visitor numbers. A DCE estimates preferences for:

- access fee;
- crowding;
- water quality;
- seabed type;
- distance from accommodation;
- availability of parking.

A conventional mixed logit can estimate WTP for lower crowding and improved ecological quality. ML can help by:

- predicting attendance under different quota rules;
- detecting nonlinear crowding thresholds;
- segmenting visitors by experience and trip purpose;
- using image-based measures of beach crowding;
- identifying interactions between crowding and environmental quality.

But the welfare effect of the quota must account for both improved on-site quality and reduced access for excluded visitors. A classifier that predicts who still visits is not enough.

19.10.2 Biodiversity conservation programme

A DCE presents households with alternative conservation projects. Attributes include annual tax, protected species, habitat area, local employment, and access restrictions.

A choice model can estimate WTP for biodiversity and compare total WTP with programme cost. ML can support:

- flexible modelling of heterogeneous preferences;
- text analysis of open-ended survey comments;
- validation of attribute non-attendance;
- identification of subgroups with negative WTP;
- survey design improvements.

The final cost-benefit analysis should still be based on welfare-consistent estimates or clearly labelled as predictive/descriptive.

19.10.3 Dynamic electricity tariffs

A utility wants to introduce dynamic electricity pricing. Smart-meter data provide high-frequency demand observations. ML can forecast load and predict demand response. Causal ML can estimate heterogeneous treatment effects of information nudges or tariff trials.

Welfare analysis requires more: household utility from consumption, discomfort from shifting use, income effects, and distributional impacts. A highly accurate load forecast does not directly tell us whether households are better off.

19.11 Worked R lab: MNL welfare vs ML counterfactual prediction

This lab simulates an environmental DCE. We estimate an MNL model and a random forest classifier. The MNL model is used for WTP and logsum welfare. The random forest is used for predictive counterfactuals and finite-difference sensitivity.

19.11.1 Simulate the data

```
library(tidyverse)
library(mlogit)
library(tidymodels)

set.seed(1901)

N  <- 600      # respondents
Tn <- 8        # tasks per respondent
alts <- c("A", "B", "SQ")

beta_cost <- -0.06
beta_bio  <-  0.55
beta_water <- 0.40
beta_crowd <- -0.35
asc <- c(A = 0.05, B = 0.03, SQ = 0.00)

long <- expand_grid(
  id = 1:N,
  task = 1:Tn,
  alt = alts
) %>%
  mutate(
    chid = paste(id, task, sep = "_"),
    cost = if_else(alt == "SQ", 0,
                   sample(c(10, 25, 50, 75), n(), replace = TRUE)),
    biodiversity = if_else(alt == "SQ", 0,
                           sample(0:3, n(), replace = TRUE)),
    water_quality = if_else(alt == "SQ", 0,
                            sample(0:3, n(), replace = TRUE)),
    crowding = if_else(alt == "SQ", 2,
                       sample(0:3, n(), replace = TRUE)),
    V = asc[alt] +
      beta_cost * cost +
      beta_bio * biodiversity +
      beta_water * water_quality +
      beta_crowd * crowding
  ) %>%
  group_by(chid) %>%
  mutate(
    p = exp(V) / sum(exp(V)),
    chosen_alt = sample(alt, size = 1, prob = p),
```

```
    choice = as.integer(alt == chosen_alt)
  ) %>%
  ungroup()
```

19.11.2 Estimate MNL

```
mdata <- mlogit.data(
  long,
  choice = "choice",
  shape = "long",
  chid.var = "chid",
  alt.var = "alt",
  id.var = "id"
)

mnl <- mlogit(
  choice ~ cost + biodiversity + water_quality + crowding | 0,
  data = mdata
)

summary(mnl)
coef(mnl)
```

19.11.3 Compute WTP

```
b <- coef(mnl)

wtp <- tibble(
  attribute = c("biodiversity", "water_quality", "crowding"),
  wtp = c(
    -b["biodiversity"] / b["cost"],
    -b["water_quality"] / b["cost"],
    -b["crowding"] / b["cost"]
  )
)

wtp
```

Interpretation:

- A positive WTP for biodiversity means households are willing to pay for biodiversity improvement.
- A negative WTP for crowding means households require compensation to accept more crowd-

ing, or equivalently are willing to pay to reduce crowding.
- These WTP values are meaningful because they are ratios of utility coefficients under a random utility framework.

19.11.4 Logsum welfare for a policy scenario

Suppose policy 1 improves biodiversity and water quality by one level for alternatives A and B, and increases cost by 10 EUR.

```
policy <- long %>%
  mutate(
    cost_1 = if_else(alt == "SQ", cost, cost + 10),
    biodiversity_1 = if_else(alt == "SQ", biodiversity,
                             pmin(biodiversity + 1, 3)),
    water_quality_1 = if_else(alt == "SQ", water_quality,
                              pmin(water_quality + 1, 3)),
    crowding_1 = crowding
  )

b_cost <- b["cost"]

welfare <- policy %>%
  mutate(
    V0 = b["cost"] * cost +
      b["biodiversity"] * biodiversity +
      b["water_quality"] * water_quality +
      b["crowding"] * crowding,
    V1 = b["cost"] * cost_1 +
      b["biodiversity"] * biodiversity_1 +
      b["water_quality"] * water_quality_1 +
      b["crowding"] * crowding_1
  ) %>%
  group_by(chid) %>%
  summarise(
    logsum0 = log(sum(exp(V0))),
    logsum1 = log(sum(exp(V1))),
    delta_cs = -(logsum1 - logsum0) / b_cost,
    .groups = "drop"
  )

summary(welfare$delta_cs)
mean(welfare$delta_cs)
```

The object `delta_cs` is the estimated welfare change per choice task in money units. In a real study, you would carefully define the temporal scale: per visit, per household per year, per programme,

or per trip.

19.11.5 Fit an ML classifier

To use a standard ML classifier, convert the data to one row per choice task.

```
wide <- long %>%
  select(id, task, chid, alt, cost, biodiversity,
         water_quality, crowding, chosen_alt) %>%
  pivot_wider(
    names_from = alt,
    values_from = c(cost, biodiversity, water_quality, crowding)
  ) %>%
  distinct(id, chid, .keep_all = TRUE) %>%
  mutate(chosen_alt = factor(chosen_alt))

set.seed(1902)
split <- group_initial_split(wide, group = id, prop = 0.8)
train <- training(split)
test  <- testing(split)

folds <- group_vfold_cv(train, group = id, v = 5)

rf_spec <- rand_forest(
  trees = 800,
  mtry = tune(),
  min_n = tune()
) %>%
  set_engine("ranger", importance = "permutation") %>%
  set_mode("classification")

rf_wf <- workflow() %>%
  add_formula(chosen_alt ~ . - id - task - chid) %>%
  add_model(rf_spec)

rf_grid <- grid_regular(
  mtry(range = c(2, 10)),
  min_n(range = c(2, 30)),
  levels = 4
)

rf_tuned <- tune_grid(
  rf_wf,
  resamples = folds,
```

```
  grid = rf_grid,
  metrics = metric_set(accuracy, mn_log_loss)
)

best_rf <- select_best(rf_tuned, metric = "mn_log_loss")

rf_fit <- finalize_workflow(rf_wf, best_rf) %>%
  fit(train)

pred <- predict(rf_fit, test) %>%
  bind_cols(test %>% select(chosen_alt))

metrics(pred, truth = chosen_alt, estimate = .pred_class)
```

The grouped split is essential because repeated choices from the same respondent must not leak across training and test sets.

19.11.6 Counterfactual prediction with ML

Now create a counterfactual version of the test data.

```
test_policy <- test %>%
  mutate(
    cost_A = cost_A + 10,
    cost_B = cost_B + 10,
    biodiversity_A = pmin(biodiversity_A + 1, 3),
    biodiversity_B = pmin(biodiversity_B + 1, 3),
    water_quality_A = pmin(water_quality_A + 1, 3),
    water_quality_B = pmin(water_quality_B + 1, 3)
  )

p0 <- predict(rf_fit, test, type = "prob")
p1 <- predict(rf_fit, test_policy, type = "prob")

market_shift <- bind_rows(
  baseline = colMeans(p0),
  policy = colMeans(p1),
  .id = "scenario"
)

market_shift
```

This is a predicted market-share effect. It is not a welfare effect. You can say:

> Under the trained random forest, the policy scenario is predicted to increase the prob-

ability of choosing programme A and B.

You should not say:

The random forest implies a welfare gain of X EUR.

unless the model has been built with a valid utility and money-metric structure.

19.12 Computing finite-difference elasticities from any probability model

The following function computes an arc elasticity for a selected variable in a fitted tidymodels workflow.

```r
arc_elasticity <- function(fit, data, var, outcome_level, h = 0.01) {
  data1 <- data
  data1[[var]] <- data1[[var]] * (1 + h)

  p0 <- predict(fit, data, type = "prob")[[paste0(".pred_", outcome_level)]]
  p1 <- predict(fit, data1, type = "prob")[[paste0(".pred_", outcome_level)]]

  mean((p1 - p0) / (h * p0), na.rm = TRUE)
}

arc_elasticity(rf_fit, test, var = "cost_A", outcome_level = "A")
```

For a choice model, a comparable finite-difference approach can be used by predicting probabilities under baseline and perturbed scenarios. This is often safer in complex models than relying on analytic derivatives.

19.13 Python sketch: automatic differentiation for structured utility

The following sketch shows the logic of a differentiable utility model. It is not a full production model, but it illustrates how marginal WTP can be computed when utility is explicitly modelled.

```python
import torch
import torch.nn as nn

class UtilityNet(nn.Module):
    def __init__(self, p):
        super().__init__()
        self.hidden = nn.Sequential(
            nn.Linear(p, 32),
            nn.ReLU(),
```

```
            nn.Linear(32, 16),
            nn.ReLU()
        )
        self.env_part = nn.Linear(16, 1)
        self.cost_weight_raw = nn.Parameter(torch.tensor(0.0))

    def forward(self, x_noncost, cost):
        h = self.hidden(x_noncost)
        env_utility = self.env_part(h)
        beta_cost = -torch.nn.functional.softplus(self.cost_weight_raw)
        return env_utility + beta_cost * cost

model = UtilityNet(p=5)

x = torch.randn(1, 5, requires_grad=True)
cost = torch.tensor([[20.0]], requires_grad=True)

V = model(x, cost)

# Marginal utility of the first non-cost attribute
mu_x = torch.autograd.grad(V, x, retain_graph=True)[0][0, 0]
mu_c = torch.autograd.grad(V, cost)[0][0, 0]

mwtp = -mu_x / mu_c
print(mwtp.item())
```

This code enforces a negative cost coefficient using `softplus`. However, this is still only one condition. A full model would also need a choice probability layer, training by likelihood or cross-entropy on choice data, validation, uncertainty estimation, and careful checks of economic plausibility.

19.14 Reporting uncertainty

WTP and welfare estimates are functions of estimated parameters. They require uncertainty intervals.

Common approaches include:

1. **Delta method**: approximate the variance of a nonlinear function using a first-order Taylor expansion.
2. **Krinsky-Robb simulation**: draw parameters from the asymptotic multivariate normal distribution and recompute WTP or welfare.
3. **Bootstrap**: resample individuals or choice tasks appropriately, re-estimate the model, and recompute outputs.
4. **Bayesian posterior simulation**: compute welfare for each posterior draw.

5. **Sensitivity analysis**: report how results change under alternative model specifications, cost distributions, or sample restrictions.

For repeated-choice DCE data, bootstrap resampling should usually be at the **respondent level**, not the row level.

19.14.1 Krinsky-Robb simulation in R

```
library(MASS)

B <- 2000
mu <- coef(mnl)
Sigma <- vcov(mnl)

draws <- MASS::mvrnorm(B, mu = mu, Sigma = Sigma)

wtp_draws <- tibble(
  biodiversity = -draws[, "biodiversity"] / draws[, "cost"],
  water_quality = -draws[, "water_quality"] / draws[, "cost"],
  crowding = -draws[, "crowding"] / draws[, "cost"]
)

wtp_draws %>%
  summarise(across(everything(),
                   list(mean = mean,
                        lwr = ~quantile(.x, 0.025),
                        upr = ~quantile(.x, 0.975))))
```

Check the simulated cost coefficients. If many draws have positive cost coefficients, WTP intervals may be unstable. This is not a computational nuisance; it is a substantive warning about the model.

19.15 Distributional welfare and equity

Environmental policy often has distributional consequences. Mean WTP can hide important heterogeneity.

Report welfare by:

- income group;
- age group;
- rural/urban residence;
- exposure to environmental risk;
- prior experience with the resource;
- distance to the site;

- vulnerability group;
- stated environmental attitudes;
- region or community.

A welfare-improving policy on average may harm some groups. In environmental justice applications, this is not a side note. It may be the main policy issue.

ML can help identify distributional patterns, but the analyst must avoid treating discovered subgroups as automatically causal. Use ML discovery as a guide, then confirm with theory, robustness checks, and, where possible, causal identification.

19.16 Checklist: Is this model welfare-ready?

Use this checklist before reporting WTP or consumer surplus from a model that uses ML.

Question	Why it matters
Does the model represent utility, not just class labels?	WTP is a utility trade-off.
Is there a monetary attribute with correct sign?	Money-metric welfare requires a marginal utility of money.
Is cost monotonic?	Higher cost should not increase utility, all else equal.
Is the cost effect common or interpretable across alternatives?	Logsum welfare requires clear monetary scaling.
Are target attributes excluded from black-box residuals unless intended?	Otherwise WTP ratios may be partial or misleading.
Is the model valid under choice-set changes?	Policy often adds or removes alternatives.
Is the counterfactual inside the support of training data?	Extrapolation risk can dominate results.
Are repeated observations split by person in validation?	Prevents leakage and overoptimistic performance.
Are elasticities and signs plausible?	Sanity checks reveal misspecification.
Is uncertainty reported?	Welfare is not a point estimate only.
Are causal assumptions explicit?	Policy effects require identification, not just association.

If several answers are no, report the model as predictive or exploratory, not welfare-ready.

19.17 Common mistakes

Mistake 1: Treating probability changes as welfare changes

A probability increase is not a welfare gain. A policy may increase adoption because it removes alternatives, raises constraints, or shifts default options. Welfare requires a valuation of the change.

Mistake 2: Computing WTP from a classifier without utility

Feature importance is not marginal utility. SHAP values are not WTP. Partial dependence curves are not consumer surplus.

Mistake 3: Ignoring scale

Choice coefficients are confounded with scale. Comparing coefficients across datasets, models, or RP/SP sources without accounting for scale can mislead WTP interpretation.

Mistake 4: Forgetting the outside option

Welfare changes depend on the full choice set. Removing or misrepresenting the status quo can seriously distort WTP and consumer surplus.

Mistake 5: Reporting mean WTP only

Environmental policies often create winners and losers. Report distributions, quantiles, and subgroup effects.

Mistake 6: Validating prediction but not behaviour

A model can predict well and still imply positive cost effects, implausible substitution patterns, or unstable WTP. Behavioural diagnostics are part of model validation.

19.18 Concept checks

Try to answer without looking back.

1. Why can a random forest elasticity be useful but not welfare-sufficient?
2. What assumptions are needed to interpret $-\beta_x/\beta_c$ as WTP?
3. What is the logsum formula used for?
4. Why can WTP ratios be unstable when the cost coefficient is close to zero?
5. How does a WTP-space model differ from a preference-space model?
6. What is the difference between a counterfactual prediction and a causal policy effect?
7. Why should DCE bootstrap resampling often occur at the respondent level?
8. What does monotonicity in cost guarantee, and what does it not guarantee?

19.19 Applied exercises

Exercise 1: WTP from an MNL model

Using your own DCE data or the simulated data in this chapter:

1. Estimate an MNL model.
2. Compute WTP for each environmental attribute.
3. Use Krinsky-Robb simulation to construct 95 percent intervals.

4. Discuss whether the signs and magnitudes are plausible.

Exercise 2: Logsum welfare

Define a policy scenario that improves an environmental attribute but increases cost.

1. Compute baseline and policy logsums.
2. Compute individual-level welfare changes.
3. Plot the distribution of welfare changes.
4. Identify whether any respondents or segments lose from the policy.

Exercise 3: ML prediction versus welfare

Estimate a random forest or XGBoost classifier for the same choice data.

1. Compare out-of-sample log loss with MNL.
2. Simulate the same policy scenario.
3. Report predicted market-share changes.
4. Explain why you cannot directly report WTP from the classifier.

Exercise 4: Monotonicity check

For your ML model:

1. Increase cost for one alternative by 1 percent, 5 percent, and 20 percent.
2. Plot the predicted probability of choosing that alternative.
3. Check whether the model ever predicts an increase in choice probability when cost rises.
4. Discuss whether this is a problem for prediction, welfare, or both.

Exercise 5: Research extension

Design a hybrid model for an environmental DCE in which:

- cost and main environmental attributes enter an interpretable utility component;
- demographics and attitudes enter a flexible ML component;
- the model is constrained so that cost has a negative marginal effect;
- WTP can still be derived for the environmental attributes.

Write a one-page research proposal explaining the model, the data, the validation strategy, and the welfare outputs.

19.20 Summary

This chapter has developed the boundary between prediction and welfare.

The main lessons are:

1. Choice models are valuable because they connect choices to utility, utility to money, and money to welfare.

2. ML models can improve prediction, discover heterogeneity, support specification search, and compute flexible sensitivity patterns.
3. Generic ML outputs are not automatically WTP or consumer surplus.
4. Elasticities and marginal effects from ML are useful descriptive sensitivity measures, but they require caution in policy interpretation.
5. Welfare-ready ML must be structured around utility, money, monotonicity, and credible counterfactual assumptions.
6. In environmental economics, where policies often involve non-market goods and distributional consequences, this distinction is essential.

The best applied work does not ask whether ML or choice modelling is superior in general. It asks what the research task requires. For pure prediction, ML may dominate. For welfare, WTP, and counterfactual policy appraisal, economic structure remains indispensable. The frontier is to combine both.

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Chapter 20

Reproducible Applied Workflow

From Research Question to Data, Models, Counterfactuals, and a Reusable Paper

Chapter purpose

This final chapter turns the previous methodological chapters into a practical research system. The goal is not only to run models, but to build an auditable workflow that another researcher - or your future self - can rerun from raw inputs to final tables, figures, welfare measures, and policy simulations.

Learning objectives

- Design a reproducible project structure for machine learning, environmental economics, and choice modelling.
- Separate raw data, derived data, exploratory scripts, production functions, model objects, and reporting outputs.
- Use version control, package management, seeds, pipelines, and literate programming to make analyses inspectable and rerunnable.
- Connect R-based workflows for discrete choice models with ML workflows in tidymodels, mlr3, Python, Apollo, Biogeme, and Quarto.
- Create an applied checklist for validation, leakage prevention, welfare calculations, and policy counterfactuals.
- Plan a final PhD-paper workflow that is publishable, archivable, and extensible.

20.1 Why workflow is a scientific method, not an administrative detail

A reproducible workflow is the bridge between statistical theory and credible empirical research. In environmental economics and choice modelling, a typical project includes survey design files, spatial data, remote-sensing layers, raw survey exports, data cleaning rules, experimental-design metadata, multiple model families, hyperparameter tuning, welfare calculations, policy scenarios, and manuscript tables. Without an explicit workflow, the analysis becomes a sequence of memory-dependent manual actions. That is risky because small changes in recoding, filtering, train-test splitting, or cost normalization can alter coefficients, predictive performance, WTP estimates, and policy conclusions.

The problem is amplified when machine learning enters the project. ML workflows usually add preprocessing recipes, resampling folds, tuning grids, performance metrics, trained objects, model explanation outputs, and sometimes large tensor or embedding files. Choice modelling adds its own structure: alternatives belong to choice sets, repeated tasks belong to respondents, and welfare

outputs depend on signs, scales, and model assumptions. Reproducibility therefore requires more than sharing a final script. It requires a system that keeps every dependency visible.

The safest principle is simple: every number in the paper should be traceable back to raw inputs, code, model settings, and a clearly documented decision. This does not mean that all data can always be public. Many projects use restricted household, GPS, smartphone, or administrative data. But even when raw data cannot be shared, the cleaning logic, synthetic examples, model code, session information, and derivation of final outputs should be as transparent as possible.

Key idea

Reproducibility is not only about rerunning code. It is about making the chain of scientific decisions visible: what was measured, how it was transformed, how the validation strategy was chosen, how models were compared, and how policy quantities were derived.

20.2 The end-to-end applied research cycle

A strong applied project can be organized as a cycle rather than a straight line. The cycle below is useful for PhD work because it forces you to separate substantive choices from technical steps.

Stage	Main question	Typical outputs	Risks if undocumented
Research design	What behavioural, environmental, or policy question is being answered?	Research question, estimand, target population, conceptual framework	Model becomes detached from the policy question; prediction and welfare aims become confused
Data acquisition	What observations are valid for the target question?	Raw data, metadata, survey instrument, design matrix, spatial/temporal joins	Sample construction becomes impossible to audit
Data processing	How are variables, alternatives, and choice sets built?	Clean data, codebook, exclusion log, transformed features	Silent recoding errors; inconsistent alternative-specific variables
Validation design	What does out-of-sample performance mean in this context?	Grouped folds, temporal split, spatial blocks, test set, pre-analysis plan	Data leakage, overoptimistic ML performance, invalid comparison with DCMs
Model estimation	Which model family answers which task?	MNL, MXL, LCCM, causal ML, random forest, XGBoost, neural models	Overfitting, unidentified choice models, arbitrary tuning, failed convergence
Behavioural checks	Do signs, elasticities, WTP, and substitution patterns make sense?	Coefficient tables, elasticities, WTP distributions, partial dependence or ALE plots	Good accuracy but implausible policy conclusions
Counterfactuals	Which intervention is being simulated, and within what support?	Scenario definitions, prediction intervals, welfare changes, uncertainty summaries	Extrapolation beyond observed support; unsupported welfare claims
Reporting and archiving	Can the project be rerun and understood?	Quarto paper, tables, figures, README, lockfile, repository, archive	Irreproducible paper; lost model objects; unclear provenance

20.3 Five workflow principles for ML and choice modelling

Principle 1: Separate raw inputs from generated outputs
Raw data should be immutable. Never overwrite original survey exports, GPS traces, stated-choice design files, or satellite downloads. Derived files should be generated by code. If a derived dataset is wrong, fix the code and rebuild it.

Principle 2: Keep the unit of analysis explicit
Choice data are hierarchical. Alternatives sit inside choice tasks; tasks sit inside respondents; respondents may sit inside households, neighbourhoods, beaches, routes, or countries. Validation and clustering must respect this structure.

Principle 3: Estimate preprocessing inside resampling
Scaling, imputation, feature selection, text embeddings, PCA, class balancing, and tuning must be learned inside the training fold and then applied to the assessment fold. Otherwise, future information leaks into model training.

Principle 4: Match workflow to estimand
A pure prediction workflow is not the same as a welfare workflow. A DCE paper that reports WTP needs cost coefficients, scale handling, uncertainty propagation, and welfare formulas; a forecasting paper needs calibrated probabilities and robust external validation.

Principle 5: Automate what must be repeated; document what must be judged
Cleaning, model fitting, figure generation, and table production should be scripted. Judgement calls - exclusions, priors, constraints, class number, scenario design - should be logged and justified.

20.4 A project structure that works for PhD research

The following structure is intentionally conservative. It works for stated preference, revealed preference, environmental valuation, causal ML, and hybrid ML-choice projects.

```
project-name/
  README.md
  project-name.Rproj
  renv.lock
  _targets.R                # pipeline definition, optional but recommended
  _quarto.yml               # book / paper / website configuration, optional


  data/
    raw/                    # never edit manually
    intermediate/           # generated by cleaning scripts
    processed/              # analysis-ready data generated by pipeline
    external/               # shapefiles, raster layers, weather, census, etc.
    restricted/             # not committed if confidential


  design/
    dce_design.xlsx
    survey_instrument.pdf
    attribute_dictionary.csv


  R/
    00_setup.R
    clean_choice_data.R
    build_features.R
    estimate_mnl.R
    estimate_mxl.R
```

```
    train_ml_models.R
    simulate_counterfactuals.R
    make_tables.R
    make_figures.R


  python/
    embeddings.py
    neural_choice_model.py
    biogeme_models.py


  outputs/
    models/
    tables/
    figures/
    diagnostics/
    counterfactuals/


  reports/
    paper.qmd
    appendix.qmd
    slides.qmd


  docs/
    codebook.md
    exclusion_log.md
    model_registry.csv
    scenario_registry.csv
```

Practical rule

A good repository should answer four questions quickly: Where did the data come from? Which script creates this table? Which model object created this counterfactual? Which package versions were used?

20.5 Version control with Git

Version control is essential because empirical work changes continuously. You change filters, add robustness checks, revise covariates, correct coding errors, and receive referee requests. Without version control, it becomes impossible to know which code produced which table. Git is the standard tool for tracking changes, branching experimental work, and recovering previous states of a project. The official Git documentation and the freely available Pro Git book are good entry points for new users.

For PhD work, the minimum useful Git habit is to commit after each coherent change. A commit should represent a meaningful step: import raw data, fix cleaning rule, add grouped cross-validation, estimate MNL baseline, add random forest benchmark, or regenerate WTP tables. Avoid commits called "final", "really final", or "changes". Commit messages are a lab notebook.

```
# One-time setup inside the project folder
git init
git add README.md R/ reports/ design/ docs/
git commit -m "Initialize reproducible choice-modelling project"


# Daily workflow
git status
git add R/clean_choice_data.R R/estimate_mnl.R reports/paper.qmd
git commit -m "Estimate MNL baseline and add coefficient table"
```

```
# Branch for experimental ML models
git checkout -b xgboost-benchmark
# ... edit code ...
git add R/train_ml_models.R
git commit -m "Add grouped CV XGBoost benchmark"
```

A project also needs a good .gitignore. The aim is to track source code and lightweight metadata, but not confidential raw data, huge model objects, or system-specific files.

```
# .gitignore
.Rhistory
.RData
.Rproj.user/
__pycache__/
.ipynb_checkpoints/


# Large or confidential data
data/raw/
data/restricted/
outputs/models/*.rds
outputs/models/*.pkl
outputs/models/*.pt


# Generated outputs that can be rebuilt
outputs/tables/
outputs/figures/
outputs/diagnostics/
```

20.6 Reproducible computational environments

Code that runs today may fail in two years because package versions change. Reproducibility therefore requires recording the computational environment. In R, renv is a practical default: it creates project-local libraries, writes package versions to a lockfile, and can later restore the same package environment. In Python, common options include virtual environments, conda, pip requirements files, uv, or Poetry. For long-lived and multi-language projects, Docker can record system-level dependencies as well, although containers do not replace good documentation and careful version pinning.

For most PhD projects in R, start with renv. For projects using R plus Python plus system libraries, add a Dockerfile after the analysis stabilizes. For confidential data, ensure the container or repository never includes restricted raw files.

```
# R environment management with renv
install.packages("renv")
renv::init()


# Install packages as usual
install.packages(c("tidyverse", "tidymodels", "mlogit", "apollo", "targets"))


# Record exact package versions in renv.lock
renv::snapshot()


# On another machine, restore the environment
renv::restore()
```

```
# Always record session information in final outputs
sessionInfo()
```

```
# Minimal Python requirements file for an ML / choice project
pandas
numpy
scikit-learn
matplotlib
shap
biogeme
torch
jupyter
quarto
```

```
# Minimal Dockerfile sketch for a mixed R + Python project
FROM rocker/r-ver:4.5.0


RUN apt-get update && apt-get install -y \
    python3 python3-pip git libcurl4-openssl-dev libssl-dev libxml2-dev \
    && rm -rf /var/lib/apt/lists/*


WORKDIR /project
COPY renv.lock renv.lock
COPY .Rprofile .Rprofile
RUN R -e "install.packages('renv'); renv::restore()"


COPY requirements.txt requirements.txt
RUN pip3 install -r requirements.txt


COPY . .
CMD ["R", "-e", "targets::tar_make()"]
```

20.7 Data management, privacy, and FAIR principles

Environmental-economics projects often combine data from multiple sources: survey responses, experimental designs, geographical layers, weather records, administrative records, remote sensing, and possibly qualitative text. Good data management is therefore part of the research design. The FAIR principles - findable, accessible, interoperable, and reusable - are a useful benchmark for scientific data management. They do not imply that sensitive data must be public; rather, they require clarity about metadata, access conditions, formats, and reuse.

For choice modelling, the design file is part of the data. A DCE dataset is incomplete without the attribute definitions, levels, coding choices, choice-set construction, randomization blocks, opt-out/status-quo definitions, and any restrictions used in the design. For revealed-preference or GPS data, metadata should describe sampling, device rules, trip segmentation, mode detection, spatial joins, and exclusions.

When using personal data, location traces, household data, or potentially identifiable environmental justice variables, reproducibility must be balanced with privacy. A common compromise is to share synthetic data, schema, cleaning code, codebook, and aggregate diagnostics while keeping restricted raw data under controlled access.

Object	What to document	Choice-modelling example

Raw data	Source, date, licence, access restrictions, file checksum	Qualtrics export, GPS traces, land-cover raster, census table
Design metadata	Attributes, levels, alternatives, blocks, randomization, constraints	DCE design matrix and status-quo definition
Cleaning decisions	Exclusion rules, missing data treatment, recoding, joins	Dropped speed outliers; recoded protest responses; joined weather by trip date
Analysis data	Unit of analysis, row count, grouping variables, weights	One row per alternative per task; respondent ID; choice-set ID
Model outputs	Model formula, seed, package version, convergence status, tuning grid	MXL draws, class number, random parameters, XGBoost grid
Policy scenarios	Counterfactual values, support checks, uncertainty method	Increase entrance fee, reduce crowding, restore seabed habitat

20.8 Pipelines: make the paper rebuild itself

A pipeline is a machine-readable map of the analysis. It says which steps depend on which inputs. When raw data or code changes, the pipeline reruns only what is necessary. In R, the targets package is a strong option for empirical research because it treats datasets, model objects, tables, and figures as explicit targets. For a thesis or paper, this prevents the common problem of stale outputs: a table in the manuscript that was created from an older dataset or model.

The key design rule is to write functions first and scripts second. Functions live in the R/ folder; the pipeline calls them. This makes the workflow testable, modular, and easier to debug.

```
# _targets.R: simplified structure for an environmental DCE + ML project
library(targets)
library(tarchetypes)


tar_option_set(
  packages = c("tidyverse", "mlogit", "tidymodels", "ranger", "xgboost", "apollo")
)


source("R/clean_choice_data.R")
source("R/build_features.R")
source("R/estimate_mnl.R")
source("R/train_ml_models.R")
source("R/simulate_counterfactuals.R")
source("R/make_tables.R")
source("R/make_figures.R")


list(
  tar_target(raw_survey, readr::read_csv("data/raw/survey.csv"), format = "file"),
  tar_target(raw_design, readxl::read_excel("design/dce_design.xlsx"), format = "file"),
  tar_target(choice_long, clean_choice_data(raw_survey, raw_design)),
  tar_target(choice_features, build_features(choice_long)),


  tar_target(mnl_model, estimate_mnl(choice_long)),
  tar_target(ml_benchmark, train_ml_models(choice_features, group_var =
    "respondent_id")),


  tar_target(counterfactuals, simulate_counterfactuals(mnl_model, choice_long)),
  tar_target(table_models, make_model_table(mnl_model, ml_benchmark)),
  tar_target(fig_wtp, make_wtp_figure(mnl_model)),
```

```
  tar_render(paper, "reports/paper.qmd")
)
```

Pipeline discipline

Do not let the manuscript contain manually pasted results. In a reproducible workflow, tables and figures are generated from model objects. The manuscript imports them, or the pipeline renders the manuscript after generating them.

20.9 Literate programming with Quarto

Literate programming combines prose, code, and results in one document. For this textbook, Quarto is the natural default because it supports R, Python, Julia, and Observable, and can render to HTML, PDF, Word, presentations, websites, and books. For applied papers, Quarto helps you avoid the dangerous gap between analysis scripts and the manuscript.

A good Quarto workflow does not mean putting all analysis code in one paper file. Heavy computation should usually live in functions and pipelines. The paper should load already validated tables and figures, display concise code only when pedagogically useful, and report session information in an appendix.

````
---
title: "Machine Learning and Choice Modelling for Coastal Valuation"
author: "Your Name"
format:
  pdf:
    toc: true
    number-sections: true
    fig-cap-location: bottom
execute:
  echo: false
  warning: false
  message: false
bibliography: references.bib
---


```{r}
library(tidyverse)
model_table <- readr::read_csv("outputs/tables/model_table.csv")
```


# Results


Table @tbl-models reports the main econometric and machine-learning benchmarks.


```{r}
#| label: tbl-models
#| tbl-cap: "Model comparison across choice and ML specifications."
knitr::kable(model_table)
```
````

20.10 R and Python software stack

The best software stack depends on the task. The table below gives a practical orientation for PhD projects that combine environmental economics, choice modelling, and ML.

Task	R default	Python / other option	Notes
General data work	tidyverse, data.table, arrow, sf, terra	pandas, geopandas, xarray, rasterio	Use columnar formats such as Parquet for large intermediate files.
ML workflow	tidymodels, mlr3	scikit-learn, xgboost, lightgbm	Use resampling objects that respect respondents, time, or space.
Causal ML	grf, DoubleML, causalTree, econml through Python	econml, causalml, doubleml	Identification still matters; ML estimates nuisance functions or heterogeneity.
Discrete choice	mlogit, gmnl, logitr, apollo	Biogeme, PyLogit, xlogit, custom JAX/PyTorch	Apollo is especially flexible for complex R-based choice models. Biogeme is strong for Python MLE choice workflows.
Neural models	torch, keras3, mlr3torch	PyTorch, TensorFlow, JAX	Use when data volume and research question justify the flexibility.
Pipelines	targets, drake, Make	Snakemake, Make, Luigi, Prefect	Targets is often the most convenient R-native pipeline tool.
Reporting	Quarto, R Markdown	Jupyter, Quarto	Quarto is useful for mixed R/Python papers and books.
Environment	renv, Docker	venv, conda, uv, Poetry, Docker	Record package versions and system dependencies.

20.11 Validation protocols for environmental and choice data

Validation is where many ML-choice comparisons become unreliable. The systematic review literature on ML for mode choice identifies common methodological limitations, and the experimental review by Hillel shows that trip-wise sampling in travel-diary panel data can produce substantial data leakage. The lesson generalizes: when repeated observations are clustered by person, household, route, beach, region, or time period, validation must respect the cluster that defines future generalization.

The validation design should be chosen before model tuning. In a DCE, the most common split is by respondent, not by choice row. In GPS data, the split may be by person, day, or city depending on the target deployment. In spatial environmental prediction, random pixel-level splitting may be misleading because nearby pixels are highly correlated. In policy evaluation, time-based validation may be more informative than random validation.

Data setting	Bad split	Better split	Reason
Repeated DCE tasks	Random alternative rows	Respondent-level split	Rows from the same respondent share unobserved preferences.
Travel diary / GPS trips	Trip-wise split	Person-level or person-time split	Flexible ML can learn individual signatures.
Beach recreation site choices	Random site-choice rows	Respondent or trip-group split	Same person and trip context may appear repeatedly.
Satellite pixels	Random pixels	Spatial block split	Nearby pixels are spatially autocorrelated.
Energy demand time series	Random hours	Forward-chaining time split	Future prediction should not use future information.
Policy targeting	Random cross-validation only	External holdout, region holdout, or period holdout	Deployment often differs from the training sample.

```
# Respondent-level split with tidymodels
library(tidymodels)


set.seed(2026)
split <- group_initial_split(choice_wide, group = respondent_id, prop = 0.80)
train <- training(split)
test  <- testing(split)


folds <- group_vfold_cv(train, group = respondent_id, v = 5)


# Preprocessing is estimated inside resampling through a recipe/workflow.
rec <- recipe(choice ~ ., data = train) |>
  update_role(respondent_id, choice_set_id, new_role = "id") |>
  step_dummy(all_nominal_predictors()) |>
  step_normalize(all_numeric_predictors())


rf_spec <- rand_forest(trees = 500, mtry = tune(), min_n = tune()) |>
  set_engine("ranger") |>
  set_mode("classification")


wf <- workflow() |>
  add_recipe(rec) |>
  add_model(rf_spec)


res <- tune_grid(wf, resamples = folds, grid = 20,
                 metrics = metric_set(accuracy, mn_log_loss))
```

20.12 Production workflow for choice modelling

Choice models require additional workflow objects beyond ordinary ML models. At minimum, store the long-format choice data, the model specification, the estimation settings, convergence diagnostics, variance-covariance matrix, WTP formulas, and policy scenario definitions. For mixed logit, store draws, number of draws, distributional assumptions, seeds, starting values, and whether common random numbers were used. For latent class models, store class counts, starting values, label conventions, and class-membership variables.

A good model registry is one of the simplest ways to keep a PhD project under control. It prevents the common situation where many models are estimated but only the researcher remembers which one is which.

Field	Example
model_id	mxl_2026_07_16_cost_lognormal_bio_normal
purpose	Main welfare model for coastal DCE
data_version	processed_choice_v04
software	apollo 0.x / R 4.x
choice_format	long; one row per alternative per task
utility	ASC + cost + crowding + water_quality + natural_reef + interactions
random_parameters	crowding, water_quality, reef
cost_treatment	fixed negative or lognormal negative transformation
validation	respondent-level fivefold CV + holdout
welfare_outputs	WTP for quality, reef, crowding reduction; compensating variation scenarios
status	accepted main model / robustness / rejected
notes	Rejected model had positive cost tail; see diagnostics/model_id.html

```
# Simplified model registry in R
model_registry <- tibble::tribble(
  ~model_id, ~family, ~purpose, ~data_version, ~validation, ~status,
  "mnl_01", "MNL", "baseline", "processed_choice_v04", "respondent CV", "reported",
  "mxl_02", "MXL", "main welfare", "processed_choice_v04", "holdout + sign checks",
    "reported",
  "rf_01", "Random forest", "prediction benchmark", "choice_wide_v04", "grouped CV",
    "appendix",
  "xgb_01", "XGBoost", "prediction benchmark", "choice_wide_v04", "grouped CV",
    "appendix"
)
readr::write_csv(model_registry, "docs/model_registry.csv")
```

20.13 Apollo and Biogeme in a reproducible workflow

Apollo and Biogeme are particularly important for this book because they make advanced choice modelling feasible in applied research. Apollo is a flexible R package for choice model estimation and application; it allows users to write custom model functions or combine predefined components. Biogeme is an open-source Python package for maximum-likelihood estimation of parametric models with a special emphasis on discrete choice models. A reproducible workflow should treat these models as production steps, not isolated scripts.

For Apollo projects, keep model-control settings, validation inputs, probability functions, random-parameter definitions, and output processing in separate scripts. For Biogeme projects, keep database preparation, model expression, estimation, simulation, and reporting separated. In both cases, write

wrappers that return standard objects or standardized output tables, so that model comparison and reporting do not depend on manual copying.

```
# R pseudo-wrapper for Apollo models
estimate_apollo_model <- function(data, model_spec, seed = 1234) {
  set.seed(seed)


  # 1. validate input structure
  stopifnot(all(c("choice", "chid", "alt", "respondent_id") %in% names(data)))


  # 2. source specification-specific utility and probability functions
  source(model_spec$probability_file)


  # 3. estimate model
  # apollo_estimate(...)


  # 4. collect standardized outputs
  list(
    model_id = model_spec$model_id,
    coefficients = "coef_table_placeholder",
    vcov = "vcov_placeholder",
    diagnostics = "diagnostics_placeholder",
    seed = seed,
    spec = model_spec
  )
}
```

```
# Python pseudo-wrapper for Biogeme models
from pathlib import Path
import pandas as pd



def estimate_biogeme_model(data_path: str, spec_name: str, out_dir: str):
    out = Path(out_dir)
    out.mkdir(parents=True, exist_ok=True)


    df = pd.read_parquet(data_path)


    # Define database, variables, utility, and likelihood here.
    # Keep these in functions so alternative specifications are comparable.


    # results = biogeme.estimate()
    # results.get_estimated_parameters().to_csv(out / f"{spec_name}_coef.csv")


    return {
        "spec_name": spec_name,
        "data_path": data_path,
        "output_dir": str(out)
    }
```

20.14 Integrating ML benchmarks with choice models

The strongest applied papers do not present ML and choice models as enemies. They assign each model a role. A multinomial logit or mixed logit model may be the welfare model. A random forest or XGBoost model may be the predictive benchmark and diagnostic tool. A latent class model may be the

interpretable segmentation model. A neural model may be used when the research question requires flexible representation of images, text, context, or high-dimensional passive data.

The ML-choice literature uploaded for this book repeatedly supports this complementary view. The discussion-paper literature emphasizes cross-fertilization between ML and choice modelling; assisted-specification papers use ML to help find better utility structures; neural latent-class and context-aware mixed-logit papers use ML components while preserving behavioural structure; and benchmark papers warn that predictive superiority depends on validation design, data context, and research task.

Research goal	Recommended main model	Recommended ML role	Report
WTP for biodiversity	MNL, MXL, WTP-space MXL, LCCM	Specification support, robustness, nonlinear diagnostics	WTP with confidence intervals; sign and support checks
Mode-choice prediction	MNL/NL/MXL baseline	RF/XGBoost/DNN benchmark	Grouped test performance, calibration, interpretability
Preference heterogeneity	MXL, LCCM, HCM/ICLV	Clustering, embeddings, neural latent constructs	Class profiles, random-parameter distributions, external validity
Image-based choice	DCM with semantic features if available	CNN or embeddings for raw images	Predictive performance plus interpretable image-feature effects
Policy targeting	Causal design + DML/GRF	Nuisance estimation and heterogeneity discovery	Identification assumptions, CATE uncertainty, targeting rule
Large RP dataset	Scalable DCM or hybrid model	Stochastic optimization, ML benchmarks	Efficiency, validation, behavioural diagnostics

20.15 Reproducible welfare and counterfactual analysis

Welfare calculations are especially sensitive to workflow errors. A change in cost coding, scale, utility normalization, or status-quo definition can change WTP and compensating variation. Therefore, welfare functions should be scripted, tested on simple cases, and separated from model estimation.

For each policy scenario, create a scenario registry. The registry should record exactly which attributes change, whether the scenario is inside observed support, which model is used, which uncertainty method is used, and which population is being simulated. Never define scenarios only in prose.

```
# Scenario registry for a coastal DCE
scenario_registry <- tibble::tribble(
  ~scenario_id, ~description, ~changed_attributes, ~support_check, ~model_id,
  "S0", "Current access and ecosystem quality", "none", "observed", "mxl_02",
  "S1", "Improve water quality from medium to high", "water_quality", "observed",
    "mxl_02",
  "S2", "Restore natural reef and reduce crowding", "reef; crowding", "partly observed",
    "mxl_02",
  "S3", "Raise beach fee by 5 euros", "cost", "observed", "mxl_02"
)
```

```
readr::write_csv(scenario_registry, "docs/scenario_registry.csv")
```

```
# Simple WTP function with basic safety checks
compute_wtp <- function(coefs, attribute, cost_name = "cost") {
  stopifnot(attribute %in% names(coefs), cost_name %in% names(coefs))
  if (coefs[[cost_name]] >= 0) {
    warning("Cost coefficient is non-negative. WTP interpretation may be invalid.")
  }
  -coefs[[attribute]] / coefs[[cost_name]]
}

# Example
coefs <- c(cost = -0.08, biodiversity_high = 0.55, crowding_low = 0.90)
compute_wtp(coefs, "biodiversity_high")
compute_wtp(coefs, "crowding_low")
```

Counterfactual warning

Do not let a flexible ML model silently extrapolate policy effects outside the support of the data. For welfare analysis, the model must respect economic meaning: cost monotonicity, relevant substitution patterns, plausible marginal effects, and uncertainty about unobserved heterogeneity.

20.16 Testing, diagnostics, and failure logs

Empirical workflows need tests. Not every PhD project needs a formal software-engineering test suite, but every project needs automated checks for the assumptions that would invalidate the results if broken. These checks should run before model estimation and before final rendering of the paper.

Choice data are especially suitable for structural tests: each choice set should have exactly one chosen alternative; cost should be nonnegative; each respondent should appear in only one validation fold; all alternatives should have valid attributes; and status-quo alternatives should be coded consistently.

```
# Minimal data checks for long-format choice data
check_choice_data <- function(df) {
  required <- c("respondent_id", "choice_set_id", "alt", "choice", "cost")
  missing <- setdiff(required, names(df))
  if (length(missing) > 0) stop("Missing columns: ", paste(missing, collapse = ", "))

  chosen_per_set <- df |>
    dplyr::group_by(choice_set_id) |>
    dplyr::summarise(n_chosen = sum(choice), .groups = "drop")

  if (any(chosen_per_set$n_chosen != 1)) {
    stop("Each choice set must contain exactly one chosen alternative.")
  }

  if (any(df$cost < 0, na.rm = TRUE)) {
    stop("Negative costs detected. Check coding.")
  }

  TRUE
}
```

Diagnostic	Choice-model version	ML version
Signs	Cost negative; quality positive; travel time negative	Partial dependence/ALE monotonicity; constrained models if necessary
Scale	Check scale differences across SP/RP or treatments	Check calibration and log loss across groups
Heterogeneity	Random-parameter distributions; class profiles	Subgroup performance; SHAP/ALE by group
Validation	Holdout likelihood, predictive log likelihood, hit rate	Grouped CV, external test, calibration, log loss
Robustness	Alternative specifications, priors, class counts, WTP-space	Alternative algorithms, tuning ranges, seeds, feature sets
Policy use	Elasticities, WTP, welfare formulas, scenario uncertainty	Support checks, counterfactual plausibility, constrained effects

20.17 The research compendium

A research compendium is a packaged version of the project: code, documentation, metadata, and enough data or data access instructions to reproduce the analysis. For thesis work, a compendium can live in a private repository during development and be archived publicly or under controlled access after publication.

At minimum, the compendium should contain a README, data dictionary, run instructions, environment lockfile, analysis scripts or pipeline, manuscript source, and a statement about data availability. For journals, this is increasingly expected. For your own career, it is invaluable: it makes extensions, revisions, and replication easier.

File	Purpose
README.md	Explains the research question, repository structure, run order, and contact information.
CODEBOOK.md	Defines variables, alternatives, attributes, units, transformations, and missing-data codes.
DATA_AVAILABILITY.md	States which data are public, restricted, synthetic, or unavailable and why.
renv.lock / requirements.txt	Records software dependencies.
_targets.R	Defines the reproducible pipeline.
reports/paper.qmd	Contains manuscript source.
docs/model_registry.csv	Tracks model specifications, purpose, status, and outputs.
docs/scenario_registry.csv	Tracks policy counterfactual definitions.
LICENSE	Defines reuse conditions for code and documentation.
CITATION.cff	Tells others how to cite the code or compendium.

20.18 Worked mini-project: from raw DCE data to paper table

This mini-project is deliberately simple. Its purpose is to show how the pieces fit together. In a real project, each function would be longer and more carefully tested.

```
# R/clean_choice_data.R
clean_choice_data <- function(raw_survey, raw_design) {
  raw_survey |>
    dplyr::left_join(raw_design, by = c("block", "task", "alt")) |>
    dplyr::mutate(
      choice = as.integer(chosen_alt == alt),
      respondent_id = as.character(respondent_id),
      choice_set_id = paste(respondent_id, task, sep = "_"),
      cost = as.numeric(cost)
    ) |>
    dplyr::select(respondent_id, choice_set_id, task, alt, choice,
                  cost, biodiversity, crowding, water_quality, income, age)
}
```

```
# R/estimate_mnl.R
estimate_mnl <- function(choice_long) {
  library(mlogit)
  mdata <- mlogit.data(choice_long,
                       choice = "choice",
                       shape = "long",
                       chid.var = "choice_set_id",
                       alt.var = "alt",
                       id.var = "respondent_id")

  mlogit(choice ~ cost + biodiversity + crowding + water_quality | 0,
         data = mdata)
}
```

```
# R/make_tables.R
make_model_table <- function(mnl_model, ml_benchmark) {
  coef_table <- broom::tidy(mnl_model) |>
    dplyr::mutate(model = "MNL baseline")

  ml_table <- ml_benchmark$metrics |>
    dplyr::mutate(model = "ML benchmark")

  list(coefficients = coef_table, ml_metrics = ml_table)
}
```

20.19 Common failure modes and how to prevent them

Failure mode	Why it matters	Prevention
Manual spreadsheet editing	Impossible to audit; can change values silently	Write cleaning scripts; keep raw files read-only
Random row split in panel choice data	Inflates ML performance through leakage	Split by respondent, household, route, region, or time as appropriate
Global preprocessing before CV	Assessment fold influences training transformations	Use recipes/workflows or pipelines fitted inside folds
Untracked model versions	Tables cannot be linked to model	Maintain model registry and saved

	specifications	output metadata
Unclear DCE design metadata	Cannot reconstruct attributes or alternatives	Archive design matrix, survey instrument, and attribute dictionary
Positive cost coefficient ignored	Invalid WTP and welfare interpretation	Run sign checks; consider constrained or WTP-space models
Only accuracy reported	Poor probability calibration and welfare outputs hidden	Report log loss, calibration, likelihood, elasticities, WTP, and subgroup performance
No seed or stochastic settings	Model results vary without explanation	Set seeds; record draws, folds, starting values, tuning grids
Large files in Git	Repository becomes unusable or leaks confidential data	Use data access instructions, Git LFS, archives, or external storage
Paper not rebuilt from code	Final numbers may be stale	Use Quarto and targets to render outputs from current objects

20.20 Blueprint for your dissertation paper

The following blueprint can be copied into a PhD project and adapted. It is designed for an environmental DCE paper with ML benchmarks, but the logic extends to transport, energy, conservation, land use, health, and other applied-choice settings.

Week / phase	**Main tasks**	**Deliverable**
Phase 1: research design	Define estimand, target population, behavioural model, policy scenarios, and validation target	Design memo and scenario registry draft
Phase 2: data structure	Create repository, import raw data, archive design files, define codebook	README, codebook, raw-data checks
Phase 3: cleaning	Write cleaning functions, build long and wide datasets, check choice-set integrity	Processed data and exclusion log
Phase 4: baselines	Estimate MNL and simple ML baselines with grouped validation	Baseline table and validation report
Phase 5: advanced models	Estimate MXL/LCCM/HCM and tuned ML models	Model registry and diagnostics
Phase 6: interpretation	Compute WTP, elasticities, class profiles, feature effects, calibration	Tables, figures, and behavioural notes
Phase 7: counterfactuals	Run policy scenarios and uncertainty analysis	Counterfactual report
Phase 8: paper	Render Quarto manuscript and appendix from pipeline	PDF, replication package, archive

20.21 Final reproducibility checklist

- Raw data are stored separately and never manually edited.

- Every exclusion rule is scripted and logged.
- The unit of analysis and grouping variables are documented.
- Choice-set integrity checks pass: one chosen alternative per task, valid alternatives, valid costs.
- Train/test and cross-validation splits respect respondent, time, spatial, or panel structure.
- Preprocessing is estimated inside resampling, not before resampling.
- All random seeds, simulation draws, and starting values are recorded.
- The baseline choice model, advanced choice model, and ML benchmark each have a defined purpose.
- Model comparison reports task-relevant metrics, not only accuracy.
- WTP and welfare functions are scripted and tested on simple examples.
- Policy scenarios are stored in a scenario registry.
- All tables and figures are generated from code.
- The manuscript is rendered from source, preferably through a pipeline.
- Package versions are recorded with renv, requirements.txt, conda environment, or Docker.
- A README explains how to rerun the project.
- Data availability, licences, and privacy restrictions are explicitly documented.

20.22 Exercises

Exercise 1: repository design
Create a repository structure for your current paper. Include data, design, R, Python, outputs, reports, and docs folders. Write a README that explains how to rebuild the project.

Exercise 2: leakage audit
Take one previous ML analysis and identify every point where leakage could occur: splitting, imputation, scaling, feature selection, hyperparameter tuning, and repeated observations.

Exercise 3: model registry
Create a model_registry.csv for at least six models: MNL baseline, MXL, latent class, random forest, XGBoost, and one neural or hybrid model. Define each model purpose.

Exercise 4: scenario registry
Write a scenario_registry.csv for an environmental policy intervention. Include status quo, at least three policy scenarios, support checks, and the welfare model used.

Exercise 5: Quarto replication
Create a Quarto document that imports one model table and one figure produced by scripts. Render it to PDF without manually copying results.

Exercise 6: pipeline extension
Write a minimal _targets.R pipeline that imports data, cleans it, estimates one model, creates one table, and renders one report.

Exercise 7: reproducibility appendix
Draft a one-page reproducibility appendix for your PhD paper. It should explain data access, software, validation, seeds, model objects, and where replication materials are stored.

20.23 Chapter summary

This chapter closes the book by turning methods into a research system. Machine learning, AI, causal inference, and advanced choice modelling can only improve environmental economics if the workflow is rigorous enough to support trust. A reproducible workflow makes the analysis rerunnable, inspectable, and extensible. It also clarifies the role of each model: prediction, interpretation, heterogeneity discovery, welfare estimation, or counterfactual policy analysis.

The main message is practical: build your project so that the paper can rebuild itself. Keep raw data immutable, document design metadata, use Git, record environments, validate without leakage, automate model pipelines, store model and scenario registries, and render the manuscript from code. This is not extra work at the end. It is the infrastructure that makes advanced empirical research possible.

Final takeaway

The frontier is not a single algorithm. It is a disciplined empirical workflow where ML, AI, econometrics, environmental valuation, and choice theory each play the role they are best suited to play.

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